



Affirmed Networks Gx Interface Specification

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Inside ...

- Interface Overview
- Message Flows
- Configuring
- Monitoring
- Failure Handling
- Message Definitions
- AVPs

Affirmed Networks Inc.

35 Nagog Park
Acton, MA 01720
USA

Tel: +1-978-268-0800

www.affirmednetworks.com

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Preface

This specification contains information on the Gx interface used for policy and charging management.

Note: *The Affirmed Networks Mobile Content Cloud™ is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide) working with the Gx interface. The audience should have basic knowledge of Policy and Charging Rules Function (PCRF), Offline Charging Server (OFCS), Online Charging Server (OCS), and GTP' protocol. The audience should also be familiar with 3G or 4G networks.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

Read the following chapters and appendices in the order in which they appear:

Chapter/Appendix	Describes
Chapter 1, Gx Interface	The message flows, message definitions, failure handling, and result codes used to control and monitor the Gx interface.
Chapter A, Diameter AVP Descriptions	The attribute value pair definitions for the Gx interface.

References and Specifications

The system supports the following references and specifications for the Gx interface:

- 3GPP TS 23.003 V12.0.0 (2013-09): *Technical Specification Group Core Network and Terminals; Numbering, addressing and identification*
- 3GPP TS 29.212 V10.0.0 (2010-09): *Technical Specification Group Core Network and Terminals; Policy and Charging Control (PCC); Reference points*
- 3GPP TS 29.212 V11.7.0 (2012-12): *Technical Specification Group Core Network and Terminals; Policy and Charging Control (PCC); Reference points*
- 3GPP TS 29.212 V12.7.0 (2014-12): *Technical Specification Group Core Network and Terminals; Policy and Charging Control (PCC); Reference points*
- 3GPP TS 29.212 V15.1.0 (2017-12): *Technical Specification Group Core Network and Terminals; Policy and Charging Control (PCC); Reference points*
- 3GPP TS 29.212 V15.3.0 (2018-06): *Technical Specification Group Core Network and Terminals; Policy and Charging Control (PCC); Reference points*
- 3GPP TS 29.214 V10.0.0 (2010-09): *Technical Specification Group Core Network and Terminals; Policy and Charging Control over Rx reference point*
- 3GPP TS 29.214 V15.1.0 (2017-09): *Technical Specification Group Core Network and Terminals; Policy and Charging Control over Rx reference point*
- 3GPP TS 29.229 V10.0.0 (2011-03): *Cx and Dx interfaces based on the Diameter protocol; Protocol detail*
- 3GPP TS 32.299 V10.7.0 (2012-09): *Telecommunication management; Charging management; Diameter charging application*
- RFC 3588: *Diameter Based Protocol*
- RFC 3539: *Authentication, Authorization and Accounting (AAA) Transport Profile*

- RFC 4006: *Diameter Credit-Control Application*
- RFC 4282: *The Network Access Identifier*
- RFC 6733: *Diameter Base Protocol*

Features in this Release

This release includes the following new features. Note the following:

- For a complete list of the features supported in this release, see the *Software Release Notes*.
- For a list of additional Affirmed Networks documentation, see the *Affirmed Networks Documentation Library*.
- For a comprehensive list of terms and acronyms used in Affirmed Networks documentation, see *Affirmed Networks Terms and Acronyms*.

Feature	Description
Added support for the TMO-Virtual-Gi-ID VSA on RADIUS, Gx, and Gy Interfaces See TMO-Virtual-Gi-ID (page A-128) and the <i>Affirmed Networks RADIUS Interface Specification</i> .	This feature adds support for the TMO-Virtual-Gi-ID vendor-specific attribute (VSA) on RADIUS, Gx, and Gy interfaces. This feature also adds two new configuration parameters under the APN object to specify whether to process the TMO-Virtual-Gi-ID VSA when received in the Access-Accept message from the RADIUS server. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> vsa-virtual-gi-id-enabled [true false] vsa-virtual-gi-id <default-value></pre> ■ vsa-virtual-gi-id-enabled – true – If the MCC gateway receives the TMO-Virtual-Gi-ID VSA in the Access-Accept message, it sends the VSA in Accounting-Request messages (Start/Interim/Stop) on the RADIUS interface and in CCR-I messages on the Gx and Gy interfaces. – false – (default) If the MCC gateway receives the TMO-Virtual-Gi-ID VSA in the Access-Accept message, it does not send the VSA in Accounting-Request messages on the RADIUS interface or in CCR-I messages on the Gx and Gy interfaces. ■ vsa-virtual-gi-id – Specifies a default value for the TMO-Virtual-Gi-ID VSA. If the vsa-virtual-gi-id-enabled parameter is set to true but the MCC gateway does not receive the TMO-Virtual-Gi-ID VSA in the RADIUS Access-Accept message, the MCC gateway uses this default value to fill the TMO-Virtual-Gi-ID VSA sent in RADIUS Accounting-Request messages (Start/Interim/Stop) and in CCR-I messages on the Gx and Gy interfaces. This parameter is only visible in the CLI when the vsa-virtual-gi-id-enabled parameter is set to true.

Feature	Description
Added support for the TMO-Virtual-Gi-ID VSA on RADIUS, Gx, and Gy Interfaces <i>Continued</i> See TMO-Virtual-Gi-ID (page A-128) and the <i>Affirmed Networks RADIUS Interface Specification</i>.	You can configure whether to include or exclude the TMO-Virtual-Gi-ID VSA in RADIUS Access-Accept and Accounting-Request (Start/Stop/Interim) messages and Gx/Gy CCR-I messages. <pre>an3000-mcm-slot7cpu1 (config) # service-construct data-record-template pgw auth radius access-accept <template-name> tmo_virtual_gi_id presentFlag [conditional no yes] an3000-mcm-slot7cpu1 (config) # service-construct data-record-template pdngw accounting radius [start interim stop] <template-name> tmo_virtual_gi_id presentFlag [conditional no yes] an3000-mcm-slot7cpu1 (config) # service-construct data-record-template pgw [pcrf ocs] diameter start <template-name> tmovirtualgiid presentFlag [yes no conditional]</pre>
Added support for dynamic rating groups (quota-id not pre-configured on the MCC) and the provisioning of rating sub-groups and a new parameter to configure a dynamic rating group template See Configuring Rating Sub-Groups (page 1-111) and Configuring a Dynamic Rating Group Template (page 1-124).	Added the ability to dynamically allocate a rating group when the quota-id received in the Charging-Rule-Definition AVP is not pre-configured on the MCC. When the PCRF dynamically allocates a rating group, the MCC uses the Online and Offline AVPs received in the Charging-Rule-Definition AVP to enable the corresponding charging method for the rating group/service rule. This feature also adds the ability to configure a list of separate rating sub-groups under the same rating group (quota-id), provided the <code>charging-method</code> for the rating group is set to <code>subgroup</code>. A rating sub-group can have any of the following charging methods: ■ Only online charging ■ Only offline charging ■ Both offline and online charging ■ No charging This feature also adds a new configuration parameter to specify a rating group template to use when a dynamic rating group is created because of a charging rule install from the PCRF. <pre>an3000-mcm-slot7cpu1 (config) # workflow data-profile <data-profile-name> dynamic-rg-template <rating-group-name></pre> dynamic-rg-template – Specifies the name of a rating group to be associated with the data profile. The rating group must already be configured and will be used as a template when a dynamic rating group is created because of a charging rule install from the PCRF. The associated rating group must be configured with a <code>charging-method</code> of <code>subgroup</code>.

Feature	Description
Enhanced the provisioning of the default charging method See Configuring the Default Charging Method (page 1-112).	This feature enhances the provisioning of the default charging method for a session. When the Online/Offline AVPs are included in the CCR-Initial (CCR-I) message at the command level, these AVPs indicate the default charging method for the session based on the configuration at the PCEF. This feature also adds support for the Online and Offline AVPs at the RAR command-level on the Gx interface. Currently, the <code>user-subscription-charging-type</code> parameter configured under the control-profile object is used to advertise the default charging method (online/offline) supported on the MCC. With this feature, the default charging method can also be configured under the APN object and/or the crbn-profile object and used to advertise the session-wide default charging method in the Online/Offline AVPs in the CCR-I message at the command level. However, the default charging method provided by the PCRF in the CCA-Initial (CCA-I) at the command level takes precedence over any pre-configured default charging method on the MCC gateway (PCEF). The MCC also now supports a PCRF-initiated mid-session change to the session-wide default charging method by including the Online/Offline AVPs in the RAR message at the command level.
Added a new configuration parameter to control the charging method for a service rule based on both the session-wide default charging method and the service-rule default charging method. See Configuring the Default Charging Method (page 1-112) and the <i>Affirmed Networks Gy Interface Specification</i>.	This feature adds a new configuration parameter under the gateway-common profile to control the charging method for a service rule based on both the session-wide default charging method and the service-rule default charging method. <pre>an3000-mcm-slot7cpu1(config)# zone default gateway profile gateway-common <name> use-session-wide-default-charging-state-as-gating [enable disabled]</pre> ■ enabled – A rule will be charged using the online/offline charging interface if the Online/Offline AVP is enabled at both the session level and the service-rule level. Otherwise, the rule is not charged. ■ disabled – (default) A rule will be charged using the online/offline charging interface if the Online/Offline AVP is enabled at the service-rule level. This feature also adds a new configuration parameter to specify whether multiple Service-Identifiers can be included within the Multiple-Services-Credit-Control AVP on the Gy interface. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface online-charging <interface-name> avp-customization mscc-serviceid num-instances [multiple single]</pre>

Feature	Description
Added ability to select a new control profile and online/offline interface using the CRBN from the PCRF See Using the CRBN AVP to Select a New Control Profile (page 1-68).	This feature adds the ability to select a new control profile and online/offline interface using the Charging-Rule-Base-Name (CRBN) AVP received from the PCRF. When the MCC gateway receives a CCA-I message from the PCRF containing the CRBN AVP, the MCC gateway can use the CRBN as a subscriber analyzer key to select a new subscriber analyzer. In doing so, the subscriber sessions use the control profile defined for the new subscriber analyzer. When a new control profile is selected using this method, the new control profile can only be used to make the following online/offline interface changes: ■ Online – Switch from a control profile with no online interface to a control profile with an online interface. ■ Offline – Switch to a control profile with a different offline interface. Only the following parameters can differ between the old offline interface and the new offline interface. – data-volume-limit – interim-update-interval – cdr-close-time
Added new configuration parameters to control whether the Destination-Host and Destination-Realm AVPs are updated based on information received in the CCA-I/CCA-U/RAR messages See Sending Policy and Charging Records to the PCRF (page 1-42).	With this feature, you can control whether the Destination-Host/Destination-Realm AVPs are updated in CCR-U/CCR-T messages on the Gx/Gy interfaces based on information received in the CCA-I/CCA-U/RAR messages. The default is enabled for the CCA-I/CCA-U messages and disabled for the RAR message. <pre>an3000-mcc-slot7cpu1(config)# service-construct interface [pcrf-charging online-charging] <interface-name> allow-dest-host-realm-change [cca-i cca-u rar] [enabled disabled]</pre> ■ enabled – If the received Origin Host/Origin Realm in the CCA-I/CCA-U/RAR is different from the current destination host and realm, the MCC uses the Origin-Host/Origin-Realm AVPs received in the CCA-I/CCA-U/RAR message to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages. ■ disabled – The MCC gateway continues to use the existing values to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages and does not update the Destination-Host/Destination-Realm AVPs with the values received in the CCA-I/CCA-U/RAR messages. Note: The MCC gateway always sends the RAA message with the Destination Host and Realm values of the Origin Host and Realm received in the RAR message. Note: For the CCA-I/CCA-U, the MCC will only update the Destination-Host and Destination-Realm in CCR-U/CCR-T messages if the result code received in the CCA-I/CCA-U is DIAMETER_SUCCESS (2001).

Feature	Description
Added support for overriding the static configuration of the destination-host and destination-realm on the Gx interface See Overriding the Destination-Host and Destination-Realm Static Configuration (page 1-114).	The Diameter node on the MCC gateway communicates with statically configured Diameter peers via the Gx interface. When configuring a Diameter peer, you specify the peer host name and realm, which are used in the Destination-Host and Destination-Realm AVPs in Diameter request messages by default. You can now override these default settings on the Gx interface by configuring the <code>override-dest-hostname</code> and <code>override-destination-realm</code> parameters. When these override parameters are configured, the MCC gateway sends the CCR-Initial (CCR-I) to the selected Diameter peer but includes the configured override destination host/realm in the Destination-Host and Destination-Realm AVPs. The destination host/realm override configuration only applies to the Destination-Host and Destination-Realm AVPs in CCR-I messages. In CCR-U/CCR-T/ASA/RAA messages, the MCC gateway uses the values from the Origin-Host/Origin-Realm AVPs received in the CCA-Initial (CCA-I) message to fill the Destination-Host/Destination-Realm AVPs. If the override hostname is configured and if the CCA-I response does not have a different Origin-Host, the MCC gateway continues using the override hostname. Use the following CLI command to override the static configuration for the Destination-Host AVP sent in CCR-I messages on the Gx interface: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface aaa-interface-group diameter node-admin <node-name> peer <peer-host-name> override-dest-hostname <destination-host></pre> Use the following CLI command to override the static configuration for the Destination-Realm AVP sent in CCR-I messages on the Gx interface: <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> override-destination-realm <destination-realm></pre>
Added the ability for the MCC to establish/continue a session when the PCRF returns at least one valid Charging-Rule-Base-Name AVP in the CCA-I/CCA-U/RAR See PCC Rules (page 1-5).	Previously, if the PCRF returned multiple Charging-Rule-Base-Name AVPs and only one was predefined on the MCC, the MCC would terminate the session. With this feature, the MCC will establish/continue a session as long as the PCRF returns at least one predefined Charging-Rule-Base-Name AVP in the CCA-I/CCA-U/RAR message. Note: <i>Applicable only when the <code>crbn-mode</code> parameter is set to <code>data-profile</code>.</i>
Added support for modifying QCI/ARP values of the EPS dedicated bearer if the PCRF uses the Charging-Rule-Install AVP to modify QCI/ARP values See QCI/ARP Modifications of EPS Dedicated Bearer (page 1-18).	Each unique QoS Class Identifiers/Allocation and Retention Priority (QCI/ARP) combination identifies a bearer on the PGW. This feature adds support for the following two scenarios if the PCRF uses the Charging-Rule-Install AVP to modify QCI/ARP values. ■ If there are other charging rules on the existing dedicated bearer, the PGW keeps the existing dedicated bearer but removes the particular charging rule for which QCI/ARP values have been modified. The PGW creates a new dedicated bearer with the new QCI/ARP values to carry the traffic for this charging rule. ■ If there is already an existing dedicated bearer with the same QCI/ARP values as those in the charging rules being modified, the PGW moves those modified charging rules to the existing bearer and deletes the old dedicated bearer if no charging rules are left on that bearer.

Feature	Description
Added support for processing the Gx-TMO-Redirect-Server AVP when received from the PCRF in CCA-I/U and RAR messages See Processing the Gx-TMO-Redirect-Server VSA (page 1-115).	This feature adds support for the Gx-TMO-Redirect-Server AVP when received in the Charging-Rule-Definition AVP from the PCRF in the CCA-Initial (CCA-I), CCA-Update (CCA-U), or RAR message. When the PGW receives the Gx-TMO-Redirect-Server AVP within the Charging-Rule-Definition AVP from the PCRF, the MCC will perform a redirection for dynamic or externally-activated predefined rules. The redirection is only valid for HTTP URLs. This feature applies to online, offline, and metering rating groups. The Gx-TMO-Redirect-Server AVP is configurable in Gx CCA data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the field is not processed when received in the CCA response. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template pgw pcrf diameter cca <template-name> tmoredirectserver presentFlag [yes no]</pre> This feature also adds a new configuration parameter under the advice-of-charge object to specify whether to include the MS IP address in the redirect URL. <pre>an3000-mcm-slot7cpu1(config)# service-construct advice-of-charge <name> http include-msip [true false]</pre>
Added support for a predefined service rule to be activated under a dynamic rating group and for a predefined service rule under a rating sub-group to be moved to a dynamic rating group based on PCRF messaging. See PCC Rules (page 1-5).	This feature allows for a predefined service rule to be activated under a dynamic rating group. For example, if the quota-id received in the Charging-Rule-Definition AVP is not pre-configured on the MCC, the MCC will dynamically allocate a rating group with the received quota-id. If the Charging-Rule-Definition AVP also contains the name of a predefined service rule, the MCC will activate the predefined service rule under the dynamic rating group. This feature also allows for a predefined service rule under a rating sub-group to be moved to a dynamic rating group based on PCRF messaging. Note: Ensure the dynamic rating group's quota ID is not configured on the MCC.

Feature	Description
Added ability to provide variable QoS for traffic based on information retrieved from a manifest server and traffic detection using ADC over the Gx interface See Retrieving Flow Information From a Manifest Server (page 1-116).	This feature adds the ability to provide variable QoS for traffic based on information retrieved from a manifest server and traffic detection using ADC over the Gx interface. This feature adds support for the acwsponsorurl vendor-specific attribute (VSA) on the Gx interface. If the PCRF sends the acwsponsorurl VSA along with the Charging-Rule-Definition, TDF-Application-Identifier and Event-Trigger APPLICATION_START and APPLICATION_STOP in the CCA/RAR message, the MCC gateway will retrieve flow information from a manifest server and deliver this information to the PCRF via a CCR-U message. This feature adds a new parameter to configure a manifest interface profile. <pre>an3000-mcm-slot7cpu1(config)# service-construct manifest-interface-profile <manifest-interface-profile-name> admin-state <enabled/disabled> certificate-server ca-certificate-list <ca-certificate-list name> certificate-server certificate-url <cert server url/ip> certificate-server certificate-file-extension <1 to 255 char string> certificate-server network-context <network-context> certificate-server loopback-ip <local IP loopback> certificate-server crl-url <crl server url/ip> manifest manifest-max-size <5 to 512000> manifest manifest-max-size-unit <bytes/kbytes> certificate-verification admin-state <enabled/disabled></pre> The new manifest interface profile must be associated with a Workflow Control Profile. <pre>an3000-mcm-slot7cpu1(config)# workflow control-profile <control-profile-name> manifest-interface-profile <manifest-interface-profile-name></pre> This feature adds support for processing the Custom-Mute VSA received in CCA/RAR messages on the Gx interface. The Custom-Mute VSA is used to limit APPLICATION_START/STOP notifications sent to the PCRF. This feature also adds support for sending the Sponsored-Connectivity-Data AVP in the CCR-U message on the Gx interface. If the manifest response comes with a validity time, the MCC includes the validity time in the CC-Time AVP within the sponsored connectivity data and starts the validity timer for the manifest refresh.

Feature	Description
Added ability to provide variable QoS for traffic based on information retrieved from a manifest server and traffic detection using ADC over the Gx interface <i>Continued</i> See Retrieving Flow Information From a Manifest Server (page 1-116).	Use the following command to configure the Sponsored-Connectivity-Data AVP in CCR-U data record templates. The default <code>presentFlag</code> setting is <code>no</code>, indicating the AVP is not sent in CCR-U messages. <pre>an3000-mcm-slot7cpu1(config)# service-construct data-record-template [pgw ggsn] pcrf diameter interim <template-name> sponsoredconnectivitydata presentFlag [yes no conditional]</pre> <i>Note:</i> Ensure that the Application-Detection-Information AVP (<code>applicationdetectioninfo</code>) is set to <code>yes</code> in the Gx CCR-U data record templates along with the Sponsored-Connectivity-Data AVP (<code>sponsoredconnectivitydata</code>) for this feature. Use the following CLI command to view the number of manifest requests sent to the MFI agent and the number of responses received at the cluster/grid/node level. <pre>an3000-mcm-slot7cpu1# show zone default gateway statistics cluster-level call-performance pgw include manifest</pre> Use the following commands to view the number of manifest retrieval statistics. <pre>an3000-mcm-slot7cpu1# show service-construct manifest-interface-profile statistics cluster-level [summary detail] <cluster-id></pre> To view these statistics in the Acuitas Service Management System, see the <i>Acuitas Performance Statistics</i>.
Added ability to override the charging-characteristics (CC) value received in the Create Session Request message for a subscriber based on the subscriber type See Overriding the CC Value Received in the Create Session Request (page 1-123).	This feature adds the ability to override the charging-characteristics (CC) value received in the Create Session Request message for a subscriber based on the subscriber type. Set parameters using the following commands. 1 Configure a charging-characteristics profile with a configured CC value for each subscriber type. <pre>an3000-mcm-slot7cpu1(config)# service-construct charging-characteristics-profile <cc-profile-name> [home-subscriber-cc-value roaming-subscriber-cc-value visiting-subscriber-cc-value] <value></pre> 2 Associate this charging-characteristics profile with a Charging-Rule-Base-Name (CRBN) profile, APN and/or gateway service. <pre>an3000-mcm-slot7cpu1(config)# services crbn-profile <crbn-profile-name> charging-characteristics-profile <cc-profile-name></pre> <pre>an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name> charging-characteristics-profile <cc-profile-name></pre> <pre>an3000-mcm-slot7cpu1(config)# zone default gateway pgw <pgw-service-name> charging-characteristics-profile <cc-profile-name></pre>
Increase the number of supported service-rules	Increased the number of supported service-rules from 4096 to 16000. Increased the dynamic-pcc-rule range from 0x80000000 to 0xFFFFFFFF.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management .
Variable parameters for which you supply values	<i><courier bold italic blue font></i>	configure_dr convert slave --masterhost <i><host IP></i>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.		Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<i><host name></i>
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Mobile Content Cloud CLI Reference Guide</i> . <i>Workflow</i> is an application running on the system.



ESD: Static sensitive equipment. Wear a properly grounded wrist strap.



CAUTION: Proceed carefully to avoid possible equipment damage or data loss.



WARNING: Proceed carefully to avoid possible personal injury.

Note: Additional information that may apply to the subject text.

Tip: Helpful suggestions.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the latest version of Adobe Reader[®]. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server.

See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:



E-mail: prodsupport@affirmednetworks.com



24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838

Gx Interface

This section contains the following information used to control and monitor the Gx interface:

- [Interface Overview \(page 1-2\)](#)
- [Message Flows \(page 1-19\)](#)
- [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#)
- [Optional Gx/Gxa Interface Configuration Tasks \(page 1-58\)](#)
- [Monitoring the Gx Interface \(page 1-125\)](#)
- [Failure Handling \(page 1-129\)](#)
- [Message Definitions \(page 1-167\)](#)
- [Result Codes \(page 1-194\)](#)

See [Appendix A, Diameter AVP Descriptions](#) for a description of the Attribute Value Pairs (AVPs) supported on the Gx interface.

Interface Overview

Gx Interface

The Gx interface, defined in 3GPP TS 29.212, is the policy and charging control interface between the Policy and Charging Enforcement Function (PCEF) and the Policy and Charging Rules Function (PCRF).

A PCRF is a server that uses a set of operator-defined policies to create Policy and Charging Control (PCC) rules for transporting subscriber traffic. PCC rules define the network resource allocation, quality-of-service (QoS) parameters, and charging scheme to apply to a service data flow. Charging control supports both offline (post-paid services) and online (pre-paid services and its authorization) charging. The PCRF also controls the applied measurement method (data volume, duration, combined or event based).

The PCEF, located in the Gateway GPRS Support Node/Packet Data Network Gateway (GGSN/PGW), is responsible for enforcing the PCC rules received from the PCRF and provides the PCRF with subscriber and network information. The Gx interface uses the Diameter protocol.

The Mobile Content Cloud (MCC) GGSN/PGW can establish a subscriber billing profile to track an individual subscriber's usage by obtaining monthly historical usage from the operator's PCRF and combining that usage with real-time usage created within the current subscriber session. The MCC GGSN/PGW interacts with the PCRF to obtain the subscriber's policy and charging subscription characteristics. Using 3G/4G LTE capabilities, the system manages usage subscription on an individual subscriber or a small group of subscribers using policy and charging functions.

The Gx interface is the online policy interface between the GGSN and the *Charging Rules Function* (CRF). It is used to provision service data flow based on charging rules.

The MCC GGSN/PGW supports the following PCRF functions:

- Charging methods configuration (online or offline charging)
- Predefined Charging Rule (Service Data Flow) activation and deactivation
- Authorized QoS
- Session level usage monitoring and reauthorization
- Event trigger reporting
- Network-initiated QoS change
- Network-initiated session termination from PCRF

- PCRF-installed PCC rules that may trigger dedicated bearer creation, modification and deletion

Gxa Interface

This document also contains information on the Gxa information between the Bearer Binding and Event Reporting Function (BBERF) and the PCRF. The BBERF, located in the MCC Trusted WLAN Access Gateway/Proxy (TWAG/TWAP), is responsible for enforcing the PCC rules received from the PCRF and provides the PCRF with subscriber and network information. The Gxa interface uses the Diameter protocol.

The Gxa interface functionality is based on the Gx interface. Therefore, the configuration model for the Gxa interface follows the model used by the Gx interface. However, the Gxa interface only supports a subset of Gx interface features.

The TWAG/TWAP supports the following PCRF functions:

- Subscriber authentication by PCRF
- Time-based metering
- QoS installation by PCRF
- Charging methods configuration (offline charging)
- Predefined charging rule activation and deactivation
- Session level usage monitoring and reauthorization
- Event trigger reporting
- PCRF-installed PCC rules

Gx Interface Support for CUPS

The MCC Control and User Plane Separation (CUPS) architecture supports the Gx interface, including predefined and dynamic Policy and Charging Control (PCC) rules and usage monitoring. The Gx interface is configured between the PGW-C and the PCRF. There is no change to the Gx interface configuration.

The PGW-C sends PCC rules and event triggers over PFCP to the PGW-U via the Sx interface using vendor-specific information elements (IEs). The PGW-U reports traffic usage to the PGW-C via the Sx interface. The PGW-C then reports the usage information to the PCRF via the Gx interface. See the *Affirmed Networks Gateway Administration Guide* for more information on the MCC CUPS architecture.

Note: *The user-plane cluster configuration must match the configuration for the control-plane cluster, including data profiles, charging-instances, billing-plans, rating-groups, service-rules, and packet-filters.*

PCC Rules

The system supports static, predefined and dynamic PCC rules.

- [Static PCC Rules](#)
- [Predefined PCC Rules](#)
- [Dynamic PCC Rules](#)

Static PCC Rules

Static PCC rules are pre-configured on the MCC and are always active for the duration of the session (service rules with `always-on` activation). The PCRF does not need to be aware of these rules.

Predefined PCC Rules

Predefined PCC rules are pre-configured on the MCC and can be activated or deactivated by the PCRF at any time (service rules with `external-activation`). You can group predefined PCC rules within the system to allow the PCRF to dynamically activate a set of PCC rules over the Gx/Gxa interface.

Charging-Rule-Install AVP – The PCRF uses the Charging-Rule-Install AVP to activate predefined PCC rules on the PCEF. The following AVPs are included within the Charging-Rule-Install AVP.

- *Charging-Rule-Name AVP* – This AVP uniquely identifies a single PCC rule within the PCEF and corresponds to the locally configured service rule name on the MCC gateway (`service-construct` → `service-rule`).
To activate multiple predefined service rules using a single Charging-Rule-Name AVP, see [Activating/Deactivating Multiple Predefined Service Rules with a Single Charging-Rule-Name \(page 1-82\)](#).
- *Charging-Rule-Base-Name AVP* – This AVP indicates the name of a predefined group of PCC rules residing at the PCEF. The MCC supports the receipt of a Charging-Rule-Install AVP containing a single Charging-Rule-Base-Name AVP (CRBN AVP). If an MCC gateway receives a Credit-Control-Answer (CCA) or Re-Auth-Request (RAR) message with the Charging-Rule-Install AVP containing more than one Charging-Rule-Base-Name AVP, the system will establish/continue a session as long as one of the received CRBNs is predefined on the MCC. The value of the Charging-Rule-Base-Name AVP corresponds to the locally configured data profile on the MCC gateway (`workflow` → `data-profile`).
To activate multiple predefined groups of PCC rules on an MCC gateway using the Charging-Rule-Base-Name AVP, see [Activating/Deactivating Multiple Predefined Groups of PCC Rules \(page 1-77\)](#).

- *Charging-Rule-Definition AVP* – When included within the Charging-Rule-Install AVP, this AVP can also be used to activate a predefined PCC rule that is pre-configured at the PCEF. The Charging-Rule-Definition AVP within the Charging-Rule-Install AVP specifies the details of each rule as follows:

- *Charging-Rule-Name AVP* – When the Charging-Rule-Definition AVP is used to activate a predefined PCC rule, MCC tries to match the PCC rule name, sent in the Charging-Rule-Name AVP, to a service rule name configured locally on the MCC gateway (`service-construct` → `service-rule`). If a match is found, the MCC activates the matching PCC rule as a predefined PCC rule.

To activate multiple predefined service rules using a single Charging-Rule-Name AVP, see [Activating/Deactivating Multiple Predefined Service Rules with a Single Charging-Rule-Name](#) (page 1-82)

- *Rating-Group AVP* – The Rating-Group AVP corresponds to the predefined quota-id CLI parameter on the MCC (`services` → `charging` → `rating-group` → `quota-id`). If the PCRF specifies a quota-id in the Rating-Group AVP, the rating group on the MCC with the matching quota-id should have the predefined service rule configured.

If the PCRF does not specify a quota-id in the Rating-Group AVP, the MCC uses the rating group with which the predefined service rule is associated.

If the PCRF specifies a quota-id that is not configured under the subscriber's billing plan, the MCC creates a dynamic rating group and associates the service rule with the dynamic rating group.

- *Monitoring-Key AVP* – The Monitoring-Key AVP sent from the PCRF as part of the Charging-Rule-Definition AVP applies to predefined PCC rules. See [Usage Monitoring and Reporting](#) (page 1-10) for more information.
- *Redirect-Information AVP* – The Redirect-Information AVP sent from the PCRF as part of the Charging-Rule-Definition AVP applies to predefined PCC rules only and is applicable to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected.
- *Service-Identifier AVP* – Contains the identifier of a service. If the MCC gateway receives a CCA message from the PCRF with the Service-Identifier AVP under the Charging-Rule-Definition AVP, the value in the Service-Identifier AVP overrides the Service-ID value (`service-construct` → `service-rule` → `service-data-flow-id`) provisioned on the gateway assuming the Charging-Rule-Name (`service-rule`) is valid.
- *Online AVP* – Controls whether to enable/disable online charging for the associated rating group. If all service rules associated with the rating group have online charging disabled, the rating group is moved to a blocked state for online charging and no longer performs online charging.
- *Offline AVP* – Controls whether to enable/disable offline charging for the associated rating group. If all service rules associated with the rating group have offline charging disabled, the rating group is moved to a blocked state for offline charging and no longer performs offline charging.

Charging-Rule-Remove AVP – The PCRF uses the Charging-Rule-Remove AVP to deactivate predefined PCC rules on the PCEF. The following AVPs are included within the Charging-Rule-Remove AVP.

- *Charging-Rule-Name AVP* – References a specific PCC rule predefined at the PCEF to be deactivated. For PCC rules predefined at the PCEF, the Charging-Rule-Name AVP sent by the PCRF corresponds to the locally configured service rule name on the MCC gateway (`service-construct` → `service-rule`).

To deactivate multiple predefined service rules using a single Charging-Rule-Name AVP, see [Activating/Deactivating Multiple Predefined Service Rules with a Single Charging-Rule-Name](#) (page 1-82).

- *Charging-Rule-Base-Name AVP* – This AVP references an active data profile for the session to be removed (`workflow` → `data-profile`). If the MCC Gateway receives an RAR or CCA message containing the Charging-Rule-Base-Name within the Charging-Rule-Remove AVP, the MCC gateway deactivates the session if the active data profile used for the session matches the received Charging-Rule-Base-Name AVP. In this case, the Delete Bearer Request sent to the MME will contain a cause code of “Reactivation Requested”.

The MCC supports the receipt of a Charging-Rule-Remove AVP containing a single Charging-Rule-Base-Name AVP. If an MCC gateway receives a CCA or RAR message with the Charging-Rule-Remove AVP containing more than one Charging-Rule-Base-Name AVP, the system removes the first one, discards the remaining ones and logs the error.

To deactivate multiple predefined groups of PCC rules on an MCC gateway using the Charging-Rule-Base Name AVP, see [Activating/Deactivating Multiple Predefined Groups of PCC Rules](#) (page 1-77)

Dynamic PCC Rules

Dynamic PCC rules are provisioned on the PCRF and sent over the Gx/Gxa interface to the GGSN/PGW. The PCRF installs, modifies, or removes charging rules on the GGSN/PGW either through the CCA or the RAR to the GGSN/PGW.

Dynamic PCC rules can be installed on both the default bearer and dedicated bearer. If the PCRF installs a dynamic PCC rule on the default bearer (the QCI and ARP values sent within the PCC rule match those of the default bearer) with a specific Service ID and the Quota ID of a locally configured offline rating group, the PGW applies zero-rating to the default bearer traffic. If the Quota ID matches a locally configured online rating group, the PGW applies online charging to the traffic matching the dynamic PCC rule.

Charging-Rule-Install AVP – The PCRF uses the Charging-Rule-Install to install or modify PCC rules on the PCEF. The following AVPs are included within the Charging-Rule-Install AVP.

- *Charging-Rule-Definition* – The Charging-Rule-Definition AVP within the Charging-Rule-Install AVP defines the dynamic PCC rule for a service data flow sent by the PCRF to the PCEF to be installed/modified. The Charging-Rule-Definition AVP within the Charging-Rule-Install AVP specifies the details of each rule as follows:
 - *Charging-Rule-Name AVP* – For dynamic PCC rules, this value uniquely identifies a PCC rule within one IP CAN session to be installed or modified on the PCEF. To avoid installing a PCC rule incorrectly, ensure that the names used for dynamic PCC rules do not match any of the locally configured service rule names.
 - *Rating-Group AVP* – The value of the Rating-Group AVP can already be configured on the MCC as the quota-id CLI parameter. (`services -> charging -> rating-group -> quota-id`) or can be dynamically allocated by the PCRF. When the PCRF dynamically allocates a rating group, the MCC uses the Online and Offline AVPs received in the Charging-Rule-Definition AVP to enable the corresponding charging method for the rating group/service rule. You can also configure a rating group template to use when a dynamic rating group is created. See [Configuring a Dynamic Rating Group Template \(page 1-124\)](#).

A predefined service rule can be activated under a dynamic rating group. For example, if the quota-id received in the Charging-Rule-Definition AVP is not pre-configured on the MCC, the MCC will dynamically allocate a rating group with the received quota-id. If the Charging-Rule-Definition AVP also contains the name of a predefined service rule, the MCC will activate the predefined service rule under the dynamic rating group. However, you need to ensure the dynamic rating group's quota ID is not configured on the MCC.

Furthermore, a predefined service rule under a rating sub-group can be moved to a dynamic rating group based on PCRF messaging.

- *Monitoring-Key AVP* – The Monitoring-Key AVP sent from the PCRF as part of the Charging-Rule-Definition AVP applies to dynamic PCC rules. See [Usage Monitoring and Reporting \(page 1-10\)](#) for more information.
- *Redirect-Information AVP* – The Redirect-Information AVP sent from the PCRF as part of the Charging-Rule-Definition AVP is not supported for dynamic PCC rules.
- *Service-Identifier AVP* – Contains the identifier of a service. In the case of dynamic PCC rules, the Service-Identifier AVP sent from the PCRF as part of the Charging-Rule-Definition AVP does not need to be unique. Multiple Charging-Rule-Definition AVPs can have the same Service-Identifier AVP value.
- *Flow-Status AVP* – The Flow-Status AVP defines whether the service data flow is enabled or disabled.
- *Online AVP* – Controls whether to enable/disable online charging for the associated rating group. If all service rules associated with the rating group have online charging disabled, the rating group is moved to a blocked state for online charging and no longer performs online charging.
- *Offline AVP* – Controls whether to enable/disable offline charging for the associated rating group. If all service rules associated with the rating group have offline charging disabled, the rating group is moved to a blocked state for offline charging and no longer performs offline charging.

Charging-Rule-Remove AVP – Used to remove PCC rules from an IP CAN session as instructed from the PCRF to the PCEF.

- *Charging-Rule-Name AVP* – The Charging-Rule-Name AVP within the Charging-Rule-Remove AVP uniquely identifies a PCC rule, previously installed by the PCRF within one IP CAN session, to be removed.

Usage Monitoring and Reporting

Usage monitoring is used to monitor subscriber usage during an IP-CAN session. The MCC supports usage monitoring over the Gx/Gxa interface for both predefined PCC rules and dynamic PCC rules. The MCC supports usage monitoring on an individual rule (PCC rule level) or on all rules activated/installed during an IP-CAN session (session level).

Each predefined or dynamic PCC rule is associated with a monitoring key that identifies the usage monitoring control instance. When the PCRF activates/installs a PCC rule using the Charging-Rule-Install → Charging-Rule-Definition AVP and requests usage monitoring for that rule, the PCRF includes the associated PCC rule usage monitoring key in the Monitoring-Key AVP (within the Usage-Monitoring-Information AVP). Predefined rules and dynamic PCC rules can share the same monitoring key, which allows for usage monitoring of both predefined and dynamic PCC rules activated/installed by the same Charging-Rule-Definition AVP.

To request usage monitoring on all rules activated/installed during an IP-CAN session (session level), the PCRF includes the monitoring key for a metering rating group in the Monitoring-Key AVP (within the Usage-Monitoring-Information AVP). The MCC gateway first compares the value received in the Monitoring-Key AVP with all the active metering rating groups. If a match is found, the MCC gateway activates usage monitoring services on all rules associated with the metering rating group.

If there is no match, the MCC gateway compares the value received in the Monitoring-Key AVP with all the metering rating groups configured for the metering instance associated with the Workflow data profile. If there is still no match found, the MCC gateway creates a dynamic metering rating group and activates usage monitoring on all rules with the same usage monitoring key as the dynamically-created metering rating group.

The MCC gateway reports the accumulated subscriber usage in a Credit-Control-Request-Update (CCR-U) to the PCRF when any of the following events occur:

- The PCRF requests a report.
- The PCRF removes all PCC rules associated with the same monitoring key.
- When the allocated quota in the Granted-Service-Unit AVP (included within the Usage-Monitoring-Information AVP) is exhausted.

Note: If the PCRF removes all PCC rules associated with a monitoring key, the gateway deactivates the metering rating group, if created dynamically.

Changing the Monitoring Key for Predefined Service Rules

You can change the monitoring key for predefined service rules associated with a predefined metering rating group using the Monitoring-Key AVP sent from the PCRF within the Charging-Rule-Install → Charging-Rule-Definition AVP. The value sent in the Monitoring-Key AVP must correspond to a predefined rating group `monitor-key` or `monitor-key-string` on the MCC (`services > charging > rating-group > monitor-key|monitor-key-string`).

The following example outlines a use case for this feature.

Use Case: Changing Monitoring Key for Predefined Rule

Initial Charging-Rule-Definition AVP	Charging-Rule-Definition AVP with different Monitoring-Key AVP (monitor-key-string)
Charging-Rule-Name AVP SR_1 Monitoring-Key AVP 100	Charging-Rule-Name AVP SR_1 Monitoring-Key AVP 109
Rating Group Configuration	Rating Group Configuration
<code>services charging rating-group Meter_RG</code> charging-method metering monitor-key-string 100 service-rule SR_1	<code>services charging rating-group Meter_RG2</code> charging-method metering monitor-key-string 109 service-rule SR_1
Service Rule Configuration	Service Rule Configuration
<code>service-construct service-rule SR_1</code> admin-state enabled priority 100 service-data-flow-id 4088 service-activation external-activation	<code>service-construct service-rule SR_1</code> admin-state enabled priority 100 service-data-flow-id 4088 service-activation external-activation

Diameter Routing Agent

A Diameter Routing Agent (DRA) provides routing capabilities between an MCC gateway and PCRF when multiple PCRFs are deployed in a Diameter realm to ensure the same PCRF manages all messages related to a particular IP-CAN session. This type of network topology uses IP-CAN/PCRF binding.

To improve the interoperability between the MCC gateway and a DRA, the MCC gateway includes the PCRF host name and realm within the Destination-Host and Destination-Realm AVPs in CCR-U/CCA-Update (CCA-U) and CCR-Terminate/CCA-Terminate (CCR-T/CCA-T) messages sent over the Gx/Gxa interface. Only the CCR-Initial (CCR-I) message carries the Destination-Host/Realm of the DRA.

Rating Group Memory Allocation

When a billing plan is applied to a session, the MCC:

- Allocates memory for all rating groups that have at least one service rule configured with the `service-activation` parameter set to `always-on`.
- Does not allocate memory for rating groups that only have service rules configured with the `service-activation` parameter set to `external-activation`.

When applying the billing plan for a Charging-Rule-Base-Name (CRBN) profile, the MCC only allocates memory for rating groups that have at least one service rule configured with the `service-activation` parameter set to `always-on` or that have a service rule that is part of the CRBN service rule list. See [Activating/Deactivating Multiple Predefined Groups of PCC Rules \(page 1-77\)](#) for more information on CRBN profiles.

If a rating group memory allocation was skipped while applying a billing plan (because all the rules in the rating group were set to `external-activation`), memory for this rating group gets allocated when the PCRF activates the first service rule in this rating group.

***Note:** The MCC does not deallocate memory for a rating group within a billing plan even if the last service rule is removed from that rating group.*

Event-Triggers

To avoid sending unnecessary CCR-U messages to the PCRF, the MCC gateway maintains a list of the event-triggers that it receives from the PCRF in CCA and RAR messages. When an event occurs that corresponds to a PCRF-enabled event-trigger in the list, the PCEF notifies the PCRF causing the PCRF to evaluate whether to update the previously provisioned PCC rules.

When the MCC gateway receives an event-trigger that does not match an event in the list, the MCC gateway does not report the event to the PCRF.

Gx Event-Triggers

The MCC GGSN/PGW supports the following PCRF event-triggers:

- `SGSN_CHANGE (0)` – Used when a Serving Node change triggers an update. The new value of the serving SGSN is indicated in either the 3GPP-SGSN-Address AVP or 3GPP-SGSN-IPv6-Address AVP.
- `QOS_CHANGE (1)` – Used when a QoS change triggers an update. The QoS-Information AVP must be included in the same request and indicates the new QoS value.

- **RAT_CHANGE (2)** – Used when a RAT Type change triggers an update. The RAT-Type AVP indicates the new RAT type.
- **PLMN_CHANGE (4)** – Used when a Public Land Mobile Network (PLMN) change triggers an update.
- **LOSS_OF_BEARER (5)** – Used in CCA and RAR commands by the PCRF to indicate that upon loss of bearer, the PCEF should inform the PCRF. When used in a CCR command, this value indicates that the PCEF generated the request because the bearer associated with the PCC rules indicated by the corresponding Charging-Rule-Report AVP was lost. The PCC-Rule-Status AVP, within the Charging-Rule-Report AVP, indicates that these PCC rules are temporarily inactive. Applies to 3G calls only.

The mechanism of indicating loss of bearer to the GW is IP-CAN access type specific. For GPRS, this is indicated by a PDP context modification request with Maximum Bit Rate (MBR) in QoS profile changed to 0 kbps. When the PCRF performs the bearer binding, the PCEF shall provide the Bearer-Identifier AVP to indicate the bearer that has been lost.

- **IP-CAN_CHANGE (7)** – Used in CCA and RAR commands by the PCRF to indicate that upon a change in the IP-CAN type, the PCEF should inform the PCRF. For example, if there is a Wi-Fi-to-LTE/LTE-to-Wi-Fi (trusted/untrusted) handover and if the PCRF has requested to be notified of this type of change in the CCA/RAR, the PGW notifies the PCRF by sending a CCR-U message to indicate an IP-CAN change. In the case of an LTE-to-Wi-Fi handover, the PGW includes the AN-Trusted AVP in the CCR-U along with the IP-CAN_Change event-trigger to indicate whether the access network is trusted or untrusted.
- **RAI_CHANGE (12)** – Used in the CCA and RAR messages to indicate that if there is a change in RAI, the PCEF should inform the PCRF. When used in a CCR command, this value indicates that the PCEF generated the request because there has been a change in the RAI. The PCEF notifies the PCRF by sending a CCR-U with the latest ULI information and the associated trigger type set.
- **USER_LOCATION_CHANGE (13)** – Used in CCA and RAR messages to indicate that upon a change in the user location, the PCEF should inform the PCRF. When used in a CCR command, this value indicates that the PCEF generated the request because there has been a change in the user location.
- **OUT_OF_CREDIT (15)** – Used in CCA and RAR messages to indicate that the PCEF should inform the PCRF about the PCC rules for which credit is no longer available, together with the applied termination action. When used in a CCR message, this value indicates that the PCEF generated the request because the PCC rules listed in the Charging-Rule-Report AVP ran out of credit and that the termination action in the Final-Unit-Indication AVP applies.

- **REALLOCATION_OF_CREDIT (16)** – Used in CCA and RAR messages to indicate that the PCEF should inform the PCRF about the PCC rules for which credit has been reallocated after the former out-of-credit indication. When used in a CCR message, this value indicates that the PCEF generated the request because the PCC rules in the Charging-Rule-Report AVP were reallocated credit after the former out-of-credit indication.
- **REVALIDATION_TIMEOUT (17)** – Used to inform the GGSN/PGW that it should trigger a new request for PCC rules when the time value in the Revalidation-Time AVP expires.
- **DEFAULT_EPS_BEARER_QOS_CHANGE (20)** – Used in CCA and RAR messages to indicate that upon a change in the default EPS Bearer QoS, the PCEF should inform the PCRF. When used in a CCR message, this value indicates that the PCEF generated the request because the default EPS Bearer QoS changed and the new value is provided in the Default-EPS-Bearer-QoS AVP. This event trigger is reported when the corresponding event occurs, even if the event trigger is not provisioned by the PCRF.
- **SUCCESSFUL_RESOURCE_ALLOCATION (22)** – Used in CCA and RAR messages to indicate that the PCEF should inform the PCRF that the resources for a rule have been successfully allocated. The affected rules are indicated within the Charging-Rule-Report AVP with the PCC-Rule-Status AVP set to ACTIVE (0).
- **UE_TIME_ZONE_CHANGE (25)** – Used in CCA and RAR messages by the PCRF to indicate that upon a change to the time zone, in which the UE is currently located, PCC rules will be requested. When used in a CCR command, this value indicates that the PCEF generated the request because the time zone, in which the UE is currently located, has changed. The new value of the UE's time zone is indicated in the 3GPP-MS-TimeZone AVP.
- **USAGE_REPORT (26) or USAGE_REPORT (33)** – Used when the PCRF requests usage monitoring. The PCRF includes the Usage-Monitoring-Information AVP(s), the Monitoring-Key AVP and the Granted-Service-Unit AVP.

MCC supports the following USAGE_REPORT trigger codes, depending on the PCRF's 3GPP TS 29.212 release compliance:

- USAGE_REPORT (26) (3GPP TS 29.212 Release 10 or earlier)
- USAGE_REPORT (33) (3GPP TS 29.212 Release 11 or later)

Specify the PCRF's 3GPP TS 29.212 release compliance when configuring the PCRF charging interface object.

```
an3000-mcm-slot7cpu1 (config) # service-construct interface
pcrf-charging <interface-name> pcrf-3gpp-release-compliance
[release-10|release-10-plus]
```

- **TAI_CHANGE (26)** – Used in CCA and RAR messages to indicate that upon a change in the TAI, the PCEF should inform the PCRF. When used in a CCR message, this value indicates that the PCEF generated the request because there has been a change in the TAI. The PCEF notifies the PCRF by sending a CCR-U with the latest ULI information and the associated trigger type set.
- **ECGI_CHANGE (27)** – Used in CCA and RAR messages to indicate that upon a change in the ECGI, the PCEF should inform the PCRF. When used in a CCR message, this value indicates that the PCEF generated the request because there has been a change in the ECGI. The PCEF notifies the PCRF by sending a CCR-U with the latest ULI information and the associated trigger type set.

The TAI_CHANGE and ECGI_CHANGE event-triggers are applicable for 3GPP 29.212 V10.0.0 or higher. To ensure the PCEF receives and reports the TAI_CHANGE event-trigger with code 26 and the ECGI_CHANGE event-trigger with code 27, set the PCRF's 3GPP TS 29.212 release compliance to `release-10-plus`.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> pcrf-3gpp-release-compliance
release-10-plus
```

- **APN-AMBR_MODIFICATION_FAILURE (29)** – Used by the PCEF to inform the PCRF that APN-AMBR modifications have failed. The PCEF uses this value in a new CCR command to indicate the failure of the APN-AMBR change requested from the RAR/CCA message. This event trigger does not need a subscription.
- **DEFAULT-EPS-BEARER-QOS_MODIFICATION_FAILURE (34)** – Used by the PCEF to inform the PCRF that Default EPS Bearer QoS modifications have failed. The PCEF uses this value in a new CCR command to indicate the failure in QoS change from the RAR/CCA message. This event trigger does not need a subscription.
- **APPLICATION_START (39)** – Used in CCA and RAR messages by the PCRF to indicate that the PCEF should inform the PCRF when the start of the application traffic, defined by the TDF-Application-Identifier, has been detected, unless a request to mute such a notification (Mute-Notification AVP) is part of the corresponding Charging-Rule-Definition AVP. When used in a CCR message, this value indicates that the PCEF identified the start of the corresponding application's traffic.
- **APPLICATION_STOP (40)** – Used in CCA and RAR messages by the PCRF to indicate that the PCEF should inform the PCRF when the stop of the application traffic, defined by the TDF-Application-Identifier, has been detected, unless a request to mute such a notification (Mute-Notification AVP) is part of the corresponding Charging-Rule-Definition AVP. When used in a CCR message, this value indicates that the PCEF identified the stop of the corresponding application's traffic.

- **CS_TO_PS_HANOVER (42)** – Used in CCA and RAR messages by the PCRF to indicate that upon a CS-to-PS handover, the PCEF informs the PCRF. When used in a CCR message, this value indicates that the PCEF generated the request because there is a CS-to-PS handover. Applicable only to CS-to-PS Single Radio Voice Call Continuity (SRVCC) functionality feature (rSRVCC).
- **ACCESS_NETWORK_INFO_REPORT (45)** – Used in CCA and RAR messages by the PCRF to request access network information from the PCEF. When used in a CCR message, this value indicates that the PCEF generated the request because the PCEF reports the corresponding access network information to the PCRF as requested.

The system also supports the Cisco-Event-Trigger VSA on the Gx interface, which is used by the PCRF to indicate that the PCEF should inform the PCRF when an OCS failure event occurs. See [Notifying the PCRF When an OCS Failure Occurs \(page 1-85\)](#) for more information.

Gxa Event-Triggers

The TWAG/TWAP supports the following PCRF event-triggers:

- **QOS_CHANGE (1)** – Used when a QoS change triggers an update. The QoS-Information AVP must be included in the same request and indicates the new QoS value.
- **REVALIDATION_TIMEOUT (17)** – Used to inform the TWAG that it should trigger a new request for PCC rules when the time value in the Revalidation-Time AVP expires.
- **USAGE_REPORT (26) or USAGE_REPORT (33)** – Used when the PCRF requests usage monitoring. The PCRF includes the Usage-Monitoring-Information AVP(s), the Monitoring-Key AVP and the Granted-Service-Unit AVP.

MCC supports the following USAGE_REPORT trigger codes, depending on the PCRF's 3GPP TS 29.212 release compliance:

- **USAGE_REPORT (26)** (3GPP TS 29.212 Release 10 or earlier)
- **USAGE_REPORT (33)** (3GPP TS 29.212 Release 11 or later)

Specify the PCRF's 3GPP TS 29.212 release compliance when configuring the PCRF charging interface object. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#) for more information.

Policy Control for VoLTE

VoLTE Bearer Setup with Dual RAR

This feature enables the PGW to handle back-to-back RAR messages from the PCRF on the Gx interface during a dedicated bearer setup. A Voice Over LTE (VoLTE) call setup might require an initial RAR to setup a dedicated bearer with QCI 1 and then another RAR to update the QoS of that bearer. If a PGW receives a second RAR before the first RAR procedure is completed, the PGW does not process the second RAR until after the first bearer treatment procedure has completed.

If the first RAR procedure fails, the second RAR might be impacted, depending on the reason for the failure.

- **Create Bearer Request fails with GTPv2-C cause code 110 and retry is successful** – If the first RAR procedure fails with a GTPv2-C Cause code 110 and the retry is successful, the PGW then processes the second RAR.
- **Create Bearer Request fails with GTPv2-C cause code 110 and retry fails** – If the first RAR procedure fails with a GTPv2-C cause code 110 and the retry also fails, the PGW drops the second RAR then sends a CCR-U to the PCRF indicating the Create Bearer Request failure and sends another CCR-U to the PCRF indicating the failure of the second RAR.
- **Create Bearer Request fails with cause code other than GTPv2-C cause code 110** – If the first RAR procedure fails with a cause code other than GTPv2-C cause code 110, the PGW drops the second RAR then sends a CCR-U to the PCRF indicating the Create Bearer Request failure and sends another CCR-U to the PCRF indicating the failure of the second RAR. There is no retry in this scenario.
- **Create Bearer Request fails with GTPv2-C Cause code 110 for LTE to Wi-Fi handovers or handovers with RAT change** – If the first RAR procedure fails with GTPv2-C Cause code 110 and the PGW detects that a LTE to Wi-Fi handover (or vice-versa) or a handover with RAT change is in progress, the PGW drops the second RAR then sends a CCR-U to the PCRF indicating the Create Bearer Request failure. The PGW also sends another CCR-U to the PCRF indicating the failure of the second RAR.
- **Create Bearer Request fails and subset of PCC rules fail in second RAR** – If the first RAR procedure fails, the PGW sends a CCR-U to the PCRF indicating the Create Bearer Request failure. While processing the second RAR, the PGW detects that a subset of rules failed during the first RAR and processes only the remaining PCC rules. After processing the second RAR, the PGW sends another CCR-U to the PCRF indicating the failed rules.

Handling Partial Bearer Success in CBR/UBR

The PGW sends a CCR-U to the PCRF indicating the failed PCC rules in the Charging-Rule-Report AVP for the following scenarios:

- When a Create Bearer Response (CBR) is received with partial success during:
 - a Network Initiated Dedicated Bearer Creation procedure.
 - a Network Initiated Dedicated Bearer Creation procedure due to 3GPP <=> Non-3GPP handover.
- When an Update Bearer Response (UBR) is received with partial success during a Network Initiated Bearer Modification procedure.
- When a CBR is received with partial success for existing bearers during a Network Initiated Dedicated Bearer Modification procedure due to 3GPP <=> Non-3GPP handover.

QCI/ARP Modifications of EPS Dedicated Bearer

Each unique QoS Class Identifiers/Allocation and Retention Priority (QCI/ARP) combination identifies a bearer on the PGW. The following scenarios are supported if the PCRF uses the Charging-Rule-Install AVP to modify QCI/ARP values.

- If the PCRF modifies the QCI/ARP values for all charging rules on an existing dedicated bearer, the MCC gateway updates the QCI/ARP value for this dedicated bearer and sends an Update Bearer Request to the MME.
- If there are other charging rules on the existing dedicated bearer, the PGW keeps the existing dedicated bearer but removes the particular charging rule for which QCI/ARP values have been modified. The PGW creates a new dedicated bearer with the new QCI/ARP values to carry the traffic for this charging rule.
- If there is already an existing dedicated bearer with the same QCI/ARP values as those in the charging rules being modified, the PGW moves those modified charging rules to the existing bearer and deletes the old dedicated bearer if no charging rules are left on that bearer.

Message Flows

This section provides examples of the following PCRF message flows:

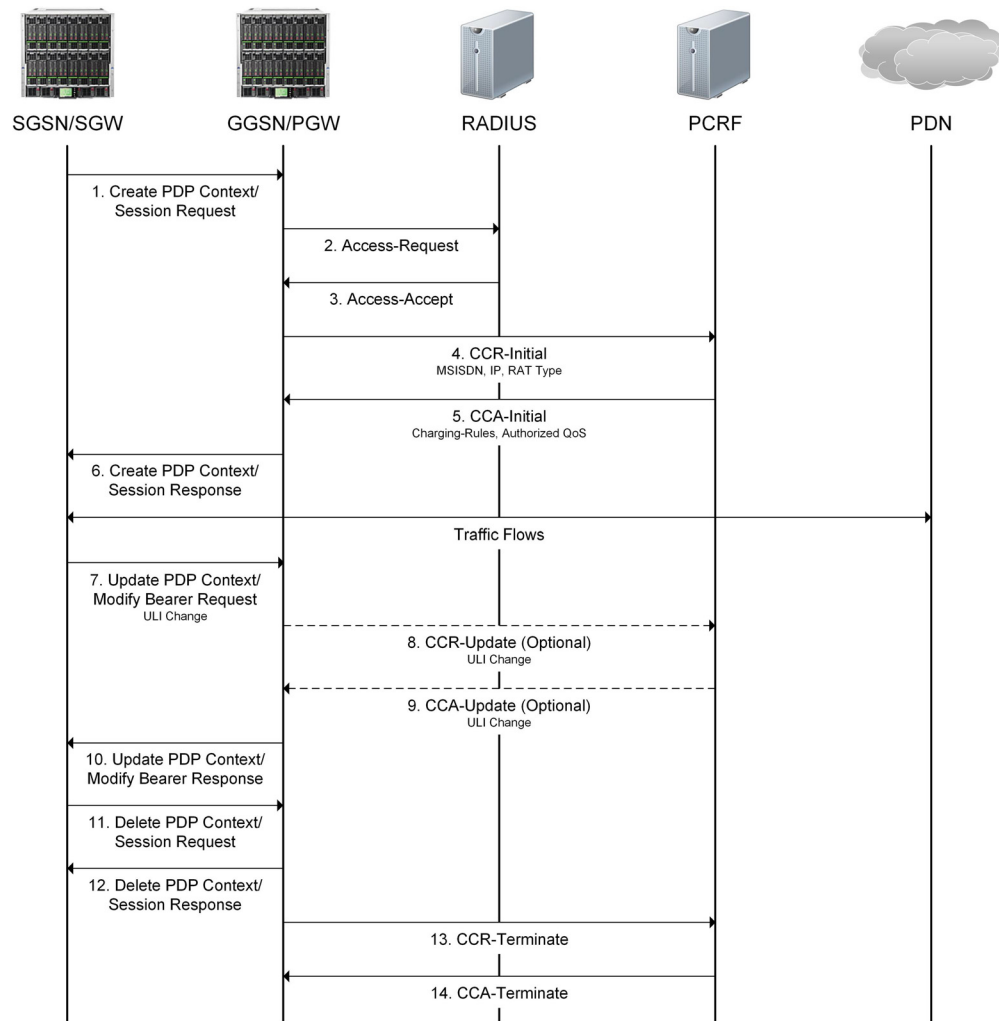
- [Typical PCRF Session \(page 1-20\)](#)
- [Network-Initiated QoS Change \(page 1-21\)](#)
- [Network-Initiated Termination \(page 1-22\)](#)
- [Network-Initiated Dedicated Bearer Activation \(through CCA\) \(page 1-23\)](#)
- [PCRF Initiated Dedicated Bearer Activation \(through RAR\) \(page 1-25\)](#)
- [PCC Rule Removal \(page 1-26\)](#)
- [Charging Rule Removal \(Session Deactivation\) \(page 1-28\)](#)
- [Dynamic Policy Control \(page 1-30\)](#)
- [Operator-Initiated Dynamic Policy Change \(page 1-32\)](#)
- [Revalidating Policy Information Using REVALIDATION_TIMEOUT \(page 1-34\)](#)
- [Gxa Session Set Up \(page 1-36\)](#)
- [Gxa Session PCC Rule Activation and Usage Reporting \(page 1-38\)](#)
- [Gxa Session Termination \(page 1-40\)](#)

Typical PCRF Session

Figure 1-1 illustrates the message flow for a typical PCRF session where the PCRF successfully grants the requested service units and user traffic commences between the User Equipment (UE) and Public Data Network (PDN).

This figure also illustrates the message flow for cases such as event-trigger reporting, User Location Information (ULI) changes, and predefined service rule activation. While Figure 1-1 uses the ULI change as an example, the message flow remains the same in all cases.

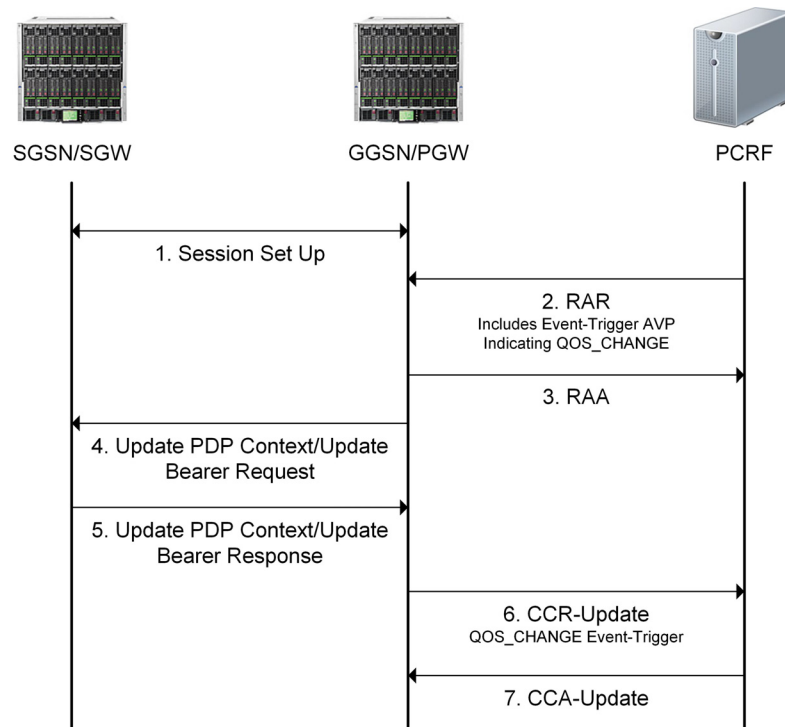
Figure 1-1 Typical PCRF Session



Network-Initiated QoS Change

Figure 1-2 illustrates the message flow for a network-initiated QoS change for which the PCRF sends an RAR to the GGSN/PGW and the GGSN/PGW responds with a Re-Auth-Answer (RAA) indicating that the QoS change was successful.

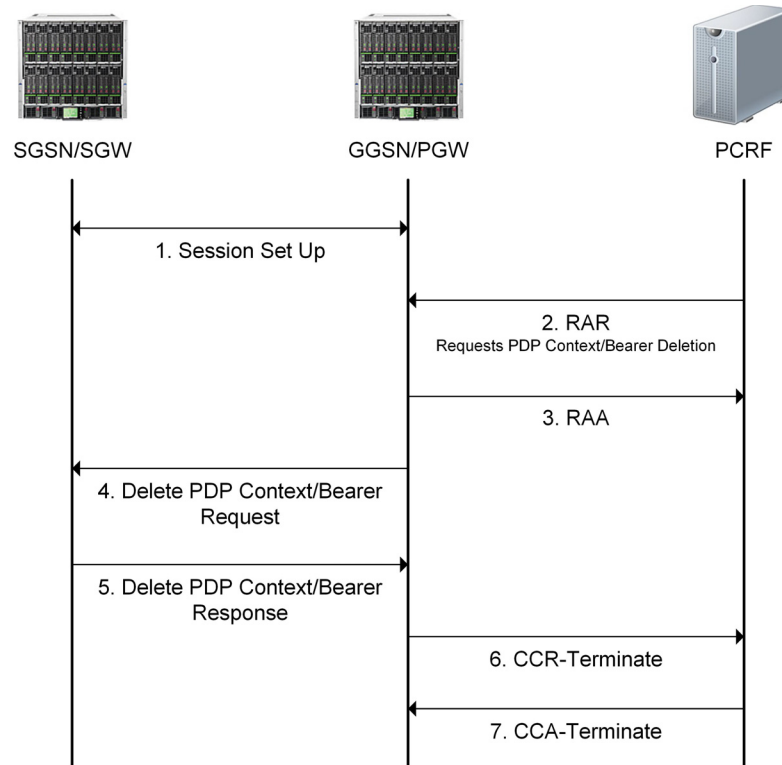
Figure 1-2 Network-Initiated QoS Change



Network-Initiated Termination

Figure 1-3 illustrates the message flow for a network-initiated termination.

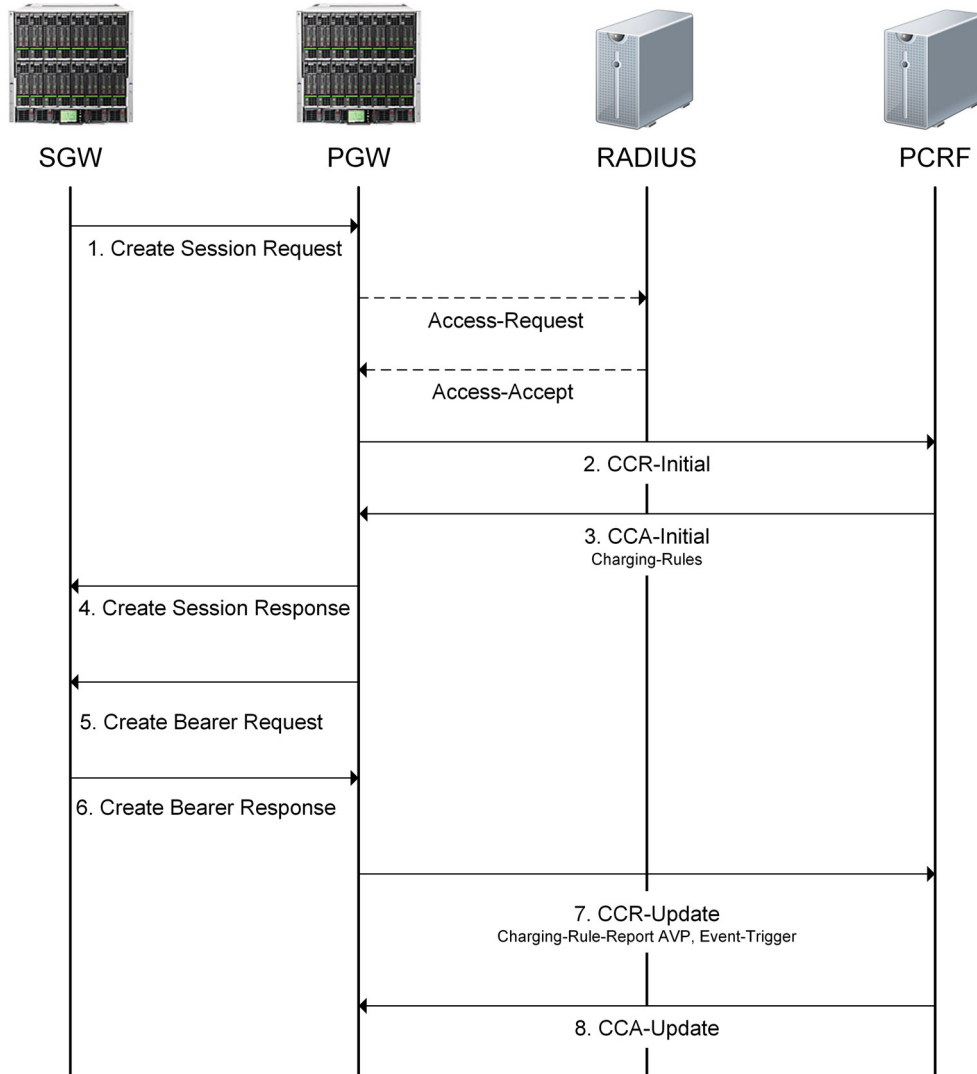
Figure 1-3 Network-Initiated Session Termination



Network-Initiated Dedicated Bearer Activation (through CCA)

Figure 1-4 illustrates an example of the message flow for activating a dedicated bearer request using dynamic PCC rules. In this example, the PCRF initiates a bearer request through the CCA message in response to a CCR from the PGW.

Figure 1-4 PCRF-Initiated Bearer Activation (through CCA)



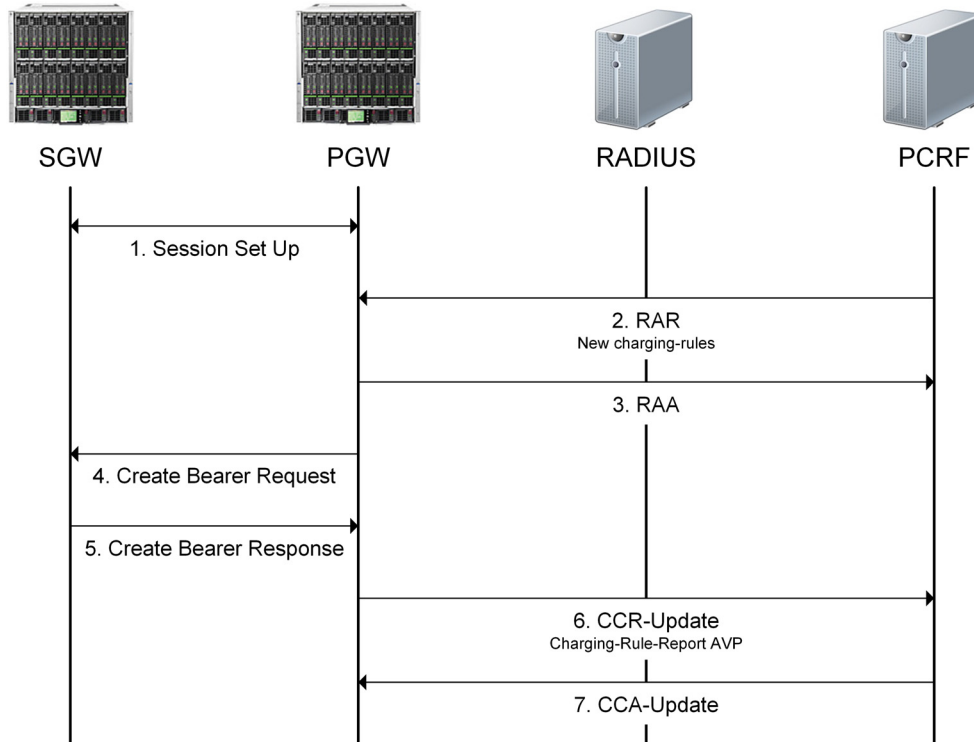
- 1 The SGW sends a Create Session Request to the PGW.
- 2 The PGW starts the session setup process by sending a CCR-I message to the PCRF.
- 3 The PCRF responds to the PGW with a CCA message containing charging rule definitions.
- 4 The PGW sends a Create Session Response to the SGW and completes the session setup.

- 5 Based on the information received in the Charging-Rule-Definition AVP included in the CCA message, the PGW determines a dedicated bearer is required. The PGW sends a Create Bearer Request to the SGW containing the QoS and packet filters received in the Charging-Rule-Definition AVP.
- 6 The SGW responds with a Create Bearer Response message and the PGW completes the dedicated bearer creation.
- 7 The PGW sends the PCRF a CCR-U message containing the Charging-Rule-Report AVP to inform the PCRF of the status of the PCC rules just installed.
- 8 The PCRF responds with a CCA message.

PCRF Initiated Dedicated Bearer Activation (through RAR)

Figure 1-5 illustrates an example of the message flow for activating a dedicated bearer request using dynamic PCC rules. In this example, the PCRF initiates a bearer request through the RAR message.

Figure 1-5 PCRF Initiated Bearer Activation (through RAR)

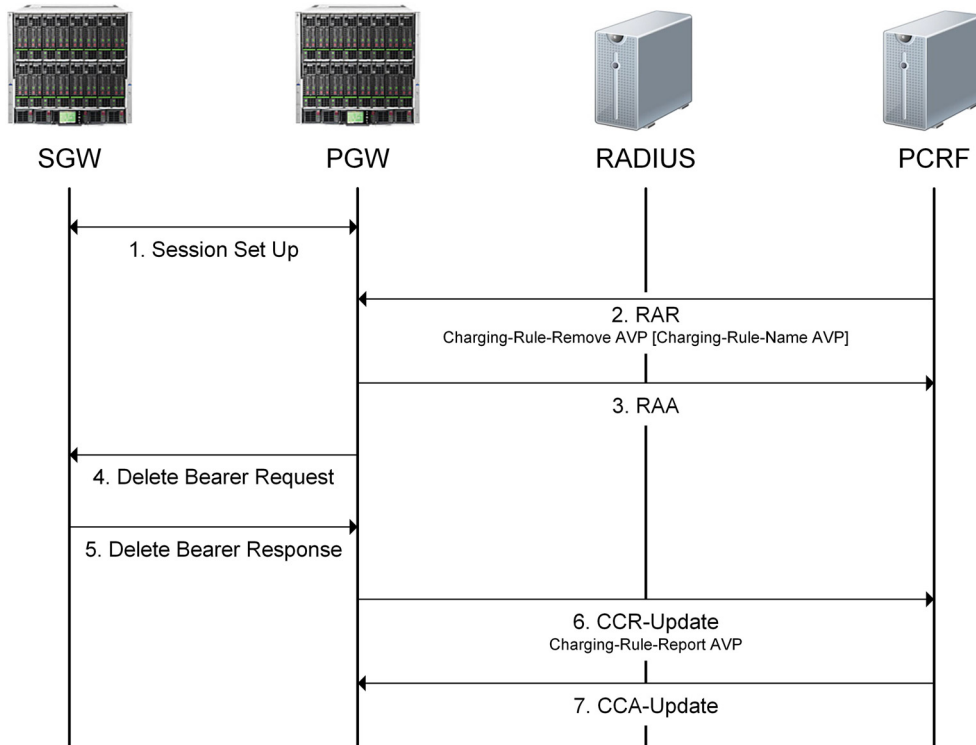


- 1 A session is set up.
- 2 During the session, the PCRF sends an RAR to the PGW containing the Charging-Rule-Install AVP.
- 3 The PGW responds to the PCRF with an RAA message to acknowledge the RAR.
- 4 Based on the information received in the Charging-Rule-Definition AVP included in the RAR message, the PGW determines a dedicated bearer is required. The PGW sends a Create Bearer Request to the SGW containing the QoS and packet filters received in the Charging-Rule-Definition AVP.
- 5 The SGW responds with a Create Bearer Response message and the PGW completes the dedicated bearer creation.
- 6 The PGW sends the PCRF a CCR-U message containing the Charging-Rule-Report AVP to inform the PCRF of the status of the PCC rules just installed.
- 7 The PCRF responds with a CCA message.

PCC Rule Removal

Figure 1-6 illustrates an example of the message flow for removing a PCC rule previously installed.

Figure 1-6 PCC Rule(s) Removal



- 1 A session is set up.
- 2 During the session, the PCRF sends an RAR to the PGW containing the Charging-Rule-Remove AVP that identifies the PCC rule to be removed.
- 3 The PGW responds to the PCRF with an RAA message to acknowledge the RAR. Based on the information received in the RAR message, the PGW removes the requested PCC rule(s).
- 4 If all PCC rules associated with a dedicated bearer are deleted as a result of the PCC removal, the bearer is deleted. The PGW sends a Delete Bearer Request to the SGW.

Note: If as a result of the PCC rule removal, the bearer QoS parameters or packet filters changed, the PGW initiates the bearer modification process.

- 5 The SGW responds with a Delete Bearer Response message and the PGW completes the dedicated bearer deletion.
- 6 The PGW sends the PCRF a CCR-U message containing the Charging-Rule-Report AVP to inform the PCRF of the status of the PCC rules just removed.

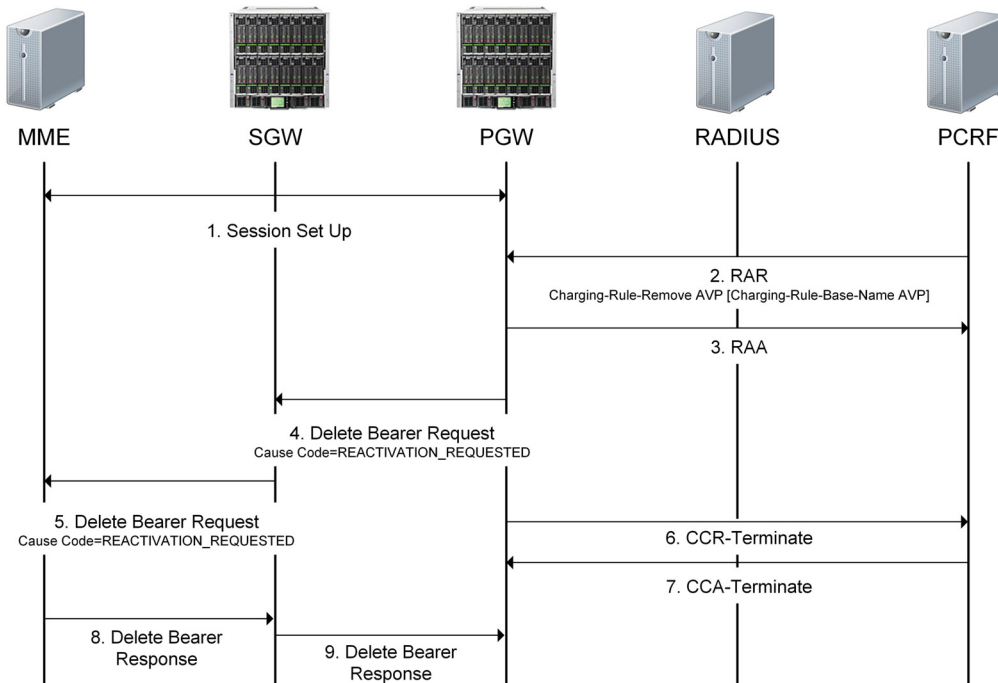
- 7 The PCRF responds with a CCA message.

Charging Rule Removal (Session Deactivation)

A PCRF can deactivate a session by providing the Charging-Rule-Base-Name AVP within the Charging-Rule-Remove AVP in the RAR or CCA-U message. The MCC PGW deactivates the session if the active data profile used for the session matches the received Charging-Rule-Base-Name AVP. In this case, the Delete Bearer Request sent to the MME will contain a cause code of “Reactivation Requested”.

Figure 1-7 illustrates an example of the message flow for deactivating a session using the Charging-Rule-Base-Name AVP within the Charging-Rule-Remove AVP.

Figure 1-7 Session Deactivation using Charging-Rule-Base-Name



- 1 A session is set up.
- 2 The PCRF sends an RAR to the PGW containing the Charging-Rule-Base-Name AVP within the Charging-Rule-Remove AVP that identifies the active data profile to be removed.
- 3 The PGW responds to the PCRF with an RAA message to acknowledge the RAR.
- 4 The PGW sends a Delete Bearer Request to the SGW with cause code “REACTIVATION_REQUESTED”.
- 5 The SGW processes the Delete Bearer Request and forwards the message to the MME for further processing.
- 6 Without waiting for a response to the Delete Bearer Request, the PGW deactivates the session by sending a CCR-T message to the PCRF.
- 7 The PCRF responds with a CCA-T message.

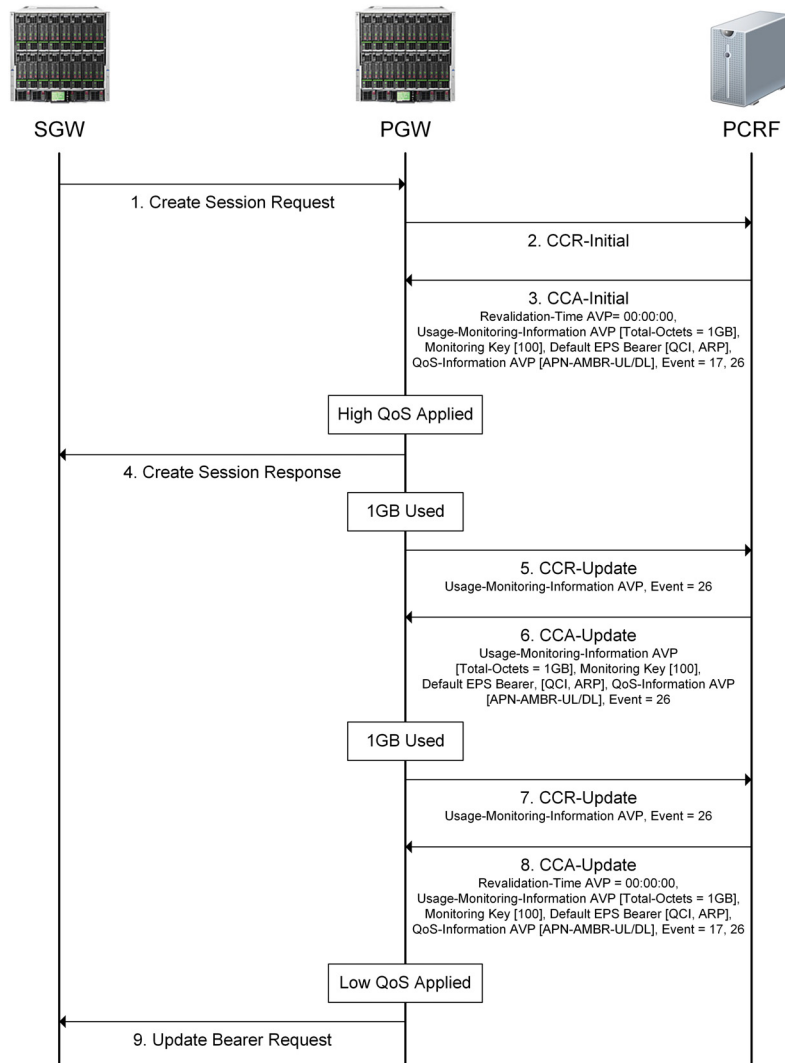
- 8 The MME responds with a Delete Bearer Response message.
- 9 The SGW responds with a Delete Bearer Response message to the PGW.

Dynamic Policy Control

Figure 1-8 illustrates an example of dynamic policy control where the PCRF installs PCC rules on the PCEF (PGW) and the PGW monitors and reports usage information to the PCRF based on these PCRF-provisioned PCC rules.

In this example, the PCRF assigns a monthly predefined usage limit for the subscriber and communicates this limit to the PGW in the form of PCC rules. The PGW notifies the PCRF when the subscriber exceeds the predefined limit and the PCRF reduces the subscriber's QoS for the remainder of the month.

Figure 1-8 Dynamic Policy Control



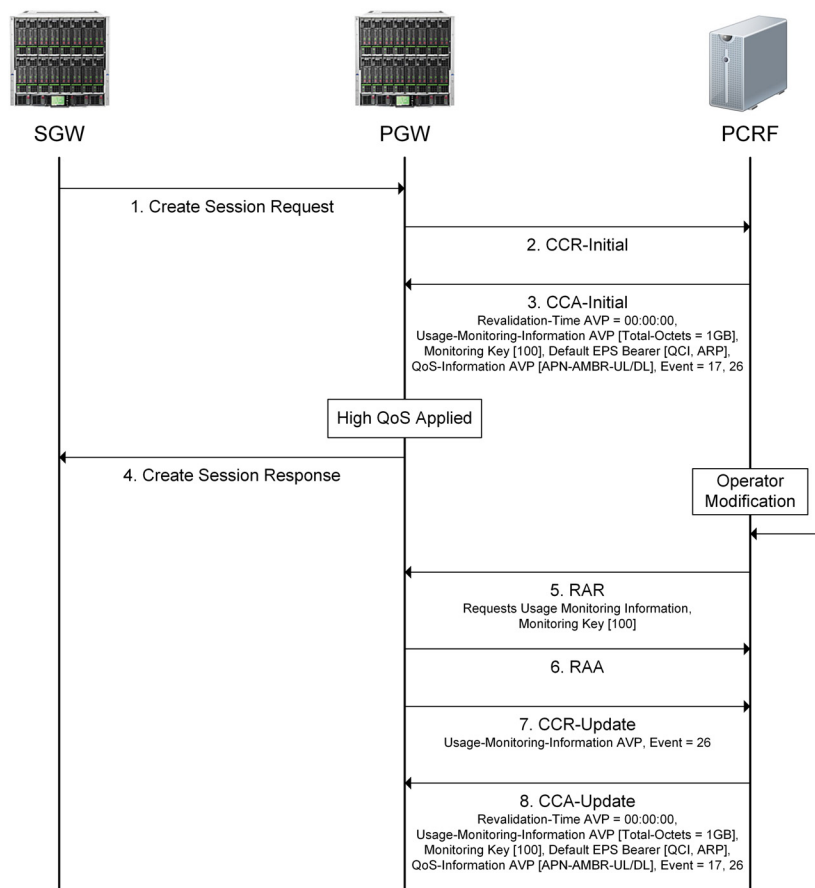
- 1 As part of the subscriber session setup, the PGW receives a Create Session Request.
- 2 The PGW processes the Create Session Request message and sends a CCR-I message to the PCRF.
- 3 The PCRF determines the appropriate subscriber usage and subscription plan and sends a CCA-I message to the PGW containing the corresponding policy rules. The PCRF also sends the usage monitoring key to identify the service data that it would like to monitor for this subscriber.
- 4 The PGW installs the QoS parameters, policy rules and usage monitoring information received from the PCRF and sends a successful Create Session Response message to the SGW. In this case, the PCRF sets the usage monitoring threshold at 1GB.
- 5 If the subscriber exceeds the 1GB usage threshold, the PGW sends a CCR-U message to the PCRF containing the subscriber's current usage and includes the Event-Trigger AVP indicating USAGE_REPORT (26).
- 6 The PCRF computes the total monthly usage for this subscriber and sends a CCA-U message to the PGW granting the subscriber another 1GB of usage. The PGW uses this grant to update the predefined PCC rules related to usage monitoring.
- 7 If the subscriber exceeds the 1GB usage threshold again, the PGW sends a CCR-U message to the PCRF containing the subscriber's current usage and includes the Event-Trigger AVP indicating USAGE_REPORT (26).
- 8 The PCRF computes the subscriber's total monthly usage and realizes that the subscriber has exceeded the monthly usage limit. The PCRF sends a CCA-U message to the PGW containing policy rules to lower the subscriber's QoS.
- 9 Upon receiving the CCA-U from the PCRF, the PGW initiates a new network QoS modification procedure to reduce the subscriber's QoS for the remainder of the month.

Operator-Initiated Dynamic Policy Change

Figure 1-9 illustrates an example of how an Operator can initiate a modification to a particular subscriber's policy using dynamic PCC rules.

In this example, the PCRF assigns a monthly predefined usage limit for the subscriber and communicates this limit to the PGW in the form of PCC rules. As illustrated in Figure 1-9, the PGW notifies the PCRF when the subscriber exceeds the predefined limit and the PCRF reduces the subscriber's QoS for the remainder of the month. In this case, the Operator wants to modify the subscriber's QoS policy to ensure the subscriber's QoS remains high.

Figure 1-9 Operator-Initiated Dynamic Policy Change



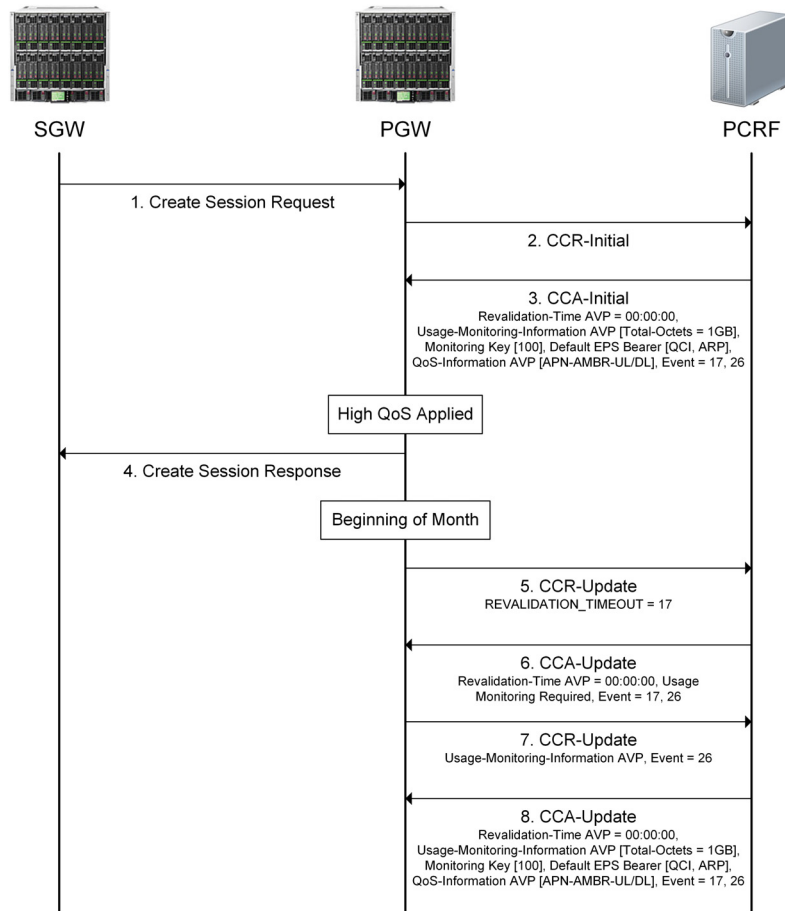
- 1 As part of the subscriber session setup, the PGW receives a Create Session Request.
- 2 The PGW processes the Create Session Request message and sends a CCR-I message to the PCRF.
- 3 The PCRF determines the appropriate subscriber usage and subscription plan and sends a CCA-I message to the PGW containing the corresponding policy rules. The PCRF also sends the usage monitoring key to identify the service data that it would like to monitor for this subscriber.
- 4 The PGW installs the QoS parameters, policy rules and usage monitoring information received from the PCRF and sends a successful Create Session Response message to the SGW. In this case, the PCRF sets the usage monitoring threshold at 1GB.
- 5 The Operator modifies the user's subscription plan and the PCRF sends an RAR message to the PGW requesting usage monitoring information and the usage monitoring key.
- 6 The PGW responds with a successful RAA and initiates the usage report procedure.
- 7 The PGW sends a CCR-U message to the PCRF containing the subscriber's current usage information.
- 8 Based the subscriber's current usage information, the PCRF computes new PCC rules and sends a CCA-U message to the PGW containing the new PCC rules and new usage monitoring grant information.
- 9 Upon receiving the CCA-U from the PCRF, the PGW installs the new policies and applies the new grant.

Revalidating Policy Information Using REVALIDATION_TIMEOUT

Figure 1-10 illustrates an example of how a subscriber's policy information is revalidated at the beginning of every month using the REVALIDATION_TIMEOUT event trigger.

In this example, the PGW uses the REVALIDATION_TIMEOUT event trigger to issue a new request for PCC rules from the PCRF.

Figure 1-10 Re-validating Policy Information

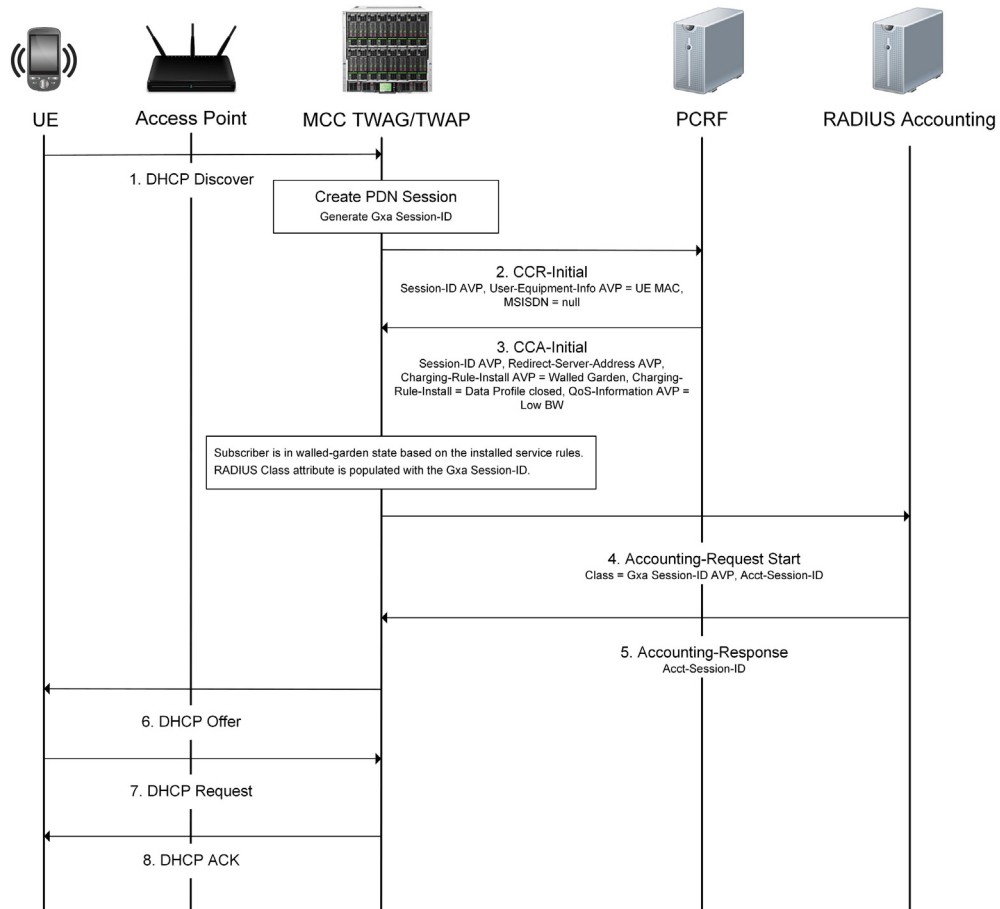


- 1 As part of the subscriber session setup, the PGW receives a Create Session Request.
- 2 The PGW processes the Create Session Request message and sends a CCR-I message to the PCRF.
- 3 The PCRF determines the appropriate subscriber usage and subscription plan and sends a CCA-I message to the PGW containing the corresponding policy rules, REVALIDATION_TIMEOUT event trigger and Revalidation-Time AVP. The PCRF also sends the usage monitoring key to identify the service data that it would like to monitor for this subscriber.
- 4 The PGW installs the QoS parameters, policy rules and usage monitoring information received from the PCRF and sends a successful Create Session Response message to the SGW. In this case, the PCRF sets the usage monitoring threshold at 1GB.
- 5 When the time value in the Revalidation-Time AVP expires at the beginning of the month, the PGW sends a CCR-U message to the PCRF containing the Event-Trigger AVP indicating REVALIDATION_TIMEOUT (17).
- 6 Upon receiving the CCR-U from the PGW, the PCRF realizes that this is the beginning of the month and sends the PGW a CCA-U message requesting the subscriber's current usage information.
- 7 The PGW sends a CCR-U message to the PCRF containing the subscriber's current usage and includes the Event-Trigger AVP indicating USAGE_REPORT (26).
- 8 Based on the subscriber's current usage information, the PCRF defines new policy parameters, QoS values and usage grants for the subscriber. The PCRF sends a CCA-U message to the PGW containing the new PCC rules, including a new time value for the Revalidation-Time AVP.
- 9 Upon receiving the CCA-U from the PCRF, the PGW installs the new QoS parameters, policy rules and usage monitoring information received from the PCRF and restarts the Revalidation Timer.

Gxa Session Set Up

Figure 1-11 illustrates the message flow for setting up a Gxa session between the TWAG and PCRF.

Figure 1-11 Gxa Session Set Up



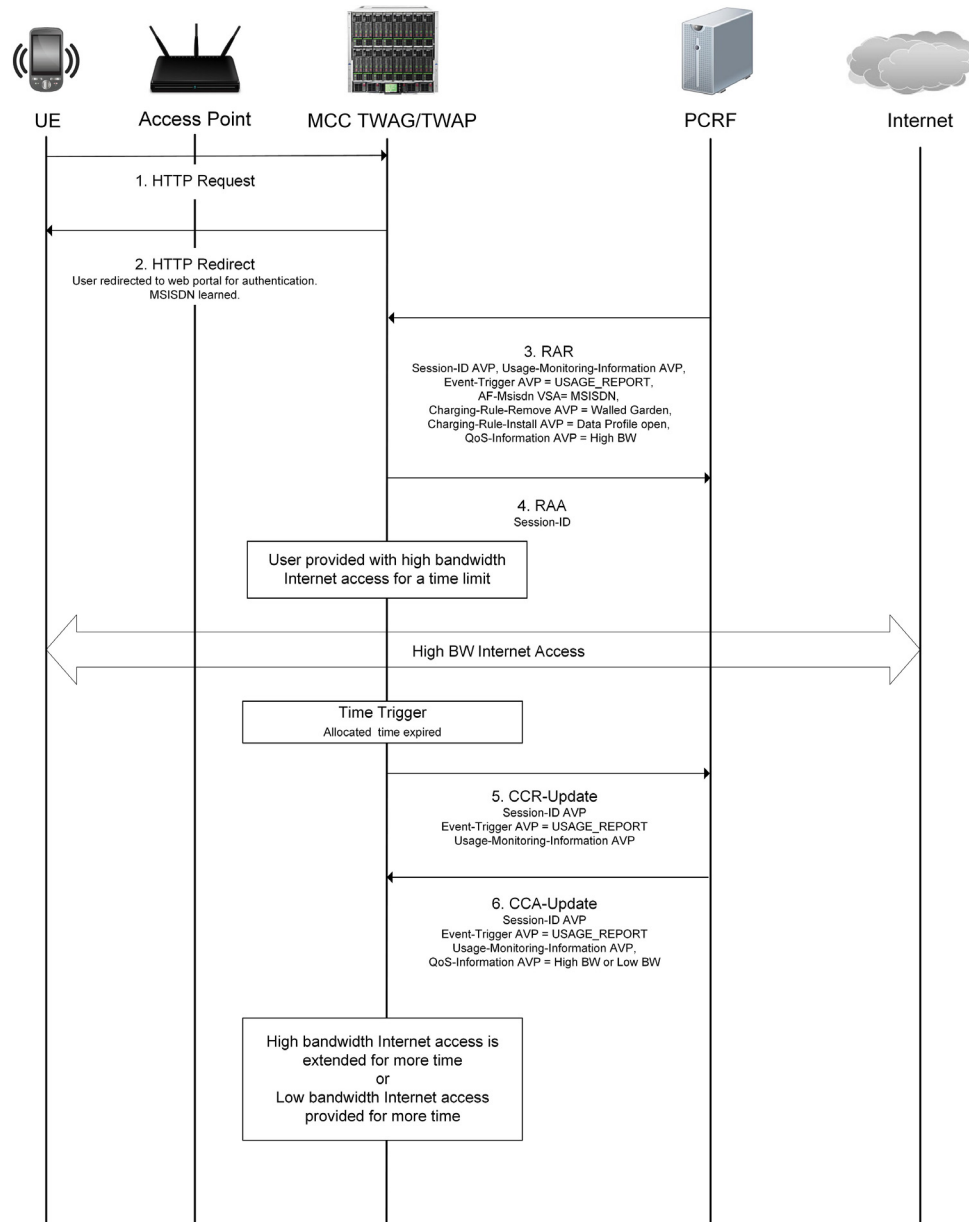
- 1 The TWAG receives a DHCP Discover message for session set up. The MCC TWAG creates a PDN session and generates a Gxa session ID. The subscriber session is in an open state, in which full Internet access is allowed, but the UE has not been allocated an IP address.
- 2 The TWAG sends a CCR-I to the PCRF containing the Session-ID AVP and the User-Equipment-Info AVP, which contains the UE MAC address.
- 3 The PCRF responds with a CCA-I containing the Session-ID AVP, Redirect-Server-Address AVP, and places the subscriber in a walled-garden state (no access to the Internet) based on the service rule installed via the Charging-Rule-Install AVP. The TWAG has the service rule predefined in its database and installs the rule.
- 4 The TWAG starts an accounting session with the remote RADIUS server and sends an Accounting-Request containing the RADIUS Class attribute, which is populated with the Gxa Session-ID, and the Acct-Session-ID attribute.

- 5 The remote RADIUS server responds with an Accounting-Response message.
- 6 The TWAG sends the allocated IP address to the UE in the DHCP Offer message.
- 7 The UE sends the DHCP Request message to the TWAG.
- 8 The TWAG responds with a DHCP ACK message.

Gxa Session PCC Rule Activation and Usage Reporting

Figure 1-12 illustrates the message flow for cases such as event-trigger reporting, predefined service rule activation and usage reporting where the TWAG monitors and reports usage information to the PCRF based on predefined PCC rules.

Figure 1-12 Gxa Session Authentication



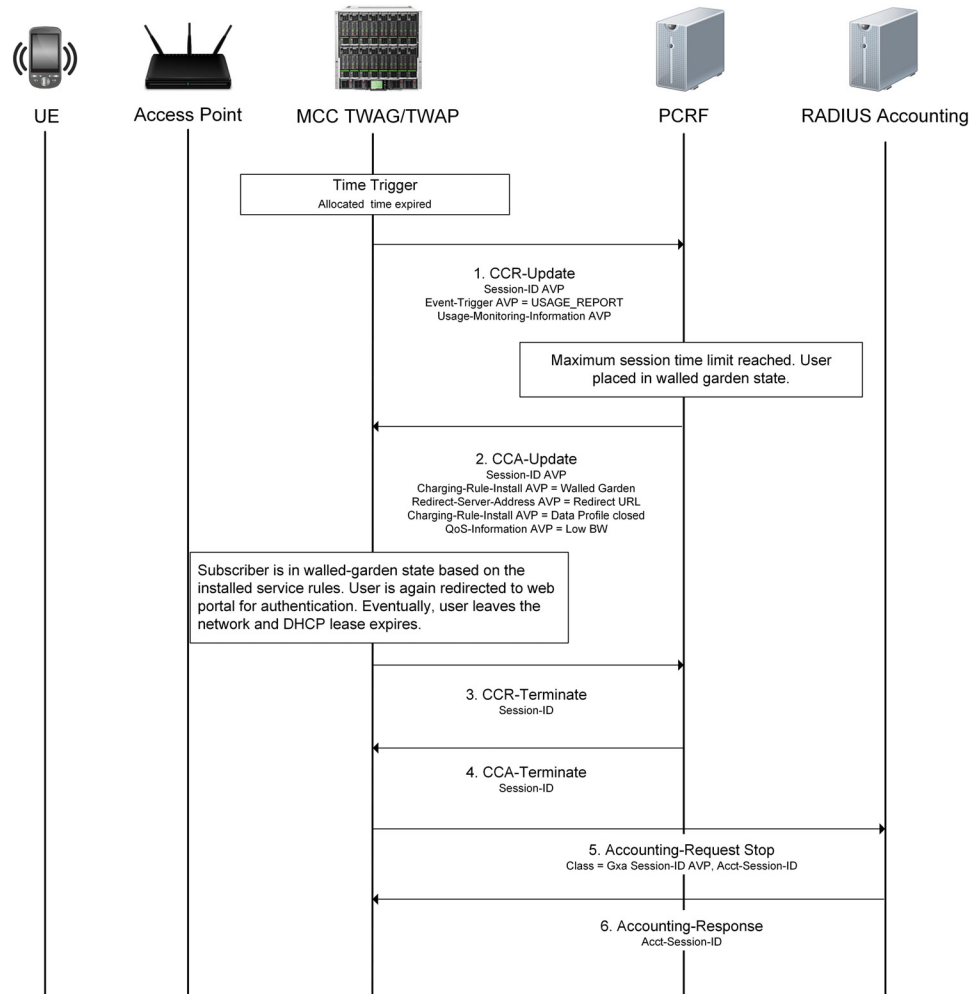
- 1 The Gxa session has been set up and the TWAG receives an HTTP Request from the UE.
- 2 Based on the installed service rules, the TWAG redirects the user to a web portal for further authentication. During the user authentication process, the MSISDN is learned.

- 3 Once the user has been authenticated with web portal, the PCRF sends an RAR to the TWAG containing the following information:
 - Session-ID AVP – Contains the Gxa session Identifier.
 - Usage-Monitoring-Information AVP – Used by the PCRF to request usage monitoring. Contains the Granted-Service-Unit AVP.
 - Event-Trigger AVP – Set to USAGE_REPORT.
 - AF-Msisdn VSA – Contains the MSISDN.
 - Charging-Rule-Remove AVP – Removes the walled-garden service rule, in which all traffic is dropped except for HTTP and DNS.
 - Charging-Rule-Install AVP – Identifies the active data profile and places the user in an open state, in which the user is allowed full access to the Internet.
 - QoS-Information AVP – Identifies the QoS parameters for the session.
- 4 The TWAG responds with an RAA and the user is provided with high bandwidth Internet access for a specified time limit.
- 5 When the time limit expires, the TWAG sends a CCR-U to the PCRF notifying the PCRF that the time limit expired.
- 6 The PCRF responds with a CCA-U and, depending on the policy of the service provider, directs the TWAG to either grant more time for high bandwidth Internet access or grant more time for Internet access but at a lower bandwidth.

Gxa Session Termination

Figure 1-13 illustrates the message flow for terminating a Gxa session between the TWAG and PCRF.

Figure 1-13 Gxa Session Termination



- 1 User is provided with Internet access for a specified time limit. When the time limit expires, the TWAG sends a CCR-U to the PCRF notifying the PCRF that the time limit expired.
- 2 The PCRF determines that the user has reached the maximum session time and responds with a CCA-U placing the subscriber in a walled-garden state (no access to the Internet) based on the service rule installed via the Charging-Rule-Install AVP. The user is redirected once again to the web portal for authentication. Eventually, the user leaves the network and the DHCP lease expires.
- 3 The TWAG sends a CCR-T to the PCRF to terminate the Gxa session.
- 4 The PCRF responds with a CCA-T.

- 5 The TWAG sends the RADIUS server an Accounting-Request Stop message to terminate the accounting session.
- 6 The RADIUS server responds with an Accounting-Response message.

Sending Policy and Charging Records to the PCRF

This section provides an example of the configuration steps for sending policy and charging records to the PCRF via the Gx/Gxa interface.

Use the following sequence to configure the system to send policy and charging records to the PCRF via the Gx/Gxa interface.

- 1 Configure the local diameter node.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
```

where **<node-name>** specifies the name of the Diameter service profile on the gateway.

Note: The maximum number of supported Diameter server groups is 128 and the maximum number of supported diameter peers is 32 peers (32 Geo-active and 32 Geo-standby).

The following table describes the configurable parameters.

Parameter	Description
<code>cea-timeout</code>	Specifies the wait time (in seconds) for receiving a Capability-Exchange-Answer (CEA).
<code>cer-origin-state-id-mode</code>	Specifies whether to increase the value of the Origin-State-Id AVP, sent in CER messages, for every new Diameter connection. ■ static – (default) The MCC updates the value of the Origin-State-Id AVP sent in CER messages with the static value, configured using the <code>cer-origin-state-id-static-value</code> parameter. ■ timestamp – For every new Diameter connection, the MCC updates the value of the Origin-State-Id AVP sent in CER messages with the current timestamp. The Diameter peer that receives the message can infer that sessions associated with a lower Origin-State-ID value are no longer active.
<code>cer-origin-state-id-static-value</code>	Specifies the static Origin-State-Id value sent in CER messages. The default value is 25. Note: This parameter only applies when <code>cer-origin-state-id-mode</code> is set to <i>static</i>.
<code>connection-attempt-count</code>	Specifies the maximum number of connection retry attempts from the PGW to the PCRF.
<code>connection-attempt-timeout</code>	Specifies the Tc timer, in seconds, as described in RFC 6733. Specifies how frequently to send transport connection attempts to a peer with whom no active transport connection exists.

Parameter	Description
dscp-value	Specifies the Differentiated Services Code Point (DSCP) value with which to mark all IP packets on this Diameter server group. For more information, see Configuring DSCP Values for Control Plane Traffic (page 1-65)
dwr-retry-count	Specifies the maximum number of Device-Watchdog-Request (DWR) attempts to the diameter peer before declaring the peer down.
dwr-retry-timer	Specifies the number of seconds the diameter node waits for a Device-Watchdog-Answer (DWA) after retransmitting a DWR once the first dwr-timer expires. <i>Note: The dwr-retry-timer is typically set at a shorter interval than the dwr-timer.</i>
dwr-timer	Number of seconds the diameter node waits for a DWA after sending a DWR to the diameter peer. <i>Note: When the dwr-timer expires, the diameter node retransmits the DWR to the diameter peer and the dwr-retry-timer is started.</i>
geo-redundant-group	Specifies the Geographic Redundancy Group name. The Geographic Redundancy Group must already exist. For information on configuring a Geographic Redundancy Group, see Configuring Geo-Redundancy Health Checks on the Gx Interface (page 1-61) .
loopback-ip	Specifies the signaling port IP address.
network-context	Specifies the network context, selected from the VPN Group list. Restricts the inbound diameter messages by accepting only inbound messages on the IP interfaces in the interface group. The network context must already exist.
node-role	Specifies the local node role: ■ client ■ server
origin-host	Specifies the local host name in Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the request message. <i>Note: You can configure multiple Diameter server groups with the same origin host.</i>
origin-realm	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message. <i>Note: You can configure multiple Diameter server groups with the same origin realm.</i>

Parameter	Description
port	Specifies the TCP listening port number used when <code>node-role</code> is set to <code>server</code> . The default is 3868. <i>Note: This is the TCP listening port, not the Gx source port used when <code>node-role</code> is set to <code>client</code>. The Gx source port is dynamically assigned and is not configurable.</i>
product-name	Specifies the diameter product name.
send-destination-host-avp	Specifies whether the PGW sends the statically configured Destination-Host and Destination-Realm on the Gz/Rf interface in the initial Accounting-Request (ACR) message. <i>Note: Applies to the Gz/Rf interface only.</i>
transport	Specifies the transport type. ■ tcp (default) ■ sctp – To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the Gx interface, you must configure an SCTP profile and associate the profile with the local Diameter node. See Configuring an SCTP Profile (page 1-88).
vendor-id	Specifies the diameter vendor ID.
sctp-profile	Specifies the name of the SCTP profile. The SCTP profile must already exist. See Configuring an SCTP Profile (page 1-88) . <i>Note: The <code>sctp-profile</code> parameter is only visible when the local Diameter node transport type is set to SCTP.</i>
secondary-loopback-ip	Specifies a secondary loopback IP address for the local Diameter node to support SCTP multi-homing. See Configuring an SCTP Profile (page 1-88) for more information. <i>Note: The <code>secondary-loopback-ip</code> parameter is only visible when the local Diameter node transport type is set to SCTP.</i>

2 Configure the list of peers with which the diameter local node communicates.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name>
```

The following table describes the configurable parameters.

Parameter	Description
node-admin	Specifies the name of the Diameter service profile on the gateway.
peer	Specifies the peer host name. Each peer is configured with a unique peer host name. The peer host name is used by the MCC for load balancing between peers and for statistics.

Parameter	Description
admin-state	Specifies the administrative state of the diameter peer. When enabled, the system attempts to establish peer connection. ■ enabled (default) ■ disabled
backup-peer-name	Identifies the secondary peer used during a failover from the primary to the secondary peer. <i>Note: Only primary peers can be configured with a backup peer. If failover to a secondary peer is supported, this parameter must be configured with the host name of a secondary peer.</i>
override-dest-hostname	Specifies the host name used to override the configured peer host name in the Destination-Host AVP within CCR-I messages. <i>Note: For more information, see Overriding the Destination-Host and Destination-Realm Static Configuration (page 1-114).</i>
peer-identifier	Specifies the external identifier for this peer.
dest-port	Specifies the signaling port number.
dest-ip-address	Specifies the IP address of the peer (can be an IPv4 or IPv6 address).
dest-realm	Specifies the peer realm name. Realm is used in the Destination-Realm AVP in diameter requests.
origin-host	Specifies the local host name in Fully Qualified Domain Name (FQDN). The value is used in the Origin-Host AVP in the request message. See Configuring Multiple Pairs of Primary and Secondary Peers (page 1-139) for more information. <i>Note: The origin-host configured at the peer level overrides the origin-host configured at the Diameter server-group level.</i>
origin-realm	Specifies the local realm name. The value is used in the Origin-Realm AVP in the request message. See Configuring Multiple Pairs of Primary and Secondary Peers (page 1-139) for more information. <i>Note: The origin-realm configured at the peer level overrides the origin-realm configured at the Diameter server-group level.</i>
role	Specifies the type of peer. When multiple primary, active peers are configured, the system uses a round-robin approach to select peers. ■ Primary – (default) Active peers used for diameter sessions. ■ Secondary – Used for diameter sessions, only when the primary peers are down. <i>Note: If failover to a secondary peer is supported, you must enter the host name of the secondary server in the <code>backup-peer-name</code> parameter.</i>

Parameter	Description
secondary-loopback-ip	Specifies a secondary loopback IP address for the Diameter peer to support SCTP multi-homing. See Configuring an SCTP Profile (page 1-88) for more information. <i>Note: The <code>secondary-loopback-ip</code> parameter is only visible when the local Diameter node transport type is set to SCTP.</i>
weight	Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions. See Weighted Round-Robin Distribution to Multiple Diameter Peers (page 1-100) for more information. <i>Note: Applies to Gy/Gx/Gz/STa/Swm/S6b interfaces.</i>

- 3 Configure the data record templates to filter optional diameter AVPs in CCR, CCA and RAA messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] pcrf diameter [start|stop|interim|cca|raa] <template-name>
<attribute-name> presentFlag [yes|no|conditional]
```

where

Parameter	Description
[pgw ggsn]	Specifies the gateway type for the PCRF charging interface.
[start stop interim cca raa]	Specifies the type of data record template.
<template-name>	Specifies the name of the data record template.
<attribute-name>	Specifies the name of the AVP field you are modifying.
presentFlag	CCR Start/Stop/Interim and RAA templates: ■ yes – Indicates the field is sent. ■ no – Indicates the field is not sent. ■ conditional – Indicates the field is sent if it is available. CCA template (initial/update/final templates): ■ yes – Indicates the field is processed when received in the CCA response. ■ no – Indicates the field is ignored when received in the CCA response.

Use the following CLI command to list the configurable AVPs in the CCR, CCA and RAA data record templates.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] pcrf diameter [start|stop|interim|cca|raa] <template-name>
?
```

4 Configure the PCRF charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name>
```

where **<interface-name>** specifies the PCRF charging interface name.

The following table describes the configurable parameters.

Parameter	Description
admin-state	Specifies the administrative state of the PCRF charging interface. ■ enabled (default) ■ disabled
allow-crbn-dataprof-based-subscriber-analyzer-reselection	Specifies whether the MCC gateway should use the CRBN AVP received from the PCRF as a subscriber analyzer key to select a new subscriber analyzer. See Using the CRBN AVP to Modify Offline-Charging Interface Parameters (page 1-69) for more information. ■ disabled (default) ■ enabled
allow-dest-host-realm-change [cca-i cca-u rar]	Specifies whether the Destination-Host/Destination-Realm AVPs are updated in CCR-U/CCR-T messages based on information received in the CCA-I/CCA-U/RAR messages. The default is enabled for the CCA-I/CCA-U messages and disabled for the RAR message. ■ enabled – If the received Origin Host/Origin Realm in the CCA-I/CCA-U/RAR is different from the current destination host and realm, the MCC uses the Origin-Host/Origin-Realm AVPs received in the CCA-I/CCA-U/RAR message to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages. ■ disabled – The MCC gateway continues to use the existing values to fill the Destination-Host/Destination-Realm AVPs in subsequent CCR-U/CCR-T messages and does not update the Destination-Host/Destination-Realm AVPs with the values received in the CCA-I/CCA-U/RAR messages. <i>Note:</i> The MCC gateway always sends the RAA message with the Destination Host and Realm values of the Origin Host and Realm received in the RAR message. <i>Note:</i> For the CCA-I/CCA-U, the MCC will only update the Destination-Host and Destination-Realm in CCR-U/CCR-T messages if the result code received in the CCA-I/CCA-U is DIAMETER_SUCCESS (2001).

Parameter	Description
apn-reporting	<i>Applies to the PGW only.</i> Specifies the Access Point Names (APNs) to be reported over the Gx interface. ■ original – Initial APN. ■ mapped – Final/current APN. ■ prior-mapped – APN mapped prior to the mapped (final/current) APN. This setting allows APN Reporting to continue to be based on the previously-mapped APN (even following a remap to a final APN). The apn-reporting value configured at the interface level overrides the apn-reporting value configured for the Control-Profile and is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages to external devices. If the apn-reporting value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the apn-reporting value configured for the Control Profile. For more information, see Configuring APN Reporting at the Interface Level (page 1-97). <i>Note: The interface-level apn-reporting parameter does not support the custom option.</i>
avp-customization user-equipment-info- imeisv allow-zero-value	Specifies whether to enable the MCC to send the IMEISV with a value of 0 within the User-Equipment-Info AVP. <i>Applies to the SGW and PGW only.</i> ■ enabled – If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC sends the IMEISV with a value of 0 within the User-Equipment-Info AVP on the Gx interface. ■ disabled – (default) If the IMEI/IMEISV received in the Create Session Request has a value of 0, the MCC does not send the IMEISV within the User-Equipment-Info AVP on the Gx interface.
cer-auth-application- id	Specifies the Auth-Application-ID to be used in Capabilities-Exchange-Request (CER) messages within the Vendor-Specific-Application-Id AVP. The 3GPP Gx Auth-Application-Id (16777238) is mandatory in CER messages. This parameter is used to add an additional Auth-Application-Id to be used within the Vendor-Specific-Application-Id AVP in CER messages.
crbn-mode	Specifies how the MCC should interpret the Charging-Rule-Base-Name AVP received on the Gx interface. See Activating/Deactivating Multiple Predefined Groups of PCC Rules (page 1-77).

Parameter	Description
default-bearer-qos-change-indication	Specifies whether to disarm the DEFAULT_EPS_BEARER_QOS_CHANGE (change in QCI/ARP) event-trigger in Gx CCR-U messages. ■ enabled – (default) When the corresponding event occurs, such as a change in QCI or ARP, the PGW reports the event in a CCR-U to the PCRF, even if the event-trigger has not been provisioned by the PCRF in the CCA-I. ■ disabled – When the corresponding event occurs, such as a change in QCI or ARP, the PGW only reports the event in a CCR-U to the PCRF if the event-trigger has been provisioned by the PCRF in the CCA-I.
default-gx-cause-code-list	Specifies the name of the result code list. <i>Note: The result code list must already exist. For more information, see Configuring the Diameter Result Code List (page 1-135).</i>
default-gx-setup-per-cause-code	Specifies whether to enable or disable session set up using the default Gx policy based on the result code received in the CCA error response. ■ enabled – The result code returned in the CCA error message is compared to the configured result code list, and if there is a match, the session is set up using the default Gx policy. ■ disabled – (default) The result code list functionality is not used to set up the session with the default Gx policy. However, if the result code returned in the CCA error message is a 3xxx error, the default Gx policy is used to set up the session. <i>Note: Applicable only when a result code list is configured. For more information, see Configuring the Diameter Result Code List (page 1-135)</i>
diameter-pending-transaction	Specifies whether to process Diameter messages in parallel on the Gx interface. ■ enabled ■ disabled (default) <i>Note: For more information, see Configuring the Diameter Pending Transaction Feature (page 1-76).</i>
override-destination-realm	Specifies the realm name used to override the configured peer realm in the Destination-Realm AVP within CCR-I messages. The override-destination-realm only applies to the Destination-Realm AVP in CCR-I messages. <i>Note: For more information, see Overriding the Destination-Host and Destination-Realm Static Configuration (page 1-114).</i>

Parameter	Description
pcrf-3gpp-release-compliance	Specifies the PCRF's 3GPP TS 29.212 release compliance: ■ release-10 – (default) The PCRF is compliant with 3GPP TS 29.212 Release 10 or earlier. The PCEF receives and reports the USAGE_REPORT event-trigger with code 26. ■ release-10-plus – The PCRF is compliant with 3GPP TS 29.212 Release 11 or later. The PCEF receives and reports the USAGE_REPORT event-trigger with code 33. <i>Note: To ensure the PCEF receives and reports the TAI_CHANGE event-trigger with code 26 and the ECGI_CHANGE event-trigger with code 27, set the PCRF's 3GPP TS 29.212 release compliance to release-10-plus.</i>
pcrf-cca-error-send-terminate	Controls whether the GGSN/PGW sends the PCRF a CCR-T message upon receipt of a CCA-U message with a permanent failure (Result Code 5xxx). ■ true (default) ■ false
pcrf-down-default-gx-call-setup	Specifies whether the Default Gx Policy feature is enabled or disabled. ■ enabled – When no PCRF is available or the PCRF returns a temporary failure (result code 3xxx), the GGSN/PGW sets up new subscriber sessions using the information in the locally configured data profile (from the subscriber analyzer) as the default policy. ■ disabled – (default) When no PCRF is available or the PCRF returns a temporary failure (result code 3xxx), the GGSN/PGW does not set up new subscriber sessions. <i>Note: For more information, see Default Gx Policy (page 1-132).</i> <i>Note: For information on using the default Gx policy to set up sessions based on the received result code, see Configuring the Diameter Result Code List (page 1-135).</i>
pcrf-down-gx-rejoin	Specifies whether the Gx Rejoin/Replay feature is enabled or disabled. ■ enabled – When no PCRF is available or the PCRF returns a temporary failure (result code 3xxx), the GGSN/PGW caches subscriber usage information locally and starts the rejoin timer (pcrf-down-rejoin-time). If the rejoin timer expires, the GGSN/PGW tries to rejoin the PCRF. ■ disabled – (default) The GGSN/PGW does not cache subscriber usage information or try to rejoin the PCRF even if the PCRF comes back up. <i>Note: For more information, see Gx Rejoin and Gx Replay (page 1-133).</i>
pcrf-down-rejoin-time	Specifies the number of seconds to wait before attempting to connect to the non-responsive PCRF.

Parameter	Description
<code>pcrf-down-gx-replay</code>	Specifies whether to enable or disable Gx Replay. ■ enabled – If all PCRF servers go down after the Gx session is already established and the session using the default policy is terminated before the PCRF comes back up, the GGSN/PGW replays the cached session information via a CCR-T message. ■ disabled – (default) If all PCRF servers go down after the Gx session is already established and the session using the default policy is terminated before the PCRF comes back up, session information is not cached and replayed to the PCRF. <i>Note: For information on Gx Replay, see Gx Rejoin and Gx Replay (page 1-133).</i>
<code>pcrf-down-tethering-entitlement</code>	Specifies whether tethering should be applied to a session if the PCRF is down during session setup. ■ enabled – If the PCRF is down during session setup, the MCC enables tethering for the session. ■ disabled – (default) If the PCRF is down during session setup, the MCC does not enable tethering for the session and uses the configured qos-flow. <i>Note: Once tethering is applied to the session and if the PCRF fails during the session, the previously applied tethering entitlement remains valid.</i> If tethering is disabled in the middle of a session, the system disables tethering entitlement for all active connections of the session and marks them as non-tethered-entitlement connections. Also, the system does not increment the following counters: ■ <code>tethering-entitled-uplink-packets/downlink-packets</code> ■ <code>tethering-entitled-uplink-bytes/downlink-bytes</code> For example, if a session initially has five active connections for which tethering has been enabled, the system increments the following counters for each packet: ■ <code>tethering-entitled-uplink-packets/downlink-packets</code> ■ <code>tethering-entitled-downlink-byte/downlink-bytes</code> If tethering is then disabled for the session, the system disables tethering entitlement for all five active connections and stops incrementing the respective counters. <i>Note: The <code>pcrf-down-tethering-entitlement</code> parameter under the APN object takes precedence over the configuration for this parameter. For information on applying tethering to a session for an APN if the PCRF is down during session setup, see the <i>Affirmed Networks Gateway Administration Guide</i>.</i>

Parameter	Description
pcrf-send-usage-update-always	Controls whether the GGSN/PGW (PCEF) sends usage monitoring information without receiving a PCRF trigger. ■ true ■ false (default)
pcrf-update-down-avoid-pcrf	Specifies whether the MCC gateway continues the session without any further messaging (CCR-U/CCR-T) to the PCRF if the CCR-U to the PCRF fails (response times out, PCRF returns an error/Tx error, or peers are down/not available). See Continue Session for CCR-U Failure to PCRF (page 1-149) for more information. ■ true – If the CCR-U to the PCRF fails, the MCC gateway continues the session without any further messaging (CCR-U/CCR-T) to the PCRF. ■ false – (default) If the CCR-U to the PCRF fails, the MCC gateway attempts to retransmit the CCR-U to the PCRF. <i>Note: The configuration for this parameter takes precedence over the configuration for the Gx Rejoin feature. If this parameter is set to true and the CCR-U messaging fails, the MCC gateway session continues the session without any further messaging to the PCRF, even if the Gx Rejoin feature is enabled.</i>
pcrf-override-plmnid	Controls whether the GGSN/PGW overrides the PLMN-ID sent in the GTP-C messages with the PLMN-ID sent from the PCRF in the CCA message. ■ enabled – The PLMN-ID received from the PCRF overrides the PLMN-ID received in GTP-C messages and is used in subsequent messages to the OCS and in local PGW CDRs (CSV and ASN.1 formats) and GTP PGW CDRs (Ga interface). ■ disabled – (default) The GGSN/PGW ignores the PLMN-ID received from the PCRF and uses the PLMN-ID sent in GTP-C messages. <i>Note: For more information, see Overriding the PLMN-ID sent in GTP-C messages (page 1-66).</i> <i>Note: Does not apply to the TWAG.</i>

Parameter	Description
<code>pcrf-send-crr-in-raa</code>	Controls whether the PGW includes the Charging-Rule-Report AVP in the RAA message upon receiving an RAR message from the PCRF containing the Charging-Rule-Install AVP or Charging-Rule-Remove AVP. ■ true – (default) The PGW sends the Charging-Rule-Report AVP in the corresponding RAA message after processing the charging rules. ■ false – The PGW immediately sends the RAA message without the Charging-Rule-Report AVP. The PGW sends the Charging-Rule-Report AVP in a separate CCR-U message after processing the charging rules. <i>Note: In deployments where the Gx interface configuration has this parameter set to <code>true</code> and the MCC detects a CBR/UBR with CSR Collision (inter LTE/Wi-Fi handover), the PGW may send the RAA message without the Charging-Rule-Report AVP. In this case, the PGW sends the Charging-Rule-Report AVP in a separate CCR-U message after the bearer operation is completed post handover.</i> <i>Note: Does not apply to the TWAG.</i>
<code>qos-change-indication</code>	Specifies whether to disarm the QOS_CHANGE (change in APN-AMBR) event-trigger in Gx CCR-U messages. ■ enabled – (default) When the corresponding event occurs, such as a change in APN-AMBR, the PGW reports the event in a CCR-U to the PCRF, even if the event-trigger has not been provisioned by the PCRF in the CCA-I. ■ disabled – When the corresponding event occurs, such as a change in APN-AMBR, the PGW only reports the event in a CCR-U to the PCRF if the event-trigger has been provisioned by the PCRF in the CCA-I.
<code>retransmit-count</code>	Specifies the maximum number of times the gateway transmits the CCR to the PCRF.
<code>retransmit-timeout</code>	Specifies the time interval, in milliseconds, before the system retransmits the CCR to the PCRF.
<code>send-qos-info-in-all-ccr</code>	Controls whether to include the QoS-Information AVP in all CCR messages (Initial/Update/Terminate) towards the PCRF. ■ enabled – The MCC gateway includes the QoS-Information AVP in all CCR messages (Initial/Update/Terminate) towards the PCRF. ■ disabled – (default) The MCC gateway includes the QoS-Information AVP in the CCR-I message towards the PCRF. <i>Note: To use this feature, ensure the QoS-Information AVP <code>presentFlag</code> setting is set to <code>yes</code> or <code>conditional</code> in Gx CCR data record templates (start/interim/stop).</i>

Parameter	Description
subscriptionid-imsi-format-non-3gpp-access	Specifies the IMSI format sent in the Subscription-Id AVP for non-3GPP access. ■ imsiformat (default) ■ imsinaiprefix0format <i>Note: Applies to Gx and Gy interfaces only.</i>
suppress-invalid-uli	Specifies whether to suppress the User Location Information (ULI) on the Gx interface if the ULI received from the MME/SGSN is invalid. ■ true ■ false (default) <i>Note: ULI changes from valid to invalid and vice versa are considered valid ULI changes and are reported for offline/PCRF/OCS event-triggers.</i>
user-name-type	Specifies the User-Name AVP format used in CCR messages on the Gx interface if not present in the PCO. This setting overrides the user-name-type present at the APN level. However, if the Gx interface user-name-type is set to none, the APN level user-name-type is used. ■ apn ■ default-username ■ imsi ■ imsi-nai-no-prefix ■ imsi-nai-prefix-0 ■ imsi-plus-apn ■ imsi-plus-apnnionly ■ msisdn ■ msisdn-plus-apn ■ msisdn-plus-apnnionly ■ none

- 5 Associate the diameter server group and CCR/CCA/RAA data record templates with the PCRF charging interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> [pgw|ggsn] diameter server-group
<service-profile> start-template <template-name> interim-template
<template-name> stop-template <template-name> cca-initial
<template-name> cca-update <template-name> cca-final <template-name>
raa-template <template-name>
```

where

Parameter	Description
<interface-name>	Specifies the PCRF charging interface profile name.
[pgw ggsn]	Specifies the gateway type for the PCRF charging interface. <i>Note: The configuration model for the Gxa interface follows the model used by the Gx interface. Therefore, the Gxa configuration uses the PGW configuration object.</i>
server-group	Specifies the Service Profile for the selected streaming protocol.
start-template	Specifies the CCR Start template.
interim-template	Specifies the CCR Interim template.
stop-template	Specifies the CCR Stop template.
cca-initial	Specifies the CCA Initial template.
cca-update	Specifies the CCA Update template.
cca-final	Specifies the CCA Final template.
raa-template	Specifies the RAA template.

6 Associate the PCRF charging interface with a workflow control profile.

Note: You must configure a default QoS profile and charging instance for the control profile. For more information on configuring these components, see the *Affirmed Networks Gateway Administration Guide*.

```
an3000-mcm-slot7cpu1(config)# workflow control-profile
<control-profile-name> default-pcrf-interface <interface-name>
default-qos-profile <qos-profile-name>
default-home-pre-paid-charging-instance <pre-paid-charging-instance>
```

where

Parameter	Description
control-profile	Specifies the workflow control profile name.
default-pcrf-interface	Specifies the PCRF charging interface name.
default-qos-profile	The default QoS Profile to use if the user equipment does not request a particular bandwidth. The QoS Profile must already exist.
default-home-pre-paid-charging-instance	Default prepaid charging service for the home PLMN. The charging instance must already exist.

7 Associate the control and data profiles with a subscriber analyzer.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer
<subscriber-analyzer-name> priority <priority-number> key <key-name>
apn-name <apn-name> event <bearer-setup>; action data profile
<data-profile-name> control-profile <control-profile-name>
```

where

Parameter	Description
subscriber-analyzer	Specifies the subscriber analyzer name.
priority	Specifies the priority.
key	Specifies the key name.
apn-name	Specifies the Access Point Name.
event	Specifies whether to associate the workflow with bearer-setup.
data profile	Specifies the workflow data profile. The workflow data profile must already exist.

Parameter	Description
<code>control-profile</code>	Specifies the workflow control profile identifying the external identity providing the workflow data profile. The workflow control profile must already exist.

Optional Gx/Gxa Interface Configuration Tasks

This section describes optional Gx/Gxa interface configuration tasks.

[Mapping to a Virtual APN via the PCRF \(page 1-60\)](#)

[Configuring Geo-Redundancy Health Checks on the Gx Interface \(page 1-61\)](#)

[Configuring DSCP Values for Control Plane Traffic \(page 1-65\)](#)

[Overriding the PLMN-ID sent in GTP-C messages \(page 1-66\)](#)

[Switching the Control/Data Profile Mid-Session \(page 1-67\)](#)

[Using the CRBN AVP to Select a New Control Profile \(page 1-68\)](#)

[Using the CRBN AVP to Modify Offline-Charging Interface Parameters \(page 1-69\)](#)

[Override Control \(page 1-74\)](#)

[Configuring the Diameter Pending Transaction Feature \(page 1-76\)](#)

[Activating/Deactivating Multiple Predefined Groups of PCC Rules \(page 1-77\)](#)

[Activating/Deactivating Metering Services for Rating Groups and Service Rules Associated with a CRBN Profile \(page 1-80\)](#)

[Activating/Deactivating Multiple Predefined Service Rules with a Single Charging-Rule-Name \(page 1-82\)](#)

[Reporting NetLoc and RAN-NAS Cause Indication to the PCRF \(page 1-83\)](#)

[Modifying Default Diameter AVP Flags \(page 1-84\)](#)

[Notifying the PCRF When an OCS Failure Occurs \(page 1-85\)](#)

[Configuring an SCTP Profile \(page 1-88\)](#)

[Redirecting the First HTTP Request of a Session \(page 1-90\)](#)

[Activating a URL Whitelist via the Gxa Interface \(page 1-91\)](#)

[Appending the Original URL to the Redirect URL for Gxa Sessions \(page 1-96\)](#)

[Customizing the APN Reported on External Interfaces \(page 1-96\)](#)

[Configuring APN Reporting at the Interface Level \(page 1-97\)](#)

[Enabling 5G NSA QoS AVPs \(page 1-98\)](#)

[Weighted Round-Robin Distribution to Multiple Diameter Peers \(page 1-100\)](#)

[Application Detection and Control \(page 1-101\)](#)

[Moving Predefined Service Rules Across Predefined Rating Groups \(page 1-106\)](#)

[Configuring Rating Sub-Groups \(page 1-111\)](#)

[Configuring the Default Charging Method \(page 1-112\)](#)

[Overriding the Destination-Host and Destination-Realm Static Configuration \(page 1-114\)](#)

[Processing the Gx-TMO-Redirect-Server VSA \(page 1-115\)](#)

[Retrieving Flow Information From a Manifest Server \(page 1-116\)](#)

[Overriding the CC Value Received in the Create Session Request \(page 1-123\)](#)

[Configuring a Dynamic Rating Group Template \(page 1-124\)](#)

Mapping to a Virtual APN via the PCRF

MCC supports the ability to map from a single common Access Point Name (APN) to an individual specific virtual APN using information received from the PCRF. When the PGW/GGSN receives a CCA message from the PCRF containing the Mobile Virtual Network Operator (MVNO) Reseller ID and/or the MVNO Sub Class ID vendor-specific attributes (VSAs), the PGW/GGSN uses these attributes (as subscriber analyzer keys) to map the subscriber session to the specified mapped virtual APN. In doing so, the subscriber session uses the workflow control profile parameters defined for the mapped virtual APN. This feature enables Operators to provide different treatments for subscribers that share a common APN.

Two configurable subscriber analyzer keys are used to match the MVNO VSAs received from the PCRF to a subscriber analyzer.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer  
<subscriber-analyzer-name> key [mvno-identifier|mvno-sub-identifier]  
<mvno-name>
```

Configuring Geo-Redundancy Health Checks on the Gx Interface

Geographic Redundancy network health checks are supported on RADIUS, Gx, Gy and S6b interfaces and use both shared loopback IPs and private loopback IPs. Operators must enable health-check monitoring on at least one interface.

Note: It is recommended that you use the Geographic Redundancy Manager application, in the Acuitas Service Management System, to configure the Geographic Redundancy group, service endpoint, and server parameters ([step 1](#) through [step 3](#)). For more information on configuring Geographic Redundancy, see the Acuitas User's Guide.

To configure Geographic Redundancy network health checks on the Gx interface:

- 1 Configure the geographic redundancy group.

```
an3000-mcm-slot8cpu1(config)# geographic-redundancy group
<geo-redundancy-group-name> admin-state [enabled|disabled] group-id
<id> manual-state [active|standby] mode
[hot-auto|hot-manual|warm-manual] role [primary|secondary]
```

where

Parameter	Description
geographic-redundancy group	Specifies a Geographic Redundancy group name that is unique across the network.
admin-state	Specifies the administrative state: ■ enabled – Initializes and starts the redundancy mechanism (such as synchronization in hot mode). ■ disabled – Invalidates any synchronized data and sets the mechanism into a non-ready state.
group-id	Specifies the unique group ID that matches with the peer.
manual-state	Specifies the manually configured HA state (the Operator current desired state). ■ active ■ standby
mode	Specifies the instance configured HA mode. ■ warm-manual – No session synchronization occurs between nodes, failovers are Operator controlled. ■ hot-auto – Synchronization of per-session data occurs between nodes, failovers are automatically triggered based on network health monitoring results. ■ hot-manual – Synchronization of per-session data occurs.

Parameter	Description
role	Specifies the instance configured HA backup role. ■ primary – Specifies which node is “active” under normal network conditions. ■ secondary – Specifies which node is “standby” under normal network conditions.

2 Configure the service endpoint for the Geographic Redundancy server.

```
an3000-mcm-slot8cpu1(config)# network-context <network-context-name>
service-endpoint <service-endpoint-name> local-loopback-address
<ip-address> local-port-range-base <port> local-port-range-numports
<value>
```

where

Parameter	Description
network-context	Specifies a network context containing this Geographic Redundancy service endpoint.
service-endpoint	Specifies the name of the Geographic Redundancy service endpoint.
local-loopback-address	Specifies the local node address used for communication between primary and secondary nodes.
local-port-range-base	Specifies the base UDP port number used for communication between primary and secondary nodes. Must be the initial value of a contiguous block of UDP ports.
local-port-range-numports	Specifies the size of the contiguous block of ports.

3 Configure the Geographic Redundancy server parameters.

```
an3000-mcm-slot8cpu1(config)# service-construct geo-redundancy-server
<geo-redundancy-group-name> network-context <network-context-name>
service-endpoint <service-endpoint-name> remote-ip-address
<ip-address> remote-base-port <port> connectivity-monitoring-params
[monitor-AAA|monitor-OCS|monitor-PCRF|monitor-S6B] admin-state
[enabled|disabled]
```

where

Parameter	Description
geo-redundancy-server	Specifies the Geographic Redundancy group name to which this Geographic Redundancy server belongs.
network-context	Specifies the network context containing this Geographic Redundancy service endpoint.

Parameter	Description
service-endpoint	Specifies the service endpoint for this Geographic Redundancy server.
remote-ip-address	Specifies the IP address for the remote peer in the group.
remote-base-port	Specifies the port number for the remote peer in the group
connectivity-monitoring-params	Specifies network connectivity monitoring for automatic failover. Configure the parameters as follows: ■ monitor-AAA – Set to enabled to monitor the RADIUS Accounting Authentication Authorization (AAA) interface to govern fault detection and automatic failover. ■ monitor-PCRF – Set to enabled to monitor the Gx interface to govern fault detection and automatic failover. ■ monitor-OCS – Set to enabled to monitor the Gy interface to govern fault detection and automatic failover. ■ monitor-S6B – Set to enabled to monitor the S6B interface to govern fault detection and automatic failover.

- 4 Associate the Geographic Redundancy group with the local diameter node associated with the interface and configure health-check parameters on the interface.

```
an3000-mcm-slot8cpu1 (config) # service-construct interface
aaa-interface-group diameter node-admin <diameter-node-name>
geo-redundant-group <geo-redundancy-group-name>
net-health-report-enabled [true|false] link-type [private|shared]
```

where

Parameter	Description
node-admin	Specifies the local diameter node associated with the interface. The local diameter node must already exist. See Sending Policy and Charging Records to the PCRF (page 1-42) .
geo-redundant-group	Specifies the Geographic Redundancy group name.
net-health-report-enabled	Specifies whether to monitor network health towards the interface: ■ true ■ false
link-type	Specifies whether the link used to monitor neighboring network elements is a private (local) or shared IP address among the primary and secondary network elements: ■ private ■ shared

- 5 Specify a default Gx interface in the workflow control profile.

```
an3000-mcm-slot8cpu1(config)# workflow control-profile <profile-name>  
default-pcrf-interface <interface-name>
```

Note: The interface must already exist. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#).

- 6 Use the following command to view the parameters of the diameter node:

```
an3000-mcm-slot8cpu1# show service-construct aaa-interface-group  
diameter node-admin <diameter-node-name>
```

Configuring DSCP Values for Control Plane Traffic

Operators can assign different levels of service to control plane traffic on Diameter interfaces using Differentiated Services Code Point (DSCP).

Diameter interfaces (S6b, Gx/Gxa, Gy, Gz/Rf) – DSCP value configured at the Diameter server group level.

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
aaa-interface-group diameter node-admin <name> dscp-value <value>
```

Operators can also configure a DSCP value at the network-context level applicable to end marker packets, error indication messages, DNS queries and GTP-U Echo messages.

```
an3000-mcm-slot7cpu1(config)# network-context <name>  
control-dscp-value <value>
```

Note: The DSCP value configured at the Diameter server group level overrides the network-context DSCP value.

Overriding the PLMN-ID sent in GTP-C messages

Operators can configure the MCC GGSN/PGW to override the PLMN-ID sent in the GTP-C messages on the S4/S11/S5/S8/Gn/Gp interfaces. If the override PLMN-ID feature is enabled and the PCRF sends the CCA message containing the PLMN-ID within the 3GPP-SGSN-MCC-MNC AVP, the GGSN/PGW replaces the PLMN-ID received in GTP-C messages with the PLMN-ID received from the PCRF.

The GGSN/PGW uses the PLMN-ID received from the PCRF in subsequent messages to the OCS and in local PGW CDRs (CSV and ASN.1 formats) and GTPP CDRs (Ga interface). Furthermore, the PLMN-ID received from the PCRF is also used if no PLMN-ID is received from MME/SGSN.

Operators can enable/disable the override PLMN-ID feature at the PCRF interface level using the following CLI command:

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
pcrf-charging <name> pcrf-override-plmnid [enabled|disabled]
```

- **enabled** – The PLMN-ID received from the PCRF overrides the PLMN-ID received in GTP-C messages and is used in subsequent messages to the OCS and in local PGW CDRs (CSV and ASN.1 formats) and GTPP PGW CDRs (Ga interface).
- **disabled** – (default) The GGSN/PGW ignores the PLMN-ID received from PCRF and uses the PLMN-ID sent in GTP-C messages.

Switching the Control/Data Profile Mid-Session

When the PGW/GGSN receives an RAR message from the PCRF containing the MVNO Reseller Id and/or the MVNO Sub Class Id VSAs, the PGW/GGSN uses these VSAs (as keys) to choose a new subscriber analyzer. In doing so, these subscriber sessions use the workflow control profile parameters defined for the new subscriber analyzer, including APN mapping, if required. In addition, if the data profile configured for the new subscriber analyzer is different from the one currently used, the data profile is switched. Once the profile change occurs and if online charging is used, the PGW/GGSN includes the MVNO Reseller Id and/or the MVNO Sub Class Id VSAs in subsequent CCR messages with the OCS.

The data profile can also be switched mid-session when the PGW/GGSN receives an RAR message from the PCRF containing the Charging-Rule-Base-Name AVP within the Charging-Rule-Install AVP. However, if the RAR contains both the MVNO VSAs and Charging-Rule-Base-Name AVP, the PGW/GGSN changes the data profile using the information sent in the Charging-Rule-Base-Name AVP.

- 1 Configure the subscriber analyzer keys used to match the MVNO VSAs received from the PCRF to a subscriber analyzer.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer
<subscriber-analyzer-name> key [mvno-identifier|mvno-sub-identifier]
<mvno-name>
```

- 2 Enable the MVNO Reseller Id and/or the MVNO Sub Class Id VSAs in CCR messages with the OCS.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pgw ocs diameter [start|interim|stop|event] <template-name>
[affmnvostringid|affmvnosubclassid] presentFlag yes
```

- MVNO Reseller Id = `affmnvostringid`
- MVNO Sub Class Id = `affmvnosubclassid`

- 3 Specify whether to update the prior mapped APN when APN mapping is performed.

```
an3000-mcm-slot7cpu1(config)# workflow control-profile
<control-profile-name> update-prior-mapped-apn [true|false]
```

- `true` – (default) If APN mapping is configured on this control profile, the APN mapped prior to the current APN is populated in the prior mapped APN.
- `false` – If APN mapping is configured on this control profile, the APN mapped prior to the current APN is not populated in the prior mapped APN.

Note: The default Offline/Online/RADIUS interfaces are not switched when there is a mid-session change of the workflow control profile based on information received in the RAR message from the PCRF.

Using the CRBN AVP to Select a New Control Profile

You can use the Charging-Rule-Base-Name (CRBN) AVP from the PCRF to select a new control profile and online/offline interface. When the MCC gateway receives a CCA-I message from the PCRF containing the CRBN AVP, the MCC gateway can use the CRBN as a subscriber analyzer key to select a new subscriber analyzer. In doing so, the subscriber sessions use the control profile defined for the new subscriber analyzer.

When a new control profile is selected using this method, the new control profile can only be used to make the following online/offline interface changes:

- Online – Switch from a control profile with no online interface to a control profile with an online interface.
- Offline – Switch to a control profile with a different offline interface. Only the following parameters can differ between the old offline interface and the new offline interface.
 - data-volume-limit
 - interim-update-interval
 - cdr-close-time

Set parameters for this feature using the following commands.

- 1 Ensure the `cbrn-mode` parameter for the `pcrf-charging` interface is set to `data-profile`.
- 2 Enable the MCC gateway to use the CRBN AVP received from the PCRF as a key to select a new subscriber analyzer.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> cbrn-mode data-profile
```

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name>
allow-cbrn-dataprof-based-subscriber-analyzer-reselection enabled
```

- 3 Configure a subscriber analyzer with a list of CRBNs (locally configured data-profiles) as one of the key attributes and associate a control profile with this subscriber analyzer. The control profile must already exist.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer
<subscriber-analyzer-name> analyzer-type workflow-profile action
control-profile <control-profile-name> key <key-name>
cbrn-dataprof-name-list values <list-of-data-profiles> match-type
[match-any|match-none]
```

Using the CRBN AVP to Modify Offline-Charging Interface Parameters

This feature adds the ability to modify the following offline-charging interface parameters at any time during a session for certain subscribers within a selected control profile using the Charging-Rule-Base-Name (CRBN) AVP received from the PCRF.

- `data-volume-limit`
- `interim-update-interval`
- `cdr-close-time`

With this feature, Operators configure a Workflow Override Profile. The Override Profile defines the override values for each offline-charging interface. The Override Profile gets associated with a subscriber analyzer, which can be configured with a list of CRBNs (data-profiles) as one of the key attributes and a Control Profile.

When this feature is enabled on the `pcrf-charging` interface and the PCRF sends a CCA-I, CCA-U, or RAR containing the CRBN AVP, the MCC gateway uses the value received in the CRBN AVP as a key to select a subscriber analyzer.

- If the new subscriber analyzer is different from the existing subscriber analyzer and has an Override Profile associated with it, the MCC gateway overrides the offline-charging parameters with the values defined in the Override Profile while keeping the same control profile.
- If there is no difference between the new and existing subscriber analyzer, the existing offline-charging threshold values remain in use for the session. Once the offline-charging threshold values have been updated, the MCC gateway does not revert back to the old values.
- If the existing subscriber analyzer has an Override Profile associated with it and the new subscriber analyzer does not, the MCC gateway reverts back to the values configured in the corresponding offline-charging interface.

Set parameters for this feature using the following commands:

- 1 Ensure the `cbrn-mode` parameter for the `pcrf-charging` interface is set to `data-profile`.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> cbrn-mode data-profile
```

- 2 Specify whether the MCC gateway should use the CRBN AVP received from the PCRF as a key to select a new subscriber analyzer.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name>
allow-cbrn-dataprof-based-subscriber-analyzer-reselection
[disabled|enabled]
```

where

Parameter	Description
pcrf-charging	Specifies the PCRF charging (Gx) interface name.
allow-crbn-dataprof-based-subscriber-analyzer-reselection	Specifies whether the MCC gateway should use the CRBN AVP received from the PCRF as a subscriber analyzer key to select a new subscriber analyzer. ■ disabled (default) ■ enabled

- 3 Configure the Workflow Override Profile to define the override values for each offline-charging interface.

```
an3000-mcm-slot7cpu1(config)# workflow override-profile
<override-profile-name> control-profile offline-interface
[local|gtp|diameter] [data-volume-limit|interim-update-interval|
cdr-close-time] <value>
```

where

Parameter	Description
override-profile	Specifies the Override Profile name.
offline-interface	Specifies the offline-charging interface type. ■ local – Configure override parameter values for the local offline-charging interface. ■ gtp – Configure override parameter values for the Ga interface. ■ diameter – Configure override parameter values for the Gz/Rf interface.

Parameter	Description
<code>[data-volume-limit interim-update-interval cdr-close-time]</code>	Defines the override values for each offline-charging interface. ■ data-volume-limit – Specifies the maximum number of kilobytes that the gateway maintains across all service containers (for a bearer) before generating a new CDR. The volume usage counts are reset when a CDR is closed and sent. By default, this parameter is not configured. A value of 0 indicates disabled. When this value is updated with the override value for an offline-charging interface and if the volume threshold changes, CDRs for the subscriber sessions are closed with the Management Intervention (20) cause code. ■ interim-update-interval – Specifies the time interval, in seconds, after which a new service container or CDR is generated. The timer resets when a CDR is sent. When this time interval expires, the MCC reports the current usage statistics (up to when the timer expired) in the CDR. By default, this parameter is not configured. A value of 0 indicates disabled. When this value is updated with an override value for an offline-charging interface and if the timer is already running, CDRs for the subscriber sessions are closed with the Management Intervention (20) cause code and the timer resets. ■ cdr-close-time – If <code>interim-updates</code> are enabled on the interface, specifies the number of seconds for storing closed service containers before transferring service containers in an interim CDR to the offline server. By default, this parameter is not configured. A value of 0 indicates disabled. When this value is updated with an override value for an offline-charging interface, the new value takes effect immediately. The <code>cdr-close-time</code> timer is canceled (if it was already running) and is restarted with the updated value.

- 4 Configure a subscriber analyzer with a list of CRBNs (data-profiles) as one of the key attributes and associate the Override Profile with this subscriber analyzer.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer
<subscriber-analyzer-name> action control-profile
<control-profile-name> action override-profile <override-profile-name>
key <key-name> crbn-dataprof-name-list values <list-of-data-profiles>
match-type [match-any|match-none]
```

where

Parameter	Description
subscriber-analyzer	Specifies the subscriber analyzer name.
control-profile	Specifies the Control Profile name to associate with this subscriber analyzer.

Parameter	Description
override-profile	Specifies the Override Profile name to associate with this subscriber analyzer. <i>Note: The override-profile parameter is only configurable if a Control Profile is specified.</i>
key	Specifies the name of the subscriber analyzer key list.
crbn-dataprof-name-list	Defines a list of locally configured Data Profiles. ■ values – List of CRBNs (data profile) names. ■ match-type – Matching type for the list. – match-any – match-none

Message Flows for Using the Override Profile

This section provides examples of message flows when using the Override Profile to modify offline-charging parameters.

Figure 1-14 illustrates the message flow for overriding offline-charging parameters when provisioned by the PCRF in the CCA-I.

Figure 1-14 Overriding Offline-Charging Parameters (through CCA-I)

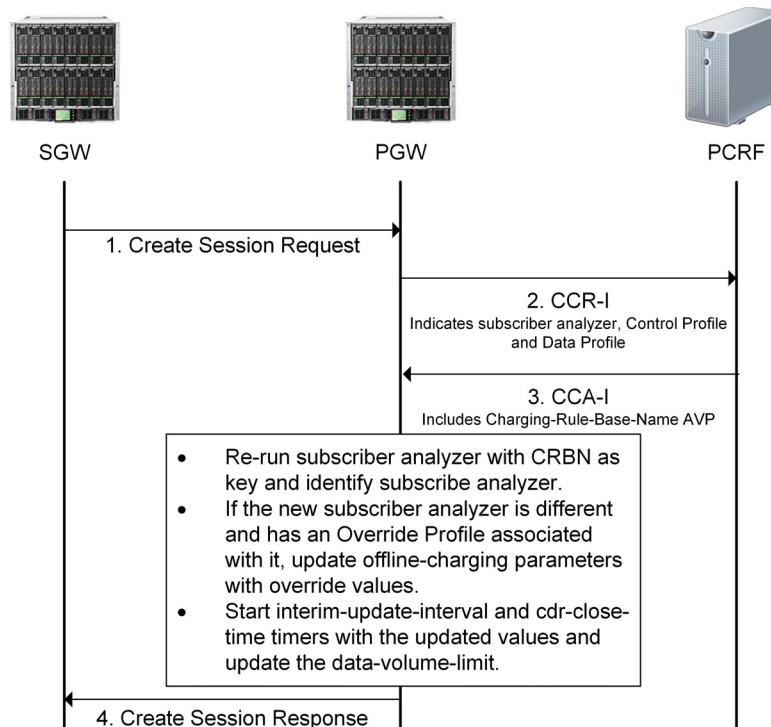
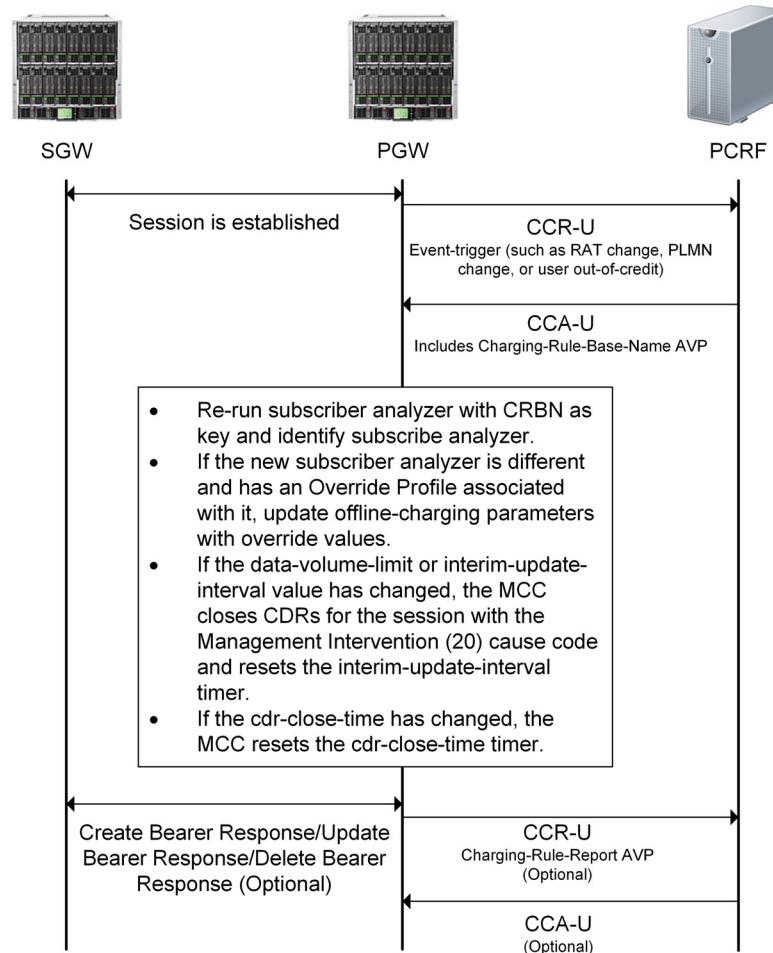


Figure 1-15 illustrates the message flow for overriding offline-charging parameters when provisioned by the PCRF in the CCA-U. While Figure 1-15 illustrates an example for overriding offline-charging parameters through the CCA-U, the message flow remains the same when overriding offline-charging parameters through the RAR.

Figure 1-15 Overriding Offline-Charging Parameters (through CCA-U)



Override Control

The Override-Control feature adds the ability to override pre-configured charging parameters for static (service rules with `always-on` activation) and predefined (service rules with `external-activation`) service rules using the Override-Control Vendor-Specific Attribute (VSA) sent in CCA-I/CCA-U and RAR messages from the PCRF. The Override-Control feature provides the ability to dynamically override the following charging parameters based on information sent from the PCRF.

- Change the rating group for all static/predefined service rules associated with a PDN session. The rating group identifier sent in the Override-Control VSA must correspond to a predefined rating group quota-id on the MCC (`services` → `charging` → `rating-group` → `quota-id`).
- Change the Online value (enable/disable) configured for a static/predefined rule. If the Online value is disabled for a service rule associated with an online rating group, the service rule usage is not used towards online charging.
- Change the Offline value configured for a static rule. If the Offline value is disabled for a service rule associated with an online rating group, the service rule usage is not counted towards offline charging.

If the override rating group received from the PCRF points to an offline rating group, all service rules (static/predefined) are associated with that offline rating group. In this case, changing the Online charging properties (enable/disable) for a service rule associated with the override offline rating group is not applicable because offline rating groups do not use the OCS.

The override charging parameters applies to all static/predefined rules for a PDN session except for the service rules specified in the Override-Charging-Action-Exclude-Rule VSA. Service rules that are specified in the Override-Charging-Action-Exclude-Rule VSA will still be associated with the rating group based on the workflow configuration. The Override-Charging-Action-Exclude-Rule VSA may be included more than once if more than one rule needs to be excluded.

When override-control is enabled, the PGW continues to associate the service rule to the QoS and metering services based on the predefined configuration.

The Override-Control VSA is grouped and the detailed syntax is as follows:

Name	Vendor ID	AVP Code	Format
[Override-Control]	9	132017	Grouped
[Override-Charging-Action-Parameters]	9	132019	Grouped
*[Override-Charging-Action-Exclude-Rule]	9	132021	Octet String
[Override-Charging-Parameters]	9	132022	Grouped
[Override-Rating-Group]	9	132024	Unsigned32
[Override-Online]	9	132026	Enumerated
[Override-Offline]	9	132027	Enumerated

Configuring the Diameter Pending Transaction Feature

You can configure the MCC GGSN/PGW to process Diameter messages in parallel on the Gx interface to avoid possible race conditions in which the PCEF (GGSN/PGW) receives a PCRF-initiated message (for example, RAR) while a PCEF-initiated message (for example, CCR-U) is still pending a response (CCA-U) from the PCRF.

When the Pending Transaction feature is enabled on the Gx interface, the GGSN/PGW informs the PCRF that it supports the Diameter Pending Transaction feature by including the Supported-Features AVP in the CCR-I message.

In the case where the GGSN/PGW receives an RAR while waiting for a CCA-U, the GGSN/PGW immediately responds with an RAA with the Experimental-Result-Code AVP set to `DIAMETER_PENDING_TRANSACTION`.

In the case where the GGSN/PGW receives a CCA-U with the Experimental-Result-Code AVP set to `DIAMETER_PENDING_TRANSACTION` from the PCRF, the GGSN/PGW immediately sends another CCR to the same peer and increments the `retransmit-count`. If the `retransmit-count` reaches the maximum configured value, the GGSN/PGW rejects the session.

Use the following CLI command to enable the Diameter Pending Transaction feature. By default, this feature is disabled.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> diameter-pending-transaction
[enabled|disabled]
```

Added statistics to track the number of pending transactions sent/received by the GGSN/PGW per PCRF peer.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level
peer-interface-type <peer-name>
```

```
num-cca-pending-transaction
```

```
number-of-raa-pending-transaction
```

Activating/Deactivating Multiple Predefined Groups of PCC Rules

Operators can configure a Charging-Rule-Base-Name (CRBN) profile to activate/deactivate multiple predefined groups of PCC rules on an MCC gateway using the Charging-Rule-Base-Name AVP within the Charging-Rule-Install/Charging-Rule-Remove AVP. The Charging-Rule-Base-Name AVP received from the PCRF corresponds to the locally configured CRBN profile. When the CRBN profile is enabled, multiple CRBN names can be received in the same PCRF message, all of which will be applied to the subscriber.

Rating groups can be shared between CRBN and data profiles or between multiple CRBN profiles. When a rating group is shared between multiple CRBN profiles or between CRBN and data profiles, Operators configure a list of service rules under the CRBN profile. The service rules in the CRBN list must be linked with a rating group associated with the same CRBN profile and they must have the `service-activation` parameter set to `external-activation`. You can configure a maximum of 128 service rules in the CRBN service rule list.

When CRBN profile names are received in the Charging-Rule-Base-Name AVP from the PCRF, the MCC gateway checks the CRBN Profile configuration and matches the name. If a matching profile configuration is found, the MCC applies the CRBN profile configuration accordingly:

- Selects the billing plan based on the charging instance linked with the CRBN profile.
- Activates all the rating groups under the billing plan with the `service-activation` parameter set to `always-on`.
- Activates all the service rules under the rating groups with the `service-activation` parameter set to `always-on`. The service rules under the rating groups with the `service-activation` parameter set to `external-activation` are activated only when they are part of the CRBN service rule list.

If a matching profile is not found, the GGSN/PGW sends a Charging-Rule-Report AVP with the UNKNOWN_RULE_NAME failure code.

Note the following guidelines when activating/deactivating a CRBN profile:

- All rating groups and service rules associated with a CRBN profile should have the `service-activation` parameter set to `always-on`. Service rules under the rating groups with the `service-activation` parameter set to `external-activation` are activated only when they are part of the CRBN service rule list.
- The QoS Flow associated with CRBN profile should have the `bearer-control` parameter configured as `default-bearer`.

- There should be only one billing plan under the charging instance associated with the CRBN Profile, and it must have wild card charging characteristics.
- If the same service rule is part of multiple CRBN Profiles, the service rule is deactivated only after all the CRBN Profiles referring to it are deactivated.
- A static service rule that is activated as part of a CRBN Profile will not be deactivated when the CRBN Profile referring to it is deactivated.

***Note:** When applying the billing plan for a CRBN profile, the MCC only allocates memory for rating groups that have at least one service rule configured with the `service-activation` parameter set to `always-on` or that have a service rule that is part of the CRBN service rule list. See [Rating Group Memory Allocation \(page 1-12\)](#).*

To activate/deactivate multiple predefined groups of PCC rules, use the following CLI commands:

- 1 Configure the CRBN Profile.

```
an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name>
charging-instance <charging-instance-name> qos-policy
<qos-policy-name>
```

where

Parameter	Description
<code>crbn-profile</code>	Specifies the name of the CRBN profile.
<code>charging-instance</code>	Specifies the Charging Instance associated with this CRBN Profile.
<code>qos-policy</code>	Specifies the QoS Profile associated with this CRBN Profile.

- 2 Specify how the MCC should interpret the Charging-Rule-Base-Name AVP received on the Gx interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> crbn-mode [data-profile|crbn-profile]
```

where

Parameter	Description
<code>pcrf-charging</code>	Specifies the PCRF charging (Gx) interface name.

Parameter	Description
crbn-mode	Specifies how the MCC should interpret the Charging-Rule-Base-Name AVP received on the Gx interface. ■ data-profile – (default) The value of the Charging-Rule-Base-Name AVP corresponds to the locally configured workflow data profile on the MCC gateway. In this case, the MCC allows only one predefined group of PCC rules to be activated/deactivated on an MCC gateway. If the MCC gateway receives more than one Charging-Rule-Base-Name AVP within the Charging-Rule-Install AVP, the system will establish/continue a session as long as one of the received CRBNs is predefined on the MCC. ■ crbn-profile – The value of the Charging-Rule-Base-Name AVP corresponds to the locally configured CRBN Profile on the MCC Gateway. In this case, the MCC supports the receipt of a Charging-Rule-Install AVP containing multiple Charging-Rule-Base-Name AVPs and allows for multiple predefined groups of PCC rules to be activated/deactivated on an MCC gateway.

3 (Optional) Configure the CRBN service rule list:

```
an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name>
service-rule <list-of-service-rules>
```

4 (Optional) Specify a default CRBN profile to be used when the PCRF is unreachable and the PGW uses the default Gx policy to set up the session.

If the PCRF is unreachable during the session setup and the default Gx policy is enabled, the default CRBN profile is activated for a subscriber. Once the PCRF comes back up and the PGW receives a CCA-I from the PCRF, the default CRBN is deactivated for the subscriber. *Applicable only when the `crbn-mode` is set to `crbn-profile`.*

```
an3000-mcm-slot7cpu1(config)# workflow data-profile
<data-profile-name> default-crbn-profile <crbn-profile-name>
```

5 To display a list of all the CRBN Profiles activated:

```
an3000-mcm-slot7cpu1# show subscriber pdn-session
```

Activating/Deactivating Metering Services for Rating Groups and Service Rules Associated with a CRBN Profile

This feature enables the PCRF to activate/deactivate metering services for rating groups and service rules associated with a CRBN profile using the Monitoring-Key AVP, included within the Usage-Monitoring-Information AVP (UMI AVP).

Use the following CLI parameter to associate a metering instance with the CRBN profile.

```
an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name>  
metering <metering-instance-name>
```

When the PCRF requests usage monitoring, the MCC gateway compares the metering instance received in the Monitoring-Key AVP (within the UMI AVP) with all the rating groups configured under the metering instance associated with active CRBN profiles.

If a matching metering instance is found under an active CRBN profile, the MCC gateway activates usage monitoring services accordingly:

- Monitors all the rating groups configured under the metering instance of an active CRBN profile with the `service-activation` parameter set to `external-activation`.
- Monitors all the service rules configured under the metering rating groups with the `service-activation` parameter set to `always-on`. The service rules configured under the metering rating groups with the `service-activation` parameter set to `external-activation` are monitored only when they are part of the CRBN service rule list.

The gateway reports the accumulated usage in the CCR-U when the allocated quota in the Granted-Service-Unit AVP (within the UMI AVP) is exhausted.

When the PCRF deactivates a CRBN profile, the gateway deactivates usage monitoring services if all the service rules related to the monitoring-key are removed as part of the CRBN profile deactivation. However, the gateway will only send a final usage report to the PCRF if usage monitoring for the monitoring-key is disabled via the UMI AVP.

Separate services for a service rule can be enabled/disabled independently using different CRBN profiles. For example, if a service rule, shared between two CRBN profiles, has charging services enabled under one CRBN profile and has metering services enabled under another CRBN profile, the charging or metering services for the rule can be enabled/disabled independently by activating/deactivating the appropriate CRBN profile.

To activate/deactivate metering services for rating groups and service rules associated with a CRBN Profile, use the following CLI commands:

- 1 Configure the CRBN Profile and associate a metering instance with the CRBN profile.

```
an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name>
metering <metering-instance-name>
```

where

Parameter	Description
crbn-profile	Specifies the name of the CRBN profile.
metering-instance	Specifies the metering instance associated with this CRBN Profile.

- 2 Specify how the MCC should interpret the Charging-Rule-Base-Name AVP received on the Gx interface. In this case, select **crbn-profile**.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> crbn-mode [data-profile|crbn-profile]
```

where

Parameter	Description
pcrf-charging	Specifies the PCRF charging (Gx) interface name.
crbn-mode	Specifies how the MCC should interpret the Charging-Rule-Base-Name AVP received on the Gx interface. ■ data-profile – (default) The value of the Charging-Rule-Base-Name AVP corresponds to the locally configured workflow data profile on the MCC gateway. ■ crbn-profile – The value of the Charging-Rule-Base-Name AVP corresponds to the locally configured CRBN Profile on the MCC Gateway.

- 3 (Optional) Configure the CRBN service rule list. The service rules configured under the metering rating groups with the **service-activation** parameter set to **external-activation** are monitored only when they are part of the CRBN service rule list.

```
an3000-mcm-slot7cpu1(config)# services crbn-profile <profile-name>
service-rule <list-of-service-rules>
```

Activating/Deactivating Multiple Predefined Service Rules with a Single Charging-Rule-Name

This feature adds the `pcc-rule-name` parameter for service-rule configuration when the `service-activation` parameter is set to `external-activation`. This configuration option allows for the configuration of multiple predefined service rules with the same name but with different priorities and different correlation IDs.

For example, you can apply the same `pcc-rule-name` to a group of predefined service rules, some with packet filters, some with HTTP rule groups, and some with TLS groups, which can all be activated/deactivated by a single Charging-Rule-Name within the Charging-Rule-Install/Charging-Rule-Remove AVP from the PCRF.

You can activate a list of service rules that share the same `pcc-rule-name` and apply the configuration to the following objects:

- Online/offline rating group
- Metering rating group
- QoS flow

You can associate a maximum of 32 service rules with one service rule group.

```
an3000-mcm-slot7cpu1(config)# service-construct service-rule <name>  
pcc-rule-name <name>
```


Reporting NetLoc and RAN-NAS Cause Indication to the PCRF

If Network Location (NetLoc) and RAN-NAS reporting is enabled on the MCC gateway, NetLoc and RAN-NAS Release Cause is included in the Supported-Features AVP of the CCR-I message to indicate that the reporting of access network information and RAN/NAS-Cause indication is supported.

The PCRF sends the ACCESS_NETWORK_INFO_REPORT (45) event trigger within the Event-Trigger AVP to request access network information from the PCEF. The Required-Access-Info AVP is sent from the PCRF to request that the PCEF report the required access network information for that Application Function (AF) session.

The MCC gateway reports the following information in CCR-U/CCR-T messages to the PCRF:

- Access Network Information from the access network in the following AVPs:
 - 3GPP-User-Location-Info AVP
 - 3GPP-SGSN-MCC-MNC AVP
 - 3GPP-MS-TimeZone AVP
 - User-Location-Info-Time AVP
- RAN-NAS-Cause indication – Reports detailed release or failure code information in the RAN-NAS-Release-Cause AVP.

When the PGW receives an Update Bearer Response with a RAN-NAS Cause indication, the PGW reports detailed release or failure code information in the Charging-Rule-Report → RAN-NAS-Release-Cause AVP in a CCR-U to the PCRF.

To configure this feature, you must first enable RAN-NAS reporting in the enhanced-feature-list associated with the gateway APN:

- 1 Enable NetLoc and RAN-NAS reporting. By default, the parameters in this list are disabled:

```
an3000-mcm-slot7cpu1(config)# zone default gateway
enhanced-features-list <list-name> <net-loc|ran-nas>
<enabled|disabled>
```

- 2 Map the configured enhanced-features-list to an APN:

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name>
enhanced-features <list-name>
```

Modifying Default Diameter AVP Flags

Operators can modify the default AVP flag values for Diameter AVPs on a specific Diameter interface (Gx/Gxa, Gy, Gz, SWm, STa, and S6b). Diameter AVPs have flags that indicate to the receiver how each AVP must be handled in Diameter messages. An AVP can have one or both of these flags set.

- Mandatory “M” flag – Indicates that support for this AVP is required. If either the AVP or its value is not recognized, then the message is not processed.
- Vendor-specific “V” flag – Indicates that this AVP has the optional Vendor ID field and when set, the AVP belongs to the specific vendor code address space.

For example, if the RAT-Type AVP has the Mandatory flag and Vendor-Specific flag set by default, all Diameter messages must have this AVP, otherwise the message is not processed. Also, when this AVP is sent, the Vendor ID field will be populated with the value associated with the specific Vendor.

With this feature, if the Operator changes the RAT-Type AVP Flag setting so that only the Vendor-Specific flag is set for the Gx interface, then subsequent Gx messages may or may not have this AVP. However, the Vendor ID field will still be populated with the value associated with the specific Vendor when this AVP is sent.

Use the following CLI command to modify the default Diameter AVP flag values.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter avp-flags <entry-name> attribute-id
<attribute-id> vendor-id <vendor-id> [gx|gy|gz|swm|sta|s6b] flag-value
[none|mandatory|vendor-specific|mandatory-and-vendor-specific]
```

where

Parameter	Description
avp-flags	Specifies the configured name for the AVP flag entry. The entry name is a custom name used for the AVP flag entry and is not necessarily the name of the Diameter AVP.
attribute-id	Specifies the Diameter AVP attribute ID. <i>Note: The attribute ID and vendor ID are mandatory fields. This feature will only modify the AVP flag value if the attribute ID and vendor ID are correctly provided. If there is no AVP found with the values provided, the configuration will have no effect.</i>
vendor-id	Specifies the Diameter AVP vendor ID. <i>Note: The attribute ID and vendor ID are mandatory fields. This feature will only modify the AVP flag value if the attribute ID and vendor ID are correctly provided. If there is no AVP found with the values provided, the configuration will have no effect.</i>
[gx gy gz swm sta s6b]	Specifies the Diameter interface.

Parameter	Description
<code>flag-value</code>	Specifies the flag value for the Diameter AVP. ■ If an AVP has a non-zero Vendor ID, the possible flag values are <code>vendor-specific</code> and <code>mandatory-and-vendor-specific</code>. ■ If an AVP has a Vendor ID of zero, the possible flag values are <code>none</code> and <code>mandatory</code>.

Notifying the PCRF When an OCS Failure Occurs

When the PGW receives a CCA-I/CCA-U/RAR message with the Cisco-Event-Trigger-Type VSA set to CREDIT_CONTROL_FAILURE (5), the PGW notifies the PCRF when an OCS failure event occurs by sending a CCR-U message containing the Cisco-Event VSA indicating the OCS failure result code.

The PGW also includes the affected service rules in the Charging-Rule-Report → PCC-Rule-Status AVP set to ACTIVE_WITHOUT_CREDIT_CONTROL(10) within the CCR-U to the PCRF and the PGW takes action based on the Gy failure handling configuration.

The MCC can process the Cisco-Event-Trigger-Type VSA if received in the CCA message as a command-level VSA or if received in the CCA within the Cisco-Event-Trigger VSA.

The Cisco-Event-Trigger VSA and Cisco-Event-Trigger-Type VSA are configurable in CCA data record templates on the Gx interface. To ensure the PGW processes the Cisco-Event-Trigger-Type VSA received in CCA messages, set the `presentFlag` parameter to `yes`. The default `presentFlag` setting is `no`, indicating the field is ignored when received in the CCA response.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] pcrf diameter cca <template-name>
[ciscoeventtrigger|ciscoeventtriggertype] presentFlag yes
```

The Cisco-Event VSA is configurable in CCR interim data record templates on the Gx interface. Set the `presentFlag` parameter to `yes` to send the Cisco-Event VSA in CCR interim messages. The default `presentFlag` setting is `no`, indicating the field is not sent in the CCR message.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] pcrf diameter interim <template-name> ciscoevent
presentFlag yes
```

The following examples outline use cases for this feature:

	Configuration	Description
Use Case 1	Default Gy Quota = enabled Cisco-Event-Trigger-Type VSA = CREDIT_CONTROL_FAILURE (5)	When a Gy failure occurs, the PGW allocates the default Gy quota and sends the CCR-U to the PCRF with the affected service rules. When the PCRF responds with a CCA/RAR message containing the Online AVP set to disable, the PGW checks whether the respective rating group is allocated the default quota and if so, then blocks those rating groups. When the PCRF sends the PGW an RAR/CCA message containing the Online AVP set to enable, the PGW sends new session requests to the recovered OCS and continues to process the sessions using the locally configured default quota until the sessions are complete. Existing sessions using the default quota do not rejoin the OCS unless the Gy Rejoin feature is enabled.
Use Case 2	Default Gy Quota = disabled Cisco-Event-Trigger-Type VSA = CREDIT_CONTROL_FAILURE (5)	When a Gy failure occurs, the PGW sends the CCR-U to the PCRF with the affected service rules. When the PCRF responds with a CCA/RAR message containing the Online AVP set to disable, the PGW blocks the online rating groups with which the affected service rules are associated. If all the rating groups for that subscriber are set to DISABLE_ONLINE, the PGW terminates the Gy session, otherwise the PGW keeps the Gy session. When the PCRF sends the PGW an RAR/CCA message containing the Online AVP set to enable, the PGW sends new session requests to the recovered OCS.
Use Case 3	Default Gy Quota = enabled or disabled Cisco-Event-Trigger-Type VSA = NO_CISCO_EVENT_TRIGGERS (0)	When a Gy failure occurs, the PGW does not inform the PCRF. If the default Gy quota is enabled, the PGW allocates the quota, otherwise the PGW terminates the session.
Use Case 4	Default Gy Quota = enabled Gy Rejoin = enabled Cisco-Event-Trigger-Type VSA = CREDIT_CONTROL_FAILURE (5)	When a Gy failure occurs, the PGW sends the CCR-U to the PCRF with the affected service rules. If the PCRF does not respond with a CCA-U message containing the Online AVP set to disable, the PGW does not block the online rating groups for that subscriber and when the Gy rejoin timer expires, the PGW resends the CCR-U message to the OCS. If the CCR-U message to the OCS fails again, the PGW does not notify the PCRF of this failure because it has already informed the PCRF.

Failure Handling When Reporting Gy Failure Codes to the PCRF

If the PCRF does not respond to a CCR-U from the PGW that is reporting Gy failure codes and failed service rules with the Cisco-Event VSA and if the Gx Rejoin feature is enabled, the PGW starts the Gx rejoin timer (`pcrf-down-rejoin-time`). When this timer expires, the PGW retransmits the Gy failure codes and failed service rules to the PCRF in subsequent CCR-U messages until it receives a CCA-U response from the PCRF. For more information on the Gx Rejoin feature, see [Gx Rejoin and Gx Replay \(page 1-133\)](#).

The following examples outline use cases for this feature:

- **Another OCS transaction fails for other rules before the Gx rejoin timer expires** – In this case, when the Gx rejoin timer expires, the PGW retransmits the Gy failure codes and failed service rules for all failed OCS transactions in the rejoin CCR-U sent to the PCRF.
- **Another event triggers a CCR-U (such as a QoS or RAT change) before the Gx rejoin timer expires** – In this case, the PGW includes the Gy failure codes and the failed service rules for all failed OCS transactions in this CCR-U to the PCRF. When the Gx rejoin timer expires, the PGW does not report any Gy failure codes or failed service rules with the CISCO event-trigger in the rejoin CCR-U sent to the PCRF.
- **The PGW receives a successful response from the OCS for a particular rating group before the Gx Rejoin timer expires** – In this case, the PGW does not report the failed service rules associated with that rating group in the rejoin CCR-U sent to the PCRF.

Configuring an SCTP Profile

To use Stream Control Transmission Protocol (SCTP) as a transport protocol on the Gx interface, you must configure an SCTP profile and associate the profile with the local Diameter node.

You also have the option of configuring a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. SCTP multi-homing enables the MCC to switchover to an alternate path in the case of a failure on the primary path. The MCC will revert back to the primary path once the primary connection becomes available. The primary and secondary loopback IP addresses can be IPv4 or IPv6. MCC supports socket creation to be associated to a single network context. SCTP multi-homing will be supported in a single network context.

***Note:** If using a mix of IPv4/IPv6, ensure the primary loopback IP address is IPv4 and the secondary is IPv6. SCTP Multi-homing will not work if the primary loopback IP address is IPv6 and the secondary is IPv4. For this reason, Affirmed Networks recommends not using a mix of IPv4/IPv6 addresses for SCTP Multi-homing.*

- 1 Create an SCTP profile.

```
an3000-mcm-slot7cpu1(config)# service-construct sctp-profile
<profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
<code>cookie-life</code>	Specifies the lifetime of the SCTP cookie, in 100 milliseconds.
<code>heartbeat-interval</code>	Specifies the periodic time interval (in milliseconds) at which the SCTP heart beat messages are sent. Value zero disables SCTP heart beats.
<code>init-retry</code>	Specifies the maximum number of attempts to retransmit the INIT message after which the attempted association is terminated.
<code>path-max-retransmit</code>	Specifies the maximum number of consecutive retransmissions over a destination transport address of a peer endpoint before it is marked as inactive.
<code>max-in-streams</code>	Specifies the maximum number of incoming SCTP streams.
<code>max-out-streams</code>	Specifies the number of outgoing SCTP streams.
<code>path-mtu</code>	Specifies the maximum transmission unit (MTU), in bytes, for SCTP streams. Value 0 enables auto-discovery of the path MTU.

Parameter	Description
<code>rto-initial</code>	Specifies the initial retransmission timeout (RTO) value, in milliseconds. Subsequent SCTP RTO will be recomputed starting from this value.
<code>rto-max</code>	Specifies the maximum RTO value (in milliseconds) that any subsequent recomputed RTO may never exceed.
<code>rto-min</code>	Specifies the minimum RTO value (in milliseconds) that any subsequent recomputed RTO may never go below.
<code>sack-period</code>	Specifies the delay before sending the Selective Acknowledgment (SACK), in milliseconds. Value 0 disables the delay value.

- 2 Specify SCTP as the transport type for the local Diameter node and associate the SCTP profile.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> transport sctp
sctp-profile <profile-name>
```

- 3 (Optional) Configure a secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing. The secondary loopback IP address is only configurable for the SCTP transport type.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name>
secondary-loopback-ip <loopback-name>

an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> secondary-loopback-ip <loopback-name>
```

Note: The `secondary-loopback-ip` parameter is only visible when the local Diameter node transport type is set to SCTP.

Redirecting the First HTTP Request of a Session

This feature adds the ability to redirect the first HTTP request of a session based on information received from the PCRF as part of the Charging-Rule-Name (CRN) AVP via the CCA-I, CCA-U (including rejoin), or RAR message. Receiving the CRN as part of the Charging-Rule-Definition AVP is also supported.

When the PGW receives the CRN AVP within the CCA or RAR message, the PGW performs a one-time HTTP redirection to the URL configured in the MCC as part of the rating group configuration. The CRN AVP references a predefined service rule configured under the metering rating group and any traffic matching this rule must be redirected to the URL configured in the advice-of-charge (AoC) that is associated with the metering rating group. The redirection is only valid for the first HTTP request of the session and is not valid for subsequent HTTP transactions. Non-HTTP traffic should pass as usual.

Use the following CLI command to enable one-time-redirection on a metering rating group. The default is `disabled`.

```
an3000-mcm-slot7cpu1(config)# services charging rating-group
<rating-group-name> one-time-redirection [enabled|disabled]
```

Note the following when configuring a metering rating group for one-time HTTP redirection:

- The `one-time-redirection` parameter is only visible when the `charging-method` parameter for the rating group is set to `metering` and the `advice-of-charge` is configured for the rating group.
- The `service-activation` parameter for the rating group must be set to `external-activation`.
- The service rules associated with the metering rating group used for one-time HTTP redirection must be configured with a high priority to redirect the HTTP traffic.
- A one-time HTTP redirection service rule cannot be associated with multiple rating groups. However, the PGW can support multiple one-time-redirection metering rating groups each with its own set of service rules and AoC.
- The PGW will not remove the service rule that triggered the one-time HTTP redirection or install the same service rule again, if received from the PCRF, until it receives the Charging-Rule-Remove → Charging-Rule-Name AVP from the PCRF.

For example, the metering rating group configuration would be similar to the following:

```
an3000-mcm-slot7cpu1(config)# services charging rating-group
metering_rg charging-method metering one-time-redirection enabled
advice-of-charge one_time_aoc service-activation external-activation
service-rule One_time_http_rule
```


Activating a URL Whitelist via the Gxa Interface

This feature adds support for using predefined PCC rules to activate a whitelist of allowed HTTP URLs (URLs that users can always access) on the TWAG/TWAP via the Gxa interface. Predefined PCC rules are pre-configured on the TWAG/TWAP and can be activated or deactivated by the PCRF at any time.

To activate a URL whitelist, the PCRF sends a CCA-I message to the TWAG/TWAP containing a whitelist service rule in the Charging-Rule-Install AVP. The whitelist Service Rule is a predefined service rule configured on the TWAG/TWAP that references a list of allowed URLs. In a separate Charging-Rule-Install AVP, the PCRF sends a redirect service rule, which is also predefined on the TWAG/TWAP. The TWAG/TWAP allows users to access URLs in the whitelist and redirects all other URL requests based on the parameters configured for the redirect service rule. The whitelist service rule must be configured with a higher priority than the redirect service rule.

Note: The service rule configured for redirection must already exist. See the *Affirmed Networks Gateway Administration Guide* for more information on configuring service rules.

Use the following steps to configure a URL whitelist and associate it with a Service Rule on the TWAG/TWAP.

- 1 Add the URL list.

```
an3000-mcm-slot7cpu1(config)# service-construct url-list
<url-list-name> [import-from-local-file|import-from-server] filename
<filename>
```

where

Parameter	Description
<code>url-list</code>	Specifies the URL list name. The following guidelines apply to the URL list file format: ■ The file should have a Fully Qualified Domain Name (FQDN) for each line for HTTP request URL prefix or exact-match: – The protocol schema is ignored and anything before “//” and including “//” is skipped. Only the URL following “//” is used. – If line contains a star “*”, the entry is added in the database using the string before “*” for URL prefix-match. Otherwise, the entry is added in the database for URL exact-match.

Parameter	Description
import-from-local-file	References a file name for importing the file on the local directory: /public/content/policylists/url/userlists/ The url-list database is built after an import action. If you make changes to the file, you must import the file to rebuild the database for the new file to take effect. Operators can upload the local file through Acuitas or using fault-management clear to SCP the file to /public/content/policylists/url/userlists/ on the TWAG/TWAP. There is no scheduled automatic import for this mode.
filename	Specifies the filename containing the URL list (such as folder/file-\$(DATE:yyyymmdd).txt).

- 2 Configure an HTTP Rule Group and associate the URL list with the HTTP rule group. The HTTP Rule Group function defines a set of rules for matching header attributes in HTTP request and response headers. This object is referred to by a service rule. When the system finds a match, a specific action or service(s) is applied based on where the service rule is used. For example, the system may terminate, block, redirect, or allow an action for the configured service.

```
an3000-mcm-slot7cpu1(config)# service-construct http-rule-group
<rule-group-name> rule <rule-name> url-list <url-list-name>
```

where

Parameter	Description
http-rule-group	Specifies the HTTP rule group name.
rule	Specifies the rule condition name.
url-list	Specifies the URL list for matching against an HTTP request's URL. The URL list supports 100,000 or more entries without impacting performance.

- 3 Configure the whitelist URL service rule and associate the HTTP rule group with the service rule. The service rule name is sent by the PCRF in the Charging-Rule-Install AVP.

```
an3000-mcm-slot7cpu1(config)# service-construct service-rule
<service-rule-name> admin-state [enabled|disabled] priority
<priority-number> service-data-flow-id <sdf-numerical-id>
http-rule-group <rule-group-name> tcp-filter all service-activation
external-activation install-default-bearer-packet-filters-on-ue
[enabled|disabled]
```

where

Parameter	Description
<code>service-rule</code>	Specifies the service rule name.
<code>admin-state</code>	Specifies whether to enable or disable the administrative state for this service rule. ■ <code>enabled</code> ■ <code>disabled</code>
<code>priority</code>	Specifies the service rule priority that allows service flows to select a service rule from a list of service rules based on priority. <i>Note: The whitelist service rule must be configured with a higher priority than the redirect service rule.</i>
<code>service-data-flow-id</code>	Specifies the numerical identifier for the service rule.
<code>http-rule-group</code>	Specifies the HTTP rule group name associated with this service rule.
<code>tcp-filter</code>	Specifies the TCP accounting category of the service rule. In this case, select <code>all</code> , which accounts for all traffic.
<code>service-activation</code>	Specifies when to use or skip the service flow for packet classification. In this case, select <code>external-activation</code> to allow the PCRF to activate/deactivate service rules.
<code>install-default-bearer-packet-filters-on-ue</code>	Specifies whether the TFT packet filters defined for a service rule should be sent to the UE. ■ <code>enabled</code> – The PGW sends TFT packet filters included in the service rule to the UE. ■ <code>disabled</code> – (default) The PGW does not send TFT packet filters included in the service rule to the UE.

- 4 Configure the Rating Group and associate the service rule with the Rating Group. The Rating Group defines the basis for charging service data flows.

```
an3000-mcm-slot7cpu1(config)# services charging rating-group
<rating-group-name>
```

The following table describes the configurable parameters.

Parameter	Description
<code>admin-state</code>	Specifies the administration state of this Rating Group. ■ <code>enabled</code> ■ <code>disabled</code>

Parameter	Description
charging-method	Specifies the charging method. In this case, select nometering. ■ local ■ offline ■ online ■ nocharge – Subscriber traffic that uses this Rating Group is either not charged or CDRs are not generated for this Rating Group. This feature is useful when an Operator wants to exclude certain flows such as DNS requests or portal requests used to charge subscribers. ■ metering ■ nometering
home-subscriber-charging	Specifies whether to use the Rating Group for home subscriber charging. In this case, select disabled. ■ enabled ■ disabled
roamer-subscriber-charging	Specifies whether to use the Rating Group for roaming subscriber charging. In this case, select disabled. ■ enabled ■ disabled
service-activation	Specifies when to use or skip the service flow for packet classification. In this case, select always-on. ■ always-on – Always use this service flow for classification. ■ external-activation – Use the activation state of the service flow to determine whether to use or skip the service flow for packet classification. This setting allows the gateway to dynamically activate/deactivate Rating Groups.
measurement-method	Specifies the measurement method. In this case, select volume. ■ time ■ volume ■ timeAndVolume ■ event – Only applies when charging-method is set to offline or local.
multiplier	Specifies the multiplier applied to time, volume and event charging.
priority	Specifies the priority. This value resolves conflicts in the case of multiple service instances. Priority is hierarchical in that the priority of the higher-level object takes precedence over the lower-level object. Thus, the Rating Group priority takes precedence over the associated service rule priority.
quota-hold-time	Specifies the hold time after inactivity, in seconds.

Parameter	Description
service-rule	Specifies the service rule(s) to link with the Rating Group.
service-type	Specifies the service type. In this case, select <code>data</code> .
tcp-retransmission	Inclusion or exclusion of TCP retransmissions in volume counts. In this case, select <code>charge</code>. ■ charge ■ noCharge <i>Note: Applies when <code>measurement-method</code> is set to <code>volume</code> or <code>timeAndVolume</code>.</i>
volume-measurement-count	Specifies the volume measurement count. In this case, select <code>totalOctets</code>. ■ inputOctets ■ outputOctets ■ totalOctets <i>Note: Applies when <code>measurement-method</code> is set to <code>volume</code> or <code>TimeAndVolume</code>.</i>
volume-measurement-layer	Specifies the volume measurement layer. In this case, select <code>layer3</code>. ■ layer3 – Counts the entire IP datagram. ■ layer4 – Excludes the IP header. ■ layer7 – Excludes the IP header and the UDP/TCP/SCTP header. <i>Note: Applies when <code>measurement-method</code> is set to <code>volume</code> or <code>timeAndVolume</code>.</i>

- 5 Configure the metering service instance and associate the Rating Group with the metering service instance.

```
an3000-mcm-slot7cpu1(config)# services metering metering-instance
<metering-instance-name> admin-state enabled rating-group
<rating-group-name>
```

where

Parameter	Description
metering-instance	Specifies the metering service instance name.
admin-state	Specifies whether to enable or disable the administrative state for this metering service instance. ■ enabled ■ disabled
rating-group	Specifies the name of the Rating Group associated with this metering service instance.

- 6 In the Workflow Data Profile, specify the metering service instance. The Workflow Data Profile must already exist.

```
an3000-mcm-slot7cpu1(config)# workflow data-profile
<data-profile-name> metering-service <metering-instance-name>
```

Appending the Original URL to the Redirect URL for Gxa Sessions

The TWAG appends the original URL to the redirect URL when the TWAG receives the redirect URL (within the Redirect-Information AVP) from the PCRF and the subscriber session requires redirection. The TWAG obtains the Advice of Charge (AoC) ID from the workflow control profile corresponding to the session. If the TWAG receives a redirect URL from the PCRF and a redirect URL is also configured under the AoC object, the redirect URL from the PCRF takes precedence.

The `include-original-url` parameter under the AoC configuration object must be set to `true` to ensure the TWAG appends the original URL to the redirect URL for Gx sessions.

```
an3000-mcm-slot7cpu1(config)# service-construct advice-of-charge
<name> http include-original-url true
```

Customizing the APN Reported on External Interfaces

Operators can customize the APN reported on external interfaces. The customized APN is used in the Called-Station-ID AVP and sent in messages on the following interfaces:

- Access-Request on RADIUS
- Accounting-Request on Gz and RADIUS
- CCR on Gx and Gy interfaces

For Ga (GTPP) and local interfaces, the customized APN-reporting value is filled in the Access Point Name Network Identifier CDR field.

Set parameters for this feature using the following commands:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile <name>
apn-reporting custom custom-reporting-apn-name <apn-name>
```

To enable this feature, set the `apn-reporting` parameter to `custom`. The value set in the `custom-reporting-apn-name` parameter is used to fill the Called-Station-Id value in messages to external interfaces.

***Note:** The `custom-reporting-apn-name` parameter is visible only if `apn-reporting` is set to `custom`.*

Use the following command to display the reported APN name
(`reporting-apn-name`) sent as the Called-Station-Id AVP on external interfaces.

```
an3000-mcm-slot7cpu1# show subscriber pdn-session <session-id>
```

Configuring APN Reporting at the Interface Level

You can configure `apn-reporting` at the interface level to allow for different APNs to be reported over different external interfaces. The `apn-reporting` value configured at the interface level is used to fill the Called-Station-ID AVP and User-Name AVP sent in messages on the following interfaces:

- Access-Request on RADIUS
- Accounting-Request on Gz and RADIUS
- CCR on Gx and Gy interfaces

For Ga (GTPP) and local interfaces, the `apn-reporting` value configured at the interface level is filled in the Access Point Name Network Identifier CDR field.

The `apn-reporting` value configured at the interface level overrides the `apn-reporting` value configured for the Control Profile. If the `apn-reporting` value is not configured at the interface level, the Called-Station-ID AVP and User-Name AVP will be filled based on the `apn-reporting` value configured for the Control Profile.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <name> apn-reporting [original|mapped|prior-mapped]
```

Note: The interface-level `apn-reporting` parameter does not support the `custom` option and does not apply to the S6b interface.

Note: This feature applies to the PGW only.

Enabling 5G NSA QoS AVPs

The existing QoS AVPs defined on the Gx interface represent values in bits-per-second (bps) with a 32-bit integer data type. However, the 5G non-standalone (NSA) solution requires higher QoS bit rate values (up to 4 Tbps).

To accommodate the higher 5G NSA throughput rates, the following AVPs are supported within the QoS-Information AVP on the Gx interface. These QoS-Information AVPs represent values in kilobits-per-second (Kbps).

- Extended-APN-AMBR-DL
- Extended-APN-AMBR-UL
- Extended-GBR-DL
- Extended-GBR-UL
- Extended-Max-Requested-BW-DL
- Extended-Max-Requested-BW-UL

During session setup, the PCEF and PCRF exchange the Supported-Features AVP to negotiate the list of supported features that will be used during the session. One of these features, as defined in 3GPP 29.212 version 15.1.0, is “Extended-BW-NR”. The “Extended-BW-NR” feature is sent in the Features-List-ID (2), within the Supported-Features AVP, to indicate support for the extended QoS-Information AVP values in Kbps.

Enabling the “Extended-BW-For-NR” Flag

To enable the “Extended-BW-For-NR” flag in the Supported-Features AVP on the MCC, Operators configure an enhanced-features-list and enable/disable the “Extended-BW-NR” feature for the list. The enhanced-feature-list is then associated with an APN.

- 1 Configure the enhanced-features-list. When enabled, the MCC enables the “Extended-BW-NR” flag in the Supported-Features AVP sent in the CCR-I to the PCRF. The default is `disabled`.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway
enhanced-features-list <list-name> extended-bw-nr [enabled|disabled]
```

- 2 Associate the enhanced-features-list with an APN.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
enhanced-features <list-name>
```

When “Extended-BW-NR” is configured as a supported feature and associated with an APN (which is associated with a PGW service) and if the PGW and PCRF validate this support through the exchange of the Supported-Features AVP, the following occurs:

- For bandwidth values higher than $2^{32}-1$ bps, the MCC will use the QoS AVPs, representing bit rates in Kbps.
- For bandwidth values lower or equal to $2^{32}-1$ bps, the MCC will continue to use the QoS AVPs representing bit rates in bps.

When sent from the PCRF to the PCEF, the extended QoS-Information AVPs are independent of the existing AMBR, MBR, and GBR AVPs. If both exist, the extended QoS-Information AVPs take precedence over the existing AMBR, MBR, and GBR AVPs.

When the QoS AVPs are sent from the PCEF to the PCRF and if the bandwidth values are higher than $2^{32}-1$, the values of the AVPs that are in bps are set to $2^{32}-1$ and the extended AVPs are filled with the Kbps values.

Weighted Round-Robin Distribution to Multiple Diameter Peers

Operators can configure the MCC to use a weighted round-robin algorithm to distribute initial session requests among the available Diameter peers within a Diameter server group based on the configured weight assigned to each peer. The Diameter peer with the highest weight (with 10 being the highest and 1 being the lowest) receives the most distribution of new session requests. The MCC only uses the configured weight for the initial peer selection (for example, CCR-I).

For example, if Peer A is set to 4 and Peer B is set 2, the MCC will distribute the initial session requests as ABABAA. In this example, the MCC alternates between both peers starting with Peer A because it has the highest weight.

When a Diameter peer is configured with a weight of 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions.

If the Diameter peer with the highest weight is unavailable, the request is sent to the peer with the next highest weight in the same Diameter server group. If multiple Diameter peers are assigned the highest weight (10), the MCC uses a round-robin approach to select from among those peers for new requests.

Operators can configure the MCC to use a weighted round-robin algorithm to select a Diameter peer from among a list of primary peers or from among a list of secondary peers, but not *across* primary and secondary peers.

Use the following command to assign a weight to a Diameter peer:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
aaa-interface-group diameter node-admin <node-name> peer
<peer-host-name> weight <value>
```

- **weight** – Specifies the weight allocated to this Diameter peer. The Diameter peer with the highest weight (10) receives the most distribution of new session requests. The default is 1 (lowest weight). When set to 0, the MCC places the peer in “maintenance mode” and stops sending new session requests (for example, CCR-I) to this peer, but continues to send requests (for example, CCR-U/CCR-T) for existing sessions. *Applies to Gy/Gx/Gz/Sta/Swm/S6b interfaces.*

Application Detection and Control

The Gx interface supports Application Detection and Control (ADC), as defined in 3GPP TS 29.212 Release 15.3.0. The ADC function enables the PGW to detect applications and send notifications to the PCRF with application identifiers, flow information, and application Start/Stop events.

Based on the PGW-reported information, the PCRF can install, remove, or modify PCC or ADC rules, and the PGW can perform service control, charging, and bearer management (bearer creation, update, and deletion) based on new PCC/ADC rules.

The TDF-Application-Identifier AVP identifies the PCC/ADC rule and enables the service rule for reporting of application Start/Stop notifications to the PCRF. The PCRF can instruct the PGW to detect applications using static, predefined rules or dynamic rules:

- Static rules – The PCRF enables APPLICATION_START and APPLICATION_STOP triggers for all static service rules on the session that have the TDF Application ID configured. Static rules are pre-configured on the MCC and are always active for the duration of the session (service rules with *always-on* activation).
- Predefined rules
 - The PCRF can use the Charging-Rule-Install or ADC-Rule-Install AVP in the CCR message to activate a predefined service rule on the PGW.
 - The PCRF enables APPLICATION_START and APPLICATION_STOP triggers for all service rules on the session that have the TDF Application ID configured.
- Dynamic rules – The PCRF includes the TDF-Application-Identifier AVP within the Charging-Rule-Definition or ADC-Rule-Definition AVP in CCA or RAR messages.

For static, predefined and dynamic PCC/ADC rules, the TDF-Application-Identifier is included within the Application-Detection-Information AVP in CCR messages when the PGW reports application Start/Stop notifications for traffic matching these rules. If the TDF-Application-Identifier is not configured or sent by the PCRF, the application Start/Stop notifications are not enabled for the service rule.

The Application-Detection-Information AVP also contains the TDF-Application-Instance-Identifier AVP, which is dynamically assigned by the PGW to correlate the application Start/Stop events.

If any rule activation fails, the PGW generates a CCR message with the Charging-Rule-Report/ADC-Rule-Report AVP, depending on whether the rule install came through the Charging-Rule-Install/ADC-Rule-Install AVP.

The PCRF can also mute application Start/Stop notifications about a specific detected application by providing the Mute-Notification AVP within the corresponding Charging-Rule-Definition or ADC-Rule-Definition AVP.

***Note:** The TDF-Application-Identifier sent from the PCRF is applicable only to dynamic rules. Static and predefined rules must use the configured TDF Application ID associated with the static or predefined service rule.*

Use the following command to configure the TDF-Application-Identifier for a static or predefined service rule:

```
an3000-mcm-slot7cpu1(config)# service-construct service-rule <name>
tdf-application-id <name>
```

Use the following commands to include/exclude the ADC-Rule-Report, Application-Detection-Information, and Flow-Information AVPs in CCR/RAA messages using data record templates.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pgw pcrf diameter [start|stop|interim|raa] <template-name>
adcrulereport presentFlag [yes|no|conditional]
```

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
pgw pcrf diameter [stop|interim] <template-name>
[applicationdetectioninfo|flowinfo] presentFlag [yes|no|conditional]
```

If the PCRF goes down or is unreachable after the session is already established, the PGW will not replay application Start/Stop notifications.

- If the PCRF is down, the PGW will drop the Stop notification even if the Start notification was sent.
- If the PCRF comes back up, the PGW will send the Stop notification even if the Start notification was dropped.

If the Subscriber Services Module (SSM) restarts, Fastpath will send another Start notification for active flows, which will cause the PGW to send an application Start notification to the PCRF with a new TDF-Application-Instance-Identifier. In this case, the PGW will not send an application Stop notification for the previous TDF-Application-Instance-Identifier.

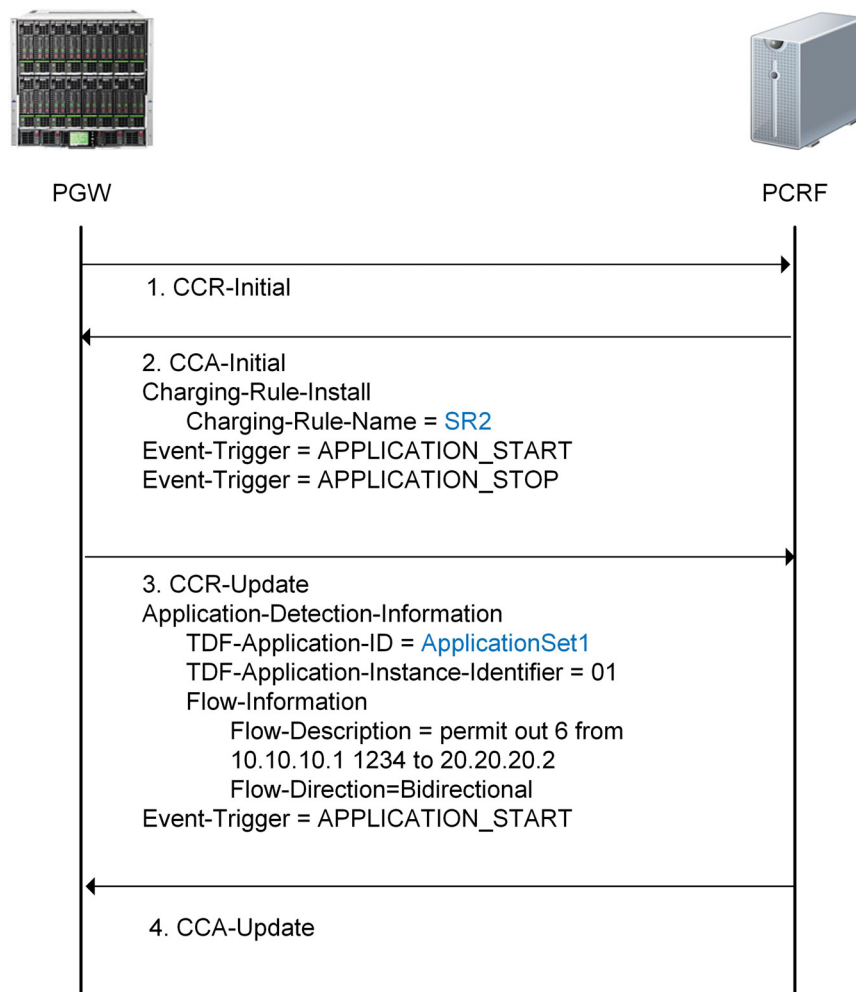
Activating a Predefined Service Rule for ADC

Figure 1-16 illustrates an example of ADC where the PCRF activates the predefined rule (SR2) and enables APPLICATION_START and APPLICATION_STOP event-triggers for all PCC rules enabled on the session, but only sessions that have the TDF Application ID generate the APPLICATION_START /STOP event-triggers.

In this example, the PGW is configured with a predefined service rule that has a TDF Application ID configured. While Figure 1-16 uses the Charging-Rule-Install AVP as an example, the message flow remains the same if the ADC-Rule-Install AVP is used.

```
an3000-mcm-slot7cpu1(config)# service-construct service-rule SR2
admin-state enabled priority 100 service-data-flow-id 66
http-rule-group HGR service-activation external-activation
tdf-application-id ApplicationSet1
```

Figure 1-16 Predefined Service Rule Activation for ADC



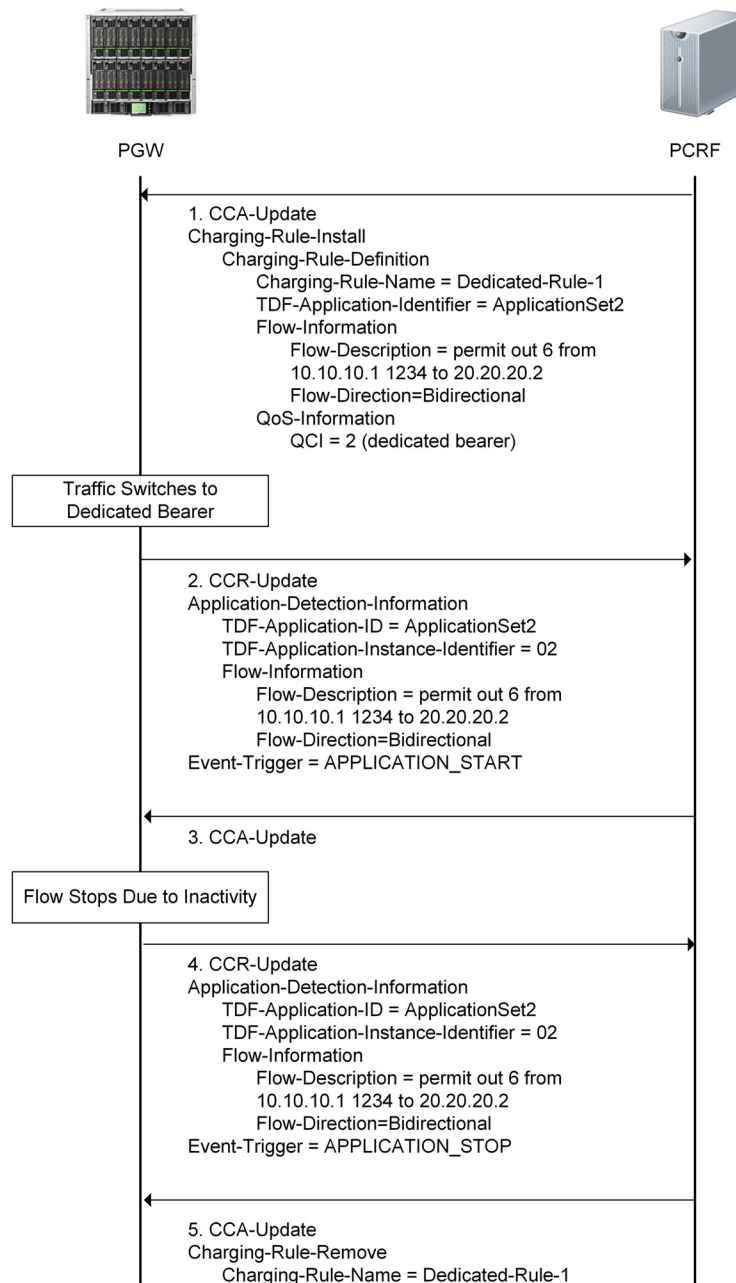
- 1 The PGW starts the session setup process by sending a CCR-I message to the PCRF.

- 2 The PCRF responds to the PGW with a CCA message to activate the predefined rule and enable APPLICATION_START and APPLICATION_STOP triggers.
- 3 When application traffic matches the service rule with a TDF Application ID, the PGW sends the CCR-U message to the PCRF with the APPLICATION-START notification, the flow information and the TDF Application Instance Identifier.
- 4 The PCRF responds with a CCA-U message.

Installing a Dynamic Service Rule for ADC

Figure 1-17 illustrates an example of ADC where, during an established session, the PCRF installs a dynamic service rule to redirect traffic over a dedicated bearer with different QoS Policies. When the PGW detects the flow has stopped due to inactivity, it sends the APPLICATION_STOP notification to the PCRF and the PCRF sends a Charging-Rule-Remove in the CCA to remove the dynamic PCC rule and delete the dedicated bearer.

Figure 1-17 Dynamic Service Rule Installation for ADC



Moving Predefined Service Rules Across Predefined Rating Groups

This feature adds the ability to move predefined service rules (service rules with `external-activation`) from one predefined rating group to another using the Rating-Group AVP sent from the PCRF within the Charging-Rule-Install → Charging-Rule-Definition AVP.

- The rating group identifier sent in the Rating-Group AVP must correspond to a predefined rating group `quota-id` on the MCC (`services > charging > rating-group > quota-id`).
- The predefined service rules on the MCC must be associated with all the predefined rating groups across which they can be moved.
- Predefined service rules can be moved from an offline to online rating group and vice versa, from an online to another online rating group or from an offline to another offline rating group, but these rating groups must all be predefined on the MCC.

When the MCC gateway receives a CCA-I from the PCRF with the Rating-Group AVP, the MCC tries to match the value received in the Rating-Group AVP to a locally configured `quota-id`.

- If a match is found, the MCC associates the service rule(s) with the corresponding rating group. If the Rating-Group AVP is received in a CCA-U or RAR and a match is found, the MCC moves the associated service rule(s) from the initial predefined rating group to the corresponding rating group configured with the matching `quota-id`.
- If no match is found, the MCC ignores the value received in the Rating-Group AVP.

The following examples outline use cases for this feature.

Use Case 1: Online Rating Group to Offline Rating Group

If the rule is the last rule in the online rating group, the MCC gateway:

- Dynamically reports the Used-Service-Units (USU) for the rating group in the CCR-U/CCR-T to the OCS.
- Creates a service data container (SDC) for the service rule/rating group.

Initial Charging-Rule-Definition AVP	Charging-Rule-Definition AVP with different Rating-Group AVP (quota-id) and Online AVP
Charging-Rule-Name AVP SR_1	Charging-Rule-Name AVP SR_1
Rating-Group AVP 9999	Rating-Group AVP 99
Online AVP ENABLE_ONLINE	Online AVP DISABLE_ONLINE
Offline AVP ENABLE_OFFLINE	Offline AVP ENABLE_OFFLINE
Rating Group Configuration	Rating Group Configuration
services charging rating-group RG9_Online	services charging rating-group RG9_Offline
charging-method online	charging-method offline
quota-id 9999	quota-id 99
service-rule SR_1	service-rule SR_1
Service Rule Configuration	Service Rule Configuration
service-construct service-rule SR_1	service-construct service-rule SR_1
admin-state enabled	admin-state enabled
priority 100	priority 100
service-data-flow-id 4088	service-data-flow-id 4088
service-activation external-activation	service-activation external-activation

Use Case 2: Offline Rating Group to Online Rating Group

In this case, if the rule is the first rule in the online rating group, the MCC gateway:

- Dynamically sends the Requested-Service-Units (RSU) for the rating group in the CCR-I/CCR-U to the OCS.
- Creates a SDC for the service rule/rating group.

Initial Charging-Rule-Definition AVP	Charging-Rule-Definition AVP with different Rating-Group AVP (quota-id) and Online AVP
Charging-Rule-Name AVP SR_1	Charging-Rule-Name AVP SR_1
Rating-Group AVP 99	Rating-Group AVP 9999
Online AVP DISABLE_ONLINE	Online AVP ENABLE_ONLINE
Offline AVP ENABLE_OFFLINE	Offline AVP ENABLE_OFFLINE
Rating Group Configuration	Rating Group Configuration
services charging rating-group RG9_Offline	services charging rating-group RG9_Online
charging-method offline	charging-method online
quota-id 99	quota-id 9999
service-rule SR_1	service-rule SR_1
Service Rule Configuration	Service Rule Configuration
service-construct service-rule SR_1	service-construct service-rule SR_1
admin-state enabled	admin-state enabled
priority 100	priority 100
service-data-flow-id 4088	service-data-flow-id 4088
service-activation external-activation	service-activation external-activation

Use Case 3: Online Rating Group to Different Online Rating Group

If the rule is the last rule in the initial online rating group, the MCC gateway:

- Dynamically reports the USU for the rating group in the CCR-U/CCR-T to the OCS.
- Creates a SDC for the service rule/rating group.

If the rule is the first rule in the new online rating group, the MCC gateway:

- Dynamically sends the RSU for the rating group in the CCR-I/CCR-U to the OCS.
- Creates a SDC for the service rule/rating group.

Initial Charging-Rule-Definition AVP	Charging-Rule-Definition AVP with different Rating-Group AVP (quota-id) and Offline AVP
Charging-Rule-Name AVP SR_1	Charging-Rule-Name AVP SR_1
Rating-Group AVP 9999	Rating-Group AVP 999
Online AVP ENABLE_ONLINE	Online AVP ENABLE_ONLINE
Offline AVP ENABLE_OFFLINE	Offline AVP DISABLE_OFFLINE
Rating Group Configuration	Rating Group Configuration
services charging rating-group RG9_Online	services charging rating-group RG9_Online2
charging-method online	charging-method online
quota-id 9999	quota-id 999
service-rule SR_1	service-rule SR_1
Service Rule Configuration	Service Rule Configuration
service-construct service-rule SR_1	service-construct service-rule SR_1
admin-state enabled	admin-state enabled
priority 100	priority 100
service-data-flow-id 4088	service-data-flow-id 4088
service-activation external-activation	service-activation external-activation

Use Case 4: Offline Rating Group to Different Offline Rating Group

The MCC gateway creates a SDC for the service rule/rating group.

Initial Charging-Rule-Definition AVP	Charging-Rule-Definition AVP with different Rating-Group AVP (quota-id)
Charging-Rule-Name AVP SR_1	Charging-Rule-Name AVP SR_1
Rating-Group AVP 99	Rating-Group AVP 9
Online AVP DISABLE_ONLINE	Online AVP DISABLE_ONLINE
Offline AVP ENABLE_OFFLINE	Offline AVP ENABLE_OFFLINE
Rating Group Configuration	Rating Group Configuration
services charging rating-group RG9_Offline	services charging rating-group RG9_Offline2
charging-method offline	charging-method offline
quota-id 99	quota-id 9
service-rule SR_1	service-rule SR_1
Service Rule Configuration	Service Rule Configuration
service-construct service-rule SR_1	service-construct service-rule SR_1
admin-state enabled	admin-state enabled
priority 100	priority 100
service-data-flow-id 4088	service-data-flow-id 4088
service-activation external-activation	service-activation external-activation

Configuring Rating Sub-Groups

This feature also adds the ability to configure a list of separate rating sub-groups under the same rating group (quota-id), provided the `charging-method` for the rating group is set to `subgroup`. A rating sub-group can have any of the following charging methods:

- Only online charging
- Only offline charging
- Both offline and online charging
- No charging

Use the following commands to configure a rating group with rating sub-groups.

- 1 Configure a rating sub-group.

```
an3000-mcm-slot7cpu1(config)# service-construct rating-sub-group
<sub-group-name>
```

The following table describes the configurable parameters.

Parameter	Description
<code>offline-enabled</code>	Specifies whether to enable/disable offline charging for the rating sub-group. ■ <code>true</code> (default) ■ <code>false</code>
<code>online-enabled</code>	Specifies whether to enable/disable online charging for the rating sub-group. ■ <code>true</code> (default) ■ <code>false</code>
<code>offline-volume-threshold</code>	Specifies the volume threshold, in kilobytes, after which a new service-rule level service data container is generated for rules associated with this rating sub-group.
<code>offline-time-threshold</code>	Specifies the time threshold, in seconds, after which a new service-rule level service data container is generated for rules associated with this rating sub-group.
<code>service-rule</code>	Specifies the service rule(s) to associate with this rating sub-group. A service rule can only be associated with one rating sub-group within the same rating group. A service rule can be associated with two rating sub-groups as long as the two rating sub-groups are not associated with the same rating group. The service rule must already exist.

- 2 Associate the rating sub-groups with a rating group.

```
an3000-mcm-slot7cpul(config)# services charging rating-group
<rating-group> charging-method subgroup rating-sub-group
<list-of-sub-groups>
```

If the `charging-method` for the rating group is set to `subgroup`, you cannot configure a service rule directly under the rating group. Service rules must be associated with a rating sub-group.

***Note:** The Override Control feature and synchronizing Gz/Rf and Gy Charging Data do not apply to rating sub-groups.*

Configuring the Default Charging Method

When the Online/Offline AVPs are included in the CCR-I message at the command level, these AVPs indicate the default charging method for the session based on the configuration at the PCEF.

The default charging method can be configured under the `crbn-profile` object, the APN object, and/or the control-profile object. The MCC gateway uses these configured values to advertise the session-wide default charging method (online/offline) supported on the MCC. However, the default charging method provided by the PCRF in the CCA-I at the command level takes precedence over any pre-configured default charging method on the MCC gateway (PCEF).

After processing the CCA-I, the MCC will derive the session-wide default charging method using the following order of priority.

Priority	Default Charging Method Configuration
1	Online/Offline AVP provided by the PCRF in the CCA-I at the command level.
2	Online/Offline configuration under the first activated CRBN profile in the CCA-I. The default is true. an3000-mcm-slot7cpul(config)# services crbn-profile <crbn-profile-name> charging-methods offline-enabled [true false] online-enabled [true false]
3	Online/Offline configuration under the APN. The default is true. an3000-mcm-slot7cpul(config)# zone default gateway apn <apn-name> charging-methods offline-enabled [true false] online-enabled [true false]
4	Online/Offline configuration under the Control Profile. The default is no-subscription, indicating the MCC sends the Online and Offline AVP as enabled. an3000-mcm-slot7cpul(config)# workflow control-profile <control-profile-name> user-subscription-charging-type [prepaid-subscription postpaid-subscription no-subscription]

Mid-Session Change to the Default Charging Method

The MCC supports a PCRF-initiated mid-session change to the session-wide default charging method by including the Online/Offline AVPs in the RAR message at the command level.

- If the PCRF changes online charging from enabled to disabled mid-session, the MCC reports all current outstanding usage to the OCS and disables online charging for the entire session. When online-charging is changed from disabled to enabled, the MCC starts tracking usage from that moment and reports the usage to the OCS.
- If the PCRF changes offline charging from enabled to disabled mid-session, the MCC reports all current outstanding usage in service-rule level service data containers and disables offline charging for the entire session. When offline-charging is changed from disabled to enabled, the MCC starts tracking usage from that moment and reports the usage in offline CDRs.

Controlling the Charging Method for a Service Rule

You can control the charging method for a service rule based on both the session-wide default charging method and the service-rule default charging method.

```
an3000-mcm-slot7cpu1(config)# zone default gateway profile
gateway-common <name>
use-session-wide-default-charging-state-as-gating [enabled|disabled]
```

- **enabled** – A rule will be charged using the online/offline charging interface if the Online/Offline AVP is enabled at both the session level and the service-rule level. Otherwise, the rule is not charged.
- **disabled** – (default) A rule will be charged using the online/offline charging interface if the Online/Offline AVP is enabled at the service-rule level.

Overriding the Destination-Host and Destination-Realm Static Configuration

The Diameter node on the MCC gateway communicates with statically configured Diameter peers via the Gx interface. When configuring a Diameter peer, you specify the peer host name and realm, which are used in the Destination-Host and Destination-Realm AVPs in Diameter request messages by default.

You can override these default settings on the Gx interface by configuring the `override-hostname` and `override-destination-realm` parameters. When these override parameters are configured, the MCC gateway sends the CCR-I to the selected Diameter peer but includes the configured override destination host/realm in the Destination-Host and Destination-Realm AVPs.

The destination host/realm override configuration only applies to the Destination-Host and Destination-Realm AVPs in CCR-I messages.

In CCR-U/CCR-T/ASA/RAA messages, the MCC gateway uses the values from the Origin-Host/Origin-Realm AVPs received in the CCA-I message to fill the Destination-Host/Destination-Realm AVPs. If the `override-hostname` is configured and if the CCA-I response does not have a different Origin-Host, the MCC gateway continues using the `override-hostname`.

Use the following CLI command to override the static configuration for the Destination-Host AVP sent in CCR-I messages on the Gx interface:

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
aaa-interface-group diameter node-admin <node-name> peer  
<peer-host-name> override-hostname <destination-host>
```

Use the following CLI command to override the static configuration for the Destination-Realm AVP sent in CCR-I messages on the Gx interface:

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
pcrf-charging <interface-name> override-destination-realm  
<destination-realm>
```


Processing the Gx-TMO-Redirect-Server VSA

The MCC can be configured to process the Gx-TMO-Redirect-Server VSA when received in the Charging-Rule-Definition AVP from the PCRF in the CCA-Initial (CCA-I), CCA-Update (CCA-U), or RAR message.

When the PGW receives the Gx-TMO-Redirect-Server VSA within the Charging-Rule-Definition AVP from the PCRF, the MCC will perform a redirection for dynamic or externally-activated predefined rules. The redirection is only valid for HTTP URLs. This feature applies to online, offline, and metering rating groups.

The following subfields are supported under the Gx-TMO-Redirect-Server VSA.

■ Gx-TMO-Redirect-Address-Type	
■ Gx-TMO-Redirect-Server-Address	
■ Gx-TMO-Append-Original-URL	If enabled, the MCC appends the original URL to the redirect URL.
■ Gx-TMO-Append-MSISDN	If enabled, the MCC appends the MSISDN to the redirect URL.
■ Gx-TMO-Append-IMSI	If enabled, the MCC appends the IMSI to the redirect URL.
■ Gx-TMO-Append-IMEI	If enabled, the MCC appends the IMEI to the redirect URL.
■ Gx-TMO-Append-MSIP	If enabled, the MCC appends the Mobil Station (MS) IP address to the redirect URL. ■ If both IPv4 and IPv6 MS IP addresses are allocated, the MCC appends the active IP address to the redirect URL. ■ For IPv6 IP addresses, the MCC uses the preferred format.
■ Gx-TMO-Deactivate-By-Redirect	If enabled, the MCC performs a one-time redirection. After the one-time redirection, the PGW will deactivate the rule.

Note: If both the Redirect-Information AVP and Gx-TMO Redirect-Server VSA are received, the MCC will use the Gx-TMO Redirect-Server VSA for redirection and will ignore the Redirect-Information AVP provided the Gx-TMO-Redirect-Server VSA is enabled in the CCA data record template.

The Gx-TMO-Redirect-Server VSA is configurable in Gx CCA data record templates. The default `presentFlag` setting is `no`, indicating the field is not processed when received in the CCA response. To configure the MCC to process this VSA, set the `presentFlag` setting to `yes`.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template  
pgw pcrf diameter cca <template-name> tmoredirectserver presentFlag  
[yes|no]
```

This feature also adds a new configuration parameter under the advice-of-charge object to specify whether to include the MS IP address in the redirect URL.

```
an3000-mcm-slot7cpu1(config)# service-construct advice-of-charge  
<name> http include-msip [true|false]
```

Retrieving Flow Information From a Manifest Server

Operators can configure a Manifest Interface Profile to retrieve flow information from the manifest server over the Gx interface.

Interaction Between the PCRF and MCC Gateway

If the PCRF sends the `acwsponsorurl` VSA along with the Charging-Rule-Definition, TDF-Application-Identifier and Event-Trigger `APPLICATION_START` and `APPLICATION_STOP` in the CCA/RAR message, the MCC gateway will retrieve flow information from the configured manifest server and deliver this information to the PCRF via a CCR-U message. If there are multiple Charging-Rule-Definitions in the CCA-I with different TDF-Application-Identifiers, the PGW will send the flow information matching each TDF-Application-Identifier in separate CCR messages.

After receiving the Flow-Descriptions from the manifest server, the PCRF can then generate unique TDF-Application-Identifiers for each Flow-Description received and install these PCC rules on the MCC gateway with the Event-Trigger `APPLICATION_START` and `APPLICATION_STOP`. When traffic matches these PCC rules, the MCC gateway will send the CCR-U to the PCRF with an Event-Trigger `APPLICATION_START` event-trigger and the Application-Detection-Information AVP with flow-information. The PCRF can then re-install the PCC rule for the detected TDF-Application-Identifier with modified QoS/QCI/ARP.

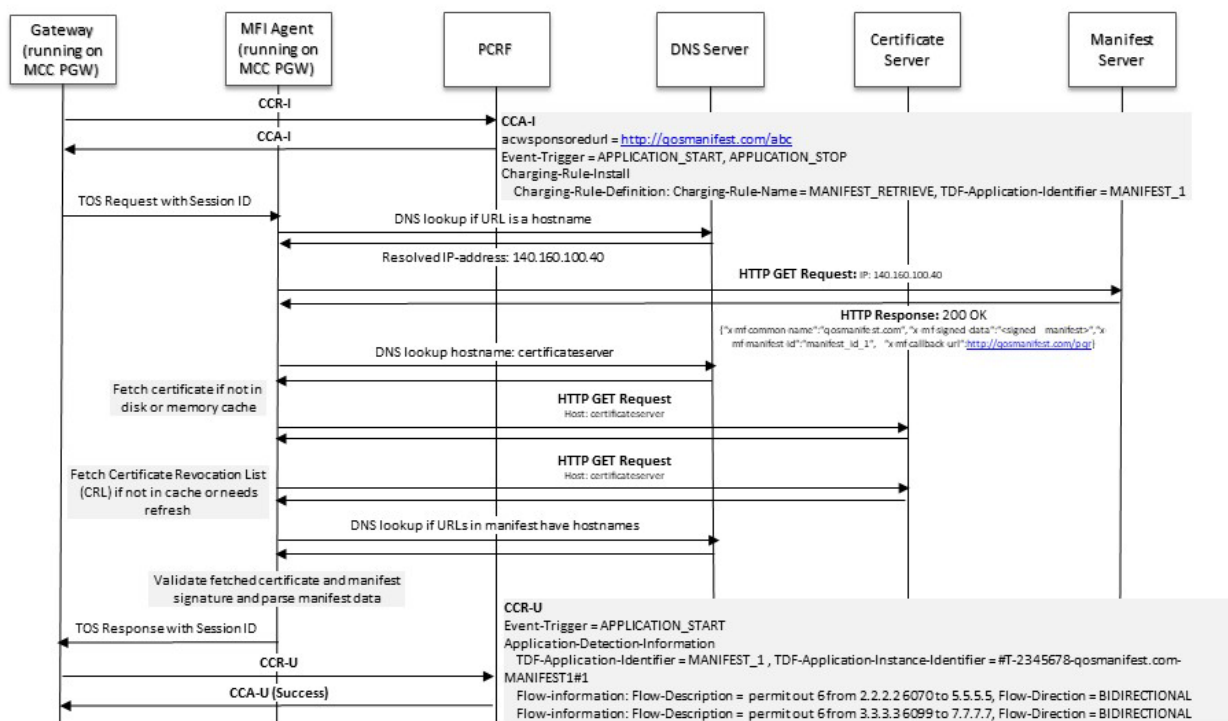
Interaction Between the MFI Agent and Manifest Server

Upon receiving the URL for a manifest server from the PCRF, the gateway component on the MCC sends this URL to the Manifest Interface (MFI) agent (running on the MCC). If there are multiple manifest server URLs inside the request message, the MFI agent only processes the first URL. If the URL received from the PCRF is a host name, the MFI agent issues a DNS lookup to obtain the IP-address.

Upon receiving the manifest from the manifest server, the MFI agent validates the manifest signature by loading the manifest certificate from the disk, if it is not in memory already. If the certificate is not on disk, the MFI agent issues a DNS lookup to obtain the IP address of the certificate server. The MFI agent then downloads the certificate from the certificate server, stores it in memory, and validates the certificate and manifest signature. After successfully validating the manifest signature, the MFI agent parses the manifest data and validates it before sending the manifest information to the PCRF via the Gx interface.

Figure 1-18 illustrates the message flow for a PCRF-triggered manifest retrieval.

Figure 1-18 PCRF-Triggered Manifest Retrieval Message Flow



Custom-Mute VSA

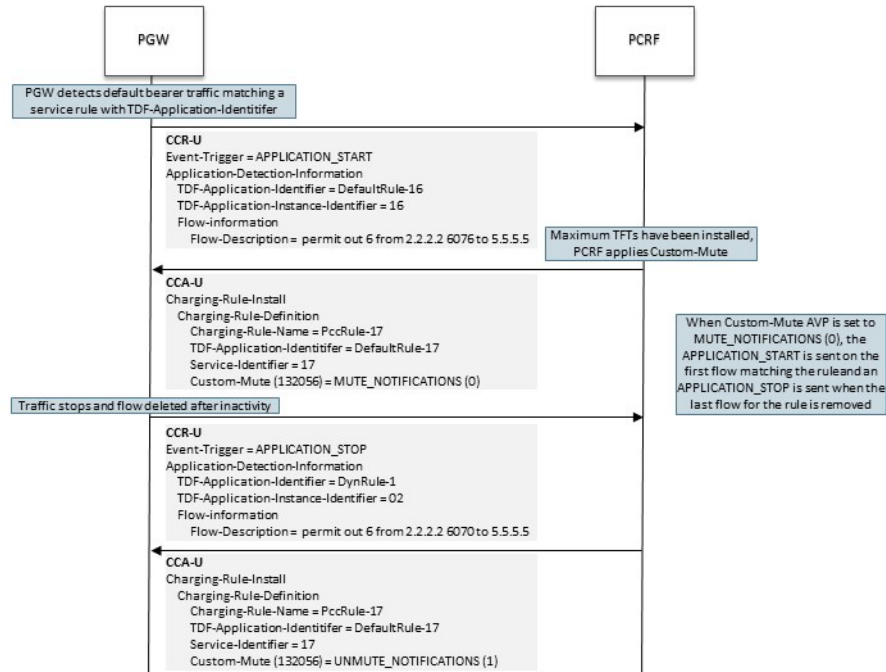
By default, all unique 5-tuple flows when detected will generate APPLICATION_START/STOP notifications to the PCRF. Once the maximum number of dedicated bearers has been created, the PCRF can limit subsequent APPLICATION_START notifications by sending the Custom-Mute VSA with the value MUTE_NOTIFICATIONS (0) for the default bearer PCC rules. Once traffic on the dedicated bearers has stopped and the MCC gateway deletes the dedicated bearers due to an idle-timeout, the MCC will send the PCRF a CCR-U with the APPLICATION_STOP event-trigger. The PCRF can then re-activate APPLICATION_START/STOP notifications by re-installing the PCC rules with the Custom-Mute VSA set to value UNMUTE_NOTIFICATIONS (1). [Figure 1-19 on page 1-119](#) illustrates the message flow for using the Custom-Mute VSA.

The Custom-Mute VSA is sent at the PCC rule level in CCA-U or RAR messages and only applies to the PCC rules on which it is enabled. By default, the Custom-Mute functionality is disabled. If the Custom-Mute VSA is not present in the Charging-Rule-Install AVP, then it will be set to disabled. The Custom-Mute VSA has two values:

- MUTE-NOTIFICATIONS (0) – If the MCC gateway receives a PCC rule with the Custom-Mute VSA set to MUTE_NOTIFICATIONS (0), the MCC PGW will only send one APPLICATION_START notification for the first flow matching the PCC rule and one APPLICATION_STOP notification when the last flow matching the rule is terminated. In this case, the MCC sends the APPLICATION_STOP with the TDF-Application-Instance-Identifier set to 0. Intermediate start/stop for new flows or even flows with a source port change will not generate APPLICATION_START/STOP notifications to the PCRF.
- UNMUTE-NOTIFICATIONS (1) – If the Custom-Mute VSA with the UNMUTE_NOTIFICATION is received for any PCC rule, the MCC PGW will send APPLICATION_START/STOP notifications for every unique 5-tuple flows to the PCRF.

Figure 1-19 illustrates the message flow for using the Custom-Mute VSA.

Figure 1-19 Custom-Mute VSA Message Flow



Manifest Refresh After Validity Time Expiry

If the manifest response comes with a validity time, the MCC includes the validity time in the CC-Time AVP within the Sponsored-Connectivity-Data AVP and starts the validity timer for the manifest refresh.

When the timer expires, the MCC sends another manifest request to the MFI agent. If the new manifest response from the MFI agent contains the validity time, the MCC sends the manifest data to the PCRF with the CC-Time AVP and restarts the validity timer.

To handle multiple manifests that need to be refreshed, the MCC maintains a linked list sorted by the manifest refresh time.

- If the new manifest validity time is smaller than the first entry on the list, the timer is restarted with the smaller time.
- If the list is not empty after the manifest refresh timeout is processed, the MCC restarts the timer immediately for the next entry on the list.

Use the following command to configure the Sponsored-Connectivity-Data AVP in CCR-U data record templates. The default `presentFlag` setting is `no`, indicating the AVP is not sent in CCR-U messages.

```
an3000-mcm-slot7cpu1(config)# service-construct data-record-template
[pgw|ggsn] pcrf diameter interim <template-name>
sponsoredconnectivitydata presentFlag [yes|no|conditional]
```

Note: Ensure that the Application-Detection-Information AVP (`applicationdetectioninfo`) is set to `yes` in the Gx CCR-U data record templates along with the Sponsored-Connectivity-Data AVP (`sponsoredconnectivitydata`) for this feature.

Failure Handling

If the CCR-U to the PCRF fails, the MCC saves the flow information retrieved from the manifest server and resends it in the next CCR-U when the server connection comes back up. Validity time information is also synced with the standby so that after a PGW failover, the manifest can be requested again.

If there is an error in parsing and validating the manifest data, the MCC increments a counter in the statistics that tracks the number of failures and generates a failure log. The MCC PGW will not send a Charging-Rule-Report for this failure and the subscriber will not receive the enhanced QoS, but the session will continue with the default QoS.

Performance Statistics

Use the following command to view the number of manifest requests sent to the MFI agent and the number of responses received at the cluster/grid/node level.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw | include manifest
```

```
num-manifest-server-requests-sent 2
```

```
num-manifest-server-responses-rcvd 2
```

Use the following command to view the number of manifest retrieval statistics in summary.

```
an3000-mcm-slot7cpu1# show service-construct
manifest-interface-profile statistics cluster-level summary
<cluster-id>
```

```
MFI Request - Total Messages
```

```
MFI Request - TDF App Counts
```

```
MFI Response - Successes
```

```
MFI Response - Failures
```

Use the following command to view the number of manifest retrieval statistics in detail.

```
an3000-mcm-slot7cpu1# show service-construct  
manifest-interface-profile statistics cluster-level detail  
<cluster-id>
```

```
MFI Request - No Service Name  
MFI Request - No Session ID  
MFI Request - No TDF Application ID  
MFI Request - No MFI Profile AutoIndex  
MFI Request - No Manifest URL  
MFI Request - GW Service Not Found  
MFI Request - MFI Profile Not Found  
MFI Request - MFI Profile Disabled  
MFI Request - Ports Not Initialized  
MFI Manifest - Download Attempts  
MFI Manifest - Download Successes  
MFI Manifest - Download Errors  
MFI Manifest - Invalid  
MFI Manifest - Received  
MFI Certificate - Download Attempts  
MFI Certificate - Download Successes  
MFI Certificate - Download Errors  
MFI Certificate - Invalid  
MFI Certificate - Received  
MFI CRL - Download Attempts  
MFI CRL - Download Successes  
MFI CRL - Download Errors  
MFI CRL - Invalid  
MFI CRL - Received  
MFI Parser - Decode Errors  
MFI Parser - Signature Verification Failures  
MFI Parser - Expired Validity Time  
MFI DNS Domains Received
```

MFI DNS Queries Sent
 MFI DNS Responses Success
 MFI DNS Send Failed
 MFI DNS Responses Failed
 MFI Response - Sent Successes
 MFI Response - Send Failures
 MFI Response - Empty TFT
 MFI Response - No Service Name
 MFI Response - Processing Failures

Configuring a Manifest Interface Profile

Use the following commands to configure the MCC gateway to retrieve flow information from a manifest server for ADC reporting over the Gx interface.

- 1 Configure the manifest interface profile.

```
an3000-mcm-slot7cpu1(config)# service-construct
manifest-interface-profile <manifest-profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
admin-state	Specifies the administrative state of the manifest interface profile. ■ enabled (default) ■ disabled
certificate-server	Configure the certificate server parameters: ■ ca-certificate-list – Specifies a list of CA certificates. ■ certificate-url – Specifies the certificate server URL or IP address. ■ certificate-file-extension – Specifies the certificate file extension. ■ crl-url – Species the URL of the Certification Revocation List (CRL). ■ network-context – Specifies the network context to use to connect to the certificate server. ■ loopback-ip – Specifies the loopback IP address to use to connect to the certificate server.
certificate-verification-admin-state	Specifies whether the certificate requires verification. ■ enabled (default) ■ disabled

Parameter	Description
manifest	Configure the manifest parameters: ■ manifest-max-size – Specifies the maximum allowed size of the manifest. ■ manifest-max-size-unit – Specifies the unit for the manifest maximum size. – bytes – kbytes

- 2 Associate the manifest-interface-profile with a workflow control-profile.

```
an3000-mcm-slot7cpu1(config)# workflow control-profile
<control-profile-name> manifest-interface-profile
<manifest-profile-name>
```

Overriding the CC Value Received in the Create Session Request

This feature adds the ability to override the charging-characteristics (CC) value received in the Create Session Request message for a subscriber based on the subscriber type.

Set parameters using the following commands.

- 1 Configure a charging-characteristics profile with a configured CC value for each subscriber type.

```
an3000-mcm-slot7cpu1(config)# service-construct
charging-characteristics-profile <cc-profile-name>
[home-subscriber-cc-value|roaming-subscriber-cc-value|
visiting-subscriber-cc-value] <value>
```

- 2 Associate this charging-characteristics profile with a Charging-Rule-Base-Name (CRBN) profile, APN and/or gateway service.

```
an3000-mcm-slot7cpu1(config)# services crbn-profile
<crbn-profile-name> charging-characteristics-profile <cc-profile-name>

an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name>
charging-characteristics-profile <cc-profile-name>

an3000-mcm-slot7cpu1(config)# zone default gateway pgw
<pgw-service-name> charging-characteristics-profile <cc-profile-name>
```

If the Create Session Request message contains a CC value, the MCC will override this value based on the following order of precedence. The MCC also uses this order of precedence for selecting the CC value.

For example, if none of the CRBN profiles have a charging-characteristics-profile associated with them, the MCC will check the 3GPP-Charging-Characteristics AVP received in the RADIUS Access-Accept message. If the RADIUS Access-Accept does not provide a CC value, the MCC will check the charging-characteristics-profile associated with the APN, etc.

Table 1-1 Order of Precedence for Selecting CC Value

Priority	Description
1	The CC value will be selected based on the charging-characteristics-profile associated with a CRBN profile. If multiple CRBNs are received, the MCC selects the first CRBN in the CRBN list. ■ CRBN profile received from the PCRF ■ CRBN profile received from the RADIUS server ■ The default CRBN profile (associated with a data profile)
2	The 3GPP-Charging-Characteristics AVP received in the RADIUS Access-Accept message.
3	The charging-characteristics-profile associated with the APN.
4	The charging-characteristics-profile associated with the gateway service.
5	The CC value received in the Create Session Request/Create PDP Context Request message.
6	The default CC value configured for the Control Profile.

***Note:** If the CC value is not configured in the charging-characteristics-profile, the MCC uses the CC value received in the Create Session Request message.*

Configuring a Dynamic Rating Group Template

You can specify a rating group template to use when a dynamic rating group is created because of a charging rule install from the PCRF. If you do not configure a dynamic rating group template, the MCC uses default values for all the rating group parameters.

```
an3000-mcm-slot7cpu1(config)# workflow data-profile
<data-profile-name> dynamic-rg-template <rating-group-name>
```

dynamic-rg-template – Specifies the name of a rating group to be associated with the data profile. The rating group must already be configured and will be used as a template when a dynamic rating group is created because of a charging rule install from the PCRF. The associated rating group must be configured with a **charging-method** of **subgroup**.

Monitoring the Gx Interface

This section describes how to monitor the operational status of the PCRF via the Gx/Gxa interface and how to display statistical information for PCRF communication messages. This section also describes the alarm notification that occurs when the TCP connection between the GGSN/PGW/TWAG and the PCRF fails. For more information on statistics, see *Acuitas Performance Statistics*.

Node Status

Use the following CLI command to display the operational status of the local diameter node.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter node-status <diameter-node-name>
```

The system displays output similar to the following:

```
origin-host      client.affirmed.net
origin-realm     affirmed.net
hostIPAddress    10.48.11.2
transport        tcp
port             3868
geo-redundant-group ""
status           inservice
```

Peer Status

Use the following CLI command to display the operational status of the TCP connection between the system and the specified PCRF.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-status <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

```
realm            affirmed.com
ip-address       10.6.1.217
port             5999
transport        tcp
role             primary
backup-host-name pcrf-secondary-server.affirmed.com
status           Okay
last-state-change-timestamp 2015-06-23T12:44:08-0400
override-hostname ""
```

Peer Command-Code Statistics

Use the following CLI command to display the number of request/response messages sent per command-code between the system and the specified PCRF.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-cmd-stats <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

NODE NAME	HOST NAME	COMMAND CODE	REALM	NUM OUT REQ	NUM OUT RSP	NUM IN REQ	NUM IN RSP
PCRF	pcrf-server.affirmed.com	257	affirmed.com	98	0	0	98
PCRF	pcrf-server.affirmed.com	271	affirmed.com	8559	0	0	8559
PCRF	pcrf-server.affirmed.com	280	affirmed.com	5744	5831	5831	5743
PCRF	pcrf-server.affirmed.com	282	affirmed.com	0	20	20	0

Peer Statistics

Use the following CLI command to display the number of inbound/outbound requests/responses, protocol errors and failures between the system and the specified PCRF.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter peer-stats <diameter-node-name>
<peer-host-name>
```

The system displays output similar to the following:

realm	affirmed.com
num-protocol-error	0
num-permanent-failure	0
num-routing-failure	0
num-out-request	5844
num-out-response	5850
num-in-request	5850
num-in-response	14399

Diameter Message-Level Statistics

Use the following CLI command to display statistics about diameter messages exchanged between the local diameter node and diameter peer via the Gx interface.

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level peer
interface-type GX <diameter-node-name> <peer-host-name>
```

Note: The *num-ccr-initial-failure*, *num-ccr-update-failure* and *num-ccr-terminate-failure* statistics counters reflect the totals of the 3xxx, 4xxx, and 5xxx counts combined.

Note: For more information on Gx/Gxa interface performance statistics, see *Acuitas Performance Statistics*.

Use the following CLI command to display procedure level success/failure statistics for the PCRF Re-Auth message processing:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
[cluster-level|grid-level] procedure pgw
```

Data Usage Cumulative Statistics

Use the following CLI commands to display cumulative statistics for data usage at the interface, service rule, and bearer level.

Interface Level

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats interface-usage-stats <gw-type> <gw-name>
<interface-type> <interface-name>
```

```
total-used-bytes
```

```
total-uplink-bytes/total-downlink-bytes
```

Service-Rule Level

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats service-data-container-usage-stats <gw-type>
<gw-name> <rating-group> <service-id>
```

Note: The service-id can be the configured service-data-flow-id or the service-id received from the PCRF.

```
total-uplink-bytes/total-downlink-bytes
```

```
total-uplink-packets/total-downlink-packets
```

```
total-unreported-uplink-bytes/total-unreported-downlink-bytes
```

```
total-unreported-uplink-packets/total-unreported-downlink-packets
```

Bearer Level

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
data-usage-stats bearer-usage-stats <gw-type> <gw-name> <qci>
```

```
total-uplink-bytes/total-downlink-bytes
```

```
total-uplink-packets/total-downlink-packets
```

Alarms

A major-level trap similar to the following occurs when the TCP connection between the system and the PCRF goes down. The alarm clears after connectivity is restored.

```
anDiameterPeerLinkDown
```

Description:	Transport connection to the diameter peer is down.
Trap ID:	89
Status:	Current
Severity:	Major
Auto Clearing:	No
Potential Impact:	The Diameter Links are Down towards the OFCS server. This might result in Gz Interface not able to send the Call Data Records to Billing system.
Corrective Action:	The alarm clears when the interface becomes active. Check the routing/connectivity towards the Diameter peer. The alarm clears after the connectivity is restored. Once you have restored the IP connectivity make sure the Diameter Peer is up and routes are correctly configured and Diameter configuration is correct.

Failure Handling

The system uses the watchdog request-based mechanism defined in RFC 3588 and RFC 3539 for failure detection. By exchanging Device-Watchdog-Request (DWR) and Device-Watchdog-Answer (DWA) messages between the GGSN/PGW and PCRF, the system can detect PCRF or network-related failures and migrate sessions to an alternate PCRF, if available.

This section describes failure handling on the Gx/Gxa interface and includes the following sections.

- [Peer Selection \(page 1-130\)](#)
- [Failover Handling \(page 1-131\)](#)
- [Default Gx Policy \(page 1-132\)](#)
- [Gx Rejoin and Gx Replay \(page 1-133\)](#)
- [Configuring the Diameter Result Code List \(page 1-135\)](#)
- [Configuring Multiple Pairs of Primary and Secondary Peers \(page 1-139\)](#)
- [Configuring a Gx Hold Timer for CCR-I Failure to PCRF \(page 1-141\)](#)
- [Continue Session for CCR-U Failure to PCRF \(page 1-149\)](#)
- [Failure Handling Message Flows \(page 1-151\)](#)

Peer Selection

Diameter Node Peer List

The diameter node (GGSN/PGW/TWAG) contains a list of peers (diameter servers) with which it communicates. Each peer must be assigned one of two roles; primary or secondary. Primary peers are active peers used for diameter sessions and secondary peers are used for diameter sessions only when the active peers are unavailable or return a CCA error message with a 3xxx result code.

For new session requests, the system uses a round-robin approach to select peers from the list of primary peers. If there are no active primary peers, the system selects from the list of active secondary peers.

When the system detects that a peer is down, it removes the peer from the active primary or secondary peer list, depending on its role. When the peer comes back up (CER/CEA exchanges successfully complete), the system adds it back to the active peer list.

Backup Server

You can configure a secondary peer to act as the backup server for a primary peer. The backup configuration is a way to establish a redundancy relationship between two peers so that if the primary peer is unavailable, the system automatically sends all new and pending requests to the configured peer (backup server) instead of the next peer in the list. If the backup server fails, the system automatically fails back to the primary server. Only primary peers can be configured with a backup server. When configuring a primary peer with a backup server, enter the secondary peer host name in the `backup-peer-name` parameter. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#) for more information.

If the system determines that the peer to which the CCR message was sent is unavailable, subsequent messages are sent to the backup server. Existing sessions that were redirected to the backup server, stay on the backup server even when the primary peer recovers. Once connectivity to the primary peer is restored, the GGSN/PGW/TWAG uses the primary peer for new sessions.

For the CCR-I, if no backup server is found, then there is another lookup to send the message to a different primary/secondary peer. For all other messages, if there is no backup server found, messages are retransmitted to the *same* peer.

Failover Handling

Failover to an alternate PCRF can be triggered if one of the following events occurs:

- The PCRF returns a temporary failure (result code 3xxx).
- The PCRF does not respond to DWRs.
- The CCR timer (`retransmit-timeout`) expires. You can configure this value when you configure the PCRF charging interface. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#).
- The system detects a broken connection (packet send fails).

You can configure a secondary peer to act as the backup server for a primary peer so that when a failover event occurs, the system automatically sends all new and pending requests to the backup server. If the backup server fails, the system automatically fails over to the primary server. When configuring a secondary peer as a backup server, you must enter the host name of the secondary server in the `backup-peer-name` parameter. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#) for more information.

The system handles failovers for new and existing sessions differently:

New Sessions – When a failover event occurs, the system sends new sessions to the backup server. If no backup server is found, the system sends new session requests to another active peer from the list of primary servers. If there are no active primary servers, the system selects from the list of active secondary servers. When the primary PCRF recovers, the system sends new session requests to the primary PCRF.

Existing Sessions – When a failover event occurs, the system redirects all messages for existing sessions to the backup server. If no backup server is found, the system retransmits the CCR messages to the *same* peer. Existing sessions that were redirected to the backup PCRF, stay on the backup server even when the primary PCRF recovers.

Note: If the PCRF returns a CCA error message (result code 4xxx or 5xxx), the system terminates the session. See [Configuring the Diameter Result Code List \(page 1-135\)](#) for information on how to configure the Diameter result code list used to control failover handling on the Gx interface based on the result code received in the CCA error message.

Note: For failover information when using the Geographic Redundancy Manager, see the *Acuitas User's Guide*.

Default Gx Policy

The Default Gx Policy feature helps to avoid disruption in subscriber service by allowing the MCC gateway to set up sessions when one of the following events occurs:

- No PCRF is available.
- The PCRF returns a temporary failure (result code 3xxx).
- The CCR `retransmit-count` is reached. The `retransmit-count` specifies the maximum number of times the gateway transmits the CCR. You can configure this value when you configure the PCRF charging interface. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#).

When the Default Gx Policy feature is enabled and the PCRF is unavailable or the PCRF returns a temporary failure (result code 3xxx), the MCC gateway uses the information in the locally configured workflow data profile (from the subscriber analyzer) as the default policy to set up new subscriber sessions.

When the PCRF recovers, the MCC gateway sends new session requests to the recovered PCRF and continues to process existing sessions using the locally configured default policy until the sessions are complete. Existing sessions using the default policy do not rejoin the PCRF unless the Gx Rejoin/Replay feature is enabled. See [Gx Rejoin and Gx Replay \(page 1-133\)](#) for more information.

When there is a primary and backup PCRF configured and the primary server is unavailable or responds with a temporary failure (result code 3xxx), the MCC gateway retransmits the CCR-I request to the backup PCRF. If the backup PCRF is unavailable or responds with a temporary failure (result code 3xxx), the MCC gateway uses the default policy, if enabled.

If no backup server is configured, the system retransmits the CCR-I request to another active peer from the list of primary servers. If there are no active primary servers, the system selects from the list of active secondary servers. If there are no active primary or secondary servers or the configured CCR `retransmit-count` is reached, the MCC gateway uses the default policy, if enabled.

If the Default Gx Policy feature is *disabled*, the system terminates the session when one of the following events occurs:

- The PCRF returns a temporary failure (result code 3xxx).
- The CCR `retransmit-count` is reached. You can configure this value when you configure the PCRF charging interface. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#).

***Note:** Subscriber usage information for a session processed using the default policy is not reported to the PCRF.*

Use the following command to enable the Default Gx Policy feature:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> pcrf-down-default-gx-call-setup enabled
```

Gx Rejoin and Gx Replay

The Gx Rejoin and Gx Replay features help to ensure session continuity if one of the following events occurs during an active session:

- No PCRF is available.
- The PCRF responds with a temporary failure (result code 3xxx).
- The CCR `retransmit-count` is reached. The `retransmit-count` specifies the maximum number of times the gateway transmits the CCR. You can configure this value when you configure the PCRF charging interface. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#).

The Gx Rejoin feature allows the MCC gateway to rejoin the PCRF in the event of a PCRF failure. If Gx Rejoin is enabled and the PCRF is unavailable or the PCRF responds with a temporary failure (result code 3xxx), the GGSN/PGW starts the rejoin timer (`pcrf-down-rejoin-time`). When the rejoin timer expires, the GGSN/PGW attempts to connect to any available PCRF within the diameter server group.

Gx Replay is used if all PCRF servers go down after the Gx session is already established and the session is terminated before the PCRF becomes available. In this case, if Gx Replay is enabled, the MCC gateway replays the cached session information via a CCR-Terminate message. If a session is terminated before a PCRF becomes available, note the following:

- If a session is processed using dynamic policy rules and is terminated before a PCRF becomes available, the MCC gateway replays all cached session information in a CCR-T message to any PCRF server that becomes available in the Diameter server group.
- If a session is processed using default policy rules and is terminated before a PCRF becomes available, session information is not reported to the PCRF.

Note: If after the PCRF rejoin timer expires, the PCRF responds with a temporary failure (result code 3xxx), the PCRF rejoin timer restarts.

Note: If Gx Replay is enabled and the PCRF returns a temporary failure (result code 3xxx) in response to the CCR-T message, the system retransmits the CCR-T every 30 seconds until the PCRF sends a success code.

When there is a primary and backup PCRF configured and the primary server is unavailable or responds with a temporary failure (result code 3xxx), the MCC gateway redirects all messages for existing sessions to the backup server. If the backup PCRF is also unavailable or responds with a temporary failure (result code 3xxx), the MCC gateway uses the Gx Rejoin/Replay features, if enabled.

If no backup server is configured, the system retransmits the CCR messages to the *same* primary peer. If the configured CCR `retransmit-count` is reached, the GGSN/PGW uses the Gx Rejoin/Replay features, if enabled.

If the Gx Rejoin/Replay features are *disabled*, the system terminates the session when one of the following events occurs:

- The PCRF responds with a temporary failure (result code 3xxx).
- The CCR `retransmit-count` is reached. You can configure this value when you configure the PCRF charging interface. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#).

Use the following command to enable Gx Rejoin and Gx Replay:

```
an3000-mcm-slot7cpu1(config)# service-construct interface  
pcrf-charging <interface-name> pcrf-down-gx-rejoin enabled  
pcrf-down-rejoin-time <value> pcrf-down-gx-replay [enabled|disabled]
```

Configuring the Diameter Result Code List

This feature allows Operators to configure the Diameter credit control failover behavior on the Gx/Gxa interface based on the result code received in the CCA error message.

Operators configure a list of result code ranges (3xxx-5xxx) and when a CCA message is received with a non-success result code (other than 2001), the default Gx policy (if enabled on the interface) is used to set up the session if the received result code is present in the result code list. Also, for each result code range in the list, Operators can specify whether or not the PGW retransmits the request to a backup PCRF before applying the default Gx policy.

Note: For message flow information, see [Failure Handling – Result Code List Enabled and Retry-Secondary Disabled \(page 1-165\)](#) and [Failure Handling – Result Code List Enabled and Retry-Secondary Enabled \(page 1-166\)](#).

1 Configure the Gx/Gxa Diameter result code list.

```
an3000-mcm-slot7cpu1(config)# service-construct
diameter-result-code-policy-list <list-name> policy <policy-name>
result-code-range low <value> high <value>
```

The following table describes the configurable parameters.

Parameter	Description
<code>diameter-result-code-policy-list</code>	Specifies the name of the result code list.
<code>policy</code>	Specifies the name of the result code policy.
<code>result-code-range</code>	Specifies the low and high end of the result code range. ■ <code>low</code> ■ <code>high</code> Note: Ensure the result code range does not overlap with any previously configured ranges. Note: To configure a range with a single result code, configure the low and high value with the same code.

Parameter	Description
retry-secondary	Specifies whether the MCC gateway retransmits the request to a backup PCRF before applying the default Gx policy when the result code received in the CCA error message is within the configured range. ■ enabled – (default) When the primary server returns a result code within the configured range, the MCC gateway retransmits the request to a backup PCRF before applying the default Gx policy. ■ disabled – When the primary server returns a result code within the configured range, the MCC gateway uses the default Gx policy to set up the session and does not retransmit the request to the backup PCRF.

For example:

```
an3000-mcm-slot7cpu1(config)# service-construct
diameter-result-code-policy-list CClist policy 1 result-code-range low
3200 high 3300 retry-secondary enabled
```

***Note:** The Gy and Gx/Gxa interfaces share the same configuration for the diameter-result-code-policy-list. However, the action and subscriber-type parameters only apply to the Gy interface. The Gx/Gxa interface only uses the default options for these parameters even if they are configured differently for the Gx/Gxa Diameter result code list.*

2 Associate the result code list with the Gx/Gxa interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> default-gx-cause-code-list <list-name>
default-gx-setup-per-cause-code [enabled|disabled]
```

where

Parameter	Description
pcrf-charging	Specifies the charging interface name.
default-gx-cause-code-list	Specifies the name of the result code list.

Parameter	Description
<code>default-gx-setup-per-cause-code</code>	Specifies whether to enable or disable session set up using the default Gx policy based on the result code received in the CCA error response. ■ enabled – The result code returned in the CCA error message is compared to the configured result code list, and if there is a match, the session is set up using the default Gx policy. ■ disabled – (default) The result code list functionality is not used to set up the session with the default Gx policy. However, if the result code returned in the CCA error message is a 3xxx error, the default Gx policy is used to set up the session.

For example:

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging PCRF-1 default-gx-cause-code-list CClist
default-gx-setup-per-cause-code enabled
```

- 3 Enable the Default Gx Policy feature to ensure the default policy is applied when a PCRF responds with one of the result-code errors within the range.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> pcrf-down-default-gx-call-setup enabled
```

Diameter Result Code Statistics

Use the following commands to display cluster-level Diameter result code statistics for each message type (for example, CCA-I/CCA-U/CCA-T) for Gx peers in each node. The result code statistics for each peer will be registered dynamically as they are received from the PCRF.

To display Diameter result code statistics:

```
an3000-mcm-slot7cpu1# show service-construct interface
aaa-interface-group diameter statistics cluster-level peer
result-code-statistics <interface-type> <node-name> <peer-name>
<result-code>
```

where:

- **interface-type** – Specifies the interface type.
- **node-name** – Specifies the name of the node.
- **peer-name** – Specifies the peer name.
- **result-code** – Specifies the result code received from the PCRF.

To clear Diameter result code statistics:

```
an3000-mcm-slot7cpu1# service-construct interface aaa-interface-group  
diameter statistics cluster-level peer result-code-statistics  
<interface-type> <node-name> <peer-name> <result-code> clear
```

Note: *If a non-redundant Control/Content Services Module (CSM) reboots, the PGW may delete the result code statistics table.*

Configuring Multiple Pairs of Primary and Secondary Peers

The MCC allows for the configuration of multiple primary active peers and secondary peers on the Gx/Gxa interface within a Diameter server group. In this case, all of the primary active and secondary peers within a Diameter server group must share the same origin-host name and origin-realm (configured at the Diameter server-group level) and each peer is identified by a unique peer host name (configured at the peer level).

This feature adds two new CLI parameters that allow the origin-host and origin-realm to be configured at the peer level thereby allowing Operators to configure up to four pairs of peers (four primary peers and four secondary peers) on the Gx/Gxa interface within a Diameter server group.

Figure 1-20 shows an example of the configuration used to configure multiple pairs of primary and secondary peers on the Gx/Gxa interface. The new CLI parameters added at the peer level appear as red text. While Figure 1-20 uses the Gy interface as an example, the configuration remains the same for the Gx/Gxa interface.

Figure 1-20 Example Configuration for Multiple Pairs of Primary/Secondary Peers

Local Diameter Node Configuration

```
an3000-mcm-slot7cpu1(config)# service-construct interface aaa-interface-group diameter node-admin OCS
node-role client
origin-host defaultpeer.ocs.gwgroup
origin-realm ocs.gwgroup
network-context GI
loopback-ip OCSLOOP
transport tcp
port 3700
dscp-value dscp_0_be
connection-attempt-count 0
connection-attempt-timeout 3
cea-timeout 5
dwr-timer 15
dwr-retry-timer 1
dwr-retry-count 3
product-name AN-3000
vendor-id 10415
send-destination-host-avp true
```

Diameter Peer Configuration

Pair 1 peer OCS-Sim1.ocs.gwgroup peer-identifier 7 origin-host primarypeer1.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.148.3 dest-port 3868 transport tcp role primary backup-peer-name OCS-Sim2.ocs.gwgroup admin-state enabled override-dest-hostname PrimaryOCS.ocs.gwgroup	Pair 2 peer OCS-Sim3.ocs.gwgroup peer-identifier 8 origin-host primarypeer2.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.148.3 dest-port 3868 transport tcp role primary backup-peer-name OCS-Sim4.ocs.gwgroup admin-state enabled override-dest-hostname PrimaryOCS.ocs.gwgroup	Pair 3 peer OCS-Sim5.ocs.gwgroup peer-identifier 9 origin-host primarypeer3.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.148.3 dest-port 3868 transport tcp role primary backup-peer-name OCS-Sim6.ocs.gwgroup admin-state enabled override-dest-hostname PrimaryOCS.ocs.gwgroup	Pair 4 peer OCS-Sim7.ocs.gwgroup peer-identifier 10 origin-host primarypeer4.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.148.3 dest-port 3868 transport tcp role primary backup-peer-name OCS-Sim8.ocs.gwgroup admin-state enabled override-dest-hostname PrimaryOCS.ocs.gwgroup
peer OCS-Sim2.ocs.gwgroup peer-identifier 7 origin-host primarypeer1.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.158.3 dest-port 3869 transport tcp role secondary admin-state enabled override-dest-hostname SecondaryOCS.ocs.gwgroup	peer OCS-Sim4.ocs.gwgroup peer-identifier 8 origin-host primarypeer2.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.158.3 dest-port 3869 transport tcp role secondary admin-state enabled override-dest-hostname SecondaryOCS.ocs.gwgroup	peer OCS-Sim6.ocs.gwgroup peer-identifier 9 origin-host primarypeer3.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.158.3 dest-port 3869 transport tcp role secondary admin-state enabled override-dest-hostname SecondaryOCS.ocs.gwgroup	peer OCS-Sim8.ocs.gwgroup peer-identifier 10 origin-host primarypeer4.ocs.gwgroup origin-realm ocs.gwgroup dest-realm ocs.gwgroup dest-ip-address 10.48.158.3 dest-port 3869 transport tcp role secondary admin-state enabled override-dest-hostname SecondaryOCS.ocs.gwgroup

Figure 1-20 uses the following configuration:

- All of the primary peers are configured with the same:
 - Destination host name (`override-dest-hostname`)
 - Destination IP address (`dest-ip-address`)
 - Destination port (`dest-port`)
- All of the secondary peers are configured with the same:
 - Destination host name (`override-dest-hostname`)
 - Destination IP address (`dest-ip-address`)
 - Destination port (`dest-port`)
- Each peer is configured with a unique peer host name (`peer`). The peer host name is used by the MCC for load balancing between peers and for statistics.
- The value configured for the destination host name (`override-dest-hostname`) for each peer overrides the configured peer host name (`peer`) in the Destination-Host AVP within CCR-I messages only.
- In CCR-U/CCR-T, the gateway uses the value from the Origin-Host AVP received in the CCA-I/CCA-U message to fill the Destination-Host AVP. In RAA messages, the gateway uses the value from the Origin-Host AVP received in the RAR message to fill the Destination-Host AVP.
- Each primary and secondary peer in a pair is configured with the same:
 - Origin-host name (`origin-host`)
 - Origin-realm (`origin-realm`)

The origin-host and origin-realm configured at the peer level overrides the origin-host and origin-realm configured at the Diameter server-group level.
- Each primary peer is configured with the secondary peer host name (`peer`) in the `backup-peer-name` parameter.
- In this configuration scenario, the CCR-I messages should include the Destination-Realm AVP but not the Destination-Host AVP. Only the CCR-U and CCR-T messages will have both the Destination-Realm and Destination-Host AVPs. To ensure the Destination-Host AVP is not included in Gx/Gy CCR-I messages, set the `presentFlag` setting to `no` in the CCR start data record templates.

```
an3000-mcm-slot7cpu1(config)# service-construct
data-record-template [pgw|ggsn] [ocs|pcrf] diameter start
<template-name> destinationhost presentFlag no
```

Configuring a Gx Hold Timer for CCR-I Failure to PCRF

If the CCR-I to the PCRF fails (response times out, PCRF returns an error/Tx error, or peers are down/not available), this feature allows the MCC gateway to temporarily set up sessions using CRBN profiles received from the RADIUS server, if available, or using the default CRBN profile associated with the data profile until a configured Gx hold timer expires.

Note: To use this feature, the configuration for the *crbn-mode* parameter under the *pcrf-charging* object and *authorization-service* object must match.

Set the parameters for this feature using the following commands:

- 1 Configure the Gx hold timer and random range at the APN level or at the gateway-common level. The configuration at the APN level overrides the configuration at the gateway-common level.

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn <apn-name>
pcrf-down-session-hold-time <value>
pcrf-down-session-hold-time-adjust-range <value>
```

```
an3000-mcm-slot7cpu1(config)# zone default gateway profile
gateway-common <profile-name> pcrf-down-session-hold-time <value>
pcrf-down-session-hold-time-adjust-range <value>
```

where

Parameter	Description
apn	Specifies the name of the APN.
gateway-common	Specifies the name of the gateway common profile.
pcrf-down-session-hold-time	Specifies the Gx hold timer. The Gx hold timer indicates the amount of time (in seconds) for which the MCC gateway can allow a session to continue when the CCR-I to the PCRF fails. A value of 0 disables this parameter.
pcrf-down-session-hold-time-adjust-range	This parameter extends the Gx hold timer (<i>pcrf-down-session-hold-time</i>) after it expires to stagger the disconnection of sessions. <i>Note:</i> This parameter is only displayed when the <i>pcrf-down-session-hold-time</i> parameter has a value greater than 0.

- 2 Specify how the MCC should interpret the TMO-Charging-Rule-Base-Name/AF-Rule-Base-Name VSA received in the Access-Accept message on the RADIUS authorization interface.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
authorization-service <auth-interface-name> crbn-mode
[data-profile|crbn-profile]
```

where

Parameter	Description
authorization-service	Specifies the name of the RADIUS authorization interface.
crbn-mode	Specifies how the MCC should interpret the TMO-Charging-Rule-Base-Name/AF-Rule-Base-Name VSA received in the Access-Accept message. ■ data-profile – (default) The value of the TMO-Charging-Rule-Base-Name VSA or AF-Rule-Base-Name VSA received in the Access-Accept corresponds to the locally configured data profile on the MCC gateway. If the value received matches a local data profile, the MCC gateway applies the default CRBN profile associated with this data profile to the session. In this case, the MCC allows only one CRBN profile to be applied to a PDN session. If the MCC gateway receives more than one TMO-Charging-Rule-Base-Name VSA or AF-Rule-Base-Name VSA within the Access-Accept, the system selects the last one and ignores the other ones. If the value received does not match any local data profile, the session is terminated. ■ crbn-profile – The value of the TMO-Charging-Rule-Base-Name VSA or AF-Rule-Base-Name VSA received in the Access-Accept corresponds to the local CRBN profile on the MCC Gateway. If the value received matches a local CRBN profile, the MCC gateway applies the received CRBN profile to the session. In this case, the MCC supports the receipt of multiple TMO-Charging-Rule-Base-Name VSAs and allows for multiple CRBN profiles to be used for a PDN session. However, the MCC allows for only one CRBN profile when the AF-Rule-Base-Name VSAs is received, regardless of the configuration for this parameter. If the value received does not match any configured CRBN profile, the MCC ignores the VSA(s) and the session will continue as if they were not received. <i>Note: This parameter must match the configuration of the crbn-mode parameter under the pcrf-charging object.</i>

- 3 Ensure the Default Gx Policy feature is enabled. If the Default Gx Policy feature is disabled, the MCC does not continue the session after a CCR-I failure regardless of the configuration for the Gx hold timer (`pcrf-down-session-hold-time`).

```
an3000-mcm-slot7cpu1(config)# service-construct interface
pcrf-charging <interface-name> pcrf-down-default-gx-call-setup enabled
```

Note the following guidelines when using this feature:

- The MCC gateway sends all CRBN profiles received from the RADIUS server in the CCR-I to the PCRF.
- If the PCRF responds with a CCA-I that includes CRBNs, the MCC gateway applies the CRBNs received from the PCRF to the session.
- If the PCRF responds with a CCA-I that does not include CRBNs, the MCC gateway applies the local data profile to the session. The CRBN(s) received from the RADIUS server are ignored.
- If the Gx Rejoin feature is enabled and if the rejoin timer is less than the Gx hold timer, the MCC gateway tries to rejoin the PCRF when the rejoin timer expires. If the PCRF responds with a successful CCA-I after attempting to rejoin the PCRF, the Gx hold timer is canceled.
- If the Gx hold timer expires and there is no CCA-I response from the PCRF (even after rejoin attempts), the MCC terminates the session without any further interaction with the PCRF.

Use Cases for the Gx Hold Timer

The following message flows outline some use cases for the Gx hold timer:

- [Use Case 1: CCR-I Failure and RADIUS Returns CBRN \(page 1-144\)](#)
- [Use Case 2: CCR-I Failure, RADIUS Returns CRBNs and Gx Rejoin Enabled \(page 1-146\)](#)

Use Case 1: CCR-I Failure and RADIUS Returns CRBN

The following configuration applies to [Figure 1-21](#) and [Figure 1-22](#).

Parameter	Configuration
Gx Hold Timer (<code>pcrf-down-session-hold-time</code>)	1500 seconds (25 minutes)
RADIUS Authorization Interface <code>crbn-mode</code>	<code>crbn-profile</code>
Gx Rejoin Feature (<code>pcrf-down-gx-rejoin</code>)	disabled

[Figure 1-21](#) illustrates an example where the CCR-I fails and the RADIUS server sends multiple CRBN profiles in the Access-Accept message.

Figure 1-21 Use Case 1: CCR-I Failure with RADIUS CRBN

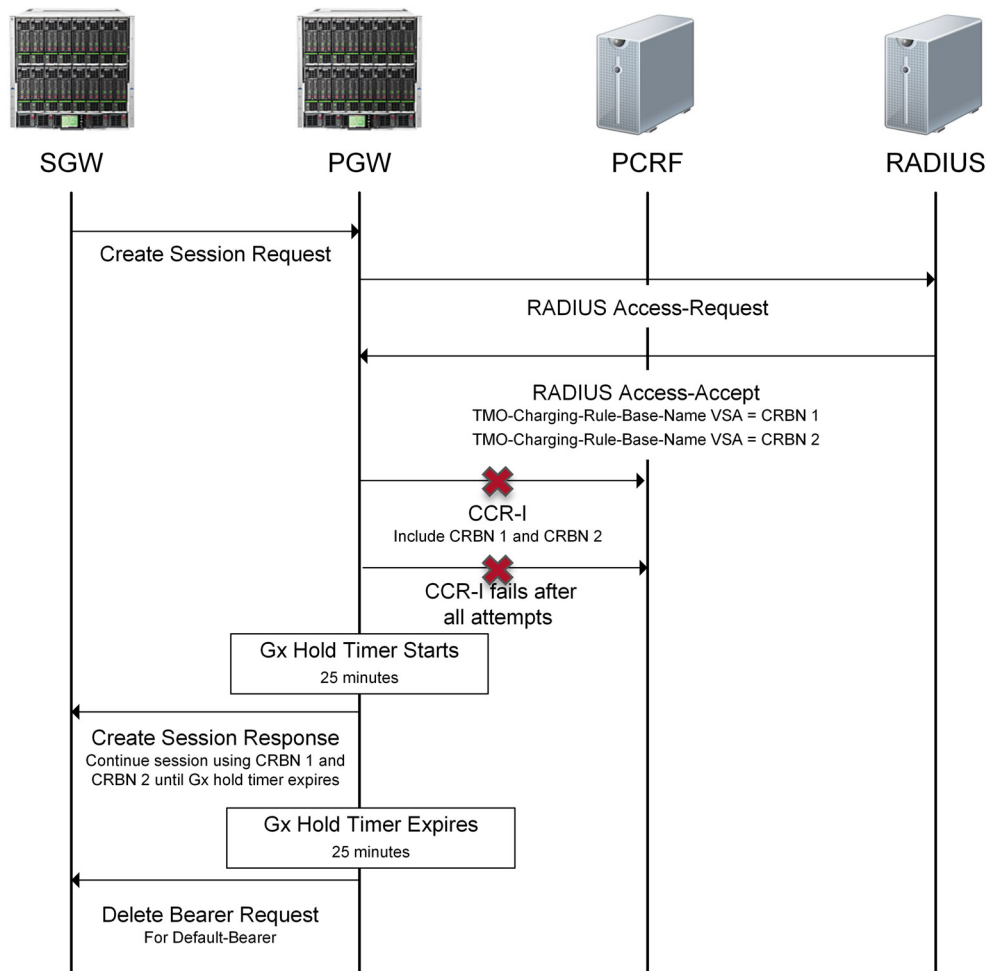
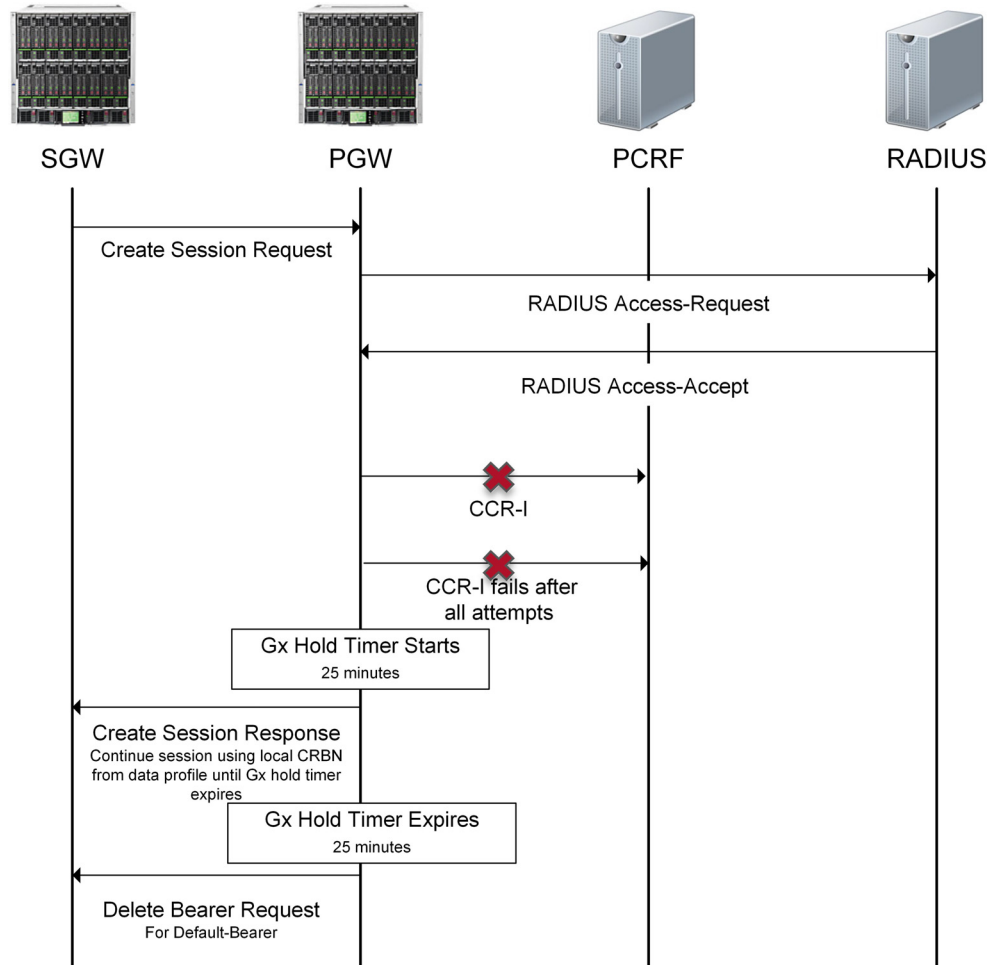


Figure 1-22 illustrates an example where the CCR-I fails and the RADIUS server does not send CRBN profiles in the Access-Accept message.

Figure 1-22 Use Case 1: CCR-I Failure Without RADIUS CRBN



Use Case 2: CCR-I Failure, RADIUS Returns CRBNs and Gx Rejoin Enabled

The following configuration applies to [Figure 1-23](#) and [Figure 1-24](#).

Parameter	Configuration
Gx Hold Timer (pcrf-down-session-hold-time)	1500 seconds (25 minutes)
RADIUS Authorization Interface crbn-mode	crbn-profile
Gx Rejoin Feature (pcrf-down-gx-rejoin)	enabled
Gx Rejoin Feature (pcrf-down-rejoin-time)	600 seconds (10 minutes)

[Figure 1-23 on page 1-147](#) illustrates an example where the CCR-I fails and the RADIUS server sends multiple CRBN profiles in the Access-Accept message. The Gx Rejoin feature is enabled and the rejoin timer is less than the Gx hold timer, so the MCC gateway tries to rejoin the PCRF when the rejoin timer expires. In this example, the PCRF does not respond with a CCA-I (even after rejoin attempts) and the Gx hold timer expires, so the MCC terminates the session without any further interaction with the PCRF.

[Figure 1-24 on page 1-148](#) illustrates an example where the CCR-I fails and the RADIUS server sends multiple CRBN profiles in the Access-Accept message. The Gx Rejoin feature is enabled and the rejoin timer is less than the Gx hold timer, so the MCC gateway tries to rejoin the PCRF when the rejoin timer expires. In this example, the PCRF responds with a successful CCA-I after attempting to rejoin the PCRF and the Gx hold timer is canceled. Furthermore, the PCRF includes CRBNs in the CCA-I response, so the MCC gateway applies the CRBNs received from the PCRF to the session.

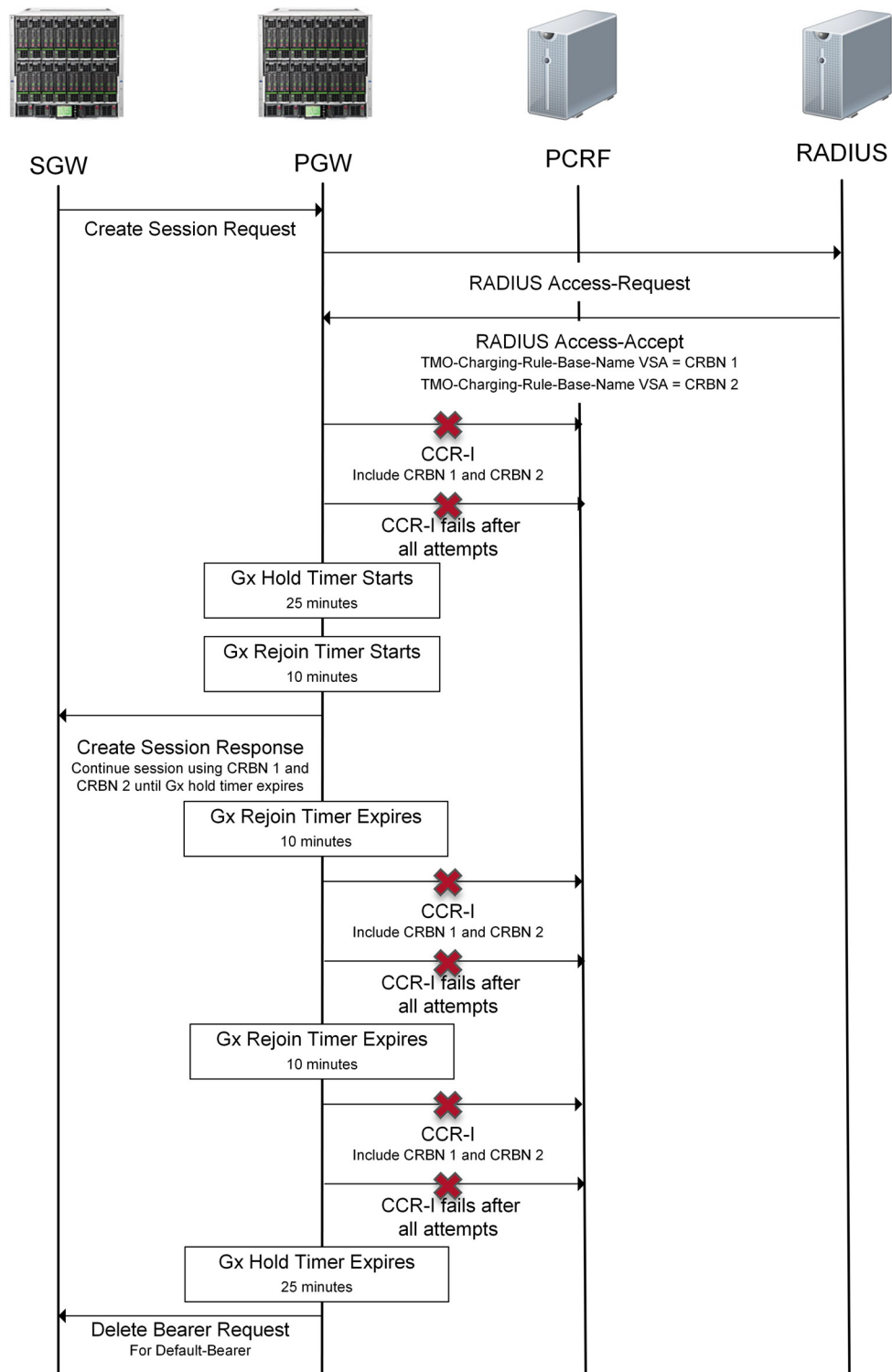
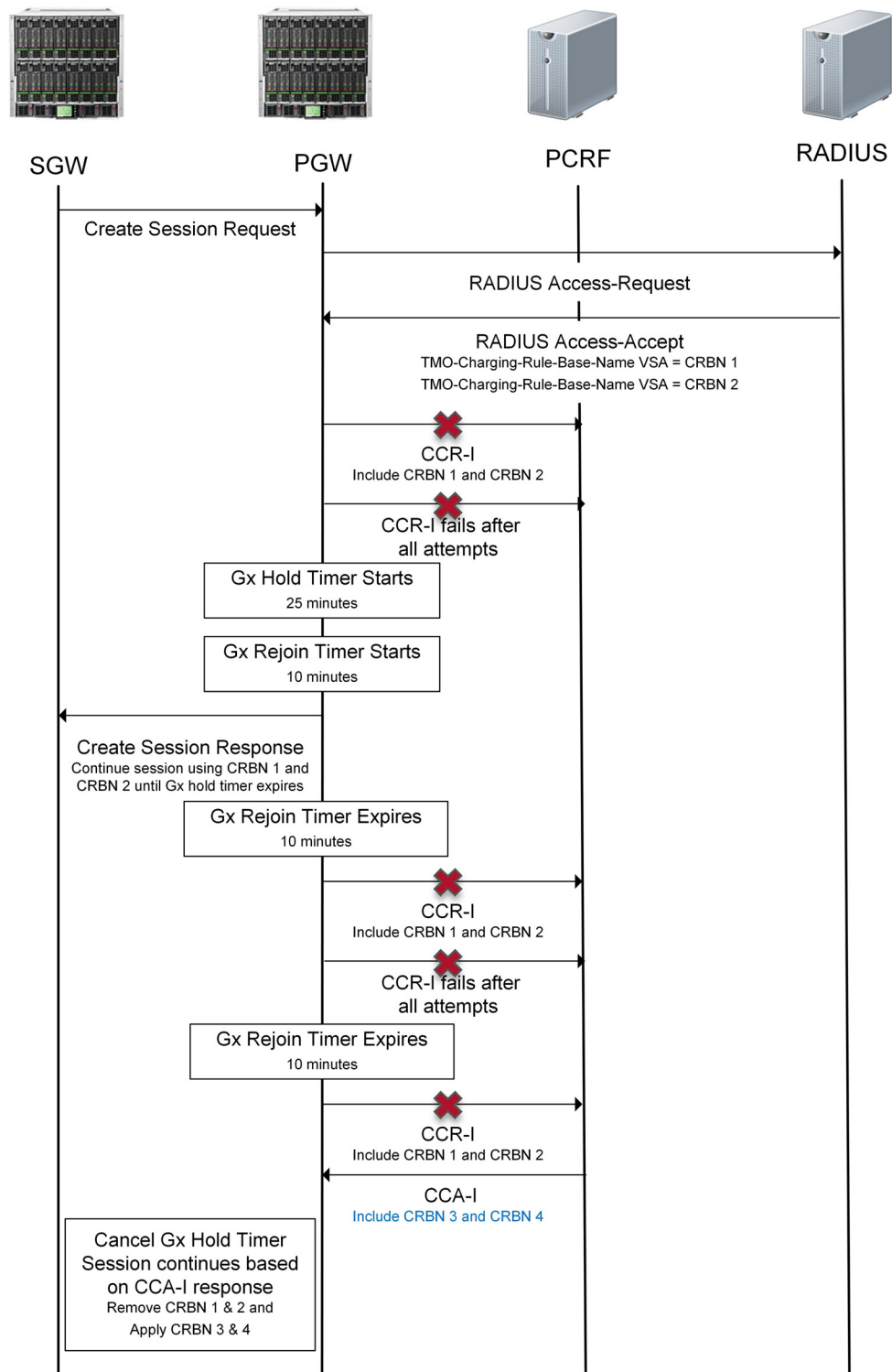
Figure 1-23 Use Case 2: CCR-I Failure with RADIUS CRBN and Gx Rejoin Fails

Figure 1-24 Use Case 2: CCR-I Failure with RADIUS CRBN and Gx Rejoin Succeeds



Continue Session for CCR-U Failure to PCRF

If the CCR-U to the PCRF fails (response times out, PCRF returns an error/Tx error, or peers are down/not available), this feature allows the MCC gateway to continue the session without any further messaging (CCR-U/CCR-T) to the PCRF.

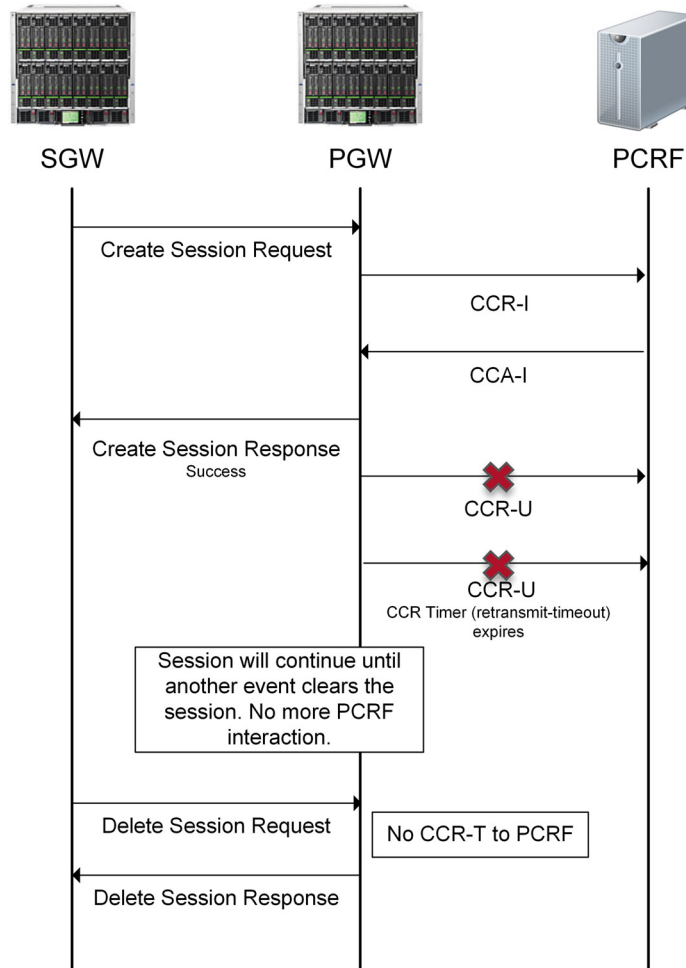
```
an3000-mcm-slot7cpu1(config)# service-construct interface  
pcrf-charging <interface-name> pcrf-update-down-avoid-pcrf  
[true|false]
```

- **true** – If the CCR-U to the PCRF fails, the MCC gateway continues the session without any further messaging (CCR-U/CCR-T) to the PCRF. After the CCR-U failure, if the MCC gateway receives an RAR during this session, it returns an RAA response with diameter result code DIAMETER_UNKNOWN_SESSION_ID (5002).
- **false** – (default) If the CCR-U to the PCRF fails, the MCC gateway attempts to retransmit the CCR-U to the PCRF.

Note: The configuration for this parameter takes precedence over the configuration for the Gx Rejoin feature. If this parameter is set to **true** and the CCR-U messaging fails, the MCC gateway session continues the session without any further messaging to the PCRF, even if the Gx Rejoin feature is enabled.

Figure 1-25 illustrates an example where the CCR-U fails and the `pcrf-update-down-avoid-pcrf` parameter is set to `true`.

Figure 1-25 Continue Session for CCR-U Failure



Failure Handling Message Flows

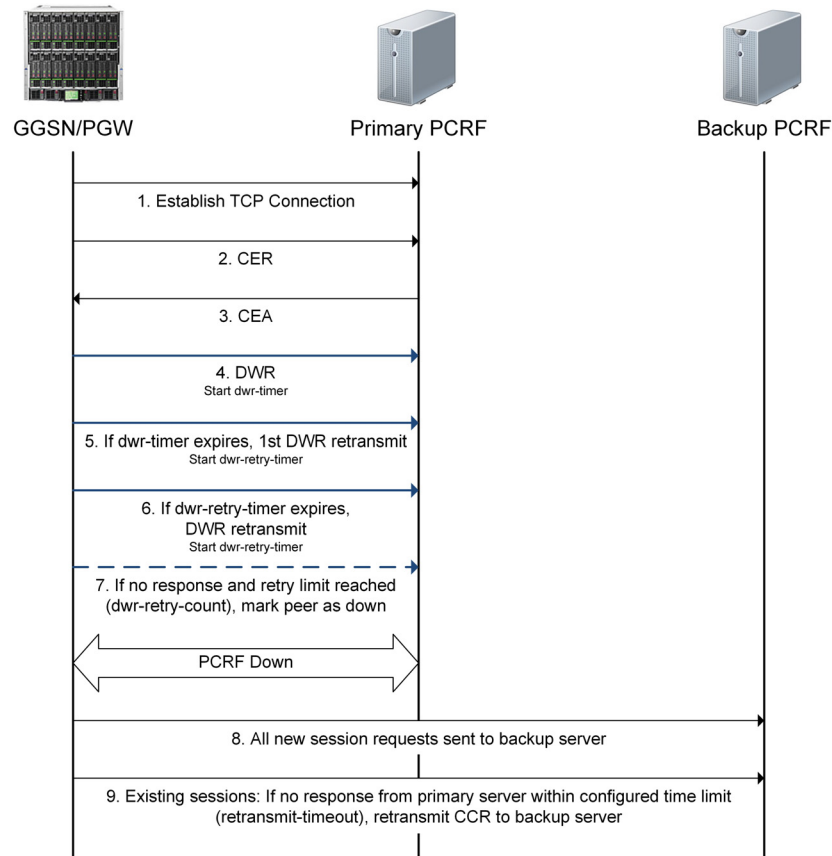
This section provides examples of the following PCRF message flows:

- [Failover to Backup PCRF \(Transport Failure\) \(page 1-152\)](#)
- [Failover to Backup PCRF \(Application Failure\) \(page 1-154\)](#)
- [Failure Handling - Default Gx Policy \(page 1-156\)](#)
- [Failure Handling - Gx Rejoin and Gx Replay \(page 1-157\)](#)
- [Failure Handling - Gx Replay \(page 1-159\)](#)
- [Failure Handling - Default Gx Policy/Gx Rejoin/Replay Enabled \(page 1-161\)](#)
- [Failure Handling - PCRF Responds with 3xxx Result Code \(page 1-163\)](#)
- [Failure Handling – Result Code List Enabled and Retry-Secondary Disabled \(page 1-165\)](#)
- [Failure Handling – Result Code List Enabled and Retry-Secondary Enabled \(page 1-166\)](#)

Failover to Backup PCRF (Transport Failure)

Figure 1-26 demonstrates the message flow during a failover from the primary to the backup PCRF when the watchdog request-based mechanism detects a failure. A failure at the transport level affects all sessions.

Figure 1-26 Failover to Backup PCRF (Transport Failure)



- 1 The GGSN/PGW establishes a TCP Connection with the PCRF.
- 2 The GGSN/PGW sends a Capabilities-Exchange-Request (CER) to the PCRF.
- 3 Assuming both diameter peers share the same capabilities, such as diameter applications and security mechanisms, the PCRF returns with a successful Capabilities-Exchange-Answer (CEA). The TCP connection is successfully established and the peers can begin servicing sessions.
- 4 Periodically, the GGSN/PGW sends DWRs to the PCRF to monitor the TCP connection.
- 5 If the PCRF does not respond to the first DWR with a Device-Watchdog-Answer (DWA) within the configured time limit (*dwr-timer*), the system retransmits the DWR and starts the DWR retry timer (*dwr-retry-timer*) with each retransmission.

Note: The `dwr-retry-timer` is typically set at a shorter interval than the `dwr-timer`.

- 6 If the PCRF does not respond within the time limit (`dwr-retry-timer`), the GGSN/PGW retransmits the DWR. This cycle continues until the GGSN/PGW receives a valid DWA response from the PCRF or the configured retry limit (`dwr-retry-count`) is reached.
- 7 If the configured retry limit (`dwr-retry-count`) is reached, the GGSN/PGW marks the primary PCRF as down.
- 8 The GGSN/PGW sends all new session requests to the backup PCRF.
- 9 For existing sessions, when the GGSN/PGW does not receive a CCA response from the primary PCRF and the CCR timer (`retransmit-timeout`) expires, it retransmits the CCR to the backup server, if available. The TCP connection with the backup PCRF is pre-configured.

Note: If a CCR comes in before the `dwr-retry-timer` expires, the CCR timer (`retransmit-timeout`) itself can expire before the watchdog mechanism detects a failure.

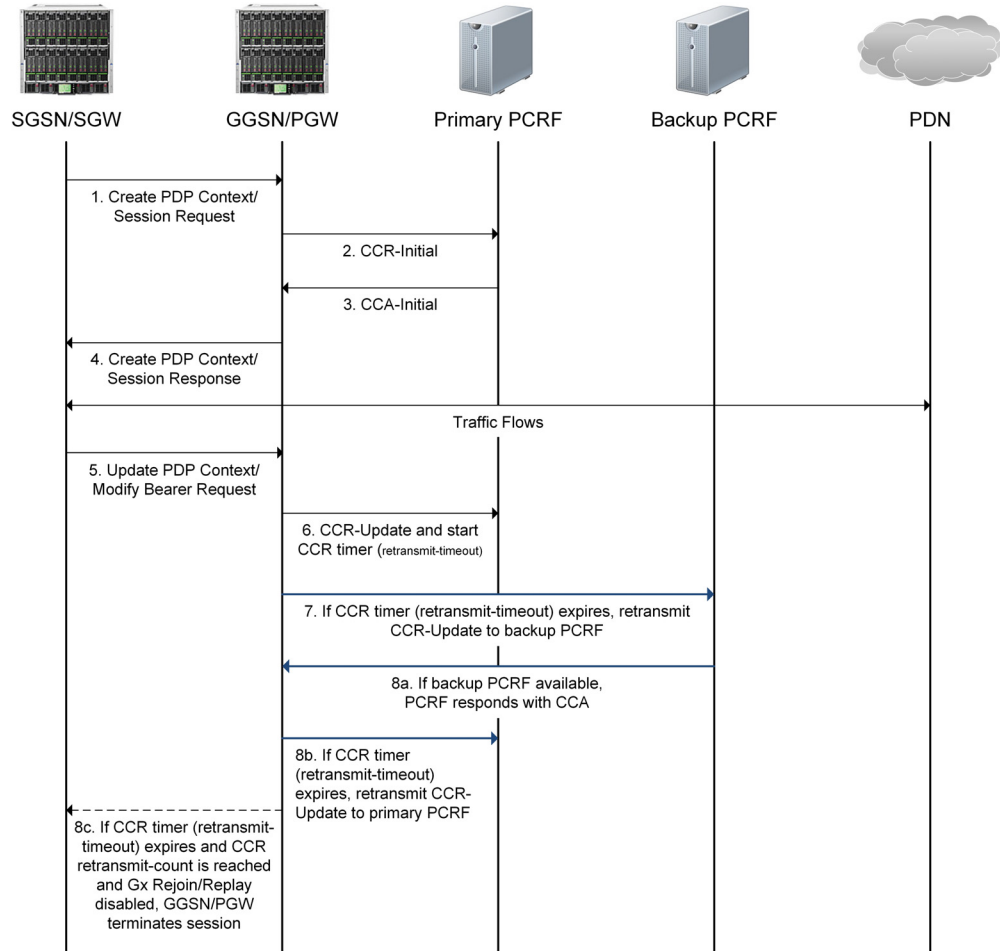
Note: If there is no available backup server and the failure occurs because the Watchdog mechanism expired, all new sessions fail to establish and all existing sessions are terminated.

Failover to Backup PCRF (Application Failure)

Figure 1-27 demonstrates the message flow during a failover from the primary to the backup PCRF when the CCR timer (`retransmit-timeout`) expires and the CCR `retransmit-count` is reached during an existing session.

While a failure at the transport level affects all sessions, a failure at the application level only affects the session on which the CCR expiration occurs.

Figure 1-27 Failover to Backup PCRF (Application Failure)

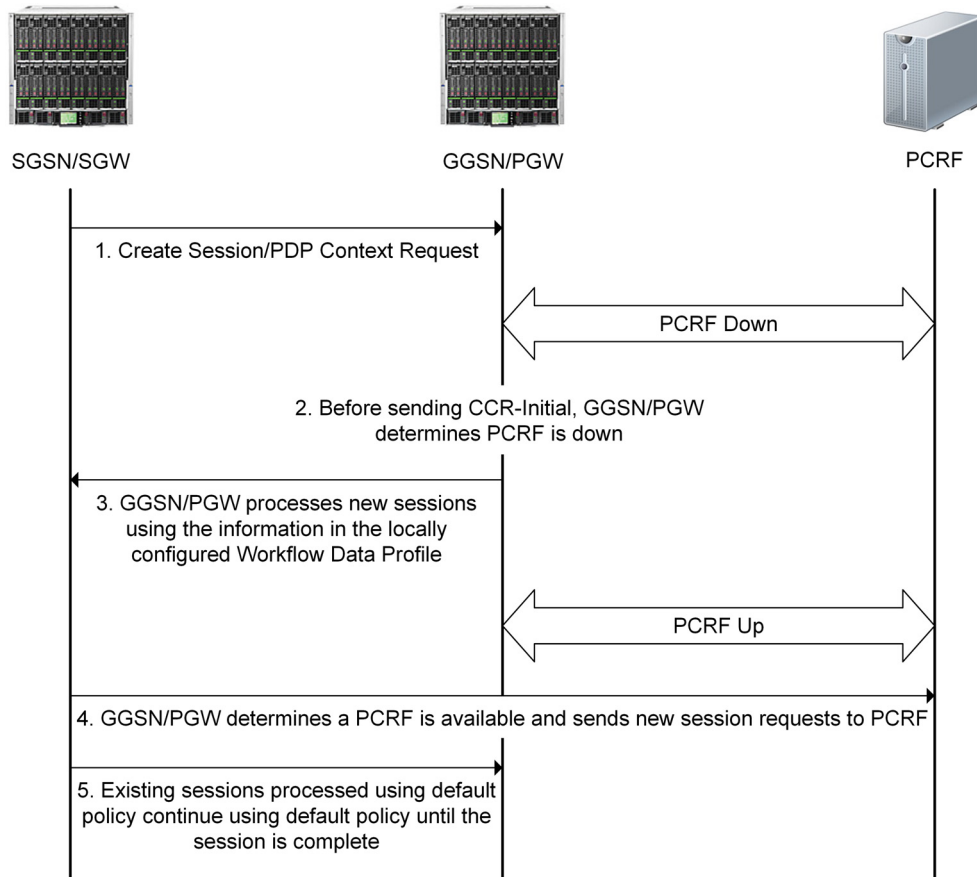


- 1 The SGSN/SGW sends a Create PDP Context/Session Request to the GGSN/PGW.
- 2 The GGSN/PGW starts the session setup process by sending a CCR-I message to the PCRF.
- 3 The PCRF responds to the GGSN/PGW with a CCA message.
- 4 The PGW completes the session setup.
- 5 During the session, the SGSN/SGW sends an Update PDP Context/Modify Bearer Response to the GGSN/PGW.
- 6 The GGSN/PGW sends a CCR-U to the primary PCRF and starts the CCR timer (`retransmit-timeout`).
- 7 If the primary PCRF does not respond to the CCR-U within the configured time limit (`retransmit-timeout`), the GGSN/PGW retransmits the CCR-U to the backup PCRF.
- 8 Depending on the response from the PCRF, one of the following actions can occur:
 - a The backup PCRF responds with a CCA message, if available.
 - b If there is no response from the backup PCRF within the configured time limit (`retransmit-timeout`), the GGSN/PGW retransmits the CCR-U to the primary server once again. This cycle continues until the PCRF responds with a CCA message or the CCR `retransmit-count` is reached.
 - c If there is no response from the backup PCRF within the configured time limit and the CCR `retransmit-count` is reached and the Gx Rejoin/Replay feature is disabled, the GGSN/PGW terminates the session.

Failure Handling - Default Gx Policy

Figure 1-28 demonstrates the message flow for setting up a session when no PCRF is available and the Default Gx Policy feature is enabled. In this case, the Gx Rejoin and Gx Replay features are disabled.

Figure 1-28 Failure Handling - Default Gx Policy Enabled

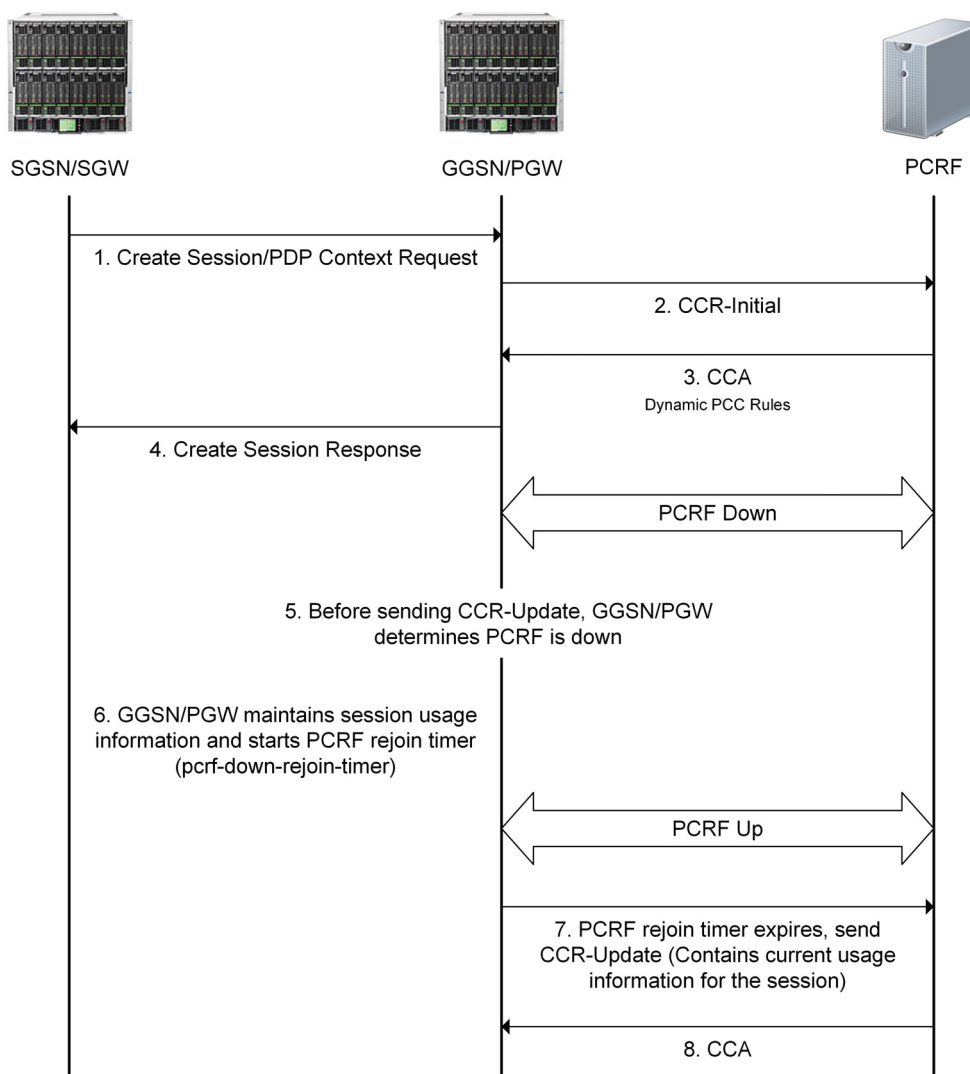


- 1 The SGSN/SGW sends a Create Session/PDP Context Request to the GGSN/PGW.
- 2 The GGSN/PGW determines all PCRFs are down.
- 3 The GGSN/PGW uses the information in the locally configured workflow data profile (from the subscriber analyzer) to set up subscriber sessions. Note that the subscriber usage consumed while a session is processed using the default policy is not reported to the PCRF.
- 4 The GGSN/PGW determines that a PCRF is available and sends new session requests to the PCRF.
- 5 The GGSN/PGW continues to process existing sessions using the default policy until the session is complete.

Failure Handling - Gx Rejoin and Gx Replay

Figure 1-29 demonstrates the message flow when the system detects a PCRF failure during an active session (processed using dynamic PCC rules) and no PCRF is available and Gx Rejoin and Gx Replay are enabled. In this case, the session is not terminated before the PCRF becomes available.

Figure 1-29 Failure Handling - Gx Rejoin and Gx Replay



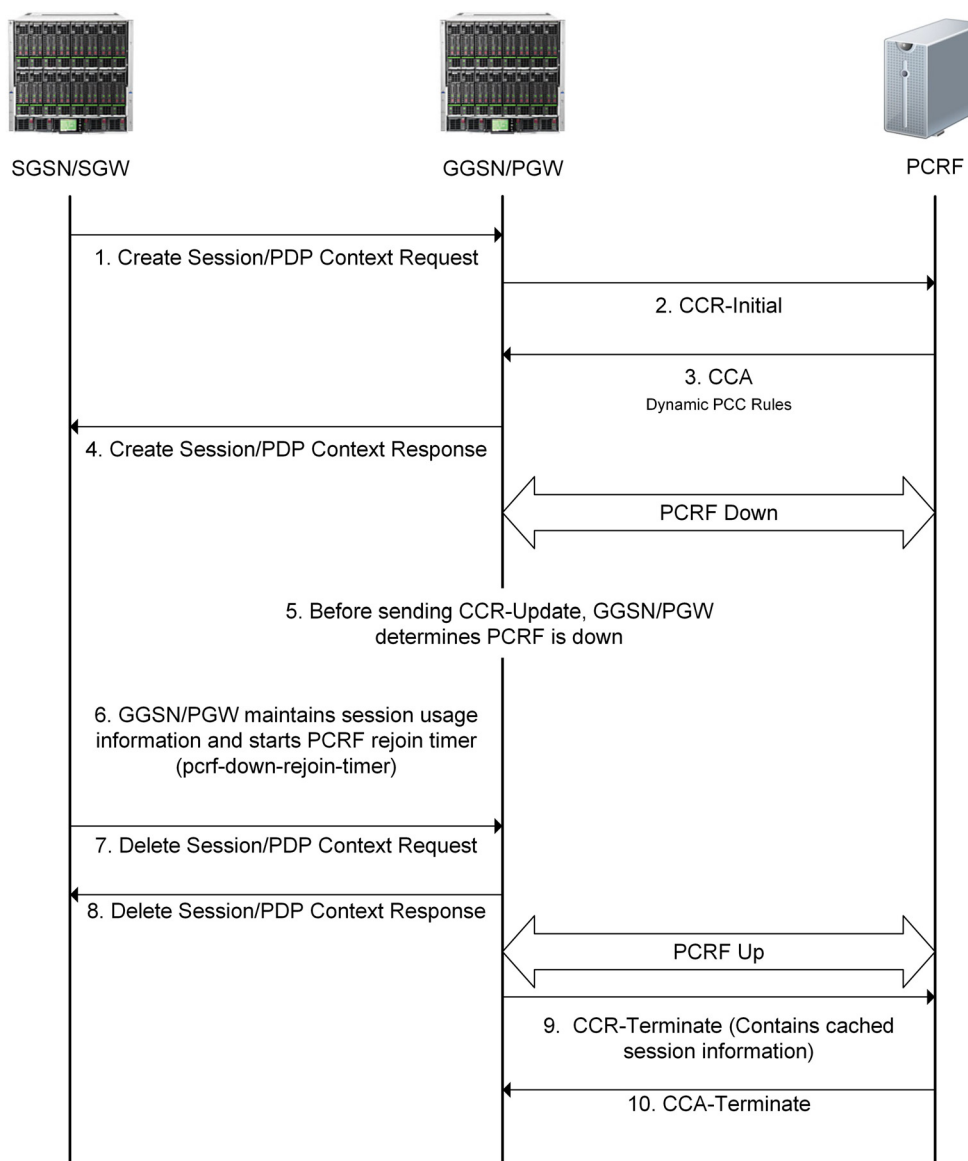
- 1 The SGW/SGSN sends a Create Session/PDP Context Request to the GGSN/PGW.
- 2 The GGSN/PGW determines the PCRF is available and sends a CCR-I message to the PCRF.
- 3 The PCRF responds with a CCA message containing dynamic policy rules.
- 4 Upon receipt of the CCA message, the GGSN/PGW returns a Create Session/PDP Context Response message to the SGW/SGSN.
- 5 Before sending a CCR-U message, the GGSN/PGW determines the PCRF status.

- 6 If the PCRF is down, the GGSN/PGW temporarily assumes the role of the PCRF:
 - Stores session usage information (including usage statistics) on the local drives.
 - Starts PCRF rejoin timer (`pcrf-down-rejoin-time`).
- 7 When the PCRF rejoin timer expires, the GGSN/PGW sends a CCR-Update message containing the current usage information for the session to any PCRF server that becomes available in the Diameter server group.
- 8 The PCRF responds with a CCA message.

Failure Handling - Gx Replay

Figure 1-30 demonstrates the message flow during a PCRF failure on an established Gx session when Gx Rejoin and Gx Replay are enabled and the session (processed using dynamic PCC rules) is terminated before the PCRF becomes available.

Figure 1-30 Failure Handling - Gx Replay



- 1 The SGW/SGSN sends a Create Session/PDP Context Request to the GGSN/PGW.
- 2 The GGSN/PGW determines the PCRF is available and sends a CCR-I message to the PCRF.
- 3 The PCRF responds with a CCA message containing dynamic policy rules.

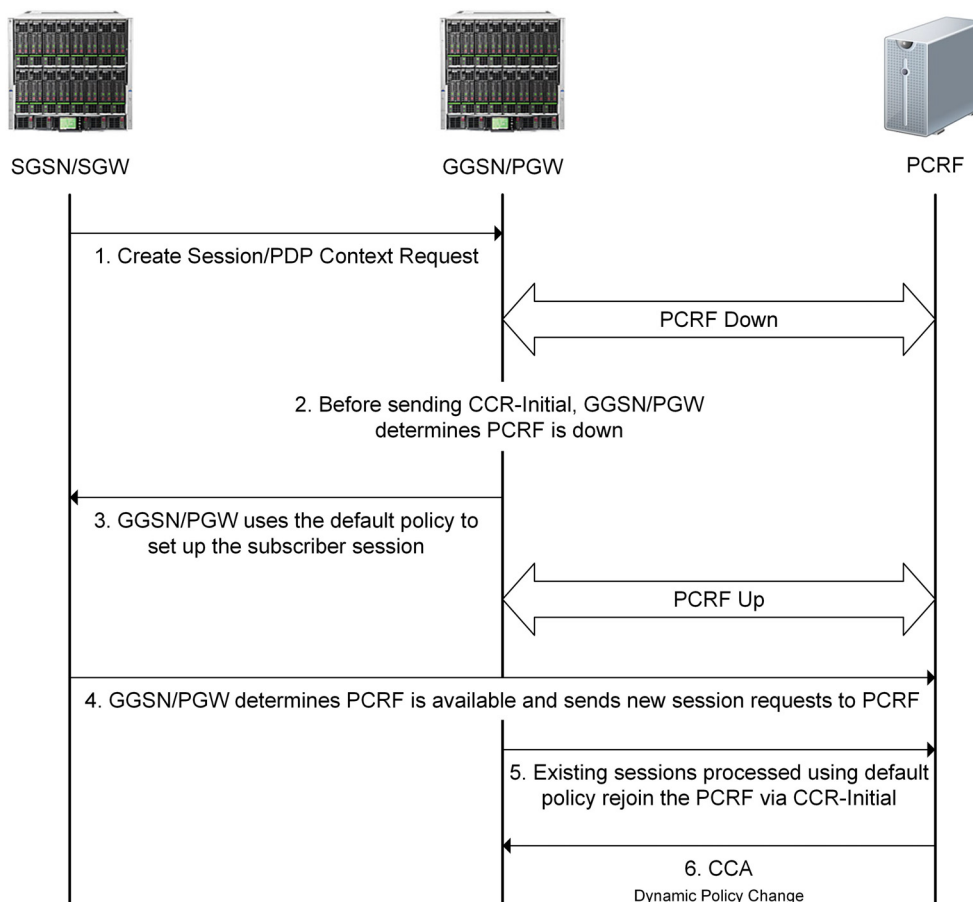
- 4 Upon receipt of the CCA message, the GGSN/PGW returns a Create Session/PDP Context Response message to the SGW/SGSN.
- 5 Before sending a CCR-U message, the GGSN/PGW determines the PCRF status.
- 6 If the PCRF is down, the GGSN/PGW temporarily assumes the role of the PCRF:
 - Stores session usage information (including usage statistics) on the local drives.
 - Starts PCRF rejoin timer (`pcrf-down-rejoin-time`).
- 7 The SGW sends a Delete Session Request to the PGW to terminate the session.
- 8 The PGW sends a Delete Session Response and the session is terminated.
- 9 The PGW sends a CCR-T message containing all cached session information to any PCRF server that becomes available in the Diameter server group.
- 10 The PCRF responds with a CCA-Terminate message.

Note: *If a session is processed using default policy rules and is terminated before a PCRF becomes available, session information is not reported to the PCRF.*

Failure Handling - Default Gx Policy/Gx Rejoin/Replay Enabled

Figure 1-31 demonstrates the message flow for handling a PCRF failure when Default Gx Policy, Gx Rejoin, and Gx Replay are enabled.

Figure 1-31 Failure Handling - Default Gx Policy/Gx Rejoin/Gx Replay Enabled



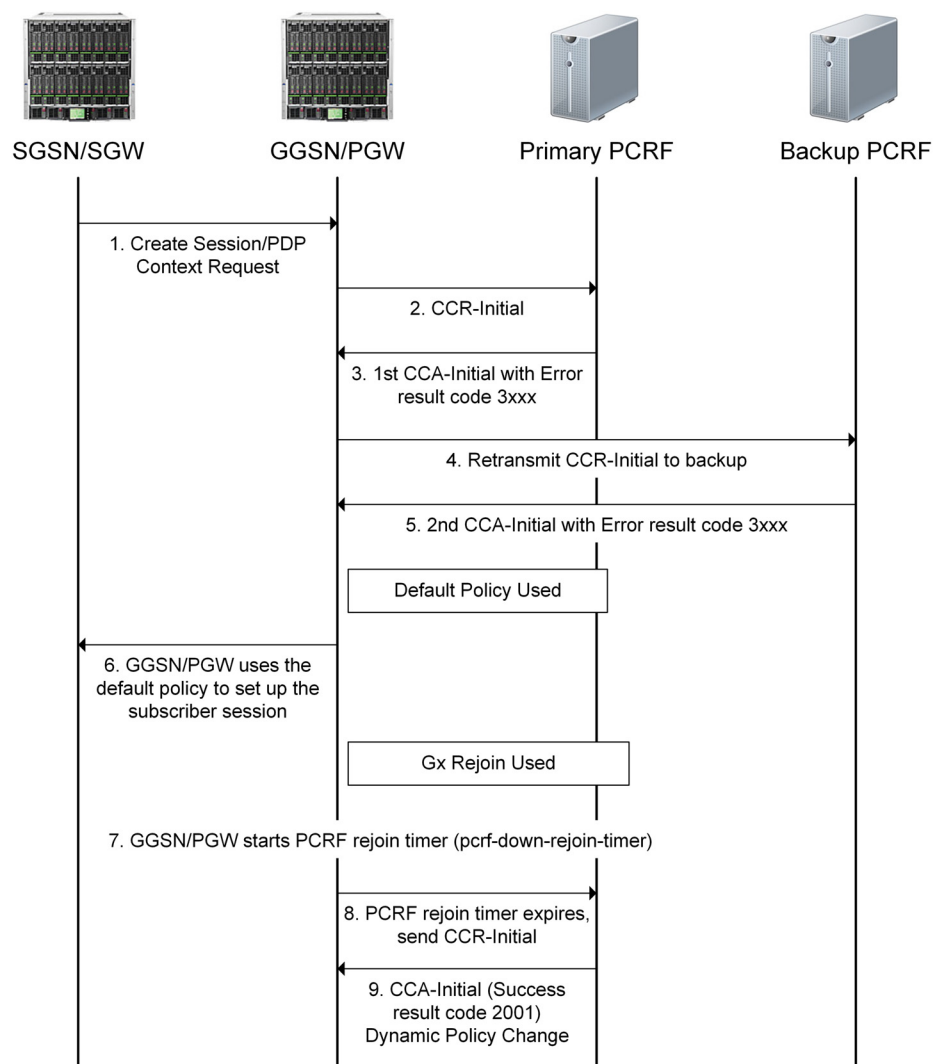
- 1 The SGSN/SGW sends a Create Session/PDP Context Request to the GGSN/PGW.
- 2 The GGSN/PGW determines all PCRFs are down.
- 3 The GGSN/PGW uses the information in the locally configured workflow data profile (from the subscriber analyzer) to set up subscriber sessions.
- 4 The GGSN/PGW determines that the PCRF is available and sends new session requests to the PCRF.
- 5 Existing sessions processed using the default policy from the locally configured data profile rejoin the PCRF via CCR-I messages. Note that the subscriber usage consumed while a session is processed using the default policy is not reported to the PCRF.
- 6 The PCRF sends a CCA message to the GGSN/PGW containing the new PCC rules for the dynamic policy change.
- 7 Upon receiving the CCA message from the PCRF, the PGW installs the new policies and processes the session using the dynamic policy.

- 8 If a session is terminated before a PCRF becomes available, note the following:
 - If a session is processed using *dynamic policy rules* and is terminated before a PCRF becomes available, the GGSN/PGW replays all cached session information in a CCR-T message to any PCRF server that becomes available in the Diameter server group.
 - If a session is processed using *default policy rules* and is terminated before a PCRF becomes available, session information is not reported to the PCRF.

Failure Handling - PCRF Responds with 3xxx Result Code

Figure 1-32 demonstrates the message flow for setting up a session when the PCRF responds to the CCR-I message with a CCA-I error message containing result code 3xxx and both Default Gx Policy and Gx Rejoin/Replay are enabled.

Figure 1-32 Failure Handling - PCRF Responds with 3xxx Result Code



- 1 The SGW/SGSN sends a Create Session/PDP Context Request to the GGSN/PGW.
- 2 The GGSN/PGW determines the primary PCRF is available and sends a CCR-I message.
- 3 The primary PCRF responds with a CCA-I error message containing result code 3xxx.
- 4 Upon receipt of the CCA-I 3xxx error message, the GGSN/PGW retransmits the CCR-I message to the backup PCRF.
- 5 The backup PCRF responds with a CCA-I error message containing result code 3xxx.
- 6 Upon receipt of the second CCA-I 3xxx error message, the GGSN/PGW sets up the session using the information in the locally configured workflow data profile (from the subscriber analyzer).

***Note:** The subscriber usage consumed while a session is processed using the default policy is not reported to the PCRF.*

- 7 The GGSN/PGW starts the PCRF rejoin timer (`pcrf-down-rejoin-time`).
- 8 When the PCRF rejoin timer expires, the GGSN/PGW attempts to connect to the primary PCRF by sending another CCR-I message.
- 9 The PCRF responds with a CCA-I success message (result code 2001) containing the new PCC rules for the dynamic policy change.
- 10 Upon receiving the CCA message from the PCRF, the GGSN/PGW installs the new policies and processes the session using the dynamic policy.
- 11 If a session is terminated before a PCRF becomes available, note the following:
 - If a session is processed using *dynamic policy rules* and is terminated before a PCRF becomes available, the GGSN/PGW replays all cached session information to any PCRF server that becomes available in the Diameter server group.
 - If a session is processed using *default policy rules* and is terminated before a PCRF becomes available, session information is not reported to the PCRF.

***Note:** When no backup PCRF is configured and the primary server returns a 3xxx error message, the GGSN/PGW retransmits the CCR-I message to another active peer from the list of primary servers. If there are no active primary servers, the system selects from the list of active secondary servers. If the GGSN/PGW receives two 3xxx error messages or the configured CCR `retransmit-count` is reached, the GGSN/PGW uses the Gx Default Policy and Gx Rejoin/Replay features, if enabled.*

Failure Handling – Result Code List Enabled and Retry-Secondary Disabled

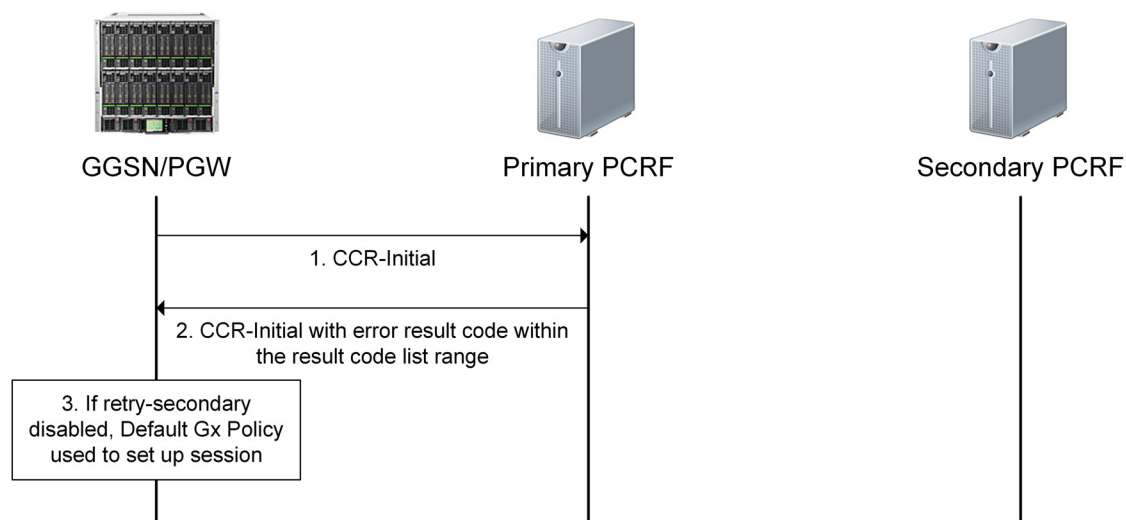
Figure 1-33 demonstrates the message flow for setting up a session using the default Gx policy when the result code error sent in the CCA-I message is present within the configured list of the result code ranges.

Note the following:

- The result code list (`default-gx-setup-per-cause-code`) and the Default Gx Policy feature (`pcrf-down-default-gx-call-setup`) are enabled on the interface.
- The `retry-secondary` parameter for the result code range is *disabled*.

For configuration information, see [Configuring the Diameter Result Code List \(page 1-135\)](#).

Figure 1-33 Failure Handling – Result Code List Enabled and Retry-Secondary Disabled



Failure Handling – Result Code List Enabled and Retry-Secondary Enabled

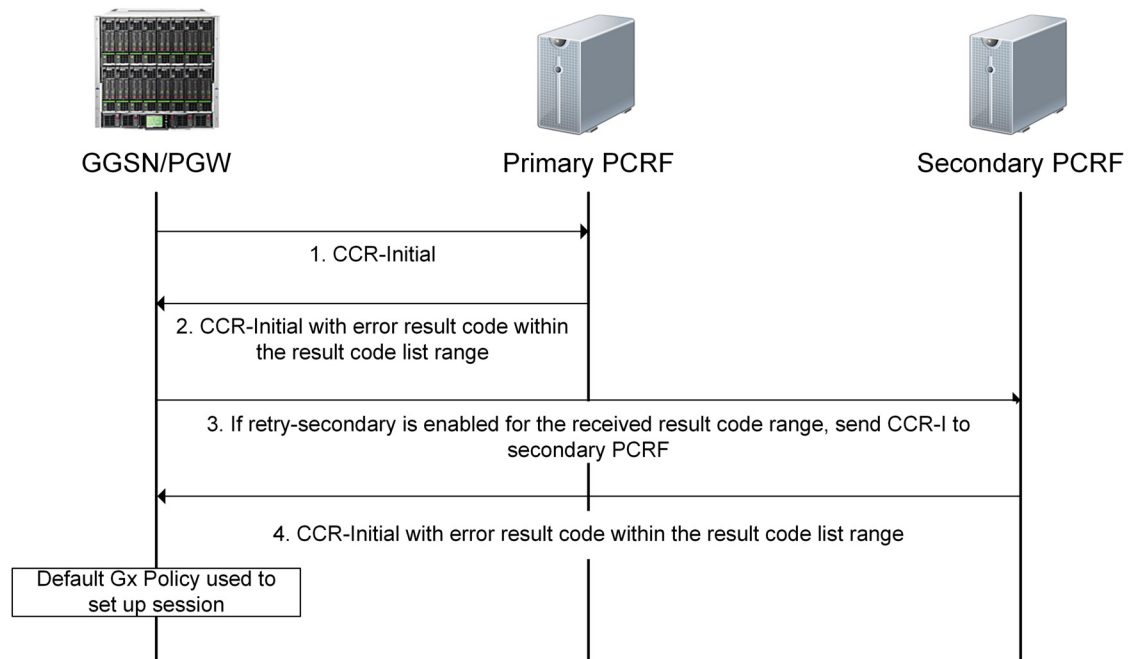
Figure 1-34 demonstrates the message flow for setting up a session using the default Gx policy when the result code error sent in the CCA-I message is present within the configured list of the result code ranges.

Note the following:

- The result code list (`default-gx-setup-per-cause-code`) and the Default Gx Policy feature (`pcrf-down-default-gx-call-setup`) are enabled on the interface.
- The `retry-secondary` parameter for the result code range is *enabled*.

For configuration information, see [Configuring the Diameter Result Code List \(page 1-135\)](#).

Figure 1-34 Failure Handling – Result Code List Enabled and Retry-Secondary Enabled



Message Definitions

Table 1-2 describes the message definitions for the Gx/Gxa interface. The Gx and Gxa interfaces support different AVPs for the following message types:

- Credit-Control-Request/Answer
- Re-Auth-Request/Answer

Table 1-2 Gx Interface Message Definitions

Command	Acronym	Command Code	Reference
Capabilities-Exchange-Request (page 1-169)	CER	257	RFC 3588
Capabilities-Exchange-Answer (page 1-169)	CEA	257	RFC 3588
Credit-Control-Request (page 1-170)	CCR	272	RFC 4006, 3GPP TS 29.212 V12.7.0
Credit-Control-Answer (page 1-177)	CCA	272	RFC 4006, 3GPP TS 29.212 V12.7.0
Disconnect-Peer-Request (page 1-184)	DPR	282	RFC 3588
Disconnect-Peer-Answer (page 1-184)	DPA	282	RFC 3588
Device-Watchdog-Request (page 1-184)	DWR	280	RFC 3588
Device-Watchdog-Answer (page 1-184))	DWA	280	RFC 3588
Re-Auth-Request (page 1-185)	RAR	258	RFC 4006, 3GPP TS 29.212 V12.7.0
Re-Auth-Answer (page 1-192)	RAA	258	RFC 4006, 3GPP TS 29.212 V12.7.0

Table 1-3 describes the special symbols in the AVP name for each message definition.

Table 1-3 AVP Name Symbols

Symbol	Description
<>	AVP is mandatory and the AVP with this symbol is at the beginning of a message.
{ }	AVP is mandatory.
[]	AVP is optional.
* []	AVP is optional and is repeated.
M	Mandatory item.
C	Optional item.
Om	Mandatory item defined by the carrier.
Oc	Optional item defined by the carrier.

Table 1-3 AVP Name Symbols

Symbol	Description
–	Not supported.

Diameter AVPs have flags that indicate to the receiver how each AVP must be handled in Diameter messages. [Table 1-4](#) describes the AVP flag symbols.

Table 1-4 AVP Flag Symbols

Symbol	Description
M	The Mandatory “M” flag indicates that support for this AVP is required. If either the AVP or its value is not recognized, the message is not processed.
V	The Vendor-specific “V” flag indicates that this AVP has the optional Vendor ID field and when set, the AVP belongs to the specific vendor code address space.
M, V	Indicates that this AVP has both the mandatory “M” flag and vendor-specific “V” flag set.

Note: Operators can modify the default AVP flag values for Diameter AVPs on a specific Diameter interface (Gx/Gxa, Gy, Gz, SWm, STa, and S6b). See [Modifying Default Diameter AVP Flags \(page 1-84\)](#) for more information.

Note: To control the sending (or not sending) of an optional diameter AVP in CCR, CCA or RAA messages, configure the PCRF data record templates. See [Sending Policy and Charging Records to the PCRF \(page 1-42\)](#) for more information.

Capabilities-Exchange-Request

The Capabilities-Exchange-Request (CER), indicated by the Command-Code set to 257 and the Command Flags' 'R' bit set, is sent to exchange local capabilities. Upon detection of a transport failure, this message must not be sent to an alternate peer.

Message Format:

```
<CER> ::= < Diameter Header: 257, REQ >
    {Origin-Host}
    {Origin-Realm}
    1*{Host-IP-Address}
    {Vendor-Id}
    {Product-Name}
    [Origin-State-Id]
    *[Supported-Vendor-Id]
    *[Auth-Application-Id]
    *[Inband-Security-Id]
    *[Acct-Application-Id]
    *[Vendor-Specific-Application-Id]
    [Firmware-Revision]
    *[AVP]
```

Capabilities-Exchange-Answer

The Capabilities-Exchange-Answer (CEA), indicated by the Command-Code set to 257 and the Command Flags' 'R' bit cleared, is sent in response to a CER message.

Message Format:

```
<CEA> ::= < Diameter Header: 257 >
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    1*{Host-IP-Address}
    {Vendor-Id}
    {Product-Name}
    [Origin-State-Id]
    [Error-Message]
    *[Failed-AVP]
    *[Supported-Vendor-Id]
    *[Auth-Application-Id]
    *[Inband-Security-Id]
    *[Acct-Application-Id]
    *[Vendor-Specific-Application-Id]
    [Firmware-Revision]
```

* [AVP]

Credit-Control-Request

The system sends the Credit-Control-Request (CCR) to the PCRF to:

- Request PCC rules for a bearer.
- Indicate bearer and PCC rule-related events.
- Terminate the IP CAN bearer and/or session.

CCR Message Format

```
<CC-Request> ::= < Diameter Header: 272, REQ, PXY >
    <Session-Id>
    {Auth-Application-Id}
    {Origin-Host}
    {Origin-Realm}
    {Destination-Realm}
    {CC-Request-Type}
    {CC-Request-Number}
    [Destination-Host]
    [User-Name]
    [Origin-State-Id]
    [Event-Timestamp]
    *[Subscription-Id]
    *[Supported-Features]
    [Network-Request-Support]
    [Bearer-Identifier]
    [Bearer-Operation]
    [Framed-IP-Address]
    [Framed-IPv6-Prefix]
    [IP-CAN-Type]
    [3GPP-RAT-Type]
    [AN-Trusted]
    [RAT-Type]
    [Termination-Cause]
    [User-Equipment-Info]
    [QoS-Information]
    [Default-EPS-Bearer-QoS]
    [AN-GW-Address]
    [3GPP-SGSN-MCC-MNC]
    [3GPP-SGSN-Address]
    [3GPP-SGSN-IPv6-Address]
```


[3GPP-User-Location-Info]
 [User-Location-Info-Time]
 [3GPP-MS-TimeZone]
 * [RAN-NAS-Release-Cause]
 [Called-Station-ID]
 [PDN-Connection-ID]
 [Bearer-Usage]
 [Online]
 [Offline]
 * [ADC-Rule-Report]
 * [Charging-Rule-Report]
 * [Application-Detection-Information]
 * [Event-Trigger]
 [Access-Network-Charging-Address]
 * [Access-Network-Charging-Identifier-Gx]
 * [Usage-Monitoring-Information]
 [Sponsored-Connectivity-Data]
 * [AVP]

Table 1-5 lists the supported standard and 3GPP AVPs for the Gx/Gxa interface. The AVPs not listed in the table are not supported. The system discards the unsupported AVPs and they are not sent. For a description of each AVP, see [Appendix A, Diameter AVP Descriptions](#). The AVPs with the vendor-specific “V” flag set include a Vendor-ID of 10415 (3GPP).

Table 1-5 CCR Message Format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
Standard Diameter AVPs					
<Session-ID>	263	M	M	M	M
{Auth-Application-Id}	258	M	M	M	M
{Origin-Host}	264	M	M	M	M
{Origin-Realm}	296	M	M	M	M
{Destination-Realm}	283	M	M	M	M
{CC-Request-Type}	416	M	M	M	M
{CC-Request-Number}	415	M	M	M	M
[Destination-Host]	293	Oc	Oc	Oc	M
[User-Name]	1	Oc	Oc	Oc	M

Table 1-5 CCR Message Format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[Origin-State-Id] ¹	278	Oc	Oc	Oc	M
[Event-Timestamp] ¹	55	Oc	-	-	M
*[Subscription-Id] ¹	443	Om	Oc	Oc	M
{Subscription-Id-Type} ¹	450	M	M	M	M
{Subscription-Id-Data} ¹	444	M	M	M	M
[Termination-Cause]	295	-	-	Oc	M
[User-Equipment-Info]	458	Oc	Oc	Oc	M
{User-Equipment-Info-Type}	459	Oc	Oc	Oc	M
{User-Equipment-Info-Value}	460	Oc	Oc	Oc	M
[Called-Station-Id]	30	Om	Oc	Oc	M
3GPP AVPs					
*[Supported-Features] ¹	628	Oc	Oc	Oc	V
{Vendor-Id} ¹	266	Oc	Oc	Oc	M
{Feature-List-ID} ¹	629	Oc	Oc	Oc	V
{Feature-List} ¹	630	Oc	Oc	Oc	V
[Network-Request-Support] ¹	1024	Oc	Oc	Oc	M,V
[Bearer-Identifier] ¹	1020	Oc	Oc	Oc	M,V
[Bearer-Operation] ¹	1021	Om	Oc	-	M,V
[Framed-IP-Address]	8	Oc	-	-	M
*[Framed-IPv6-Prefix]	97	Oc	-	-	M
[IP-CAN-Type]	1027	Om	Oc	-	M,V
[3GPP-RAT-Type]	21	Oc	Oc	Oc	M,V
[AN-Trusted]	1503	-	Oc	-	M,V
[RAT-Type]	1032	Oc	Oc	-	M,V
*[QoS-Information]	1016	Om	Oc	-	M,V
[APN-Aggregate-Max-Bitrate-DL]	1040	Oc	Oc	-	V

Table 1-5 CCR Message Format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[APN-Aggregate-Max-Bitrate-UL]	1041	Oc	Oc	-	V
[Extended-APN-AMBR-DL] ¹	2848	Oc	Oc	-	V
[Extended-APN-AMBR-UL] ¹	2849	Oc	Oc	-	V
[Default-EPS-Bearer-QoS] ¹	1049	Oc	Oc	-	V
[QoS-Class-Identifier] ¹	1028	Oc	Oc	-	M,V
[Allocation-Retention-Priority] ¹	1034	-	Oc	-	V
[Priority-Level] ¹	1046	-	Oc	-	M,V
[Pre-emption-Capability] ¹	1047	-	Oc	-	V
[Pre-emption-Vulnerability] ¹	1048	-	Oc	-	V
[AN-GW-Address] ¹	1050	Oc	Oc	-	V
[3GPP-SGSN-MCC-MNC] ¹	18	Om	Oc	Oc	M,V
[3GPP-SGSN-Address] ¹	6	Oc	Oc	Oc	M,V
[3GPP-SGSN-IPv6-Address] ¹	15	Oc	Oc	Oc	M,V
[3GPP-User-Location-Info] ¹	22	Oc	Oc	-	M,V
[User-Location-Info-Time] ¹	2812	-	Oc	Oc	V
[3GPP-MS-TimeZone] ¹	23	Oc	Oc	Oc	M,V
*[RAN-NAS-Release-Cause] ¹	2819	-	Oc	Oc	V
[PDN-Connection-ID] ¹	1065	Oc	Oc	Oc	V
[Bearer-Usage] ¹	1000	Oc	Oc	-	M,V
[Online]	1009	Oc	-	-	M,V
[Offline]	1008	Oc	-	-	M,V
*[ADC-Rule-Report] ¹	1097	-	Oc	-	M,V
*[ADC-Rule-Name] ¹	1096	-	Oc	-	M,V
*[ADC-Rule-Base-Name] ¹	1095	-	Oc	-	M,V
[PCC-Rule-Status] ¹	1019	-	Oc	-	M,V

Table 1-5 CCR Message Format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[Rule-Failure-Code] ¹	1031	-	Oc	-	M,V
[Final-Unit-Indication] ¹	430	-	Oc	1	M
*[Charging-Rule-Report]	1018	-	Oc	-	M,V
*[Charging-Rule-Name]	1005	-	Oc	-	M,V
*[Charging-Rule-Base-Name]	1004	-	Oc	-	M,V
[PCC-Rule-Status]	1019	-	Oc	-	M,V
[Rule-Failure-Code]	1031	-	Oc	-	M,V
[Final-Unit-Indication] ¹	430	-	Oc	-	M
*[RAN-NAS-Release-Cause] ¹	2819	-	Oc	Oc	V
*[Application-Detection-Information] ¹	1098	-	Oc	-	V
[TDF-Application-Identifier] ¹	1088	-	Oc	-	V
[TDF-Application-Instance-Identifier] ¹	2802	-	Oc	-	V
*[Flow-Information] ¹	1058	-	Oc	-	V
*[Event-Trigger]	1006	-	Oc	Oc	M,V
[Access-Network-Charging-Address]	501	Oc	Oc	-	M,V
*[Access-Network-Charging-Identifier-Gx]	1022	Oc	Oc	-	M,V
[Access-Network-Charging-Identifier-Value]	503	Oc	Oc	-	M,V
*[Charging-Rule-Base-Name] ¹	1004	Oc	Oc	-	M,V
*[Charging-Rule-Name]	1005	Oc	Oc	-	M,V
*[Usage-Monitoring-Information]	1067	-	Oc	Oc	V
[Monitoring-Key]	1066	-	Oc	Oc	V
[Granted-Service-Unit]	431	-	Oc	Oc	M
[CC-Time]	420	-	Oc	Oc	M
[CC-Total-Octets]	421	-	Oc	Oc	M
[CC-Input-Octets] ¹	412	-	Oc	Oc	M
[CC-Output-Octets] ¹	414	-	Oc	Oc	M

Table 1-5 CCR Message Format

AVP Name	AVP Code	Initial	Update	Terminate	AVP Flag
[Used-Service-Unit]	446	-	Oc	Oc	M
[CC-Time]	420	-	Oc	Oc	M
[CC-Total-Octets]	421	-	Oc	Oc	M
[CC-Input-Octets] ¹	412	-	Oc	Oc	M
[CC-Output-Octets] ¹	414	-	Oc	Oc	M
[Usage-Monitoring-Level]	1068	-	Oc	Oc	V
[Usage-Monitoring-Report] ¹	1069	-	Oc	Oc	V
[Usage-Monitoring-Support] ¹	1070	-	Oc	Oc	V
[Sponsored-Connectivity-Data] ¹	530	-	Oc	-	V

¹ Indicates AVPs that are not qualified on the Gxa interface.

Table 1-6 lists the vendor-specific attributes (VSAs) for CCR messages on the Gx interface. For a description of each attribute, see [Appendix A, Diameter AVP Descriptions](#). These VSAs are not qualified on the Gxa interface.

Table 1-6 CCR Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	Initial	Update	Terminate
[Origination-Time-Stamp]	12951	7102	-	Oc	-
[Max-Wait-Time]	12951	7103	-	Oc	-
[Cisco-Event]	9	131195	-	Oc	-
{Cisco-Event-Trigger-Type}	9	131192	-	Oc	-
[Cisco-CC-Failure-Type]	9	132077	-	Oc	-
[TMO-Virtual-Gi-ID]	29168	15	Oc	-	-
[Gx-TMO-Redirect-Server]	29168	111	Oc	Oc	-
[Gx-TMO-Redirect-Address-Type]	29168	112	Oc	Oc	-
[Gx-TMO-Redirect-Server-Address]	29168	113	Oc	Oc	-
[Gx-TMO-Append-Original-URL]	29168	114	Oc	Oc	-
[Gx-TMO-Append-MSISDN]	29168	115	Oc	Oc	-
[Gx-TMO-Append-IMSI]	29168	116	Oc	Oc	-

Table 1-6 CCR Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	Initial	Update	Terminate
[Gx-TMO-Append-IMEI]	29168	117	Oc	Oc	-
[Gx-TMO-Append-MSIP]	29168	118	Oc	Oc	-
[Gx-TMO-Deactivate-By-Redirect]	29168	119	Oc	Oc	-

Table 1-7 lists the vendor-specific attributes (VSAs) for CCR messages on the Gxa interface. For a description of each attribute, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-7 CCR Vendor-Specific Attributes (Gxa)

AVP Name	Vendor ID	AVP Code	Initial	Update	Terminate
[Fixed-User-Location-Info]	13019	2825	Oc	-	-
[Logical-Access-Id]	13019	302	Oc	-	-

Credit-Control-Answer

The PCRF sends a Credit-Control-Answer (CCA) in response to a CCR. It is used to provision PCC rules and event triggers for the bearer/session and to provide the selected bearer control mode for the IP-CAN session.

CCA Message Format

```
<CC-Answer> ::=  < Diameter Header: 272, PXY >
    <Session-Id>
    {Auth-Application-Id}
    {Origin-Host}
    {Origin-Realm}
    [Result-Code]
    [Experimental-Result]
    {CC-Request-Type}
    {CC-Request-Number}
    [CC-Session-Failover]
    *[Redirect-Host]
    *[Supported-Features]
    [Bearer-Control-Mode]
    *[Event-Trigger]
    [Origin-State-Id]
    [Final-Unit-Indication]
    *[ADC-Rule-Remove]
    *[ADC-Rule-Install]
    *[Charging-Rule-Remove]
    *[Charging-Rule-Install]
    [Charging-Information]
    [Online]
    [Offline]
    [QoS-Information]
    [Revalidation-Time]
    [Default-EPS-Bearer-QoS]
    [Bearer-Usage]
    *[Usage-Monitoring-Information]
    [Session-Release-Cause]
    [3GPP-SGSN-MCC-MNC]
    *[AVP]
```

Table 1-8 lists the CCA message format for the Gx/Gxa interface. For a description of each AVP, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-8 CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
Standard Diameter AVPs				
<Session-ID>	263	M	M	M
{Auth-Application-Id}	258	M	M	M
{Origin-Host}	264	M	M	M
{Origin-Realm}	296	M	M	M
[Result-Code] ¹	268	Oc	Oc	Oc
[Experimental-Result] ¹	297	Oc	Oc	Oc
{CC-Request-Type}	416	M	M	M
{CC-Request-Number}	415	M	M	M
[Origin-State-Id] ¹	278	Oc	Oc	Oc
[CC-Session-Failover] ¹	418	Oc	-	-
*[Redirect-Host] ¹	292	Oc	Oc	Oc
3GPP AVPs				
*[Supported-Features] ¹	628	Oc	Oc	Oc
{Vendor-Id} ¹	266	Oc	Oc	Oc
{Feature-List-ID} ¹	629	Oc	Oc	Oc
{Feature-List} ¹	630	Oc	Oc	Oc
[Bearer-Control-Mode] ¹	1023	Oc	Oc	Oc
*[Event-Trigger]	1006	Oc	Oc	-
[Final-Unit-Indication] ¹	430	Oc	Oc	Oc
{Final-Unit-Action} ¹	449	Oc	Oc	Oc
[Redirect-Server] ¹	434	Oc	Oc	Oc
*[ADC-Rule-Remove] ¹	1093	Oc	Oc	-
*[ADC-Rule-Name] ¹	1096	Oc	Oc	-

Table 1-8 CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
*[ADC-Rule-Base-Name] ¹	1095	Oc	Oc	-
*[ADC-Rule-Install] ¹	1092	Oc	Oc	-
*[ADC-Rule-Definition] ¹	1094	Oc	Oc	-
*[ADC-Rule-Name] ¹	1096	Oc	Oc	-
*[ADC-Rule-Base-Name] ¹	1095	Oc	Oc	-
[Rule-Activation-Time] ¹	1043	Oc	Oc	-
[Rule-Deactivation-Time] ¹	1044	Oc	Oc	-
*[Charging-Rule-Remove]	1002	Oc	Oc	-
*[Charging-Rule-Name]	1005	Oc	Oc	-
*[Charging-Rule-Base-Name]	1004	-	Oc	-
*[Required-Access-Info] ¹	536	Oc	Oc	-
*[Charging-Rule-Install]	1001	Oc	Oc	-
*[Charging-Rule-Definition]	1003	Oc	Oc	-
{Charging-Rule-Name}	1005	Oc	Oc	-
[Service-Identifier] ¹	439	Oc	Oc	-
[Rating-Group]	432	Oc	Oc	-
*[Flow-Information] ¹	1058	Oc	Oc	-
[TDF-Application-Identifier] ¹	1088	Oc	Oc	-
[Flow-Status] ¹	511	Oc	Oc	-
[QoS-Information] ¹	1016	Oc	Oc	-
[Max-Requested-Bandwidth-DL] ¹	515	Oc	Oc	-
[Max-Requested-Bandwidth-UL] ¹	516	Oc	Oc	-
[Guaranteed-Bitrate-DL] ¹	1025	Oc	Oc	-
[Guaranteed-Bitrate-UL] ¹	1026	Oc	Oc	-
[APN-Aggregate-Max-Bitrate-DL] ¹	1040	Oc	Oc	-

Table 1-8 CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
[APN-Aggregate-Max-Bitrate-UL] ¹	1041	Oc	Oc	-
[Extended-Max-Requested-BW-DL] ¹	554	Oc	Oc	-
[Extended-Max-Requested-BW-UL] ¹	555	Oc	Oc	-
[Extended-GBR-DL] ¹	2850	Oc	Oc	-
[Extended-GBR-UL] ¹	2851	Oc	Oc	-
[Extended-APN-AMBR-DL] ¹	2848	Oc	Oc	-
[Extended-APN-AMBR-UL] ¹	2849	Oc	Oc	-
[PS-to-CS-Session-Continuity] ¹	1099	Oc	Oc	-
[Reporting-Level] ¹	1011	Oc	Oc	-
[Online] ¹	1009	Oc	Oc	-
[Offline] ¹	1008	Oc	Oc	-
[Metering-Method]	1007	Oc	Oc	-
[Precedence] ¹	1010	Oc	Oc	-
[AF-Charging-Identifier] ¹	505	Oc	-	-
*[Flows] ¹	510	Oc	-	-
[Monitoring-Key]	1066	Oc	Oc	-
[Redirect-Information]	1085	Oc	Oc	-
[Mute-Notification]	2809	Oc	Oc	1
[Sponsor-Identity] ¹	531	Oc	Oc	-
[Application-Service-Provider-Identity] ¹	532	Oc	Oc	-
*[Required-Access-Info] ¹	536	Oc	Oc	-
*[Charging-Rule-Name] ¹	1005	Oc	Oc	-
*[Charging-Rule-Base-Name]	1004	Oc	Oc	-
[Rule-Activation-Time] ¹	1043	Oc	Oc	-
[Rule-Deactivation-Time] ¹	1044	Oc	Oc	-

Table 1-8 CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
[Resource-Allocation-Notification] ¹	1063	Oc	Oc	-
[Charging-Information] ¹	618	Oc	Oc	-
[Primary-Event-Charging-Function-Name] ¹	619	Oc	Oc	-
[Secondary-Event-Charging-Function-Name] ¹	620	Oc	Oc	-
[Online] ¹	1009	Oc	-	-
[Offline] ¹	1008	Oc	-	-
*[QoS-Information]	1016	Oc	Oc	-
[Max-Requested-Bandwidth-DL] ¹	515	Oc	Oc	-
[Max-Requested-Bandwidth-UL] ¹	516	Oc	Oc	-
[Guaranteed-Bitrate-DL] ¹	1025	Oc	Oc	-
[Guaranteed-Bitrate-UL] ¹	1026	Oc	Oc	-
[APN-Aggregate-Max-Bitrate-DL]	1040	Oc	Oc	-
[APN-Aggregate-Max-Bitrate-UL] ¹	1041	Oc	Oc	-
[Extended-Max-Requested-BW-DL] ¹	554	Oc	Oc	-
[Extended-Max-Requested-BW-UL] ¹	555	Oc	Oc	-
[Extended-GBR-DL] ¹	2850	Oc	Oc	-
[Extended-GBR-UL] ¹	2851	Oc	Oc	-
[Extended-APN-AMBR-DL] ¹	2848	Oc	Oc	-
[Extended-APN-AMBR-UL] ¹	2849	Oc	Oc	-
[Revalidation-Time]	1042	Oc	Oc	-
[Default-EPS-Bearer-QoS]	1049	Oc	Oc	-
[QoS-Class-Identifier]	1028	Oc	Oc	-
[Allocation-Retention-Priority] ¹	1034	Oc	Oc	-
[Pre-emption-Capability] ¹	1047	Oc	Oc	-
[Pre-emption-Vulnerability] ¹	1048	Oc	Oc	-

Table 1-8 CCA Message Format

AVP Name	AVP Code	Initial	Update	Terminate
[Bearer-Usage] ¹	1000	Oc	Oc	-
*[Usage-Monitoring-Information]	1067	Oc	Oc	-
[Monitoring-Key]	1066	Oc	Oc	-
[Granted-Service-Unit]	431	Oc	Oc	-
[CC-Time]	420	Oc	Oc	-
[CC-Total-Octets]	421	Oc	Oc	-
[CC-Input-Octets] ¹	412	Oc	Oc	-
[CC-Output-Octets] ¹	414	Oc	Oc	-
[Used-Service-Unit]	446	Oc	Oc	-
[CC-Time]	420	Oc	Oc	-
[CC-Total-Octets]	421	Oc	Oc	-
[CC-Input-Octets] ¹	412	Oc	Oc	-
[CC-Output-Octets] ¹	414	Oc	Oc	-
[Usage-Monitoring-Level]	1068	Oc	Oc	-
[Usage-Monitoring-Report]	1069	Oc	Oc	-
[Usage-Monitoring-Support] ¹	1070	Oc	Oc	-
[Session-Release-Cause]	1045	-	Oc	-
[3GPP-SGSN-MCC-MNC] ¹	18	Oc	Oc	-

¹ Indicates AVPs that are not qualified on the Gxa interface.

Table 1-9 lists the vendor-specific attributes (VSAs) for CCA messages on the Gx interface. For a description of each attribute, see [Appendix A, Diameter AVP Descriptions](#). These VSAs are not qualified on the Gxa interface.

Table 1-9 CCA Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	Initial	Update	Terminate
[Cisco-Event-Trigger-Type]	9	131192	Oc	Oc	-
[Cisco-Event-Trigger]	9	131193	Oc	Oc	-
[ACW-Entitlement]	9	131501	Oc	Oc	-
[ACW-Sponsor-URL]	9	131506	-	Oc	-
[MVNO-Reseller-Id]	9	131507	Oc	Oc	-
[QoS-Group-Rule-Install]	9	132001	Oc	Oc	-
*[QoS-Group-Rule-Definition]	9	132003	Oc	Oc	-
[QoS-Group-Rule-Remove]	9	132002	Oc	Oc	-
*[QoS-Group-Rule-Name]	9	132004	Oc	Oc	-
[MVNO-Sub-Class-Id]	9	131508	Oc	Oc	-
[Override-Control]	9	132017	Oc	Oc	-
[Override-Charging-Action-Parameters]	9	132019	Oc	Oc	-
*[Override-Charging-Action-Exclude-Rule]	9	132021	Oc	Oc	-
[Override-Charging-Parameters]	9	132022	Oc	Oc	-
[Override-Rating-Group]	9	132024	Oc	Oc	-
[Override-Online]	9	132026	Oc	Oc	-
[Override-Offline]	9	132027	Oc	Oc	-
[Trigger (VSA)]	12951	1264	Oc	Oc	-
[Trigger-Type (VSA)]	12951	870	Oc	Oc	-
[Assume-Positive-Treatment]	12951	7001	Oc	-	-

Disconnect-Peer-Request

The system sends a Disconnect-Peer-Request (DPR) to gracefully bring down the connection with the PCRF when the PCRF charging interface is taken down administratively.

Message format:

```
<Disconnect-Peer-Request> ::= <Diameter Header: 282, REQ>
    {Origin-Host}
    {Origin-Realm}
    {Disconnect-Cause}
```

Disconnect-Peer-Answer

The PCRF is expected to send a Disconnect-Peer-Answer (DPA) in response to a DPR. The system brings down the connection on receipt of the DPA.

Message format:

```
<Disconnect-Peer -Answer> ::= <Diameter Header: 282>
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    [Error-Message]
    *[Failed-AVP]
```

Device-Watchdog-Request

The system sends a Device-Watchdog-Request (DWR) to the PCRF at configured intervals.

Message format:

```
<Device-Watchdog-Request> ::= <Diameter Header: 280, REQ>
    {Origin-Host}
    {Origin-Realm}
    [Origin-State-Id]
```

Device-Watchdog-Answer

The system expects to receive a Device-Watchdog-Answer (DWA) to complete the DWR request it sends to the PCRF.

Message format:

```
<Device-Watchdog-Answer> ::= <Diameter Header: 280>
    {Result-Code}
    {Origin-Host}
    {Origin-Realm}
    [Error-Message]
    *[Failed-AVP]
    [Original-State-Id]
```

Re-Auth-Request

The PCRF sends the system a Re-Auth-Request (RAR) to provision QoS/PCC rules, Event-Triggers, and Event-Report-Indications for the session.

RAR Message format

```
<RA-Request> ::= < Diameter Header: 258, REQ, PXY >
    <Session-Id>
    {Auth-Application-Id}
    {Origin-Host}
    {Origin-Realm}
    {Destination-Realm}
    {Destination-Host}
    {Re-Auth-Request-Type}
    [Session-Release-Cause]
    [Origin-State-Id]
    *[Event-Trigger]
    *[Charging-Rule-Remove]
    *[Charging-Rule-Install]
    [Default-EPS-Bearer-QoS]
    *[QoS-Information]
    *[ADC-Rule-Remove]
    *[ADC-Rule-Install]
    [Revalidation-Time]
    *[Usage-Monitoring-Information]
    [MVNO-Reseller-Id]
    [MVNO-Sub-Class-Id]
    [online]
    [offline]
    *[AVP]
```

Table 1-10 lists the RAR message format for the Gx/Gxa interface. For a description of each AVP, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-10 RAR Message Format

AVP Name	AVP Code	AVP Type	RAR
Standard Diameter AVPs			
<Session-ID>	263	UTF8String	M
{Auth-Application-Id}	258	Unsigned32	M
{Origin-Host}	264	DiameterIdentity	M
{Origin-Realm}	296	DiameterIdentity	M
{Destination-Realm}	283	DiameterIdentity	M
{Destination-Host}	293	DiameterIdentity	M
{Re-Auth-Request-Type}	285	Enumerated	M
[Origin-State-Id] ¹	278	Unsigned32	Oc
3GPP AVPs			
[Session-Release-Cause]	1045	Enumerated	Oc
*[Event-Trigger]	1006	Enumerated	Oc
*[Charging-Rule-Remove]	1002	Grouped	Oc
*[Charging-Rule-Name]	1005	OctetString	Oc
*[Charging-Rule-Base-Name]	1004	UTF8String	Oc
*[Required-Access-Info] ¹	536	Enumerated	Oc
*[Charging-Rule-Install]	1001	Grouped	Oc
*[Charging-Rule-Definition]	1003	Grouped	Oc
{Charging-Rule-Name}	1005	OctetString	Oc
[Service-Identifier] ¹	439	Unsigned32	Oc
[Rating-Group]	432	Unsigned32	Oc
[Flow-Information] ¹	1058	Grouped	Oc
[TDF-Application-Identifier] ¹	1088	OctetString	Oc
[Flow-Status] ¹	511	Enumerated	Oc
[QoS-Information] ¹	1016	Grouped	Oc

Table 1-10 RAR Message Format

AVP Name	AVP Code	AVP Type	RAR
[Max-Requested-Bandwidth-DL] ¹	515	Unsigned32	Oc
[Max-Requested-Bandwidth-UL] ¹	516	Unsigned32	Oc
[Guaranteed-Bitrate-DL] ¹	1025	Unsigned32	Oc
[Guaranteed-Bitrate-UL] ¹	1026	Unsigned32	Oc
[APN-Aggregate-Max-Bitrate-DL] ¹	1040	Unsigned32	Oc
[APN-Aggregate-Max-Bitrate-UL] ¹	1041	Unsigned32	Oc
[Extended-Max-Requested-BW-DL] ¹	554	Unsigned32	Oc
[Extended-Max-Requested-BW-UL] ¹	555	Unsigned32	Oc
[Extended-GBR-DL] ¹	2850	Unsigned32	Oc
[Extended-GBR-UL] ¹	2851	Unsigned32	Oc
[Extended-APN-AMBR-DL] ¹	2848	Unsigned32	Oc
[Extended-APN-AMBR-UL] ¹	2849	Unsigned32	Oc
[PS-to-CS-Session-Continuity] ¹	1099	Enumerated	Oc
[Reporting-Level] ¹	1011	Enumerated	Oc
[Online] ¹	1009	Enumerated	Oc
[Offline]	1008	Enumerated	Oc
[Metering-Method]	1007	Enumerated	Oc
[Precedence] ¹	1010	Unsigned32	Oc
[AF-Charging-Identifier] ¹	505	OctetString	Oc
*[Flows] ¹	510	Grouped	Oc
[Monitoring-Key]	1066	OctetString	Oc
[Redirect-Information]	1085	Grouped	Oc
[Mute-Notification]	2809	Enumerated	Oc
*[Required-Access-Info] ¹	536	Enumerated	Oc
*[Charging-Rule-Name] ¹	1005	OctetString	Oc
*[Charging-Rule-Base-Name]	1004	UTF8String	Oc

Table 1-10 RAR Message Format

AVP Name	AVP Code	AVP Type	RAR
[Rule-Activation-Time] ¹	1043	Time	Oc
[Rule-Deactivation-Time] ¹	1044	Time	Oc
[Default-EPS-Bearer-QoS]	1049	Grouped	Oc
[QoS-Class-Identifier]	1028	Enumerated	Oc
[Allocation-Retention-Priority] ¹	1034	Grouped	Oc
[Pre-emption-Capability] ¹	1047	Enumerated	Oc
[Pre-emption-Vulnerability] ¹	1048	Enumerated	Oc
*[QoS-Information]	1016	Grouped	Oc
[Max-Requested-Bandwidth-DL] ¹	515	Unsigned32	Oc
[Max-Requested-Bandwidth-UL] ¹	516	Unsigned32	Oc
[Guaranteed-Bitrate-DL] ¹	1025	Unsigned32	Oc
[Guaranteed-Bitrate-UL] ¹	1026	Unsigned32	Oc
[APN-Aggregate-Max-Bitrate-DL]	1040	Unsigned32	Oc
[APN-Aggregate-Max-Bitrate-UL]	1041	Unsigned32	Oc
[Extended-Max-Requested-BW-DL] ¹	554	Unsigned32	Oc
[Extended-Max-Requested-BW-UL] ¹	555	Unsigned32	Oc
[Extended-GBR-DL] ¹	2850	Unsigned32	Oc
[Extended-GBR-UL] ¹	2851	Unsigned32	Oc
[Extended-APN-AMBR-DL] ¹	2848	Unsigned32	Oc
[Extended-APN-AMBR-UL] ¹	2849	Unsigned32	Oc
*[ADC-Rule-Remove] ¹	1093	Grouped	Oc
*[ADC-Rule-Name] ¹	1096	OctetString	Oc
*[ADC-Rule-Base-Name] ¹	1095	UTF8String	Oc
*[ADC-Rule-Install] ¹	1092	Grouped	Oc
*[ADC-Rule-Definition] ¹	1094	Grouped	Oc
*[ADC-Rule-Name] ¹	1096	OctetString	Oc

Table 1-10 RAR Message Format

AVP Name	AVP Code	AVP Type	RAR
*[ADC-Rule-Base-Name] ¹	1095	UTF8String	Oc
[Rule-Activation-Time] ¹	1043	Time	Oc
[Rule-Deactivation-Time] ¹	1044	Time	Oc
[Revalidation-Time]	1042	Time	Oc
*[Usage-Monitoring-Information]	1067	Grouped	Oc
[Monitoring-Key]	1066	OctetString	Oc
[Granted-Service-Unit]	431	Grouped	Oc
[CC-Time]	420	Unsigned32	Oc
[CC-Total-Octets]	421	Unsigned32	Oc
[CC-Input-Octets] ¹	412	Unsigned32	Oc
[CC-Output-Octets] ¹	414	Unsigned32	Oc
[Used-Service-Unit]	446	Grouped	Oc
[CC-Time]	420	Unsigned32	Oc
[CC-Total-Octets]	421	Unsigned32	Oc
[CC-Input-Octets] ¹	412	Unsigned32	Oc
[CC-Output-Octets] ¹	414	Unsigned32	Oc
[Usage-Monitoring-Level]	1068	Enumerated	Oc
[Usage-Monitoring-Report]	1069	Enumerated	Oc
[Usage-Monitoring-Support] ¹	1070	Enumerated	Oc
[Online]	1009	Enumerated	Oc
[Offline]	1008	Enumerated	Oc

¹ Indicates AVPs that are not qualified on the Gxa interface.

Table 1-11 lists the vendor-specific attributes (VSAs) for RAR messages on the Gx interface. For a description of each attribute, see [Appendix A, Diameter AVP Descriptions](#). These VSAs are not qualified on the Gxa interface.

Table 1-11 RAR Vendor-Specific Attributes

AVP Name	Vendor ID	AVP Code	AVP Type	RAR
[Cisco-Event-Trigger-Type]	9	131192	Enumerated	Oc
[Cisco-Event-Trigger]	9	131193	Grouped	Oc
[ACW-Entitlement]	9	131501	Octet String	Oc
[ACW-Sponsor-URL]	9	131506	UTF8String	Oc
[MVNO-Reseller-Id]	9	131507	UTF8String	Oc
[MVNO-Sub-Class-Id]	9	131508	UTF8String	Oc
[QoS-Group-Rule-Install]	9	132001	Grouped	Oc
*[QoS-Group-Rule-Definition]	9	132003	Grouped	Oc
[QoS-Group-Rule-Remove]	9	132002	Grouped	Oc
*[QoS-Group-Rule-Name]	9	132004	Octet String	Oc
[Override-Control]	9	132017	Grouped	Oc
[Override-Charging-Action-Parameters]	9	132019	Grouped	Oc
*[Override-Charging-Action-Exclude-Rule]	9	132021	Octet String	Oc
[Override-Charging-Parameters]	9	132022	Grouped	Oc
[Override-Rating-Group]	9	132024	Unsigned32	Oc
[Override-Online]	9	132026	Enumerated	Oc
[Override-Offline]	9	132027	Enumerated	Oc
[Gx-TMO-Redirect-Server]	29168	111	Grouped	Oc
[Gx-TMO-Redirect-Address-Type]	29168	112	Enumerated	Oc
[Gx-TMO-Redirect-Server-Address]	29168	113	UTF8String	Oc
[Gx-TMO-Append-Original-URL]	29168	114	Enumerated	Oc
[Gx-TMO-Append-MSISDN]	29168	115	Enumerated	Oc
[Gx-TMO-Append-IMSI]	29168	116	Enumerated	Oc
[Gx-TMO-Append-IMEI]	29168	117	Enumerated	Oc
[Gx-TMO-Append-MSIP]	29168	118	Enumerated	Oc
[Gx-TMO-Deactivate-By-Redirect]	29168	119	Enumerated	Oc

Table 1-12 lists the vendor-specific attributes (VSAs) for RAR messages on the Gxa interface. For a description of each attribute, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-12 RAR Vendor-Specific Attributes (Gxa)

AVP Name	Vendor ID	AVP Code	AVP Type	RAR
[ACW-Entitlement]	37963	26	String	Oc

Re-Auth-Answer

The system sends a Re-Auth-Answer (RAA) in response to an RAR.

RAA Message Format

```
<RA-Answer> ::= < Diameter Header: 258, PXY >
    <Session-Id>
    {Auth-Application-Id}
    {Origin-Host}
    {Origin-Realm}
    {Destination-Realm}
    [Result-Code]
    [Experimental-Result]
    [Origin-State-Id]
    [IP-CAN-Type]
    [RAT-Type]
    0*2[AN-GW-Address]
    [3GPP-SGSN-MCC-MNC]
    [3GPP-SGSN-Address]
    [3GPP-SGSN-IPv6-Address]
    [3GPP-User-Location-Info]
    [3GPP-MS-TimeZone]
    *[ADC-Rule-Report]
    *[Charging-Rule-Report]
    *[AVP]
```

Table 1-13 lists the RAA message format on the Gx/Gxa interface. For a description of each AVP, see [Appendix A, Diameter AVP Descriptions](#).

Table 1-13 RAA Message Format

AVP Name	AVP Code	AVP Type	RAR
Standard Diameter AVPs			
<Session-ID>	263	UTF8String	M
{Auth-Application-Id}	258	Unsigned32	M
{Origin-Host}	264	DiameterIdentity	M
{Origin-Realm}	296	DiameterIdentity	M
{Destination-Realm}	283	DiameterIdentity	M
[Result-Code]	268	Unsigned32	Oc
[Experimental-Result] ¹	297	Grouped	Oc
[Origin-State-Id] ¹	278	Unsigned32	Oc
3GPP AVPs			
[IP-CAN-Type]	1027	Enumerated	Oc
[RAT-Type]	1032	Enumerated	Oc
[AN-GW-Address] ¹	1050	Address	Oc
[3GPP-SGSN-MCC-MNC] ¹	18	UTF8String	Oc
[3GPP-SGSN-Address] ¹	6	OctetString	Oc
[3GPP-SGSN-IPv6-Address] ¹	15	OctetString	Oc
[3GPP-User-Location-Info] ¹	22	OctetString	Oc
[3GPP-MS-TimeZone] ¹	23	UTF8String	Oc
[ADC-Rule-Report] ¹	1097	Grouped	Oc
[Charging-Rule-Report] ¹	1018	Grouped	Oc

¹ Indicates AVPs that are not qualified on the Gxa interface.

Result Codes

The Result-Code AVP specifies whether a request has successfully completed or failed. The system interprets a request as successful if a diameter peer returns a request with the Result-Code AVP value of 2001. However, a result code other than 2001 is considered unsuccessful.

[Table 1-14](#) summarizes the result codes supported by the system that indicate a request has failed.

Table 1-14 Supported Result Codes

Procedure	Value	Name	Description	Action
Initial	3002	DIAMETER_UNABLE_TO_DELIVER	The diameter device cannot deliver the message to the desired destination because the host described in the Destination-Host AVP does not match a host within the associated realm.	See Failure Handling (page 1-129) for more information.
Initial	3004	DIAMETER_TOO_BUSY	The diameter device cannot deliver the message to the desired destination.	See Failure Handling (page 1-129) for more information.
Initial	3005	DIAMETER_LOOP_DETECTED	A diameter agent detected a loop while trying to get the message to the desired destination.	See Failure Handling (page 1-129) for more information.
Update	3002	DIAMETER_UNABLE_TO_DELIVER	The diameter device cannot deliver the message to the desired destination because the host described in the Destination-Host AVP does not match a host within the associated realm.	See Failure Handling (page 1-129) for more information.
Update	3004	DIAMETER_TOO_BUSY	The diameter device cannot deliver the message to the desired destination.	See Failure Handling (page 1-129) for more information.
Update	3005	DIAMETER_LOOP_DETECTED	A diameter agent detected a loop while trying to get the message to the desired destination.	See Failure Handling (page 1-129) for more information.
Update	Other non-successful codes	N/A	N/A	The system logs the event, updates statistics, and sets the GTP result to failure. If this occurs while processing event triggers and the quota list is not empty, the system moves the charging groups.
Terminate	3002	DIAMETER_UNABLE_TO_DELIVER	The diameter device cannot deliver the message to the desired destination because the host described in the Destination-Host AVP does not match a host within the associated realm.	See Failure Handling (page 1-129) for more information.
Terminate	3004	DIAMETER_TOO_BUSY	The diameter device cannot deliver the message to the destination.	See Failure Handling (page 1-129) for more information.

Table 1-14 Supported Result Codes

Procedure	Value	Name	Description	Action
Terminate	3005	DIAMETER_LOOP_DETECTED	A diameter agent detected a loop while trying to get the message to the desired destination.	See Failure Handling (page 1-129) for more information.

Diameter AVP Descriptions

This section describes the supported Attribute Value Pair (AVP) definitions for the Gy, Gx, and Gz/Rf Diameter interfaces and therefore the AVPs described in this section do not apply to all interfaces.

- [Standard AVP Definitions \(page A-2\)](#)
- [3GPP AVP Definitions \(page A-28\)](#)
- [NASREQ/RADIUS AVP Definitions \(page A-102\)](#)
- [Vodafone AVP Definitions \(page A-104\)](#)
- [Vendor-Specific AVP Definitions \(page A-112\)](#)

Note: Refer to the 3GPP specifications and RFCs listed in References and Specifications for more details.

Standard AVP Definitions

This section describes the following standard AVPs used in the messages for DCCA.

- Accounting-Input-Octets (page A-3)
- Accounting-Input-Packets (page A-3)
- Accounting-Output-Octets (page A-3)
- Accounting-Output-Packets (page A-3)
- Accounting-Record-Number (page A-4)
- Accounting-Record-Type (page A-4)
- Acct-Application-Id (page A-4)
- Acct-Interim-Interval (page A-4)
- Auth-Application-Id (page A-5)
- CC-Input-Octets (page A-5)
- CC-Output-Octets (page A-5)
- CC-Request-Number (page A-6)
- CC-Request-Type (page A-6)
- CC-Session-Failover (page A-6)
- CC-Service-Specific-Units (page A-7)
- CC-Time (page A-7)
- CC-Total-Octets (page A-7)
- CC-Input-Octets (page A-5)
- Credit-Control-Failure-Handling (page A-7)
- Destination-Host (page A-8)
- Destination-Realm (page A-8)
- Event-Timestamp (page A-8)
- Experimental-Result (page A-9)
- Failed-AVP (page A-9)
- Filter-ID (page A-9)
- Final-Unit-Action (page A-10)
- Final-Unit-Indication (page A-11)
- Firmware-Revision (page A-12)
- Granted-Service-Unit (page A-12)
- Host-IP-Address (page A-12)
- Inband-Security-Id (page A-12)
- Multiple-Services-Credit-Control (page A-13)
- Multiple-Services-Indicator (page A-13)
- Origin-Host (page A-13)
- Origin-Realm (page A-14)
- Origin-State-Id (page A-14)
- Product-Name (page A-14)
- Rating-Group (page A-14)
- Re-Auth-Request-Type (page A-15)
- Redirect-Address-Type (page A-15)
- Redirect-Host (page A-16)
- Redirect-Server (page A-16)
- Redirect-Server-Address (page A-17)
- Requested-Action (page A-17)
- Requested-Service-Unit (page A-17)
- Result-Code (page A-18)
- Route-Record (page A-18)
- Route-Record (page A-18)
- Service-Identifier (page A-19)
- Session-ID (page A-20)
- Subscription-Id (page A-21)
- Subscription-Id-Data (page A-22)
- Subscription-Id-Type (page A-22)
- Supported-Vendor-Id (page A-23)
- Tariff-Change-Usage (page A-23)
- Tariff-Time-Change (page A-23)
- Termination-Cause (page A-24)
- User-Equipment-Info (page A-24)
- User-Equipment-Info-Type (page A-25)
- User-Equipment-Info-Value (page A-25)
- User-Name (page A-25)
- Used-Service-Unit (page A-26)
- Validity-Time (page A-26)
- Vendor-Id (page A-26)
- Vendor-Specific-Application-Id (page A-27)

Accounting-Input-Octets

AVP Name	Accounting-Input-Octets
AVP Code	363
AVP Data Type	Integer64
Description	Together with the Accounting-Input-Packets AVP, this AVP contains the number of octets (resp packets), transmitted during the data container recording interval, reflecting the volume counts for uplink traffic for a data flow.

Accounting-Input-Packets

AVP Name	Accounting-Input-Packets
AVP Code	365
AVP Data Type	Integer64
Description	Together with the Accounting-Input-Octets AVP, this AVP contains the number of packets (resp octets), transmitted during the data container recording interval, reflecting the volume counts for uplink traffic for a data flow.

Accounting-Output-Octets

AVP Name	Accounting-Output-Octets
AVP Code	364
AVP Data Type	Integer64
Description	Together with the Accounting-Output-Packets AVP, this AVP contains the number of octets (resp packets), transmitted during the data container recording interval, reflecting the volume counts for downlink traffic for a data flow.

Accounting-Output-Packets

AVP Name	Accounting-Output-Packets
AVP Code	366
AVP Data Type	Integer64
Description	Together with the Accounting-Output-Octets AVP, this AVP contains the number of packets (resp octets), transmitted during the data container recording interval, reflecting the volume counts for downlink traffic for a data flow.

Accounting-Record-Number

AVP Name	Accounting-Record-Number
AVP Code	485
AVP Data Type	Unsigned32
Description	Contains the sequence number of the transferred messages.

Accounting-Record-Type

AVP Name	Accounting-Record-Type
AVP Code	480
AVP Data Type	Enumerated
Description	Defines the type of accounting record: ■ START – Starts an accounting session. ■ INTERIM – Updates an accounting session. ■ STOP – Ends an accounting session. ■ EVENT – Indicates a one-time accounting event.

Acct-Application-Id

AVP Name	Acct-Application-Id
AVP Code	259
AVP Data Type	Unsigned32
Description	Indicates the unique ID of the Diameter Accounting Application and is defined with the value 3.

Acct-Interim-Interval

AVP Name	Acct-Interim-Interval
AVP Code	85
AVP Data Type	Unsigned32
Description	Indicates the number of seconds between each interim update for this session.

Auth-Application-Id

AVP Name	Auth-Application-Id
AVP Code	258
AVP Data Type	Unsigned32
Description	Indicates the unique ID that is used for re-authentication or re-authorization. The AVP contains the following values: ■ Gy interface – 4 (DCCA) ■ Gx interface – 16777238 ■ S6b interface – You can specify the application ID (standard or non-standard) on the S6b interface using the following command. This is any value up to 32 bits. The default is 16777272 (default - 3GPP S6b). <pre>an3000-mcm-slot7cpu1(config)# service-construct interface s6b-interface <name> auth-application-id <value></pre>

CC-Input-Octets

AVP Name	CC-Input-Octets
AVP Code	412
AVP Data Type	Unsigned64
Description	The amount of octets to be received.

CC-Output-Octets

AVP Name	CC-Output-Octets
AVP Code	414
AVP Data Type	Unsigned64
Description	The amount of octets to be sent.

CC-Request-Number

AVP Name	CC-Request-Number
AVP Code	415
AVP Data Type	Unsigned32
Description	Indicates the identity of a certain request in a session, starting from 0.

CC-Request-Type

AVP Name	CC-Request-Type
AVP Code	416
AVP Data Type	Enumerated
Description	Indicates the reason for sending a CCR message. This AVP must be included in the CCR message. The CC-Request-Type AVP contains the following values: ■ INITIAL_REQUEST (1) – Indicates that the initial CCR is used to initiate a credit control session. The session contains the credit control information about initiating a session. ■ UPDATE_REQUEST (2) – Indicates that the update CCR contains the credit control information about the sessions of which credit control is set up. When the re-authorization needs to be initiated, an update CCR needs to be initiated. ■ TERMINATION_REQUEST (3) – Indicates an interruption to a credit control session by terminating a credit request.

CC-Session-Failover

AVP Name	CC-Session-Failover
AVP Code	418
AVP Data Type	Enumerated
Description	Indicates whether the OCS supports moving the CCR messages to a secondary server for a CC session. If the OCS supports session failover, the system sends CCR messages to the secondary server when the transport connection with the primary server goes down. The following values are defined: ■ FAILOVER_NOT_SUPPORTED (0) ■ FAILOVER_SUPPORTED (1)

CC-Service-Specific-Units

AVP Name	CC-Service-Specific-Units
AVP Code	417
AVP Data Type	Unsigned64
Description	The amount of service specific units. For example, the number of events for a rating-group.

CC-Time

AVP Name	CC-Time
AVP Code	420
AVP Data Type	Unsigned32
Description	The requested, distributed, or used time of the charging party, in seconds.

CC-Total-Octets

AVP Name	CC-Total-Octets
AVP Code	421
AVP Data Type	Unsigned64
Description	The amount of octets to be sent and received.

Credit-Control-Failure-Handling

AVP Name	Credit-Control-Failure-Handling
AVP Code	427
AVP Data Type	Enumerated
Description	The OCS uses this AVP to instruct clients how to handle the PDN session during a failure condition (transport failure etc.). The following values are defined: ■ TERMINATE (0) ■ CONTINUE (1) ■ RETRY_AND_TERMINATE (2)

Destination-Host

AVP Name	Destination-Host
AVP Code	293
AVP Data Type	DiameterIdentity
Description	Indicates the identity of the destination device. This AVP must be contained in messages that are initiated by an agent. This AVP can be contained in request messages but cannot be contained in response messages.

Destination-Realm

AVP Name	Destination-Realm
AVP Code	283
AVP Data Type	DiameterIdentity
Description	Indicates the home realm of the destination device.

Event-Timestamp

AVP Name	Event-Timestamp
AVP Code	55
AVP Data Type	Time
Description	Indicates the time of an event. The value of this AVP is generated by the DCC client. The time is converted from January 1, 1900, 00:00 UTC, in seconds. <i>Note: UTC: Coordinated Universal Time.</i> Gx Interface – Indicates the time when the Create Session Request was received. Gy Interface – Indicates the time when the quota was requested.

Experimental-Result

AVP Name	Experimental-Result
AVP Code	297
AVP Data Type	Grouped
Description	Indicates whether a particular vendor-specific request was completed successfully or whether an error occurred. The detailed syntax is as follows: <pre>Experimental-Result ::= <AVP Header: 297> {Vendor-Id} {Experimental-Result-Code}</pre>

Failed-AVP

AVP Name	Failed-AVP
AVP Code	279
AVP Data Type	Grouped
Description	Provides debugging information in cases where a request is rejected or not fully processed due to erroneous information in a specific AVP. The value of the Result-Code AVP will provide information on the reason for the Failed-AVP.

Filter-ID

AVP Name	Filter-ID
AVP Code	11
AVP Data Type	UTF8String
Description	Indicates the name of the filter list for this user. Used to restrict user access when the FUI AVP (embedded within the MSCC AVP), received in the CCA message from the OCS, contains the Final-Unit-Action AVP set to RESTRICT_ACCESS and one or more Filter-ID AVPs referencing a locally configured Rating Group with an IP address filter list. Multiple Filter-IDs are supported by configuring a Rating Group for each Filter-ID. See “Mapping a Filter-ID to a Rating-Group” in the <i>Affirmed Networks Gy Interface Specification</i>.

Final-Unit-Action

AVP Name	Final-Unit-Action
AVP Code	449
AVP Data Type	Enumerated
Description	Indicates the action to be taken once the subscriber has consumed all of the final granted units. The following values are defined: ■ TERMINATE (0) – Upon consumption of the final granted units, the session is terminated. ■ REDIRECT (1) – Upon consumption of the final granted units, the subscriber is redirected to the address specified in the Redirect-Server AVP. ■ RESTRICT-ACCESS (2) – Gy Interface – Upon consumption of the final granted units, user access is restricted according to the IP packet filters identified by the Filter-ID AVP. All of the packets that do not match the filters are dropped. For more information, see “Mapping a Filter-ID to a Rating-Group” in the <i>Affirmed Networks Gy Interface Specification</i>. – Gx Interface – If a flow is out of charging credits and when a packet for that flow arrives, the packet is dropped except when an Advice of Charge (AoC) ID is configured for the associated Rating Group on the MCC device. In this case, the subscriber is redirected to the redirect URL configured for the AoC ID.

Final-Unit-Indication

AVP Name	Final-Unit-Indication
AVP Code	430
AVP Data Type	Grouped
Description	The Final-Unit-Indication (FUI) AVP indicates that the Granted-Service-Unit AVP in the CCA contains the final units for the service. After these units have expired, the gateway executes the action indicated in the Final-Unit-Action AVP. Note the following: Gy Interface The FUI AVP is embedded within the Multiple-Services-Credit-Control (MSCC) AVP in the CCA message. The MCC supports FUI-based redirection when the MSCC AVP received in the CCA message from the OCS contains: ■ MSCC result code = 2001 (DIAMETER_SUCCESS) or 4012 (CREDIT_LIMIT_REACHED) + FUI AVP ■ MSCC result code = 2001 (DIAMETER_SUCCESS) + FUI AVP, no grant (no GSU AVP) ■ MSCC result code = 2001 (DIAMETER_SUCCESS) + FUI AVP + zero Grant (GSU AVP with zero grant / no grant info) The detailed syntax is as follows: <pre>Final-Unit-Indication ::= < AVP Header: 430 > {Final-Unit-Action} *[Filter-Id] [Redirect-Server]</pre> Gx Interface The FUI AVP is included in the CCA message at the command level. The detailed syntax is as follows: <pre>Final-Unit-Indication ::= < AVP Header: 430 > {Final-Unit-Action} [Redirect-Server]</pre> The FUI AVP is also included in the CCR message within the Charging-Rule-Report AVP to report the status of the PCC rules for which credit is no longer available or credit has been reallocated after the former out-of-credit indication. When reporting an out-of-credit condition, the Final-Unit-Indication AVP indicates the termination action the PCEF applies to the PCC rules as instructed by the OCS. The detailed syntax is as follows: <pre>Final-Unit-Indication ::= < AVP Header: 430 > {Final-Unit-Action} *[Filter-Id] [Redirect-Server]</pre>

Firmware-Revision

AVP Name	Firmware-Revision
AVP Code	267
AVP Data Type	Unsigned32
Description	Used to inform a diameter peer of the firmware revision of the issuing device.

Granted-Service-Unit

AVP Name	Granted-Service-Unit
AVP Code	431
AVP Data Type	Grouped
Description	The total service units that the DCC client can provide for end users.

Host-IP-Address

AVP Name	Host-IP-Address
AVP Code	257
AVP Data Type	Address
Description	Informs a diameter peer of the sender's IP address.

Inband-Security-Id

AVP Name	Inband-Security-Id
AVP Code	299
AVP Data Type	Unsigned32
Description	Indicates support of the security portion of the application.

Multiple-Services-Credit-Control

AVP Name	Multiple-Services-Credit-Control
AVP Code	456
AVP Data Type	Grouped
Description	Indicates all parameters for the CTF quota management and defines the quotas to allow traffic to flow. Each Multiple-Services-Credit-Control (MSCC) carries units for a single rating group.

Multiple-Services-Indicator

AVP Name	Multiple-Services-Indicator
AVP Code	455
AVP Data Type	Enumerated
Description	Indicates whether the system is capable of reporting multiple rating groups within a CC session. It is included in the CCR-Initial message. The following values are defined: ■ MULTIPLE_SERVICES_NOT_SUPPORTED (0) ■ MULTIPLE_SERVICES_SUPPORTED (1)

Origin-Host

AVP Name	Origin-Host
AVP Code	264
AVP Data Type	DiameterIdentity
Description	Contains the identification of the source point of the operation and the realm of the operation originator. Relay agents must not modify this AVP. The value of the Origin-Host AVP is guaranteed to be unique within a single host. <i>Note: The Origin-Host AVP may resolve to more than one address as the diameter peer may support more than one address (for example: host1.com).</i>

Origin-Realm

AVP Name	Origin-Realm
AVP Code	296
AVP Data Type	DiameterIdentity
Description	Indicates the home realm of the device that initiates DCC messages.

Origin-State-Id

AVP Name	Origin-State-Id
AVP Code	278
AVP Data Type	Unsigned32
Description	Carries the state associated with an Origin-Host.

Product-Name

AVP Name	Product-Name
AVP Code	269
AVP Data Type	UTF8String
Description	Contains the vendor-assigned name for the product. The Product-Name AVP should remain constant across firmware revisions for the same product.

Rating-Group

AVP Name	Rating-Group
AVP Code	432
AVP Data Type	Unsigned32
Description	The identifier of a rating group. For information on how the Rating-Group AVP is used on the Gx interface to install/activate PCC rules, see “PCC Rules”, in the <i>Affirmed Networks Gx Interface Specification</i>.

Re-Auth-Request-Type

AVP Name	Re-Auth-Request-Type
AVP Code	285
AVP Data Type	Enumerated
Description	Included in application-specific authorization answers to inform the client of the action expected upon expiration of the Authorization-Lifetime. The following values are defined: ■ AUTHORIZE_ONLY (0) – An authorization only re-authorization is expected upon expiration of the Authorization-Lifetime. This is the default value if the AVP is not present in answer messages that include the Authorization-Lifetime. ■ AUTHORIZE_AUTHENTICATE (1) – An authentication and authorization re-authorization is expected upon expiration of the Authorization-Lifetime.

Redirect-Address-Type

AVP Name	Redirect-Address-Type
AVP Code	433
AVP Data Type	Enumerated
Description	Defines the type of address given in the Redirect-Server-Address AVP. The address type can be one of the following: ■ URL (2) – The address is in the form of Uniform Resource Locator (URL). <i>Note: MCC does not support Redirect-Address-Type SIP URI.</i>

Redirect-Host

AVP Name	Redirect-Host
AVP Code	292
AVP Data Type	DiameterURI
Description	Contains the host to which the request should be forwarded. If the PGW receives a CCA (Initial, Update or Terminate) message from the PCRF and the Result-Code AVP is set to DIAMETER_REDIRECT_INDICATION, the PGW will forward the CCR message for that session to the host contained in the Redirect-Host AVP. The PCRF can send multiple Redirect-Host AVPs in the CCA message and if one fails, the MCC selects the next one in the list. The Redirect-Host AVP is configurable in CCA data record templates. The default <code>presentFlag</code> setting is <code>yes</code>, indicating the AVP is processed when received in the CCA message. <pre>an3000-mcm-slot7cpul (config) # service-construct data-record-template [pgw ggsn] pcrf diameter cca <template-name> redirecthost presentFlag [yes no]</pre>

Redirect-Server

AVP Name	Redirect-Server
AVP Code	434
AVP Data Type	Grouped
Description	Contains the address information of the redirect server (for example, HTTP redirect server) with which the end user is to be connected when the account cannot cover the service cost. This AVP must be present when the Final-Unit-Action AVP is set to REDIRECT. The detailed syntax is as follows: <pre>Redirect-Server ::= < AVP Header: 434 > {Redirect-Address-Type} {Redirect-Server-Address}</pre> Note: The MCC does not support Redirect-Address-Type SIP URI.

Redirect-Server-Address

AVP Name	Redirect-Server-Address
AVP Code	435
AVP Data Type	UTF8String
Description	Defines the address of the redirect server.

Requested-Action

AVP Name	Requested-Action
AVP Code	436
AVP Data Type	Enumerated
Description	Contains the requested action sent in the CCR where the CC-Request-Type is set to EVENT_REQUEST. The following values are defined: ■ DIRECT_DEBITING (0) – Indicates a request to decrease the end user's account according to information specified in the Requested-Service-Unit AVP and/or Service-Identifier AVP (additional rating information may be included in service-specific AVPs or in the Service-Parameter-Info AVP). The Granted-Service-Unit AVP in the CCA contains the debited units. ■ REFUND_ACCOUNT (1) – Indicates a request to increase the end user's account according to information specified in the Requested-Service-Unit AVP and/or Service-Identifier AVP (additional rating information may be included in service-specific AVPs or in the Service-Parameter-Info AVP). The Granted-Service-Unit AVP in the CCA contains the refunded units. ■ CHECK_BALANCE (2) – Indicates a balance check request. In this case, the checking of the account balance is done without any credit reservation from the account. The Check-Balance-Result AVP in the CCA contains the result of the balance check. ■ PRICE_ENQUIRY (3) – Indicates a price inquiry request. In this case, neither checking of the account balance nor reservation from the account is done. Only the price of the service is returned in the Cost-Information AVP in the CCA.

Requested-Service-Unit

AVP Name	Requested-Service-Unit
AVP Code	437
AVP Data Type	Grouped
Description	The amount of requested service units for a particular category, including monetary or non-monetary units.

Result-Code

AVP Name	Result-Code
AVP Code	268
AVP Data Type	Unsigned32
Description	A specified request is successfully fulfilled or fails.

Route-Record

AVP Name	Route-Record
AVP Code	282
AVP Data Type	DiameterIdentity
Description	Contains an identifier inserted by a relay or proxy node to identify the node from which the message is received.

Service-Context-Id

AVP Name	Service-Context-Id
AVP Code	461
AVP Data Type	UTF8String
Description	Indicates the unique ID (may be extended) of a Diameter credit-control (DCC) service-specific document that applies to the request. Use the following CLI parameter to specify the Service-Context-Id AVP sent in ACR messages via the Gz/Rf offline-charging interface to the OFCS. The default service-context-id is 9.32251@3gpp.org. <pre>an3000-mcm-slot7cpul(config)# service-construct interface offline-charging <name> service-context-id <service-context-id></pre>

Service-Identifier

AVP Name	Service-Identifier
AVP Code	439
AVP Data Type	Unsigned32
Description	Contains the identifier of a service. The specific service the request relates to is uniquely identified by the combination of the Service-Context-Id and Service-Identifier AVPs. Note the following: ■ For dynamic PCC rules, the Service-Identifier AVP sent from the PCRF as part of the Charging-Rule-Definition AVP does not need to be unique. Multiple Charging-Rule-Definition AVPs can have the same Service-Identifier AVP value. ■ For predefined PCC rules, if the MCC gateway receives a CCA message from the PCRF with the Service-Identifier AVP under the Charging-Rule-Definition AVP, the value in the Service-Identifier AVP overrides the Service-ID (<code>service-construct->service-rule->service-data-flow-id</code>) value provisioned on the gateway assuming the Charging-Rule-Name (<code>service-rule</code>) is valid. ■ If the <code>reporting-level</code> of a Rating Group is set to <code>service-rule</code>, the PGW sends the Service-Identifier AVP and Rating Group AVP (contained within the MSCC AVP) in CCR messages to the OCS. The Service-Identifier AVP only reports the Service-ID (<code>service-data-flow-id</code>) of the first service rule in the Rating Group. If the service rule list contains different Service-IDs, the MSCC AVP does not report all the Service-IDs.

Session-ID

AVP Name	Session-ID
AVP Code	263
AVP Data Type	UTF8String
Description	Identifies a session-ID that uniquely identifies a DCCA session. The default format is as follows: ■ <DiameterIdentity>;<high 32 bits>;<low 32 bits>;<optional value>. ■ <DiameterIdentity>: the same as filled in the Origin-Host AVP. ■ <high 32 bits>: contains the current time of the system. ■ <low 32 bits>: contains service instance cookie. ■ <optional value>: contains subscriber and bearer information. For example: <i>example.com;1450803134;2148532240;8;1450812272;262229</i> where ■ <i>example.com</i>: (Diameter identity) represents the Origin-host. ■ <i>1450803134</i>: (high 32 bits) represents the interface timestamp. ■ <i>2148532240</i>: (low 32 bits) represents the subscriber ID. ■ <i>8</i>: (optional value) represents the static Diameter session counter. ■ <i>1450812272</i>: (optional value) represents the session setup time in decimal format. ■ <i>262229</i>: (optional value) represents the gateway service ID (formed using Geo-ID, Block-ID, Node-ID, Bearer-ID). You can configure the format for the Session-ID AVP sent on the Gz/Rf interface using the following command. <pre>an3000-mcm-slot7cpul(config)# service-construct interface offline-charging <interface-name> session-id-format [v0 v1]</pre> ■ v0 – (default) The Session-ID AVP format represents the optional value for the session setup time in decimal format. For example: <i>example.com;1450803134;2148532240;8;1450812272;262229</i> where 1450812272 represents the session setup time in decimal format. ■ v1 – The Session-ID AVP format represents the optional value for the session setup time in hex format. This format applies to the Gz/Rf interface only. For example: 01-<i>example.com;1450803134;2148532240;8;5679A370-262229</i> where 01 is a prefix indicating that this Session-ID format is using version 1 (v1) and 5679A370 represents the session setup time in hex format. Note that the delimiter after the timestamp uses a dash (-) instead of a semi-colon (;).

Subscription-Id

AVP Name	Subscription-Id
AVP Code	443
AVP Data Type	Grouped
Description	The information used to identify the end user of the service subscription party. The detailed syntax is as follows: <pre>*Subscription-Id ::= <AVP Header: 443> {Subscription-Id-Type} {Subscription-Id-Data}</pre> The diameter proxy or agent must route based on this field. In the case of more than one Subscription-Id, the system handles only the first Subscription-Id. Gz interface Operators can suppress the Subscription-Id AVP in ACR messages based on the Subscription-Id-Type (IMSI/MSISDN) using ACR data record templates. In the ACR/CCR data record templates, the Subscription-Id options appear as <code>subscriptionidimsi</code> and <code>subscriptionidmsisdn</code>. Gx interface Operators can suppress the Subscription-Id AVP in CCR messages (CCR-I/CCR-U/CCR-T) based on the Subscription-Id-Type (IMSI/MSISDN/SIP URI) using CCR data record templates. In the CCR data record templates, the Subscription-Id options appear as the following: ■ <code>subscriptionid</code> ■ <code>subscriptionidimsi</code> ■ <code>subscriptionidmsisdn</code> ■ <code>subscriptionidsipuri</code> <i>Note: The GiGW sends a pseudo IMSI in the Subscription-ID AVP, which needs to be suppressed in PCRF CCR Start data record templates to use the Subscription-Id AVP with type SIP URI. In PCRF CCR Start data record templates, set the <code>subscriptionidimsi presentFlag</code> to <code>no</code>.</i> Gy Interface Operators can suppress the Subscription-Id AVP in CCR messages (CCR-I/CCR-U/CCR-T) based on the Subscription-Id-Type (IMSI/MSISDN/NAI) using CCR data record templates. In the CCR data record templates, the Subscription-Id options appear as the following: ■ <code>subscriptionidimsi</code> ■ <code>subscriptionidmsisdn</code> ■ <code>subscriptionidnai</code> <i>Note: For trusted non-3GPP Wi-Fi access using standard RADIUS/EAP methods for authentication, the end user's Network Access Identifier (NAI), received in the User-Name AVP of the RADIUS Access-Request, will be used in the Subscription-Id AVP in CCR messages to the OCS. In addition to the NAI, if the IMSI/MSISDN are available as part of the end user's identity, the MCC TWAG will encode multiple Subscription-Id AVPs in CCR messages on the Gy interface.</i>

Subscription-Id-Data

AVP Name	Subscription-Id-Data
AVP Code	444
AVP Data Type	UTF8String
Description	Used to identify the end user. This AVP indicates the number of the charged party when the Subscription-Id-Type is 0. Supports the following format for the mobile number: Mobile Subscriber Integrated Services Digital Network-Number (MSISDN) – Indicates the IMSI number of the charged party when the Subscription-Id-Type is 1.

Subscription-Id-Type

AVP Name	Subscription-Id-Type
AVP Code	450
AVP Data Type	Enumerated
Description	Indicates the type of user's terminal ID. The following values are defined: ■ END_USER_E164 (0) – The identifier is in international E.164 format defined according to the ITU-T E.164 number scheme. For example, MSISDN. ■ END_USER_IMSI (1) – The identifier is in international IMSI format defined according to the ITU-T E.212 number scheme. ■ END_USER_SIP_URI (2) – The identifier is in the form of a SIP URI, as defined in [SIP]. ■ END_USER_NAI (3) – The identifier is in the form of a Network Access Identifier, as defined in RFC 4282 and formatted as defined in 3GPP TS 23.003, clause 19.3.2. Used for Wi-Fi access. Use the following CLI parameter to configure the IMSI format to be sent in the Subscription-Id AVP for non-3gpp access. The default is <code>imsi format</code>. <i>Applicable to Gx and Gy interfaces.</i> <pre>an3000-mcm-slot7cpul(config)# service-construct interface [online-charging pcrf-charging] <interface-name> subscriptionid-imsi-format-non- 3gpp-access [imsi format imsinaiprefix0format]</pre>

Supported-Vendor-Id

AVP Name	Supported-Vendor-Id
AVP Code	265
AVP Data Type	Unsigned32
Description	Is used in the CER and CEA messages to inform the peer that the sender supports the vendor-specific AVPs defined by the vendor identified in this AVP.

Tariff-Change-Usage

AVP Name	Tariff-Change-Usage
AVP Code	452
AVP Data Type	Enumerated
Description	Identifies the reporting period for the used service unit. That is, before, after, or during tariff change. Omission of this AVP means that no tariff change has occurred. The following values are defined: ■ UNIT_BEFORE_TARIFF_CHANGE (0) ■ UNIT_AFTER_TARIFF_CHANGE (1) ■ UNIT_INDETERMINATE (2)

Tariff-Time-Change

AVP Name	Tariff-Time-Change
AVP Code	451
AVP Data Type	Time
Description	Is used when the OCS expects a tariff change to occur within the validity time of the quota it is granting.

Termination-Cause

AVP Name	Termination-Cause
AVP Code	295
AVP Data Type	Enumerated
Description	Shows the reason why the session of the diameter client terminated. The following values are defined: ■ DIAMETER_LOGOUT (1) – Interruption initiated by users. ■ DIAMETER_SERVICE_NOT_PROVIDED (2) – Session breaks before users receive authentication response messages. ■ DIAMETER_BAD_ANSWER (3) – The authentication answer received by the diameter client is not successfully handled. ■ DIAMETER_ADMINISTRATIVE (4) – Users are not granted the access authority or the session is disconnected due to operation management. For example, receiving an Abort-Session-Request. ■ DIAMETER_LINK_BROKEN (5) – Communication with users is suddenly disconnected. ■ DIAMETER_AUTH_EXPIRED (6) – User access ends because the authorized session time is reached. ■ DIAMETER_USER_MOVED (7) – Users are receiving the services from other diameter clients. ■ DIAMETER_SESSION_TIMEOUT (8) – User session is timed out and the service is stopped.

User-Equipment-Info

AVP Name	User-Equipment-Info
AVP Code	458
AVP Data Type	Grouped
Description	Indicates the identity and capability of the terminal that the subscriber is using for the connection to the network. The detailed syntax is as follows: <pre>User-Equipment-Info ::= < AVP Header: 458 > {User-Equipment-Info-Type} {User-Equipment-Info-Value}</pre> <i>Note: For Wi-Fi access, the User-Equipment-Info AVP contains the MAC address of the device if received.</i>

User-Equipment-Info-Type

AVP Name	User-Equipment-Info-Type
AVP Code	459
AVP Data Type	Enumerated
Description	Defines the type of user equipment information contained in the User-Equipment-Info-Value AVP. Contains the following values: ■ IMEISV (0) – The identifier contains the International Mobile Equipment Identifier and Software Version in the international IMEISV format according to 3GPP TS 23.003 [3GPPIMEI]. ■ MAC (1) – The 48-bit MAC address is formatted as described in [RAD802.1X]. ■ EUI64 (2) – The 64-bit identifier used to identify hardware instance of the product, as defined in [EUI64].

User-Equipment-Info-Value

AVP Name	User-Equipment-Info-Value
AVP Code	460
AVP Data Type	OctetString
Description	Defines which type of identifier is used. <i>Note: When the User-Equipment-Info-Type is set to IMEISV(0), the value within the User-Equipment-Info-Value is a UTF-8 encoded decimal.</i>

User-Name

AVP Name	User-Name
AVP Code	1
AVP Data Type	UTF8String
Description	The user provides the username (extracted from the Protocol Configuration Options field of the Create PDP Context Request message) as provided in the Access-Request message.

Used-Service-Unit

AVP Name	Used-Service-Unit
AVP Code	446
AVP Data Type	Grouped
Description	The amount of used non-monetary service units measured for a rating group. <i>Note: The “M” bit flag for this AVP is set to 0 when present in the Usage-Monitoring-Information AVP.</i> The detailed syntax is as follows: <pre>Used-Service-Unit ::= < AVP Header: 446 > [CC-Time] [CC-Total-Octets] [CC-Input-Octets] [CC-Output-Octets] * [AVP]</pre>

Validity-Time

AVP Name	Validity-Time
AVP Code	448
AVP Data Type	Unsigned32
Description	Sent from the credit-control server to the credit-control client. This AVP contains the validity time of the granted service units. The measurement of the Validity-Time is started upon receipt of the CCA message containing this AVP. <i>Note: The MCC starts the Validity-Time only when the Validity-time AVP is received in the CCA message with the current GSU and does not use Validity-Time (if any) from a previous grant.</i>

Vendor-Id

AVP Name	Vendor-Id
AVP Code	266
AVP Data Type	Unsigned32
Description	Contains the vendor ID of the diameter application.

Vendor-Specific-Application-Id

AVP Name	Vendor-Specific-Application-Id
AVP Code	259
AVP Data Type	Group
Description	Used to advertise support of a vendor-specific diameter application.

3GPP AVP Definitions

This section describes the 3GPP AVPs sent by the system in CCR, CCA, RAR, ACR, and/or ACA messages. The AVPs include a Vendor-ID of 10415 (3GPP).

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3GPP-Charging-Characteristics

AVP Name	3GPP-Charging-Characteristics
AVP Code	13
AVP Data Type	UTF8String
Description	The charging characteristics for this PDP context received in the Create PDP Context Request message (only available in R99 and later releases).

3GPP-Charging-Id

AVP Name	3GPP-Charging-Id
AVP Code	2
AVP Data Type	OctetString
Description	The Charging ID for this PDP Context. The Charging ID together with the GGSN-Address constitutes a unique identifier for the PDP context.

3GPP-GGSN-Address

AVP Name	3GPP-GGSN-Address
AVP Code	7
AVP Data Type	OctetString
Description	The Gateway GPRS Support Node/Packet Data Network Gateway (GGSN/PGW) IP address.

3GPP-GGSN-IPv6-Address

AVP Name	3GPP-GGSN-IPv6-Address
AVP Code	16
AVP Data Type	OctetString
Description	The GGSN IPv6 address that is used by the GTP control plane for the context.

3GPP-GGSN-MCC-MNC

AVP Name	3GPP-GGSN-MCC-MNC
AVP Code	9
AVP Data Type	UTF8String
Description	The Mobile Country Code (MCC) and Mobile Network Code (MNC) of the network to which the GGSN/PGW belongs. When the PLMN ID (MCC + MNC), received in the subscriber's IMSI, matches a PLMN ID configured in the home-plmnid-list, the configured PLMN ID is sent in the 3GPP-GGSN-MCC-MNC AVP on all external interfaces (RADIUS, Gx, Gy, Gz/Rf). If an entry is not found in the list, the first entry in the home-plmnid-list is used in the 3GPP-GGSN-MCC-MNC AVP.

3GPP-GPRS-Negotiated-QoS-Profile

AVP Name	3GPP-GPRS-Negotiated-QoS-Profile
AVP Code	5
AVP Data Type	UTF8String
Description	This AVP contains the QoS profile.

3GPP-IMEISV

AVP Name	3GPP-IMEISV
AVP Code	20
AVP Data Type	UTF8String
Description	The user International Mobile Equipment Identity of the equipment served (IMEISV).

3GPP-IMSI

AVP Name	3GPP-IMSI
AVP Code	1
AVP Data Type	UTF8String
Description	The International Mobile Subscriber Identity (IMSI) for this user.

3GPP-IMSI-MCC-MNC

AVP Name	3GPP-IMSI-MCC-MNC
AVP Code	8
AVP Data Type	UTF8String
Description	The Mobile Country Code (MCC) and Mobile Network Code (MNC) extracted from the user's IMSI (first 5 or 6 digits, as applicable from the presented IMSI).

3GPP-MS-TimeZone

AVP Name	3GPP-MS-TimeZone
AVP Code	23
AVP Data Type	UTF8String
Description	The Time Zone field and the Daylight Saving Time fields are used to indicate the offset between universal time and local time in steps of 15 minutes of where the MS currently resides.

3GPP-NSAPI

AVP Name	3GPP-NSAPI
AVP Code	10
AVP Data Type	UTF8String
Description	Identifies a particular PDP context for the associated PDN and Mobile Subscriber Integrated Services Digital Network-Number (MSISDN)/IMSI from creation to deletion.

3GPP-PDP-Type

AVP Name	3GPP-PDP-Type
AVP Code	3
AVP Data Type	Enumerated
Description	Type of PDP context. For example, IP or PPP. The PDP type may have the following values: ■ 0 = IPv4 ■ 1 = PPP ■ 2 = IPv6 ■ 3 = Ipv4v6 ■ 4 = Non-IP

3GPP-RAT-Type

AVP Name	3GPP-RAT-Type
AVP Code	21
AVP Data Type	OctetString
Description	The Radio Access Technology currently serving the User Equipment (UE).

3GPP-Selection-Mode

AVP Name	3GPP-Selection-Mode
AVP Code	12
AVP Data Type	UTF8String
Description	The selection mode for this PDP context received in the create PDP context request message.

3GPP-Session-Stop-Indicator

AVP Name	3GPP-Session-Stop-Indicator
AVP Code	11
AVP Data Type	UTF8String
Description	Indicates to the AAA server that the last PDP context of a session is released and the IP-CAN session has been terminated.

3GPP-SGSN-Address

AVP Name	3GPP-SGSN-Address
AVP Code	6
AVP Data Type	OctetString
Description	Indicates the IP address of the Serving GPRS Support Node (SGSN). This AVP can be used to identify the Public Land Mobile Network (PLMN) of the SGSN to which a user logs in.

3GPP-SGSN-IPv6-Address

AVP Name	3GPP-SGSN-IPv6-Address
AVP Code	15
AVP Data Type	OctetString
Description	Indicates the SGSN IPv6 address used by the GPRS Tunneling Protocol (GTP) control plane for the handling of control messages. It may be used to identify the PLMN to which the user is attached.

3GPP-SGSN-MCC-MNC

AVP Name	3GPP-SGSN-MCC-MNC
AVP Code	18
AVP Data Type	UTF8String
Description	Indicates the information about the international roam-in area of the charged party. On the Gx interface, the 3GPP-SGSN-MCC-MNC AVP can be used in the CCA message to override the PLMN-ID sent in GTP-C messages. See “Overriding the PLMN-ID sent in GTP-C messages” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

3GPP-User-Location-Info

AVP Name	3GPP-User-Location-Info
AVP Code	22
AVP Data Type	OctetString
Description	Indicates details of where the user equipment is currently located (for example, SAI or CGI).

Access-Network-Charging-Address

AVP Name	Access-Network-Charging-Address
AVP Code	501
AVP Data Type	Address
Description	Indicates the IP Address of the network entity within the access network performing charging (for example, the GGSN IP address).

Access-Network-Charging-Identifier-Gx

AVP Name	Access-Network-Charging-Identifier-Gx
AVP Code	1022
AVP Data Type	Grouped
Description	Contains a charging identifier within the Access-Network-Charging-Identifier-Value AVP and the related PCC rule name(s) within the Charging-Rule-Name AVP(s) and/or within the Charging-Rule-Base-Name AVP(s). If the charging identifier applies to the entire IP CAN session, neither the Charging-Rule-Name AVP or Charging-Rule-Base-Name AVP needs to be provided. The detailed syntax is as follows: <pre>Access-Network-Charging-Identifier-Gx ::= < AVP Header: 1022 > {Access-Network-Charging-Identifier-Value} * [Charging-Rule-Base-Name] * [Charging-Rule-Name]</pre>

Access-Network-Charging-Identifier-Value

AVP Name	Access-Network-Charging-Identifier-Value
AVP Code	503
AVP Data Type	OctetString
Description	Contains a charging identifier.

ADC-Rule-Base-Name

AVP Name	ADC-Rule-Base-Name
AVP Code	1095
AVP Data Type	UTF8String
Description	Indicates the name of a predefined group of Application Detection and Control (ADC) rules. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

ADC-Rule-Definition

AVP Name	ADC-Rule-Definition
AVP Code	1094
AVP Data Type	Grouped
Description	Defines the ADC rule sent by the PCRF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. Note: The Redirect-Information AVP sent from the PCRF as part of the ADC-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected. The detailed syntax is as follows: <pre>ADC-Rule-Definition ::= < AVP Header: 1094 > {ADC-Rule-Name} [TDF-Application-Identifier] *[Flow-Information] [Service-Identifier] [Rating-Group] [Reporting-Level] [Online] [Offline] [Metering-Method] [Flow-Status] [QoS-Information] [Monitoring-Key] [Redirect-Information] [Mute-Notification] *[AVP]</pre>

ADC-Rule-Install

AVP Name	ADC-Rule-Install
AVP Code	1092
AVP Data Type	Grouped
Description	Used to activate, install or modify Application Detection and Control (ADC) rules as instructed from the PCRF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>ADC-Rule-Install ::= < AVP Header: 1092 > * [ADC-Rule-Definition] * [ADC-Rule-Name] * [ADC-Rule-Base-Name] [Monitoring-Flags] [Rule-Activation-Time] [Rule-Deactivation-Time] * [AVP]</pre>

ADC-Rule-Name

AVP Name	ADC-Rule-Name
AVP Code	1096
AVP Data Type	OctetString
Description	Defines a name for an ADC rule. For ADC rules provided by the PCRF, it uniquely identifies an ADC rule within one session. For predefined ADC rules, it uniquely identifies an ADC rule within the PCEF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

ADC-Rule-Remove

AVP Name	ADC-Rule-Remove
AVP Code	1093
AVP Data Type	Grouped
Description	Used to deactivate or remove ADC rules as instructed from the PCRF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>ADC-Rule-Remove ::= < AVP Header: 1093 > * [ADC-Rule-Name] * [ADC-Rule-Base-Name] * [AVP]</pre>

ADC-Rule-Report

AVP Name	ADC-Rule-Report
AVP Code	1097
AVP Data Type	Grouped
Description	Used to report the status of the ADC rules which cannot be installed/activated or enforced. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>ADC-Rule-Report ::= < AVP Header: 1097 > * [ADC-Rule-Name] * [ADC-Rule-Base-Name] [PCC-Rule-Status] [Rule-Failure-Code] [Final-Unit-Indication] * [AVP]</pre>

Additional-Exception-Reports

AVP Name	Additional-Exception-Reports
AVP Code	3936
AVP Data Type	Enumerated
Description	Indicates whether additional exception reports are allowed when maximum rate is reached. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ NOT_ALLOWED (0)

AF-Charging-Identifier

AVP Name	AF-Charging-Identifier
AVP Code	505
AVP Data Type	OctetString
Description	Contains the AF Charging Identifier sent by the Application Function (AF). This information may be used for charging correlation with the bearer layer.

AF-Correlation-Information

AVP Name	AF-Correlation-Information
AVP Code	1276
AVP Data Type	Grouped
Description	Contains the AF charging identifier (ICID for IMS) and associated flow identifiers generated by the AF and received by PGW over the Gx interface as defined in TS 29.214 and TS 29.212. The AF-Correlation-Information is defined per Rating Group or per Rating Group and Service Identifier, when Service Identifier level reporting applies. When several AF sessions are conveyed over the same bearer, this AVP may appear several times per MSCC instance. The detailed syntax is as follows: <pre>AF-Correlation-Information:: = < AVP Header: 1276 > {AF-Charging-Identifier} * [Flows]</pre>

Allocation-Retention-Priority

AVP Name	Allocation-Retention-Priority
AVP Code	1034
AVP Data Type	Grouped
Description	Indicates the priority of allocation and retention, the pre-emption capability and pre-emption vulnerability for the SDF if provided within the QoS-Information AVP or for the EPS default bearer if provided within the Default-EPS-Bearer-QoS AVP. The detailed syntax is as follows: <pre>Allocation-Retention-Priority ::= < AVP Header: 1034 > {Priority-Level} [Pre-emption-Capability] [Pre-emption-Vulnerability]</pre>

AN-GW-Address

AVP Name	AN-GW-Address
AVP Code	1050
AVP Data Type	Address
Description	Contains the IPv4 and/or IPv6 (if available) address(es) of the SGW for 3GPP networks and access node gateway (AGW) for non-3GPP networks. For 2G/3G based sessions with an IP-CAN-Type of 3GPP-EPS, the MCC sends the IP address of the SGSN in the AN-GW-Address AVP in the CCR-Initial message to the PCRF. If both IPv4 and IPv6 addresses are provided, the MCC sends two instances of the AN-GW-Address AVP.

AN-Trusted

AVP Name	AN-Trusted
AVP Code	1503
AVP Data Type	Enumerated
Description	Indicates whether the access network is trusted or untrusted for the Non-3GPP access network. For example, if there is an LTE-to-Wi-Fi handover and if the PCRF has requested to be notified of an IP-CAN_CHANGE, the PGW includes the AN-Trusted AVP in the CCR-U along with the IP-CAN_Change event-trigger to indicate whether the access network is trusted or untrusted. The following values are defined: ■ TRUSTED (0) – This value is used when the non-3GPP access network is to be handled as trusted. ■ UNTRUSTED (1) – This value is used when the non-3GPP access network is to be handled as untrusted.

Application-Detection-Information

AVP Name	Application-Detection-Information
AVP Code	1098
AVP Data Type	Grouped
Description	Used to report application Start/Stop notifications when application traffic, defined by the TDF-Application-Identifier, has been detected and when the PCRF has subscribed for APPLICATION_START/APPLICATION_STOP Event-Triggers. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>Application-Detection-Information:: = <AVP Header: 1098> {TDF-Application-Identifier} [TDF-Application-Instance-Identifier] *[Flow-Information] *[AVP]</pre>

Application-Service-Provider-Identity

AVP Name	Application-Service-Provider-Identity
AVP Code	532
AVP Data Type	UTF8String
Description	Used for sponsored data connectivity purposes as an identifier of the application service provider.

APN-Aggregate-Max-Bitrate-DL

AVP Name	APN-Aggregate-Max-Bitrate-DL
AVP Code	1040
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in bits per second for the downlink direction across all non-GBR bearers with the same APN. When provided in a CCR message, it indicates the subscribed maximum bitrate. When provided in a CCA message, it indicates the maximum bandwidth authorized by the PCRF.

APN-Aggregate-Max-Bitrate-UL

AVP Name	APN-Aggregate-Max-Bitrate-UL
AVP Code	1041
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in bits per seconds for the uplink direction across all non-GBR bearers with the same APN. When provided in a CCR message, it indicates the subscribed maximum bitrate. When provided in a CCA message, it indicates the maximum bandwidth authorized by the PCRF.

APN-Rate-Control

AVP Name	APN-Rate-Control
AVP Code	3933
AVP Data Type	Grouped
Description	Indicates the APN Rate Control parameters applicable to data PDU transfers in the uplink and downlink direction for this APN. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>APN-Rate-Control ::= < AVP Header: 3933 > [APN-Rate-Control-Uplink] [APN-Rate-Control-Downlink]</pre>

APN-Rate-Control-Downlink

AVP Name	APN-Rate-Control-Downlink
AVP Code	3934
AVP Data Type	Grouped
Description	Indicates the APN-Rate Control parameters applicable to data PDU transfers in the downlink direction for this APN. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>APN-Rate-Control-Downlink ::= < AVP Header: 3934 > [Rate-Control-Time-Unit] [Rate-Control-Max-Rate] [Rate-Control-Max-Message-Size]</pre>

APN-Rate-Control-Uplink

AVP Name	APN-Rate-Control-Uplink
AVP Code	3935
AVP Data Type	Grouped
Description	Indicates the APN Rate Control parameters applicable to data PDU transfers in the uplink direction for this APN. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>APN-Rate-Control-Uplink ::= < AVP Header: 3935> [Additional-Exception-Reports] [Rate-Control-Time-Unit] [Rate-Control-Max-Rate]</pre>

Base-Time-Interval

AVP Name	Base-Time-Interval
AVP Code	1265
AVP Data Type	Unsigned32
Description	Contains the length of the base time interval for controlling the consumption of time quota, in seconds.

Bearer-Control-Mode

AVP Name	Bearer-Control-Mode
AVP Code	1023
AVP Data Type	Enumerated
Description	Indicates the PCRF selected bearer control mode. The following values are defined: ■ UE_ONLY (0) – Indicates that the UE may request any additional PDP Context establishment. ■ UE_NW (2) – Indicates that both the UE and PCEF may request any additional PDP Context establishment and add own traffic mapping information to a PDP Context.

Bearer-Identifier

AVP Name	Bearer-Identifier
AVP Code	1020
AVP Data Type	OctetString
Description	Indicates the bearer to which specific information refers. When present within the CCR command, subsequent AVPs within the CCR refer to the specific bearer identified by this AVP. The bearer identifier of an IP CAN bearer is unique within the corresponding IP CAN session and is selected by the PCEF.

Bearer-Operation

AVP Name	Bearer-Operation
AVP Code	1021
AVP Data Type	Enumerated
Description	Indicates the bearer event that causes a request for PCC rules. This AVP is supplied if the bearer event relates to an IP CAN bearer initiated by the UE. This AVP is only sent when the IP-CAN-Type AVP is 3GPP-GPRS (2G/3G). The following values are defined: ■ TERMINATION (0) – Indicates that a bearer is being terminated. ■ ESTABLISHMENT (1) – Indicates that a new bearer is being established. ■ MODIFICATION (2) – Indicates that an existing bearer is being modified.

Bearer-Usage

AVP Name	Bearer-Usage
AVP Code	1000
AVP Data Type	Enumerated
Description	Indicates how the bearer is being used. For example, whether it is used as a dedicated IMS signaling context. The following values are defined: ■ GENERAL (0) – Indicates that no specific bearer usage information is available. No TFT filter is associated with the PDP context. ■ IMS_SIGNALLING (1) – Indicates that the bearer is used for IMS signaling only. ■ DEDICATED (2) – Indicates that the bearer is being used for a dedicated purpose and accordingly has an associated TFT filter.

CG-Address

AVP Name	CG-Address
AVP Code	846
AVP Data Type	OctetString
Description	Contains the IP address of the charging gateway.

Change-Condition

AVP Name	Change-Condition
AVP Code	2037
AVP Data Type	Integer32
Description	Indicates the change in charging condition. The following values are defined: ■ 0 = Normal Release – Used to indicate an IP-CAN session termination, an IP-CAN bearer release or a service data flow termination. ■ 1 = Abnormal Release ■ 2 = QoS Change ■ 3 = Volume Limit ■ 4 = Time Limit ■ 8 = RAT Change ■ 14 = CGI-SAI Change ■ 15 = RAI Change ■ 16 = ECGI Change ■ 17 = TAI Change ■ 36 = Change in UP to UE <i>Note: If a single accounting record contains multiple changes (for example, an SGSN change with CGI change, RAI change, QoS change), the MCC sends multiple Change-Condition AVPs containing all of the change conditions.</i>

Change-Time

AVP Name	Change-Time
AVP Code	2038
AVP Data Type	Time
Description	Contains the time in UTC format when the volume counts, associated with the IP-CAN bearer or the service data container, are closed and reported due to a change in charging conditions.

Charging-Characteristics-Selection-Mode

AVP Name	Charging-Characteristics-Selection-Mode
AVP Code	2066
AVP Data Type	Enumerated
Description	Indicates how the applied Charging-Characteristics was selected. The following values are defined: ■ 0 = Serving-Node-Supplied ■ 1 = Subscription-specific ■ 2 = APN-specific ■ 3 = Home-Default ■ 4 = Roaming-Default ■ 5 = Visiting-Default

Charging-Information

AVP Name	Charging-Information
AVP Code	618
AVP Data Type	Grouped
Description	Contains the addresses of the charging functions. <i>Note: You can configure the PGW to extract information from the Charging-Information AVP received from the PCRF and use this information during Gy session setup. See “Using Gx Charging-Information AVP for Gy Session Setup” in the Affirmed Networks Gy Interface specification.</i> The detailed syntax is as follows: <pre>Charging-Information ::= <AVP Header : 618 10415> [Primary-Event-Charging-Function-Name] [Secondary-Event-Charging-Function-Name] * [AVP]</pre>

Charging-Rule-Base-Name

AVP Name	Charging-Rule-Base-Name
AVP Code	1004
AVP Data Type	UTF8String
Description	Indicates the name of a predefined group of PCC rules residing at the PCEF. The value of the Charging-Rule-Base-Name AVP corresponds to the locally configured data-profile on the MCC gateway (workflow → data-profile). For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i> .

Charging-Rule-Definition

AVP Name	Charging-Rule-Definition
AVP Code	1003
AVP Data Type	Grouped
Description	Defines the dynamic PCC rule for a service data flow sent by the PCRF to the PCEF or used to activate a predefined PCC rule that is pre-configured at the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i>. Note: The Redirect-Information AVP sent from the PCRF as part of the Charging-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected. The detailed syntax is as follows: <pre>Charging-Rule-Definition ::= < AVP Header: 1003 > {Charging-Rule-Name} [Service-Identifier] [Rating-Group] *[Flow-Information] [Flow-Status] [QoS-Information] [PS-to-CS-Session-Continuity] [Reporting-Level] [Online] [Offline] [Metering-Method] [Precedence] [Monitoring-Key] [Redirect-Information] [Mute-Notification] [Sponsor-Identity] [Application-Service-Provider-Identity] [Required-Access-Info] *[AVP]</pre>

Charging-Rule-Install

AVP Name	Charging-Rule-Install
AVP Code	1001
AVP Data Type	Grouped
Description	Used to activate, install, or modify PCC rules as instructed from the PCRF to the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i> .

Charging-Rule-Name

AVP Name	Charging-Rule-Name
AVP Code	1005
AVP Data Type	OctetString
Description	For PCC rules provided by the PCRF, it uniquely identifies a PCC rule within one IP CAN session. For PCC rules predefined at the PCEF, it uniquely identifies a PCC rule within the PCEF. For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i> .

Charging-Rule-Remove

AVP Name	Charging-Rule-Remove
AVP Code	1002
AVP Data Type	Grouped
Description	Used to deactivate or remove PCC rules from an IP CAN session. For information on the MCC implementation of this AVP, see “PCC Rules” in the <i>Affirmed Networks Gx Interface Specification</i>. The detailed syntax is as follows: <pre>Charging-Rule-Remove ::= < AVP Header: 1002 > * [Charging-Rule-Name] * [Charging-Rule-Base-Name]</pre>

Charging-Rule-Report

AVP Name	Charging-Rule-Report
AVP Code	1018
AVP Data Type	Grouped
Description	Reports the status of the PCC rules that have been successfully installed, modified or removed or that cannot be installed/activated or enforced at the PCEF. The Rule-Failure-Code indicates the reason that the PCC rules cannot be successfully installed/activated or enforced. The detailed syntax is as follows: <pre>Charging-Rule-Report ::= < AVP Header: 1018 > * [Charging-Rule-Name] * [Charging-Rule-Base-Name] [PCC-Rule-Status] [Rule-Failure-Code] [Final-Unit-Indication] * [RAN-NAS-Release-Cause]</pre>

CP-CIoT-EPS-Optimisation-Indicator

AVP Name	CP-CIoT-EPS-Optimisation-Indicator
AVP Code	3930
AVP Data Type	Enumerated
Description	Indicates whether Control Plane CIoT EPS optimisation is used by this PDN connection for data transfer with the UE. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ NOT_APPLY (0) ■ APPLY (1)

CSG-Access-Mode

AVP Name	CSG-Access-Mode
AVP Code	2317
AVP Data Type	Enumerated
Description	Contains the CSG cell access mode. The following values are defined: ■ CLOSED MODE (0) ■ HYBRID MODE (1)

CSG-Id

AVP Name	CSG-Id
AVP Code	1437
AVP Data Type	Unsigned32
Description	Contains the CSG identification information.

CSG-Information-Reporting

AVP Name	CSG-Information-Reporting
AVP Code	1071
AVP Data Type	Enumerated
Description	Sent from the PCRF to the PCEF to request the PCEF to report the user CSG information change to the OCS. The following values are defined: ■ CHANGE_CSG_CELL (0) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a CSG cell. ■ CHANGE_CSG_SUBSCRIBED_HYBRID_CELL (1) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a hybrid cell in which the subscriber is a CSG member. ■ CHANGE_CSG_UNSUBSCRIBED_HYBRID_CELL (2) – Indicates that the PCEF sends a credit re-authorization request to the OCS when the UE enters/leaves/accesses via a hybrid cell in which the subscriber is not a CSG member.

CSG-Membership-Indication

AVP Name	CSG-Membership-Indication
AVP Code	2318
AVP Data Type	Enumerated
Description	Indicates whether the UE is a member of the accessing CSG cell, if the access mode is Hybrid. The following values are defined: ■ NOT CSG MEMBER (0) ■ CSG MEMBER (1)

Default-EPS-Bearer-QoS

AVP Name	Default-EPS-Bearer-QoS
AVP Code	1049
AVP Data Type	Grouped
Description	Defines the QoS information for the EPS default bearer. When this AVP is sent from the PCEF to the PCRF, it indicates the subscribed QoS for the default EPS bearer. When this AVP is sent from the PCRF to the PCEF, it indicates the authorized QoS for the default EPS bearer.

Diagnostics

AVP Name	Diagnostics
AVP Code	2039
AVP Data Type	Integer32
Description	Provides more information on the reason for sending the ACR Stop message and may contain one of the following values: ■ UNSPECIFIED (0) ■ SESSION_TIMEOUT (1) ■ RESOURCE_LIMITATION (2) ■ ADMIN_DISCONNECT (3) ■ IDLE_TIMEOUT (4) ■ PCRF_UNREACHABLE (5) ■ AAA_UNREACHABLE (6) ■ AAA_INITIATED_SESSION_TERMINATION (7) ■ PCRF_INITIATED_SESSION_TERMINATION (9) ■ PCRF_INITIATED_FLOW_TERMINATION (10) ■ PCRF_ACCOUNTING_PARAMETERS_CHANGED (11) ■ GTP_INITIATED_SESSION_TERMINATION (14) ■ HANDOVER_ERROR (16) ■ REAUTHORIZATION_FAILED (18)

Downlink-Rate-Limit

AVP Name	Downlink-Rate-Limit
AVP Code	4312
AVP Data Type	Unsigned32
Description	Contains the maximum number of NAS Data PDUs per deci hour for this UE for downlink. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Dynamic-Address-Flag

AVP Name	Dynamic-Address-Flag
AVP Code	2051
AVP Data Type	Enumerated
Description	Indicates whether the PDP context/PDN address is statically or dynamically allocated. If this AVP is not present, the address is statically allocated. The following values are defined: ■ STATIC (0) ■ DYNAMIC (1)

Envelope

AVP Name	Envelope
AVP Code	1266
AVP Data Type	Grouped
Description	Reports the start and end time of a one time envelope using the Envelope-Start-Time and Envelope-End-Time AVPs. Further details of its usage are described in 3GPP 32.299. The detailed syntax is as follows: <pre>Envelope ::= < AVP Header: 1266> {Envelope-Start-Time} [Envelope-End-Time] [CC-Total-Octets] [CC-Input-Octets] [CC-Output-Octets] [CC-Service-Specific-Units] * [AVP]</pre>

Envelope-End-Time

AVP Name	Envelope-End-Time
AVP Code	1267
AVP Data Type	Time
Description	This AVP is set to the time of the end of the time envelope.

Envelope-Reporting

AVP Name	Envelope-Reporting
AVP Code	1268
AVP Data Type	Enumerated
Description	Used in the CCA-Initial message to indicate whether the client reports the start and end of each time envelope, in cases where the quota is consumed in envelopes. The following values are defined: ■ DO_NOT_REPORT_ENVELOPES (0) ■ REPORT_ENVELOPES (1) ■ REPORT_ENVELOPES_WITH_VOLUME (2) ■ REPORT_ENVELOPES_WITH_EVENTS (3) ■ REPORT_ENVELOPES_WITH_VOLUME_AND_EVENTS (4)

Envelope-Start-Time

AVP Name	Envelope-Start-Time
AVP Code	1269
AVP Data Type	Time
Description	This AVP is set to the time of the packet of user data which caused the time envelope to start.

Event-Trigger

AVP Name	Event-Trigger
AVP Code	1006
AVP Data Type	Enumerated
Description	When sent from the PCRF to the PCEF, this AVP indicates an event that causes a re-request of PCC rules. When sent from the PCEF to the PCRF, this AVP indicates that the corresponding event has occurred at the gateway (PGW). The system supports the following PCRF event-triggers: ■ SGSN_CHANGE (0) ■ QOS_CHANGE (1) ■ RAT_CHANGE (2) ■ PLMN_CHANGE (4) ■ LOSS_OF_BEARER (5) ■ IP-CAN_CHANGE (7) ■ RAI_CHANGE (12) ■ USER_LOCATION_CHANGE (13) ■ OUT_OF_CREDIT (15) ■ REALLOCATION_OF_CREDIT (16) ■ REVALIDATION_TIMEOUT (17) ■ DEFAULT_EPS_BEARER_QOS_CHANGE (20) ■ SUCCESSFUL_RESOURCE_ALLOCATION (22) ■ UE_TIME_ZONE_CHANGE (25) ■ USAGE_REPORT (26) or USAGE_REPORT (33) – MCC supports the following USAGE_REPORT trigger codes, depending on the PCRF's 3GPP TS 29.212 release compliance: – USAGE_REPORT (26) (3GPP TS 29.212 Release 10 or earlier) – USAGE_REPORT (33) (3GPP TS 29.212 Release 11 or later) Specify the PCRF's 3GPP TS 29.212 release compliance when configuring the PCRF charging interface object. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> pcrf-3gpp-release-compliance [release-10 release-10-plus]</pre>

AVP Name	Event-Trigger
Description (Continued)	■ TAI_CHANGE (26) ■ ECGI_CHANGE (27) The TAI_CHANGE and ECGI_CHANGE event-triggers are applicable for 3GPP 29.212 V10.0.0 or higher. To ensure the PCEF receives and reports the TAI_CHANGE event-trigger with code 26 and the ECGI_CHANGE event-trigger with code 27, set the PCRF's 3GPP TS 29.212 release compliance to <code>release-10-plus</code>. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface pcrf-charging <interface-name> pcrf-3gpp-release-compliance release-10-plus</pre> ■ APN-AMBR_MODIFICATION_FAILURE (29) ■ DEFAULT-EPS-BEARER-QOS_MODIFICATION_FAILURE (34) ■ APPLICATION_START (39) ■ APPLICATION_STOP (40) ■ CS_TO_PS_HANOVER (42) ■ ACCESS_NETWORK_INFO_REPORT (45)

Extended-APN-AMBR-DL

AVP Name	Extended-APN-AMBR-DL
AVP Code	2848
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in Kpbs for the downlink direction across all non-GBR bearers related with the same APN. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-APN-AMBR-UL

AVP Name	Extended-APN-AMBR-UL
AVP Code	2849
AVP Data Type	Unsigned32
Description	Indicates the maximum aggregate bit rate in Kpbs for the uplink direction across all non-GBR bearers related with the same APN. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-GBR-DL

AVP Name	Extended-GBR-DL
AVP Code	2850
AVP Data Type	Unsigned32
Description	Indicates the guaranteed bit rate (GBR) in Kpbs for a downlink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above. For example, IP, UDP, RTP and RTP payload. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-GBR-UL

AVP Name	Extended-GBR-UL
AVP Code	2851
AVP Data Type	Unsigned32
Description	Indicates the guaranteed bit rate in Kpbs for an uplink service data flow. The bandwidth contains all the overhead coming from the IP-layer and the layers above. For example, IP, UDP, RTP and RTP payload. For more details on this AVP, refer to 3GPP 29.212 Version 15.0.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Extended-Max-Requested-BW-DL

AVP Name	Extended-Max-Requested-BW-DL
AVP Code	554
AVP Data Type	Unsigned32
Description	Defines the maximum authorized bandwidth in Kpbs for downlink. For more details on this AVP, refer to 3GPP 29.214 Version 15.1.0. <i>Applies to the 5G NSA solution.</i> See “Enabling 5G NSA QoS AVPs” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Flows

AVP Name	Flows
AVP Code	510
AVP Data Type	Grouped
Description	Indicates the IP flows via their flow identifiers. The detailed syntax is as follows: <pre>Flows ::= < AVP Header: 510 > {Media-Component-Number} *[Flow-Number]</pre>

Flow-Information

AVP Name	Flow-Information
AVP Code	1058
AVP Data Type	Grouped
Description	Sent from the PCRF to the PCEF and contains the information from a single IP flow packet filter. When included in the Application-Detection-Information AVP, the Flow-Information contains the Flow-Description AVP and Flow-Direction AVP. The detailed syntax is as follows: <pre>Flow-Information ::= < AVP Header: 1058 > [Flow-Description] [Packet-Filter-Identifier] [Packet-Filter-Usage] [ToS-Traffic-Class] [Security-Parameter-Index] [Flow-Label] [Flow-Direction] [AVP]</pre>

Flow-Status

AVP Name	Flow-Status
AVP Code	511
AVP Data Type	Enumerated
Description	Defines whether the service data flow is enabled or disabled. For predefined PCC rules, the Flow-Status AVP overrides whatever is configured statically on the gateway. The following values are defined: ■ ENABLED (2) – Used to enable all associated IP flow(s) in both directions. ■ DISABLED (3) – Used to disable all associated IP flow(s) in both directions.

Guaranteed-Bitrate-DL

AVP Name	Guaranteed-Bitrate-DL
AVP Code	1025
AVP Data Type	Unsigned32
Description	Defines the guaranteed bit rate allowed for the downlink direction, in bps.

Guaranteed-Bitrate-UL

AVP Name	Guaranteed-Bitrate-UL
AVP Code	1026
AVP Data Type	Unsigned32
Description	Defines the guaranteed bit rate allowed for the uplink direction, in bps.

GGSN-Address

AVP Name	GGSN-Address
AVP Code	847
AVP Data Type	Address
Description	The Gateway GPRS Support Node/Packet Data Network Gateway (GGSN/PGW) IP address.

IMS-Information

AVP Name	IMS-Information
AVP Code	876
AVP Data Type	Grouped
Description	Provides additional IMS service-specific information elements. The detailed syntax is as follows: <pre>IMS-Information ::= < AVP Header: 876> {Node-Functionality}</pre>

IMSI-Unauthenticated-Flag

AVP Name	IMSI-Unauthenticated-Flag
AVP Code	2308
AVP Data Type	Enumerated
Description	Indicates whether the served IMSI is authenticated. If this AVP is not present, the served IMSI is authenticated. The following values are defined: ■ AUTHENTICATED (0) ■ UNAUTHENTICATED (1)

IP-CAN-Type

AVP Name	IP-CAN-Type
AVP Code	1027
AVP Data Type	Enumerated
Description	Indicates the type of Connectivity Access Network in which the user is connected. The IP-CAN-Type AVP is always present during the IP-CAN session establishment. During an IP-CAN session modification, this AVP is present when there has been a change in the IP-CAN type and the PCRF requested to be informed of this event. The Event-Trigger AVP with value IP-CAN-CHANGE is provided together with the IP-CAN-Type AVP. The following values are defined: ■ 3GPP-GPRS (0) – Indicates that the IP-CAN is associated with a 3GPP GPRS access that is connected to the GGSN based on the Gn/Gp interfaces and is further detailed by the RAT-Type AVP. The RAT-Type AVP includes applicable 3GPP values, except EUTRAN. ■ DOCSIS (1) – Indicates that the IP-CAN is associated with a DOCSIS access. ■ xDSL (2) – Indicates that the IP-CAN is associated with an xDSL access. ■ WiMAX (3) – Indicates that the IP-CAN is associated with a WiMAX access (IEEE 802.16). ■ 3GPP2 (4) – Indicates that the IP-CAN is associated with a 3GPP2 access connected to the 3GPP2 packet core as specified in 3GPP2 X.S0011 [20] and is further detailed by the RAT-Type AVP. ■ 3GPP-EPS (5) – Indicates that the IP-CAN associated with a 3GPP EPS access and is further detailed by the RAT-Type AVP. ■ Non-3GPP-EPS (6) – Indicates that the IP-CAN associated with an EPC based non-3GPP access and is further detailed by the RAT-Type AVP.

Local-Sequence-Number

AVP Name	Local-Sequence-Number
AVP Code	2063
AVP Data Type	Unsigned32
Description	Contains the service data container sequence number, increased by one for each service data container closed.

Logical-Access-Id

AVP Name	Logical-Access-Id
AVP Code	302
AVP Data Type	OctetString
Description	Contains a Circuit ID from the DHCP Option 82 field. <i>Applies to the Gxa interface only.</i>

Low-Priority-Indicator

AVP Name	Low-Priority-Indicator
AVP Code	2602
AVP Data Type	Enumerated
Description	Indicates if the PDN connection has a low priority (for example, for Machine Type Communications). <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ NO (0) ■ YES (1)

Max-Requested-Bandwidth-DL

AVP Name	Max-Requested-Bandwidth-DL
AVP Code	515
AVP Data Type	Unsigned32
Description	Indicates the maximum bit rate allowed for the downlink direction.

Max-Requested-Bandwidth-UL

AVP Name	Max-Requested-Bandwidth-UL
AVP Code	516
AVP Data Type	Unsigned32
Description	Indicates the maximum bit rate allowed for the uplink direction.

Metering-Method

AVP Name	Metering-Method
AVP Code	1007
AVP Data Type	OctetString
Description	Defines which parameters are metered for offline charging. The following values are defined: ■ VOLUME (1) – The volume of the service data flow traffic is metered. If the Metering-Method AVP is omitted but has been supplied previously, the previous information remains valid. If the Metering-Method AVP is omitted and has not been supplied previously, the metering method pre-configured at the PCEF is used as the default metering method.

Monitoring-Key

AVP Name	Monitoring-Key
AVP Code	1066
AVP Data Type	Enumerated
Description	Contains the monitoring key that may apply to the PCC rule. This AVP is used for usage monitoring control purposes as an identifier to a usage monitoring control instance. For more information, see “Usage Monitoring and Reporting” in the <i>Affirmed Networks Gx Interface Specification</i>.

Mute-Notification

AVP Name	Mute-Notification
AVP Code	2809
AVP Data Type	Enumerated
Description	Used by the PCEF to mute the notification to the PCRF of the detected application's Start/Stop for the specific PCC/ADC rule. The following values are defined: MUTE_REQUIRED (0) – Indicates that the PCEF should not inform the PCRF when the application's Start/Stop for the specific PCC/ADC rule is detected.

Network-Request-Support

AVP Name	Network-Request-Support
AVP Code	1024
AVP Data Type	Enumerated
Description	Indicates whether the UE and access network supports the network requested bearer control mode. The following values are defined: ■ NETWORK_REQUEST NOT SUPPORTED (0) – Indicates that the UE and the access network do not support the network initiated bearer establishment request procedure. ■ NETWORK_REQUEST SUPPORTED (1) – Indicates that the UE and the access network support the network initiated bearer establishment request procedure.

Node-Functionality

AVP Name	Node-Functionality
AVP Code	862
AVP Data Type	Enumerated
Description	Includes the functionality identifier of the node. The functionality identifier can be one of the following: ■ S-CSCF (0) ■ P-CSCF (1) ■ I-CSCF (2) ■ MRFC (3) ■ MGCF (4) ■ BGCF (5) ■ AS (6) ■ IBCF (7) ■ S-GW (8) ■ P-GW (9) ■ HSGW (10) ■ E-CSCF (11) ■ MME (12) ■ TRF (13) ■ TF (14) ■ ATCF (15) ■ Proxy Function (16) ■ ePDG (17)

Node-Id

AVP Name	Node-Id
AVP Code	2064
AVP Data Type	UTF8String
Description	Includes an optional, operator-configurable identifier string for the node that had generated the Accounting-Request (ACR) message.

Offline

AVP Name	Offline
AVP Code	1008
AVP Data Type	Enumerated
Description	When included within the Charging-Rule-definition AVP, this AVP indicates whether the offline charging interface from the PCEF for the associated PCC rule should be enabled. The absence of this AVP within the first provisioning of the Charging-Rule-definition AVP of a new PCC rule indicates that the pre-configured default charging method at the PCEF should be used. When included in the CCR-I message at the command level, this AVP indicates the pre-configured default charging method at the PCEF. The absence of this AVP within the CCR-I indicates that the charging method for offline pre-configured at the PCEF is not available. When included in the CCA-I message at the command level, this AVP controls whether to enable/disable offline charging at the PDN session level. The absence of this AVP within the CCA-I indicates that the pre-configured default charging method at the PCEF should be used. See “Configuring the Default Charging Method” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The default charging method provided by the PCRF takes precedence over any pre-configured default charging method at the PCEF. The following values are defined: ■ DISABLE_OFFLINE (0) – Indicates that the offline charging interface for the associated PCC rule is disabled. ■ ENABLE_OFFLINE (1) – Indicates that the offline charging interface for the associated PCC rule is enabled.

Online

AVP Name	Online
AVP Code	1009
AVP Data Type	Enumerated
Description	When included within the Charging-Rule-Definition AVP, this AVP defines whether the online charging interface from the PCEF for the associated PCC rule should be enabled. <i>Note: If all the dynamic PCC rules associated with the Rating Group have online charging disabled, the Rating Group is moved to a blocked state for online charging and no longer performs online charging. When the last active Rating Group is moved to a blocked state, the PGW terminates the Gy session by sending the CCR-T message to the PCRF.</i> When included in the CCR-I message at the command level, this AVP indicates the pre-configured default charging method at the PCEF. The absence of this AVP within the CCR-I indicates that the charging method for online pre-configured at the PCEF is not available. When included in the CCA-I message at the command level, this AVP controls whether to enable/disable online charging at the PDN session level. The absence of this AVP within the CCA-I indicates that the pre-configured default charging method at the PCEF should be used. The default charging method provided by the PCRF takes precedence over any pre-configured default charging method at the PCEF. See “Configuring the Default Charging Method” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the online charging interface for the associated PCC rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the online charging interface for the associated PCC rule is enabled.

PCC-Rule-Status

AVP Name	PCC-Rule-Status
AVP Code	1019
AVP Data Type	Enumerated
Description	Indicates the status of one PCC rule or a group of PCC rules. The following values are defined: ■ ACTIVE (0) – Used to indicate that the PCC rule(s) are successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF). ■ INACTIVE (1) – Used to indicate that the PCC rule(s) are removed (for those provisioned from PCRF) or inactive (for those pre-provisioned in PCEF). ■ TEMPORARILY INACTIVE (2) – Used to indicate that, for some reason (for example, loss of bearer), already installed or activated PCC rules are temporarily disabled. ■ ACTIVE_WITHOUT_CREDIT_CONTROL (10)

PDN-Connection-Charging-ID

AVP Name	PDN-Connection-Charging-ID
AVP Code	2050
AVP Data Type	Unsigned32
Description	Contains the charging identifier used to identify different records belonging to the same PDN connection. This AVP includes the Charging ID of the first IP-CAN bearer activated within the PDN connection. The PDN-Connection-Charging-ID together with the PGW address uniquely identifies the PDP connection. Applies to the Gy and Gz interfaces.

PDN-Connection-ID

AVP Name	PDN-Connection-ID
AVP Code	1065
AVP Data Type	OctetString
Description	Indicates the PDN connection to which specific information refers. Applies to the Gx interface.

PDP-Address

AVP Name	PDP-Address
AVP Code	1227
AVP Data Type	Address
Description	Contains the IP address associated with the IP CAN bearer session (PDP context/PDN connection).

PDP-Context-Type

AVP Name	PDP-Context-Type
AVP Code	1247
AVP Data Type	Enumerated
Description	Indicates the type of PDP context. The following values are defined: ■ PRIMARY (0) ■ SECONDARY (1)

Precedence

AVP Name	Precedence
AVP Code	1010
AVP Data Type	Unsigned32
Description	Determines the order in which the service data flow templates are applied at service data flow detection at the PCEF. A PCC rule with the Precedence AVP with a lower value is applied before a PCC rule with the Precedence AVP with a higher value.

Pre-emption-Capability

AVP Name	Pre-emption-Capability
AVP Code	1047
AVP Data Type	Enumerated
Description	Indicates whether a service data flow can get resources that were already assigned to another service data flow with a lower priority level. The following values are defined: ■ PRE-EMPTION_CAPABILITY_ENABLED (0) – Indicates that the service data flow is allowed to get resources that were already assigned to another service data flow with a lower priority level. ■ PRE-EMPTION_CAPABILITY_DISABLED (1) – Indicates that the service data flow is not allowed to get resources that were already assigned to another service data flow with a lower priority level. This is the default value applicable if this AVP is not supplied.

Pre-emption-Vulnerability

AVP Name	Pre-emption-Vulnerability
AVP Code	1048
AVP Data Type	Enumerated
Description	Indicates whether a service data flow can lose the resources assigned to it in order to admit a service data flow with higher priority level. The following values are defined: ■ PRE-EMPTION_VULNERABILITY_ENABLED (0) – Indicates that the resources assigned to the service data flow can be pre-empted and allocated to a service data flow with a higher priority level. This is the default value applicable if this AVP is not supplied. ■ PRE-EMPTION_VULNERABILITY_DISABLED (1) – Indicates that the resources assigned to the service data flow can not be pre-empted and allocated to a service data flow with a higher priority level.

Primary-Event-Charging-Function-Name

AVP Name	Primary-Event-Charging-Function-Name
AVP Code	619
AVP Data Type	DiameterURI
Description	Defines the address of the primary online charging system. The protocol definition in the DiameterURI is either omitted or supplied with a value of “Diameter”.

Priority-Level

AVP Name	Priority-Level
AVP Code	1046
AVP Data Type	Unsigned32
Description	Indicates whether a bearer establishment or modification request can be accepted or needs to be rejected in case of resource limitations (typically used for admission control of GBR traffic). The AVP can also be used to decide which existing bearers to pre-empt during resource limitations. The priority level defines the relative importance of a resource request. ■ Values 1 to 15 are defined, with value 1 as the highest level of priority. ■ Values 1 to 8 should only be assigned for services that are authorized to receive prioritized treatment within an operator domain. ■ Values 9 to 15 may be assigned to resources that are authorized by the home network and thus applicable when a UE is roaming.

PS-Furnish-Charging-Information

AVP Name	PS-Furnish-Charging-Information
AVP Code	865
AVP Data Type	Grouped
Description	Contains online charging session specific information. The detailed syntax is as follows: <pre>PS-Furnish-Charging-Information ::= < AVP Header: 865> { 3GPP-Charging-Id } { PS-Free-Format-Data } [PS-Append-Free-Format-Data]</pre>

PS-Information

AVP Name	PS-Information
AVP Code	874
AVP Data Type	Grouped
Description	Allows the transmission of additional PS service specific information elements. For a list of the supported AVPs within the PS-Information AVP on the Gz/Rf interface, see the <i>Affirmed Networks Charging Data Records Specification</i>.

PS-to-CS-Session-Continuity

AVP Name	PS-to-CS-Session-Continuity
AVP Code	1099
AVP Data Type	Enumerated
Description	Indicates whether the service data flow is a candidate for PS-to-CS session continuity as specified in 3GPP TS 23.216 [40]. The following value is defined: VIDEO_PS2CS_CONT_CANDIDATE (0) – Indicates that the service data flow carries video and is a candidate for PS to CS session continuity.

QoS-Class-Identifier

AVP Name	QoS-Class-Identifier
AVP Code	1028
AVP Data Type	Enumerated
Description	Identifies a set of IP-CAN specific QoS parameters that define the authorized QoS, excluding the applicable bit rates and ARP for the IP-CAN bearer or service flow. The allowed values for the nine standard QoS Class Identifiers (QCI) are defined in Table 6.1.7 of 3GPP TS 23.203 [7].

QoS-Information

AVP Name	QoS-Information
AVP Code	1016
AVP Data Type	Grouped
Description	Defines the QoS information for resources requested by the UE, an IP-CAN bearer, PCC rule, QCI or APN. <i>Gx-Interface</i> – When the PCEF sends this AVP to the PCRF, it indicates the requested QoS information associated with the resources requested by the UE, and IP-CAN bearer or the subscribed QoS information at the Access Point Name (APN) level.

Quota-Consumption-Time

AVP Name	Quota-Consumption-Time
AVP Code	881
AVP Data Type	Unsigned32
Description	Contains an idle traffic threshold time in seconds.

Quota-Holding-Time

AVP Name	Quota-Holding-Time
AVP Code	871
AVP Data Type	Unsigned32
Description	Contains the quota holding time in seconds and is contained in the Multiple-Services-Credit-Control AVP.

RAN-End-Timestamp

AVP Name	RAN-End-Timestamp
AVP Code	1301
AVP Data Type	Time
Description	Indicates the time, in UTC format, of the volume container reported was collected, the end time of the reported usage. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. <i>Applies to 5G NSA.</i>

RAN-NAS-Release-Cause

AVP Name	RAN-NAS-Release-Cause
AVP Code	2819
AVP Data Type	OctetString
Description	Indicates the Radio Access Network (RAN) or Non-Access Stratum (NAS) release cause code information. The RAN-NAS-Release-Cause AVP is sent from the PCEF to the PCRF.

RAN-Secondary-RAT-Usage-Report

AVP Name	RAN-Secondary-RAT-Usage-Report
AVP Code	1302
AVP Data Type	Grouped
Description	Contains the volume count as reported by the RAN for the secondary RAT (separated for uplink and downlink) including the time of the report. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. <i>Applies to 5G NSA.</i> The detailed syntax is as follows: <pre>RAN-Secondary-RAT-Usage-Report ::= < AVP Header: 1302> [Secondary-RAT-Type] [RAN-Start-Timestamp] [RAN-End-Timestamp] [Accounting-Input-Octets] [Accounting-Output-Octets]</pre>

RAN-Start-Timestamp

AVP Name	RAN-Start-Timestamp AVP
AVP Code	1303
AVP Data Type	Time
Description	Indicates the time, in UTC format, of the volume container reported was collected, the start time of the reported usage. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. <i>Applies to 5G NSA.</i>

RAT-Type

AVP Name	RAT-Type
AVP Code	1032
AVP Data Type	Enumerated
Description	The Radio Access Technology (RAT) currently serving the UE. Values 0-999 are used for generic radio access technologies that can apply to different IP-CAN types and are not IP-CAN specific. Notes: ■ Values 1000-1999 are used for 3GPP specific radio access technology types. ■ Values 2000-2999 are used for 3GPP2 specific radio access technology types. ■ The informative Annex C presents a mapping between the code values for different access network types. The following values are defined: ■ WLAN (0) – Indicates that the RAT is WLAN. ■ UTRAN (1000) – Indicates that the RAT is UTRAN. For further details, refer to 3GPP TS 29.060 [18]. ■ GERAN (1001) – Indicates that the RAT is GERAN. For further details, refer to 3GPP TS 29.060 [18]. ■ GAN (1002) – Indicates that the RAT is GAN. For further details, refer to 3GPP TS 29.060 [18] and 3GPP TS 43.318 [29]. ■ HSPA_EVOLUTION (1003) – Indicates that the RAT is HSPA Evolution. For further details, refer to 3GPP TS 29.060 [18]. ■ EUTRAN (1004) – Indicates that the RAT is EUTRAN. For further details, refer to 3GPP TS 29.274 [22]. ■ EUTRAN-NB-IoT (1005) – Indicates that the RAT is NB-IoT. For further details refer to 3GPP TS 29.274 [22]. ■ CDMA2000_1X (2000) – Indicates that the RAT is CDMA2000 1X. For further details, refer to 3GPP2 X.S0011 [20]. ■ HRPD (2001) – Indicates that the RAT is HRPD. For further details, refer to 3GPP2 X.S0011 [20]. ■ UMB (2002) – Indicates that the RAT is UMB. For further details, refer to 3GPP2 X.S0011 [20]. ■ EHRPD (2003) – Indicates that the RAT is eHRPD. For further details, refer to 3GPP2 X.S0057 [24].

Rate-Control-Max-Message-Size

AVP Name	Rate-Control-Max-Message-Size
AVP Code	3937
AVP Data Type	Unsigned32
Description	Indicates the maximum data PDU size in octets. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Rate-Control-Max-Rate

AVP Name	Rate-Control-Max-Rate
AVP Code	3938
AVP Data Type	Unsigned32
Description	Indicates the maximum number of data PDUs per time unit indicated in the Rate-Control-Time-Unit AVP. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Rate-Control-Time-Unit

AVP Name	Rate-Control-Time-Unit
AVP Code	3939
AVP Data Type	Unsigned32
Description	Indicates a time unit for rate control. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ UNRESTRICTED (0) – This value indicates the number of unrestricted data PDUs. ■ MINUTE (1) ■ HOUR (2) ■ DAY (3) ■ WEEK (4)

Redirect-Information

AVP Name	Redirect-Information
AVP Code	1085
AVP Data Type	Grouped
Description	Indicates whether the detected application traffic should be redirected to another controlled address. <i>Note: The Redirect-Information AVP sent from the PCRF as part of the Charging-Rule-Definition/ADC-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules. When redirect information is received for an online rule, all HTTP traffic matching that rule's rating group will be redirected.</i> The detailed syntax is as follows: <pre>Redirect-Information ::= < AVP Header: 1085 > [Redirect-Support] [Redirect-Address-Type] [Redirect-Server-Address] * [AVP]</pre>

Redirect-Support

AVP Name	Redirect-Support
AVP Code	1086
AVP Data Type	Enumerated
Description	Indicates whether redirection is disabled or enabled for a predefined PCC rule. <i>Note: The Redirect-Support AVP sent from the PCRF as part of the Redirection-Information AVP applies to predefined PCC rules only and is currently not supported for dynamic PCC rules.</i> The following values are defined: ■ REDIRECTION_DISABLED (0) – Indicates that redirection is disabled for a predefined PCC rule. ■ REDIRECTION_ENABLED (1) – Indicates that redirection is enabled for a predefined PCC rule. This is the default value applicable if a Redirect-Information AVP is provided for the first time and if this AVP is not supplied.

Reporting-Level

AVP Name	Reporting-Level
AVP Code	1011
AVP Data Type	Enumerated
Description	Defines on what level the PCEF reports the usage for the related PCC rule. The following values are defined: ■ SERVICE_IDENTIFIER_LEVEL (0) – Indicates that the usage is reported at the service-id and rating-group level and is applicable when the Service-Identifier and Rating-Group are provisioned within the Charging-Rule-Definition AVP. ■ RATING_GROUP_LEVEL (1) – Indicates that the usage is reported at the rating-group level and is applicable when the Rating-Group is provisioned within the Charging-Rule-Definition AVP.

Reporting-Reason

AVP Name	Reporting-Reason
AVP Code	872
AVP Data Type	Enumerated
Description	Specifies the reason for usage reporting for one or more types of quota for a particular category. It can occur directly in the Multiple-Services-Credit-Control AVP or in the Used-Service-Units AVP within a Credit Control Request command reporting credit usage.

Required-Access-Info

AVP Name	Required-Access-Info
AVP Code	536
AVP Data Type	Enumerated
Description	Used by the PCRF to request that the PCEF report the access network information required for that Application Function (AF) session. The following values are defined: ■ USER_LOCATION (0) – Indicates that the PCEF should report the user location information to the PCRF. If the user location information is available, the PCEF reports the user location information within the 3GPP-User-Location-Info AVP and the time when it was last known in the User-Location-Info-Time AVP (if available). If the user location information is not available, the PCEF provides the serving PLMN identifier within the 3GPP-SGSN-MCC-MNC AVP. ■ MS_TIME_ZONE (1) – Indicates that the PCEF should report the user timezone information to the PCRF. The PCEF reports the user timezone information within the 3GPP-MS-TimeZone AVP.

Resource-Allocation-Notification

AVP Name	Resource-Allocation-Notification
AVP Code	1063
AVP Data Type	Enumerated
Description	Included within the Charging-Rule-Install AVP and defines whether the rules included within the Charging-Rule-Install AVP need notification. The following value is defined: ■ ENABLE_NOTIFICATION (0) – Indicates that the allocation of resources for the related PCC rules are confirmed.

Revalidation-Time

AVP Name	Revalidation-Time
AVP Code	1042
AVP Data Type	Time
Description	Sent from the PCRF to the GGSN/PGW in CCA and RAR messages to indicate the NTP time before which the GGSN/PGW should issue a new request for PCC rules from the PCRF. The Revalidation-Time AVP is always sent with the REVALIDATION_TIMEOUT event trigger. <i>Note: The PCRF can disable the revalidation timer on the GGSN/PGW by not including the REVALIDATION_TIMEOUT event-trigger in the CCA or RAR message.</i>

Rule-Activation-Time

AVP Name	Rule-Activation-Time
AVP Code	1043
AVP Data Type	Time
Description	Sent from the PCRF to the PCEF in CCA and RAR messages to indicate the NTP time at which the PCC/ADC rule has to be enforced. The AVP is included in the Charging-Rule-Install AVP/ADC-Rule-Install AVP and is applicable for all the PCC/ADC rules included within the Charging-Rule-Install AVP/ADC-Rule-Install AVP.

Rule-Deactivation-Time

AVP Name	Rule-Deactivation-Time
AVP Code	1044
AVP Data Type	Time
Description	Sent from the PCRF to the PCEF in CCA and RAR messages to indicate the NTP time at which the PCEF has to stop enforcing the PCC/ADC rule. The AVP is included in Charging-Rule-Install AVP/ADC-Rule-Install AVP and is applicable for all the PCC/ADC rules included within the Charging-Rule-Install AVP/ADC-Rule-Install AVP.

Rule-Failure-Code

AVP Name	Rule-Failure-Code
AVP Code	1031
AVP Data Type	Enumerated
Description	Sent by the PCEF to the PCRF within a Charging-Rule-Report AVP to identify the reason a PCC Rule is being reported. The following values are defined: UNKNOWN_RULE_NAME (1) – Indicates that the pre-provisioned PCC rule could not be successfully activated because the Charging-Rule-Name or Charging-Rule-Base-Name is unknown to the PCEF. RATING_GROUP_ERROR (2) – Indicates that the PCC rule could not be successfully installed or enforced because the Rating-Group specified within the Charging-Rule-Definition AVP by the PCRF is unknown or invalid. SERVICE_IDENTIFIER_ERROR (3) – Indicates that the PCC rule could not be successfully installed or enforced because the Service-Identifier specified within the Charging-Rule-Definition AVP by the PCRF is invalid, unknown, or not applicable to the service being charged. GW/PCEF_MALFUNCTION (4) – Indicates that the PCC rule could not be successfully installed (for those provisioned from the PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to GW/PCEF malfunction. RESOURCES_LIMITATION (5) – Indicates that the PCC rule could not be successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to a limitation of resources at the PCEF. MAX_NR_BEARERS_REACHED (6) – Indicates that the PCC rule could not be successfully installed (for those provisioned from PCRF) or activated (for those pre-provisioned in PCEF) or enforced (for those already successfully installed) due to the fact that the maximum number of bearers has been reached for the IP-CAN session. UNKNOWN_BEARER_ID (7) – Indicates that the PCC rule could not be successfully installed or enforced at the PCEF because the Bearer-Id specified within the Charging-Rule-Install AVP by the PCRF is unknown or invalid. Applicable only for GPRS in the case the PCRF performs the bearer binding. MISSING_BEARER_ID (8) – Indicates that the PCC rule could not be successfully installed or enforced at the PCEF because the Bearer-Id is not specified within the Charging-Rule-Install AVP by the PCRF. Applicable only for GPRS in the case the PCRF performs the bearer binding. MISSING_FLOW_DESCRIPTION (9) – Indicates that the PCC rule could not be successfully installed or enforced because the Flow-Information AVP is not specified within the Charging-Rule-Definition AVP by the PCRF during the first install request of the PCC rule. RESOURCE_ALLOCATION_FAILURE (10) – Indicates that the PCC rule could not be successfully installed or maintained since the bearer establishment/modification failed, or the bearer was released.

AVP Name	Rule-Failure-Code
Description (Continued)	UNSUCCESSFUL_QOS_VALIDATION (11) – Indicates that the QoS validation has failed. PS_TO_CS_HANDOVER (13) – Indicates that the PCC rule could not be maintained because of PS to CS handover. Applicable only for 3GPP-GPRS and 3GPP-EPS.

SCS-Address

AVP Name	SCS-Address
AVP Code	3941
AVP Data Type	Address
Description	Contains the IP address of the SCS/AS. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

SCS-AS-Address

AVP Name	SCS-AS-Address
AVP Code	3940
AVP Data Type	Grouped
Description	Indicates the address of the Service Capability Server/Application Server (SCS/AS). <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>SCS-AS-Address ::= < AVP Header: 3940> [SCS-Realm] [SCS-Address]</pre>

SCS-Realm

AVP Name	SCS-Realm
AVP Code	3942
AVP Data Type	DiameterIdentity
Description	Indicates the Diameter Realm Identity of the SCS/AS. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Secondary-Event-Charging-Function-Name

AVP Name	Secondary-Event-Charging-Function-Name
AVP Code	620
AVP Data Type	DiameterURI
Description	Defines the address of the secondary online charging system for the bearer. The protocol definition in the DiameterURI is either omitted or supplied with a value of “Diameter”.

Secondary-RAT-Type

AVP Name	Secondary-RAT-Type
AVP Code	1304
AVP Data Type	OctetString
Description	Contains the Secondary RAT Type value, as provided by the RAN. The field is provided by the RAN and transferred to the SGW/PGW in the RAN Traffic Volume element. For more details on this AVP, refer to 3GPP 32.299 Version 15.2.0. <i>Applies to 5G NSA.</i> The following values are defined: ■ Reserved (0) ■ 5G NR (1)

Service-Data-Container

AVP Name	Service-Data-Container
AVP Code	2040
AVP Data Type	Grouped
Description	Provides a list of changes in charging conditions reported for flow-based charging. When there is a change in charging conditions, this container identifies the volume count (separated for uplink and downlink), elapsed time or number of events, per service data flow identified per rating group or combination of the rating group and service ID within an IP-CAN bearer. The detailed syntax is as follows: <pre>Service-Data-Container:: = < AVP Header: 2040> [AF-Correlation-Information] [Charging-Rule-Base-Name] [Accounting-Input-Octets] [Accounting-Output-Octets] [Accounting-Input-Packets] [Accounting-Output-Packets] [Local-Sequence-Number] [QoS-Information] [Rating-Group] [Change-Time] [Service-Identifier] [Service-Specific-Info] [SGSN-Address] [Time-First-Usage] [Time-Last-Usage] [Time-Usage] * [Change-Condition] [3GPP-User-Location-Info] [3GPP-RAT-Type] [Serving-PLMN-Rate-Control] [APN-Rate-Control]</pre>

Service-Information

AVP Name	Service-Information
AVP Code	873
AVP Data Type	Grouped
Description	Contains service-specific parameters. The format and the contents of the fields inside the Service-Information AVP are specified in the middle-tier documents which are applicable for the specific service.

Service-Specific-Info

AVP Name	Service-Specific-Info
AVP Code	1249
AVP Data Type	Grouped
Description	Contains service-specific data if provided by an Application Server or a PCEF only for predefined PCC rules. The detailed syntax is as follows: <pre>Service-Specific-Info ::= < AVP Header: 1249 > [Service-Specific-Data] [Service-Specific-Type]</pre>

Serving-Node-Type

AVP Name	Serving-Node-Type
AVP Code	2047
AVP Data Type	Enumerated
Description	Indicates the type of serving node. The following values are defined: ■ SGSN (0) ■ PMIPSGW (1) ■ GTPSGW (2) ■ ePDG (3) ■ hSGW (4) ■ MME (5)

Serving-PLMN-Rate-Control

AVP Name	Serving-PLMN-Rate-Control
AVP Code	4310
AVP Data Type	Grouped
Description	Indicates the Serving PLMN Rate Control used by the MME. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The detailed syntax is as follows: <pre>Serving-PLMN-Rate-Control ::= <AVP header: 4310 10415> [Uplink-Rate-Limit] [Downlink-Rate-Limit]</pre> A Downlink-Rate-Limit set to 0 indicates that the Serving PLMN Rate Control for downlink messages is deactivated in the MME. If the Serving PLMN Rate Control is activated, the value of the Downlink-Rate-Limit should not be less than 10, see 3GPP TS 23.401 [25]. An Uplink-Rate-Limit set to 0 indicates that the Serving PLMN Rate Control for uplink messages is deactivated in the MME. If Serving PLMN Rate Control is activated, the value of the Uplink-Rate-Limit should not be less than 10, see 3GPP TS 23.401 [25].

SGi-PtP-Tunnelling-Method

AVP Name	SGi-PtP-Tunnelling-Method
AVP Code	3931
AVP Data Type	Enumerated
Description	Indicates whether the SGi PtP tunnelling method is based on UDP/IP or other methods for a non-IP PDN type PDN connection. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ UDP_IP_BASED (0) ■ OTHERS (1)

Session-Release-Cause

AVP Name	Session-Release-Cause
AVP Code	1045
AVP Data Type	Enumerated
Description	Indicates the reason for terminating the IP-CAN session by the PCRF. The following values are defined: ■ UNSPECIFIED_REASON (0) – Used for unspecified reasons. ■ UE_SUBSCRIPTION_REASON (1) – Indicates that the subscription of the UE has changed (for example, has been removed) and the session needs to be terminated. ■ INSUFFICIENT_SERVER_RESOURCES (2) – Indicates that the server is overloaded and needs to abort the session.

SGSN-Address

AVP Name	SGSN-Address
AVP Code	1228
AVP Data Type	Address
Description	Indicates the IP address of the Serving GPRS Support Node (SGSN). This AVP can be used to identify the Public Land Mobile Network (PLMN) of the SGSN to which a user logs in.

SGW-Address

AVP Name	SGW-Address
AVP Code	2067
AVP Data Type	Address
Description	Indicates the IP-address of the SGW Node.

SGW-Change

AVP Name	SGW-Change
AVP Code	2065
AVP Data Type	Enumerated
Description	Indicates this is the first ACR Start message after an SGW change. If this AVP is not present in the ACR Start message, this ACR Start is not due to an SGW change. The following values are defined: ■ ACR START NOT DUE TO SGW CHANGE (0) ■ ACR START DUE TO SGW CHANGE (1)

Sponsor-Identity

AVP Name	Sponsor-Identity
AVP Code	531
AVP Data Type	UTF8String
Description	Used for sponsored data connectivity purposes as an identifier of the sponsor.

Sponsored-Connectivity-Data

AVP Name	Sponsored-Connectivity-Data
AVP Code	530
AVP Data Type	Grouped
Description	Contains the data associated with the sponsored data connectivity. The detailed syntax is as follows: <pre>Sponsored-Connectivity-Data ::= < AVP Header: 530 > [Sponsor-Identity] [Application-Service-Provider-Identity] [Granted-Service-Unit] * [AVP]</pre>

Supported-Features

AVP Name	Supported-Features
AVP Code	628
AVP Data Type	Grouped
Description	Informs the destination host about the features that the origin host requires to successfully complete this command exchange. <i>Note: When a Supported-Features AVP is used to identify features that have been defined by 3GPP, the Vendor-Id AVP contains the vendor ID of 3GPP (10415).</i> The detailed syntax is as follows: <pre>Supported-Features ::= < AVP header: 628 10415 > {Vendor-Id} {Feature-List-ID} {Feature-List}</pre>

TDF-Application-Identifier

AVP Name	TDF-Application-Identifier
AVP Code	1088
AVP Data Type	OctetString
Description	References the application detection filter to which the PCC rule for Application Detection and Control (ADC) in the PCEF applies. This AVP also identifies the application when reporting application Start/Stop notifications to the PCRF. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

TDF-Application-Instance-Identifier

AVP Name	TDF-Application-Instance-Identifier
AVP Code	2802
AVP Data Type	OctetString
Description	This value is dynamically assigned by the PCEF to correlate the application Start/Stop events for use with the Application Detection and Control (ADC) functionality. See “Application Detection and Control” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Time-First-Usage

AVP Name	Time-First-Usage
AVP Code	2043
AVP Data Type	Time
Description	Contains the time in UTC format for the first IP packet to be transmitted and mapped to the current service data container.

Time-Last-Usage

AVP Name	Time-Last-Usage
AVP Code	2044
AVP Data Type	Time
Description	Contains the time in UTC format for the last IP packet to be transmitted and mapped to the current service data container.

Time-Quota-Mechanism

AVP Name	Time-Quota-Mechanism
AVP Code	1270
AVP Data Type	Grouped
Description	The OCS may include this AVP in the Multiple-Services-Credit-Control AVP, when granting time quota. The detailed syntax is as follows: <pre>Time-Quota-Mechanism ::= < AVP Header: 1270> {Time-Quota-Type} {Base-Time-Interval}</pre>

Time-Quota-Threshold

AVP Name	Time-Quota-Threshold
AVP Code	868
AVP Data Type	Unsigned32
Description	Contains a threshold value in seconds.

Time-Quota-Type

AVP Name	Time-Quota-Type
AVP Code	1271
AVP Data Type	Enumerated
Description	Indicates the time quota consumption mechanism used for the associated rating group. The following values are defined: ■ DISCRETE_TIME_PERIOD (0) ■ CONTINUOUS_TIME_PERIOD (1)

Time-Usage

AVP Name	Time-Usage
AVP Code	2045
AVP Data Type	Time
Description	Contains the effective used time within the service data container reporting interval.

Traffic-Data-Volumes

AVP Name	Traffic-Data-Volumes
AVP Code	2046
AVP Data Type	Grouped
Description	When a change in charging conditions occurs for this IP-CAN bearer, this container reports the volume count (separated for uplink and downlink). The detailed syntax is as follows: <pre>Traffic-Data-Volumes ::= < AVP Header: 2046> [QoS-Information] [Accounting-Input-Octets] [Accounting-Input-Packets] [Accounting-Output-Octets] [Accounting-Output-Packets] [Change-condition] [Change-Time] [3GPP-User-Location-Info] [3GPP-RAT-Type] [CP-CIoT-EPS-Optimisation-Indicator] [Serving-PLMN-Rate-Control]</pre>

Trigger

AVP Name	Trigger
AVP Code	1264
AVP Data Type	Grouped
Description	Contains the trigger types. The detailed syntax is as follows: <pre>Trigger ::= < AVP Header: 1264 > *[Trigger-Type]</pre>

Trigger-Type

AVP Name	Trigger-Type
AVP Code	870
AVP Data Type	Enumerated
Description	When included in the CCA on the Gy interface, the Trigger-Type AVP indicates the events that cause the credit control client to re-authorize the associated quota. When included in the CCA on the Gx interface, the Trigger-Type VSA is used to notify the PGW when a ULI, Time Zone, or PLMN change occurs and the PGW is required to send the respective event triggers over the Gy interface to the OCS. For more information, see Trigger-Type (VSA) (page A-129).

TWAN-User-Location-Info

AVP Name	TWAN-User-Location-Info
AVP Code	2714
AVP Data Type	Grouped
Description	Contains the UE location in a Trusted WLAN Access Network (TWAN) as TWAN Identifier defined in TS 29.274 [226] (TS 32.299 v15.3). Sent from the TWAG/TWAP to the OCS via the Gy interface. The detailed syntax is as follows: <pre>TWAN-User-Location-Info ::= < AVP Header: 2714 > {SSID} [BSSID]</pre> The Twan-User-Location-Info AVP is configurable in Gy CCR data record templates and appears as psinfotwanuserlocationinfo. The RADIUS Attribute Mapping template must be configured so that the TWAG extracts the SSID/BSSID received in the Called-Station-ID AVP (within the RADIUS Access-Request) and includes it in the Twan-User-Location-Info AVP on the Gy interface. For example: 1 Configure the attributes in the incoming RADIUS messages. <pre>an3000-mcm-slot7cpu1(config)# service-construct radius-mapping template <template-name> attribute calledstationid to-format (.*):(.*) to-list [apbssid apssid]</pre> 2 Associate the RADIUS Attribute Mapping Template with the Session Learner interface. <pre>an3000-mcm-slot7cpu1(config)# service-construct interface gw-session-learner [non-3gpp-access-diameter-client radius-proxy twag- radius-client] <interface-name> radius-server-trigger radius-mapping access-request <template-name></pre> <i>Note:</i> For more information on configuring the RADIUS Attribute Mapping template, see the Affirmed Networks TWAG/TWAP Administration Guide.

UNI-PDU-CP-Only-Flag

AVP Name	UNI-PDU-CP-Only-Flag
AVP Code	3932
AVP Data Type	Enumerated
Description	Indicates whether this PDN connection is applied with the “Control Plane Only flag”. For example, using only the S11-U from the SGW when Control Plane Clot EPS Optimization is enabled. If this AVP is not present, then both the user plane and control plane can be used in UNI for the PDU transfer. For example, using both S1-U and S11-U respectively from the SGW when Control Plane Clot EPS Optimization is enabled. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i> The following values are defined: ■ UNI-PDU-both-UP-CP (0) ■ UNI-PDU-CP-Only (1)

Unit-Quota-Threshold

AVP Name	Unit-Quota-Threshold
AVP Code	1226
AVP Data Type	Unsigned32
Description	Contains a threshold value in service specific units.

Uplink-Rate-Limit

AVP Name	Uplink-Rate-Limit
AVP Code	4311
AVP Data Type	Unsigned32
Description	Contains the maximum number of NAS Data PDUs per deci hour for this UE for uplink. <i>Applies to the C-SGN as defined in 3GPP TS 32.299 V13.9.0.</i>

Usage-Monitoring-Information

AVP Name	Usage-Monitoring-Information
AVP Code	1067
AVP Data Type	Grouped
Description	Contains the usage monitoring control information. The detailed syntax is as follows: <pre>Usage-Monitoring-Information ::= < AVP Header: 1067 > [Monitoring-Key] [Granted-Service-Unit] [Used-Service-Unit] [Usage-Monitoring-Level] [Usage-Monitoring-Report] [Usage-Monitoring-Support] * [AVP]</pre>

Usage-Monitoring-Level

AVP Name	Usage-Monitoring-Level
AVP Code	1068
AVP Data Type	Enumerated
Description	Used by the PCRF to indicate whether the usage monitoring instance applies to the IP-CAN session or to one or more PCC rules. The following values are defined: ■ SESSION_LEVEL (0) – This value, if provided within an RAR or CCA command, indicates that the usage monitoring instance applies to the entire IP-CAN session. ■ PCC_RULE_LEVEL (1) – This value, if provided within an RAR or CCA command, indicates that the usage monitoring instance applies to one or more PCC rules. <i>Note: The absence of the Usage-Monitoring-Level AVP indicates the value PCC_RULE_LEVEL (1).</i>

Usage-Monitoring-Report

AVP Name	Usage-Monitoring-Report
AVP Code	1069
AVP Data Type	Enumerated
Description	Used by the PCRF to request that the PCEF report the accumulated usage for the usage monitoring key included in the Monitoring-Key AVP (included within the Usage-Monitoring-Information AVP). The following value is defined: USAGE_MONITORING_REPORT_REQUIRED (0) – This value, if provided within an RAR or CCA command, indicates that the PCEF should report accumulated usage regardless of whether a usage threshold is reached for a certain usage monitoring key.

Usage-Monitoring-Support

AVP Name	Usage-Monitoring-Support
AVP Code	1070
AVP Data Type	Enumerated
Description	Used by the PCRF to indicate whether usage monitoring is disabled for a certain Monitoring Key. The following value is defined: USAGE_MONITORING_DISABLED (0) – This value indicates that usage monitoring is disabled for a monitoring key.

User-CSG-Information

AVP Name	User-CSG-Information
AVP Code	2319
AVP Data Type	Grouped
Description	Contains the Closed Subscriber Group (CSG) information associated with CSG cell access, including the CSG ID within the PLMN, the access mode and the CSG membership indication for the user when hybrid access applies. The detailed syntax is as follows: <pre>User-CSG-Information :: = < AVP Header: 2319> {CSG-Id} [CSG-Access-Mode] [CSG-Membership-Indication]</pre>

User-Location-Info-Time

AVP Name	User-Location-Info-Time
AVP Code	2812
AVP Data Type	Time
Description	Contains the NTP time at which the UE was last known to be in the location, which is reported during bearer deactivation or IP-CAN session termination. The User-Location-Info-Time AVP is sent from the PCEF to the PCRF.

UWAN-User-Location-Info

AVP Name	UWAN-User-Location-Info
AVP Code	3918
AVP Data Type	Grouped
Description	Contains the UE location in an Untrusted Wireless Access Network (UWAN). The UWAN User Location Information includes the UE local IP address and optionally UDP source port number (if NAT is detected). The detailed syntax is as follows: <pre>UWAN-User-Location-Info ::= < AVP Header: 3918> {UE-Local-IP-Address} [UDP-Source-Port]</pre>

Volume-Quota-Threshold

AVP Name	Volume-Quota-Threshold
AVP Code	869
AVP Data Type	Unsigned32
Description	Contains a threshold value in octets.

NASREQ/RADIUS AVP Definitions

This section describes the following NASREQ/RADIUS AVPs in the CCR and CCA messages.

- [Called-Station-Id \(page A-102\)](#)
- [Framed-Interface-Id \(page A-102\)](#)
- [Framed-IP-Address \(page A-102\)](#)
- [Framed-IPv6-Prefix \(page A-103\)](#)

Called-Station-Id

AVP Name	Called-Station-Id
AVP Code	30
AVP Data Type	UTF8String
Description	The gateway support node of the GPRS. The identifier for the target network: APN.

Framed-Interface-Id

AVP Name	Framed-Interface-Id
AVP Code	96
AVP Data Type	Unsigned64
Description	The user IPv6 interface identifier.

Framed-IP-Address

AVP Name	Framed-IP-Address
AVP Code	8
AVP Data Type	OctetString
Description	The IP address allocated for this user.

Framed-IPv6-Prefix

AVP Name	Framed-IPv6-Prefix
AVP Code	97
AVP Data Type	OctetString
Description	The IPv6 address prefix allocated for this user.

Vodafone AVP Definitions

This section describes the following Vodafone AVPs in the CCR and CCA messages. The AVPs include Vendor-ID of 12645 (Vodafone).

- [vf-Base-Time-Interval \(page A-105\)](#)
- [vf-Context-Type \(page A-105\)](#)
- [vf-Envelopes \(page A-105\)](#)
- [vf-Envelope-End-Time \(page A-105\)](#)
- [vf-Envelope-Reporting \(page A-106\)](#)
- [vf-Envelope-Start-Time \(page A-106\)](#)
- [vf-Quota-Consumption-Time \(page A-106\)](#)
- [vf-Quota-Holding-Time \(page A-107\)](#)
- [vf-Radio-Access-Technology \(page A-107\)](#)
- [vf-Reporting-Reason \(page A-108\)](#)
- [vf-Rulebase-Id \(page A-108\)](#)
- [vf-Time-Of-First-Usage \(page A-109\)](#)
- [vf-Time-Of-Last-Usage \(page A-109\)](#)
- [vf-Time-Quota-Mechanism \(page A-109\)](#)
- [vf-Time-Quota-Threshold \(page A-109\)](#)
- [vf-Time-Quota-Type \(page A-110\)](#)
- [vf-Trigger \(page A-110\)](#)
- [vf-Trigger-Type \(page A-110\)](#)
- [vf-Unit-Quota-Threshold \(page A-111\)](#)
- [vf-User-Location-Information \(page A-111\)](#)
- [vf-Volume-Quota-Threshold \(page A-111\)](#)

vf-Base-Time-Interval

AVP Name	Base-Time-Interval
AVP Code	276
AVP Data Type	Unsigned32
Description	Contains the length of the base time interval, for controlling the consumption of time quota in seconds.

vf-Context-Type

AVP Name	Context-Type
AVP Code	256
AVP Data Type	Enumerated
Description	This AVP indicates whether the PDP context corresponding to the CC session is a primary or secondary. The following values are defined: ■ PRIMARY (0) ■ SECONDARY (1)

vf-Envelopes

AVP Name	Envelopes
AVP Code	269
AVP Data Type	Grouped
Description	The Envelope AVP reports the start and end time of one time envelope using the Envelope-Start-Time and Envelope-End-Time AVPs. Further details of its usage are described in 3GPP 32.299.

vf-Envelope-End-Time

AVP Name	Envelope-End-Time
AVP Code	271
AVP Data Type	Time
Description	The Envelope-End-Time AVP is set to the time of the end of the time envelope.

vf-Envelope-Reporting

AVP Name	Envelope-Reporting
AVP Code	272
AVP Data Type	Enumerated
Description	This AVP is of type Enumerated and is used in the initial CCA to indicate whether the client reports the start and end of each time envelope, in those cases in which quota is consumed in envelopes. The following values are defined: ■ DO_NOT_REPORT_ENVELOPES (0) ■ REPORT_ENVELOPES (1) ■ REPORT_ENVELOPES_WITH_VOLUME (2)

vf-Envelope-Start-Time

AVP Name	Envelope-Start-Time
AVP Code	270
AVP Data Type	Time
Description	The Envelope-Start-Time AVP is set to the time of the packet of user data which caused the time envelope to start.

vf-Quota-Consumption-Time

AVP Name	Quota-Consumption-Tim
AVP Code	257
AVP Data Type	Unsigned32
Description	This AVP contains an idle traffic threshold time in seconds. This AVP may be included within the Multiple-Services-Credit-Control AVP which also contains a Granted-Service-Units AVP containing a CC-Time AVP (for example, when the granted quota is a time quota).

vf-Quota-Holding-Time

AVP Name	Quota-Holding-Time
AVP Code	258
AVP Data Type	Unsigned32
Description	Contains the length of time (in seconds) for which the system can hold quota during a period of inactivity. A quota expires when no traffic associated with the quota is observed for the value indicated by this AVP. The system stops the timer when it sends a CCR message and resets the timer each time it receives a CCA with a Quota-Holding-Time value.

vf-Radio-Access-Technology

AVP Name	Radio-Access-Technology
AVP Code	260
AVP Data Type	Enumerated
Description	This AVP specifies the radio access technology being used by the user. The following values are defined: ■ UTRAN (1) ■ GERAN (2) ■ WLAN (3)

vf-Reporting-Reason

AVP Name	Reporting-Reason
AVP Code	261
AVP Data Type	Enumerated
Description	The Reporting-Reason specifies the reason for usage reporting for one or more types of quota for a rating group. It can occur directly in the Multiple-Services-Credit-Control AVP, or in the Used-Service-Units AVP. The following values apply to all quota types and are used in the Multiple-Services-Credit-Control AVP: ■ QHT ■ FINAL ■ VALIDITY_TIME ■ RATING_CONDITION_CHANGE The following values apply to one particular quota type and are used in the Used-Service-Units AVP: ■ THRESHOLD ■ QUOTA_EXHAUSTED ■ OTHER_QUOTA_TYPE The following values are defined for the Reporting-Reason AVP: ■ THRESHOLD 0 ■ QHT 1 ■ FINAL 2 ■ QUOTA_EXHAUSTED 3 ■ VALIDITY_TIME 4 ■ OTHER_QUOTA_TYPE 5 ■ RATING_CONDITION_CHANGE 6 ■ FORCED_REAUTHORISATION 7

vf-Rulebase-Id

AVP Name	Rulebase-Id
AVP Code	262
AVP Data Type	UTF8String
Description	Contains the identifier of the currently selected rule base. <i>Note: All charging instances configured under <code>zone <name> gateway apn <name> charging</code> are included in the CCR-I to the OCS. If no charging instances are configured and the system receives charging instances from external interfaces, those charging instances are included in the CCR-I. If no charging instances are available, the data profile charging instance is used.</i>

vf-Time-Of-First-Usage

AVP Name	Time-Of-First-Usage
AVP Code	263
AVP Data Type	Time
Description	This AVP is a time-stamp identifying the date and time of the first packet of usage of the quota being reported for the rating group.

vf-Time-Of-Last-Usage

AVP Name	Time-Of-Last-Usage
AVP Code	264
AVP Data Type	Time
Description	This AVP is a time-stamp identifying the date and time of the last packet of usage of the quota being reported for the rating group.

vf-Time-Quota-Mechanism

AVP Name	Time-Quota-Mechanism
AVP Code	274
AVP Data Type	Grouped
Description	This AVP may be used by the OCS for granting time quota. Time-Quota-Mechanism AVP takes precedence over Quota-Consumption-Time.

vf-Time-Quota-Threshold

AVP Name	Time-Quota-Threshold
AVP Code	259
AVP Data Type	Unsigned32
Description	The AVP contains a value indicating the quota threshold for time quota in seconds.

vf-Time-Quota-Type

AVP Name	Time-Quota-Type
AVP Code	275
AVP Data Type	Enumerated
Description	The Time-Quota-Type is used to indicate which time quota consumption mechanism is used for the associated rating group.

vf-Trigger

AVP Name	Trigger
AVP Code	265
AVP Data Type	Grouped
Description	This AVP is used to arm the triggers in the client and to identify the triggers which caused re-authorization in usage. On receipt of the AVP, the OCS arms all known triggers present in the AVP and resets all known triggers missing from the AVP. The presence of the Trigger AVP without any Trigger-Type AVP in a CCA allows the OCS to disable all the triggers. The presence of the Trigger AVP in the CCR identifies the event(s) triggering the CCR.

vf-Trigger-Type

AVP Name	Trigger-Type
AVP Code	266
AVP Data Type	Enumerated
Description	When included in the CCA, the Trigger-Type AVP indicates the events that cause the credit control client to re-authorize the associated quota. The system supports the following triggers between the system and the OCS. ■ CHANGE_IN_QOS (2) – Specifies whether a QoS change triggers an update. ■ CHANGE_IN_RAT (4) – Specifies whether a RAT Type change triggers an update. ■ CHANGE_IN_SERVING_NODE (61) – Specifies whether a Serving Node change triggers an update.

vf-Unit-Quota-Threshold

AVP Name	Unit-Quota-Threshold
AVP Code	273
AVP Data Type	Unsigned32
Description	This AVP contains a threshold value in service specific units. This AVP may be included within the Multiple-Services-Credit-Control AVP which also contains a Granted-Service-Units AVP containing CC-Service-Specific-Units AVP (that is, when the granted quota is service specific).

vf-User-Location-Information

AVP Name	User-Location-Information
AVP Code	267
AVP Data Type	OctetString
Description	Indicates details of where the UE is currently located (for example, SAI or CGI).

vf-Volume-Quota-Threshold

AVP Name	Volume-Quota-Threshold
AVP Code	268
AVP Data Type	Unsigned32
Description	The AVP contains a value indicating the quota threshold for volume quota in bytes.

Vendor-Specific AVP Definitions

This section describes the following supported Vendor-Specific AVPs.

- ACW-Entitlement (page A-113)
- ACW-Sponsor-URL (page A-113)
- AF-Msisdn (page A-113)
- Assume-Positive-Treatment (page A-114)
- Charging-Gateway-Function-Host (page A-114)
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ACW-Entitlement

AVP Name	acwentitlement
Vendor-ID	9
AVP Code	131501
AVP Data Type	OctetString
Description	Used to send tethering entitlement information from the PCRF to the PGW for the subscriber session. This VSA can be sent in CCA-I/CCA-U and RAR messages on the Gx interface. The following values for this VSA are defined: GTETH/MHS1/TETH/GVC/GVC1/MHSM/TOGGL/TGLSB/TGLSI. GTETH/MHS1/TETH – If one or more of these values are present in the acwentitlement VSA, the MCC will enable tethering (entitlement) for the session, irrespective of the configured qos-flows. However, the session-level AMBR will still be applied. If the VSA contains any value other than these three values or if the PGW does not receive the acwentitlement VSA in the CCR-I, the MCC will disable tethering (entitlement) for the session and the configured qos-flow will take effect. <i>Note: You can control tethering for an individual APN, based on the presence of configurable strings in the ACW- Entitlement VSA received from the PCRF. For more information, see the Affirmed Networks Gateway Administration Guide.</i>

ACW-Sponsor-URL

AVP Name	ACW-Sponsor-URL
Vendor-ID	9
AVP Code	131506
AVP Data Type	UTF8String
Description	Contains the URL of the manifest server.

AF-Msisdn

AVP Name	AF-Msisdn
Vendor-ID	37963
AVP Code	26
AVP Data Type	String
Description	Contains an MSISDN value used to set the MSISDN for the specific UE.

Assume-Positive-Treatment

AVP Name	Assume-Positive-Treatment
Vendor-ID	12951
AVP Code	7001
AVP Data Type	OctetString
Description	Used on the Gx interface in the CCA-Initial message to indicate a subscriber credit control group type (RTR, UC or Prepaid). The MCC PGW uses the credit control group received in the Assume-Positive-Treatment AVP from the PCRF as a key to select a Subscriber Analyzer. Use the following CLI command to configure the Subscriber Analyzer key for the credit control group. <pre>an3000-mcm-slot7cpul(config) # workflow subscriber-analyzer <name> key <name> credit-control-group <name></pre>

Charging-Gateway-Function-Host

AVP Name	Charging-Gateway-Function-Host
Vendor-ID	12951
AVP Code	6068
AVP Data Type	OctetString
Description	This AVP is sent from the PGW to the OFCS and contains the Fully Qualified Domain Name (FQDN) of the Charging Data Function (CDF). When the PGW receives the Charging-Gateway-Function-Host AVP via the S6b interface, the PGW includes this AVP in the Accounting-Request message on the Gz/Rf interface. <i>Supported on the Gz/Rf Interface only.</i>

Charging-Group-ID

AVP Name	Charging-Group-ID
Vendor-ID	12951
AVP Code	6069
AVP Data Type	UTF8String
Description	This AVP is sent from the PGW to the OFCS and identifies the subscriber charging group. When the PGW receives the Charging-Group-ID via the S6b interface, the PGW includes this AVP in the Accounting-Request message on the Gz/Rf interface. <i>Supported on the Gz/Rf Interface only.</i>

Cisco-CC-Failure-Type

AVP Name	Cisco-CC-Failure-Type
Vendor-ID	9
AVP Code	132077
AVP Data Type	Unsigned32
Description	Indicates the OCS failure reasons to the PCRF. <i>Supported on the Gx Interface.</i> The following values are defined: ■ CC_CONECTION_FAILURE (0) ■ CC_RESPONSE_TIMEOUT (1) ■ CC_UNKNOWN_FAILURE (5) ■ CC_RESULT_CODE – Result code value received over the Gy interface

Cisco-Event

AVP Name	Cisco-Event
Vendor-ID	9
AVP Code	131195
AVP Data Type	Grouped
Description	Indicates the OCS failure result code. <i>Supported on the Gx Interface.</i> The detailed syntax is as follows: <pre>Cisco-Event ::= < AVP Header: 131195> [Cisco-Event-Trigger-Type] [Cisco-CC-Failure-Type]</pre>

Cisco-Event-Trigger

AVP Name	Cisco-Event-Trigger
Vendor-ID	9
AVP Code	131193
AVP Data Type	Grouped
Description	Indicates the events that cause the OCS failure. <i>Supported on the Gx Interface.</i> The detailed syntax is as follows: <code>Cisco-Event-Trigger ::= < AVP Header: 131193></code> <code>[Cisco-Event-Trigger-Type]</code>

Cisco-Event-Trigger-Type

AVP Name	Cisco-Event-Trigger-Type
Vendor-ID	9
AVP Code	131192
AVP Data Type	Enumerated
Description	Indicates the events that cause the OCS failure. <i>Supported on the Gx Interface.</i> The following values are defined: ■ NO_CISCO_EVENT_TRIGGERS (0) ■ CREDIT_CONTROL_FAILURE (5)

Cumulative-Acct-Input-Octets

AVP Name	Cumulative-Acct-Input-Octets
Vendor-ID	12951
AVP Code	132044
AVP Data Type	Unsigned64
Description	Contains the total number of input octets used during the entire session. Added to the service data containers generated by the PGW for rating groups. <i>Supported on the Gz/Rf Interface.</i>

Cumulative-Acct-Output-Octets

AVP Name	Cumulative-Acct-Output-Octets
Vendor-ID	12951
AVP Code	132045
AVP Data Type	Unsigned64
Description	Contains the total number of output octets used during the entire session. Added to the service data containers generated by the PGW for rating groups. <i>Supported on the Gz/Rf Interface.</i>

Gx-TMO-Append-IMEI

AVP Name	Gx-TMO-Append-IMEI
Vendor-ID	29168
AVP Code	117
AVP Data Type	Enumerated
Description	Defines whether the IMEI of the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (default) Do not append the IMEI. ■ Enable (1) – Append the IMEI.

Gx-TMO-Append-IMSI

AVP Name	Gx-TMO-Append-IMSI
Vendor-ID	29168
AVP Code	116
AVP Data Type	Enumerated
Description	Defines whether the IMSI of the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (default) Do not append the IMSI. ■ Enable (1) – Append the IMSI.

Gx-TMO-Append-MSIP

AVP Name	Gx-TMO-Append-MSIP
Vendor-ID	29168
AVP Code	118
AVP Data Type	Enumerated
Description	Defines whether the IP address assigned to the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (default) Do not append the MS IP address. ■ Enable (1) – Append the MS IP address. If both IPv4 and IPv6 MS IP addresses are allocated, the MCC appends the active IP address to the redirect URL. For IPv6 IP addresses, the MCC uses the preferred format.

Gx-TMO-Append-MSISDN

AVP Name	Gx-TMO-Append-MSISDN
Vendor-ID	29168
AVP Code	115
AVP Data Type	Enumerated
Description	Defines whether the MSISDN of the mobile station is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (default) Do not append the MSISDN. ■ Enable (1) – Append the MSISDN.

Gx-TMO-Append-Original-URL

AVP Name	Gx-TMO-Append-Original-URL
Vendor-ID	29168
AVP Code	114
AVP Data Type	Enumerated
Description	Defines whether the original URL is appended to the redirect URL. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – (disable) Do not append the original URL. ■ Enable (1) – Append the original URL.

Gx-TMO-Deactivate-By-Redirect

AVP Name	Gx-TMO-Deactivate-By-Redirect
Vendor-ID	29168
AVP Code	119
AVP Data Type	Enumerated
Description	Defines whether the system performs a one-time redirection. After the one-time redirection, the rule is deactivated. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ Disable (0) – The rule is not deactivated once enforced. ■ Enable (1) – The rule is deactivated once enforced.

Gx-TMO-Redirect-Address-Type

AVP Name	Gx-TMO-Redirect-Address-Type
Vendor-ID	29168
AVP Code	112
AVP Data Type	Enumerated
Description	Defines the address type of the address given in the TMO-Redirect-Server-Address VSA. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The following values are defined: ■ IPv4 Address (0) ■ IPv6 Address (1) ■ URL (2)

Gx-TMO-Redirect-Server

AVP Name	Gx-TMO-Redirect-Server
Vendor-ID	29168
AVP Code	111
AVP Data Type	Grouped
Description	Contains the address information of the redirect server (for example, HTTP redirect server) to which the end user is redirected. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information. The detailed syntax is as follows: <pre>Gx-TMO-Redirect-Server ::= < AVP Header: 111> [Gx-TMO-Redirect-Address-Type] [Gx-TMO-Redirect-Server-Address] [Gx-TMO-Append-Original-URL] [Gx-TMO-Append-MSISDN] [Gx-TMO-Append-IMSI] [Gx-TMO-Append-IMEI] [Gx-TMO-Append-MSIP] [Gx-TMO-Deactivate-By-Redirect]</pre> Note: If both the Redirect-Information AVP and Gx-TMO Redirect-Server VSA are received, the MCC will use the Gx-TMO Redirect-Server VSA for redirection and will ignore the Redirect-Information AVP provided the Gx-TMO-Redirect-Server VSA is enabled in the CCA data record template.

Gx-TMO-Redirect-Server-Address

AVP Name	Gx-TMO-Redirect-Server-Address
Vendor-ID	29168
AVP Code	113
AVP Data Type	UTF8String
Description	Defines the address of the redirect server (for example, HTTP redirect server) with which the end user is to be connected. See “Processing the Gx-TMO-Redirect-Server VSA” in the <i>Affirmed Networks Gx Interface Specification</i> for more information.

Max-Wait-Time

AVP Name	Max-Wait-Time
Vendor-ID	12951
AVP Code	7103
AVP Data Type	Unsigned32
Description	Indicates the maximum amount of time (in milliseconds) that the originator is willing to wait for a response since the Origination-TimeStamp. This AVP is only provided with the initial authorization request. <i>Supported on the Gx and Gy Interfaces only.</i>

MVNO-Reseller-Id

AVP Name	MVNO-Reseller-Id (affmnvostringid)
Vendor-ID	9
AVP Code	131507
AVP Data Type	UTF8String
Description	This AVP is sent from the PCRF to the PGW/GGSN (via the Gx interface) or from the PGW/GGSN to the OCS (via the Gy interface) and contains the Mobile Virtual Network Operator (MVNO) name or identifier. <i>Supported on the Gx and Gy Interfaces only.</i>

MVNO-Sub-Class-Id

AVP Name	MVNO-Sub-Class-Id (<i>affmvnosubclassid</i>)
Vendor-ID	9
AVP Code	131508
AVP Data Type	UTF8String
Description	This AVP is sent from the PCRF to the PGW/GGSN (via the Gx interface) or from the PGW/GGSN to the OCS (via the Gy interface) and contains the Mobile Virtual Network Operator (MVNO) sub-name or sub-identifier. <i>Supported on the Gx and Gy Interfaces only.</i>

Origination-Time-Stamp

AVP Name	Origination-Time-Stamp
Vendor-ID	12951
AVP Code	7102
AVP Data Type	Unsigned64
Description	Indicates the time that the message was sent, representing the number of seconds since January 1, 1900 00:00 UTC. This AVP is only provided with the initial authorization request. <i>Supported on the Gx and Gy Interfaces only.</i>

Override-Charging-Action-Exclude-Rule

AVP Name	Override-Charging-Action-Exclude-Rule
Vendor-ID	9
AVP Code	132021
AVP Data Type	Octet String
Description	Indicates the name of the service rule to which override control will not be applied. Service rules that are specified in this VSA will still be associated with the rating group based on the workflow configuration. This VSA may be included more than once if more than one rule needs to be excluded.

Override-Charging-Action-Parameters

AVP Name	Override-Charging-Action-Parameters
Vendor-ID	9
AVP Code	132019
AVP Data Type	Grouped
Description	This AVP is sent from the PCRF to the PCEF (via the Gx interface) and is used to override charging parameters for predefined and static rules on the PCEF. The detailed syntax is as follows: <pre> Override-Charging-Action-Parameters ::= < AVP Header: 132019> *[Override-Charging-Action-Exclude-Rule] [Override-Charging-Parameters] </pre>

Override-Charging-Parameters

AVP Name	Override-Charging-Parameters
Vendor-ID	9
AVP Code	132022
AVP Data Type	Grouped
Description	Contains the override charging parameters to be applied to predefined and static rules on the PCEF. The detailed syntax is as follows: <pre> Override-Charging-Parameters ::= < AVP Header: 132022> [Override-Rating-Group] [Override-Online] [Override-Offline] </pre>

Override-Control

AVP Name	Override-Control
Vendor-ID	9
AVP Code	132017
AVP Data Type	Grouped
Description	This AVP is sent from the PCRF to the PCEF (via the Gx interface) and is used to override charging parameters for predefined and static rules on the PCEF. The detailed syntax is as follows: <pre>Override-Control ::= < AVP Header: 132017> [Override-Charging-Action-Parameters] *[Override-Charging-Action-Exclude-Rule] [Override-Charging-Parameters] [Override-Rating-Group] [Override-Online] [Override-Offline]</pre>

Override-Offline

AVP Name	Override-Offline
Vendor-ID	9
AVP Code	132027
AVP Data Type	Enumerated
Description	This VSA overrides the offline value configured for a static/predefined rule. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the offline charging interface for the associated service rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the offline charging interface for the associated service rule is enabled.

Override-Online

AVP Name	Override-Online
Vendor-ID	9
AVP Code	132026
AVP Data Type	Enumerated
Description	This VSA overrides the online value configured for a static/predefined rule. The following values are defined: ■ DISABLE_ONLINE (0) – Indicates that the online charging interface for the associated service rule is disabled. ■ ENABLE_ONLINE (1) – Indicates that the online charging interface for the associated service rule is enabled.

Override-Rating-Group

AVP Name	Override-Rating-Group
Vendor-ID	9
AVP Code	132024
AVP Data Type	Unsigned32
Description	This VSA is used to override the value of the rating group configured for a static/predefined rule. The rating group identifier sent in this VSA must correspond to a predefined rating group quota-id on the MCC (<code>services -> charging -> rating-group -> quota-id</code>).

QoS-Group-Rule-Definition

AVP Name	QoS-Group-Rule-Definition
Vendor-ID	9
AVP Code	132003
AVP Data Type	Grouped
Description	The PCRF uses this VSA to define the parameters for a pre-configured service rule whose name matches the QoS-Group-Rule-Name. For example, if the received QoS-Group-Rule-Definition VSA has a redirect server address, the MCC will apply redirection to the associated flows. <i>Note: The Redirect-Server AVP sent from the PCRF as part of the QoS-Group-Rule-Definition AVP applies to predefined PCC rules only and applies to both online and offline PCC rules.</i> The detailed syntax is as follows: <pre>QoS-Group-Rule-Definition ::= <AVP header: 132003> {QoS-Group-Rule-Name} [Monitoring-Key] [Flow-Status] [Redirect-Server]</pre>

QoS-Group-Rule-Install

AVP Name	QoS-Group-Rule-Install
Vendor-ID	9
AVP Code	132001
AVP Data Type	Grouped
Description	The PCRF uses this VSA to activate predefined service rules on the MCC gateway. The detailed syntax is as follows: <pre>QoS-Group-Rule-Install ::= <AVP header: 132001 > * [QoS-Group-Rule-Definition]</pre>

QoS-Group-Rule-Name

AVP Name	QoS-Group-Rule-Name
Vendor-ID	9
AVP Code	132004
AVP Data Type	OctetString
Description	The MCC tries to match the service rule name, received in the QoS-Group-Rule-Name VSA, to a service rule name configured locally on the MCC gateway (<i>service-construct</i> → <i>service-rule</i>). ■ If a match is found, the MCC activates the matching service rule and all services associated with the received QoS-Rule-Group-Name. ■ If a match is not found, the MCC sends the Charging-Rule-Report AVP with the QoS-Group-Rule-Name VSA to the PCRF.

QoS-Group-Rule-Remove

AVP Name	QoS-Group-Rule-Remove
Vendor-ID	9
AVP Code	132002
AVP Data Type	Grouped
Description	The PCRF uses this VSA to deactivate predefined service rules on the MCC gateway. The detailed syntax is as follows: <pre>QoS-Group-Rule-Remove ::= <AVP header: 132002 > * [QoS-Group-Rule-Name]</pre>

TMO-Transparent-Data

AVP Name	TMO-Transparent-Data
Vendor-ID	29168
AVP Code	1
AVP Data Type	Octet String
Description	Contains information from the TMO-Transparent-Data VSA received from the RADIUS server. The MCC gateway includes this VSA in the CCR message to the OCS when the RADIUS server returns a transparent data value in the Access-Accept message. If the RADIUS Access-Accept message does not return a transparent data value, you can configure the MCC gateway to assign a transparent data value and the MCC gateway will include the configured transparent data value in this VSA to the OCS. For more information on how to configure the MCC gateway to assign a transparent data value, see the <i>Affirmed Networks Gateway Administration Guide</i>.

TMO-Virtual-Gi-ID

AVP Name	TMO-Virtual-Gi-ID
Vendor-ID	29168
AVP Code	15
AVP Data Type	32 Bit Integer
Description	Contains the value of the virtual-ID. See “Processing the TMO-Virtual-Gi-ID VSA” in the <i>Affirmed Networks RADIUS Interface Specification</i> for more information.

Trigger (VSA)

AVP Name	Trigger
Vendor-ID	12951
AVP Code	1264
AVP Data Type	Grouped
Description	Contains the trigger types. The detailed syntax is as follows: <pre>Trigger ::= < AVP Header: 1264 > 1*4 [Trigger-Type]</pre>

Trigger-Type (VSA)

AVP Name	Trigger-Type
Vendor-ID	12951
AVP Code	870
AVP Data Type	Enumerated
Description	When included in the CCA on the Gx interface, the Trigger-Type AVP is used to notify the PGW when a ULI, Time Zone, or PLMN change occurs and the PGW is required to send the respective event triggers over the Gy interface to the OCS. On the Gx interface, the Trigger-Type AVP contains the following values: ■ CHANGE_IN_UE_TIMEZONE (5) – Indicates a change in the UE time zone. ■ CHANGEINLOCATION_MCC (30) – Indicates a change in the MCC of the serving network, which represents the first half of the PLMN. ■ CHANGEINLOCATION_MNC (31) – Indicates a change in the MNC of the serving network, which represents the second half of the PLMN. ■ CHANGE_IN_LOCATION (3) – Indicates a change in the user location. If this Trigger-Type is received, the PGW will send the OCS an updated ULI for all changes, including the PLMN; therefore, this value should not be used in conjunction with the MCC and MNC location change Trigger-Types. Upon receiving the Trigger AVP → Trigger-Type AVP from the PCRF and if the subscriber does not have a Gy session, the PGW sends a CCR-Event message to the OCS via the Gy interface that includes the following AVPs: CC-Request-Type: EVENT_REQUEST (4) Requested-Action: Location Update (4) Service-Information PS-Information 3GPP-MS-Timezone 3GPP-SGSN-MCC-MNC 3GPP-User-Location-Info <i>Note: If the subscriber already has a Gy session, the PGW sends a CCR-U message if a quota request is needed for any of the rating groups.</i>

